DATA SHEET

(DOC No. HX8394-F-DS)

HX8394-F

720RGB x 1280 dots, 16.7M color
TFT Mobile Single Chip Driver

Temporary Version 01 March, 2015

Himax Technologies, Inc.
http://www.himax.com.tw

Revision History

March, 2015

Version	Date	Description of changes
01	2014/09/03	1. New setup.
01	2014/12/02	1. Modify B2h description typo.
01	2014/12/09	1. Modify VPP typical voltage to 8.25V
01.01	2015/01/19	1. Add shottkey diode VSP to VGH for 3 power mode in layout recommendation
01.02	2015/01/30	1. Modify HS minimum spec =0.2us 2. Modify LDOHB voltage level in BAh description
01.03	2015/03/03	1. Modify FRM hardware pin description

Himax Confidential
Do Not Copy

List of Contents

March, 2015

1. GENERAL DESCRIPTION 11
2. FEATURES 12
2.1 Display 12
2.2 Display module 12
2.3 DC/DC booster 13
2.4 Display / Control interface 13
2.5 Input power 13
2.6 Miscellaneous 13
3. DEVICE OVERVIEW 14
3.1 Block diagram 14
3.2 Pin description 15
4. INTERFACE 18
4.1 System interface 18
4.1.1 Serial data transfer interface(MIPI DBI type C) 18
4.1.2 Serial data write mode 19
4.1.3 Serial data read mode 20
4.2 DSI system interface 21
4.2.1 DSI layer definitions 22
4.2.2 Lane states 23
4.2.3 Clock lane mode 24
4.2.4 Data lane mode 26
4.2.5 Escape mode 27
4.2.6 High speed data transmission 29
4.2.7 Bi-directional data lane turnaround 32
4.2.8 DSI protocol 33
4.2.9 Processor to peripheral (forward direction) packets data types 36
4.2.10 Peripheral to processor (reverse direction) packet data type 43
5. FUNCTION DESCRIPTION 45
5.1 Tearing effect output line 45
5.1.1 Tearing effect line timing 47
5.2 Oscillator 48
5.3 Source driver 49
5.3.1 Zig-Zag inversion 50
5.4 LCD power generation scheme 52
5.5 DC/DC converter circuit 53
5.5.1 Charge pump and step up circuit mode 53
5.5.2 Use external VSP,VSN,VGH,VGL circuit 54
5.5.3 Use PFM DC/DC converter 55
5.5.4 Use HX5186-A/B/C 58
5.5.5 Use external VSP,VSN circuit 59
5.6 8-color Idle display mode 60
5.7 1-bit GRAM Idle Display Mode 61
5.8 Gamma characteristic correction function 62
5.8.1 Gamma-characteristics adjustment register 64
5.8.2 Gray voltage generator for digital gamma correction 80
5.9 Characteristics of I/O 82
5.9.1 Output or bi-directional (I/O) pins 82
5.9.2 Input pins 82
5.10 GIP control singal 83

Himax Confidential
Do Not Copy

This information contained herein is the exclusive property of Himax and shall not be distributed, reproduced, or disclosed in whole or in part without prior written permission of Himax.

List of Contents

5.11 Sleep Out –command and self-diagnostic functions of the display module 84
5.11.1Register loading detection 84
5.11.2Functionality detection 85
5.12 Power on/off sequence 86
5.12.1VDD1/VDD3 input power sequence (PCCS[1:0]=”01”, “10”) 87
5.12.2VDD1/VSP/VSN input power sequence (PCCS[1:0]=”11”) 88
5.13 Uncontrolled power off 89
5.14 Content adaptive brightness control (CABC) function 90
5.14.1Module architectures 91
5.14.2CABC block 92
5.14.3Brightness control block 93
5.14.4Minimum brightness setting of CABC function 94
5.15 Temperature sensor 95
5.16 OTP programming 96
5.16.1OTP table 96
5.16.2OTP programming flow 108
5.16.3Programming sequence 109
5.16.4OTP programming example of VCOM setting VCMC 110
5.16.5OTP programming example of ID1, ID2 and ID3 111
5.16.6OTP read example of 0x00h (ID1) 112
5.17 FRM (Free running mode) 113
5.18 Command list 114
5.18.1Standard command 114
5.18.2User define command list table 116
5.18.3NOP (00h) 127
5.18.4Software reset (01h) 128
5.18.5Read Display Identification Information (04h) 129
5.18.6RDNUMPE: Read number of the parity errors (05h) 130
5.18.7Get_red_channel (06h) 131
5.18.8Get_green_channel (07h) 132
5.18.9Get_blue_channel (08h) 133
5.18.10Read Display Status (09h) 134
5.18.11Get_power_mode (0Ah) 138
5.18.12Read display MADCTL (0Bh) 140
5.18.13Get_pixel_format (0Ch) 142
5.18.14Get_display_mode (0Dh) 144
5.18.15Get_signal_mode (0Eh) 146
5.18.16Get_diagnostic_result (0Fh) 147
5.18.17Enter_sleep_mode (10h) 148
5.18.18Exit_sleep_omde (11h) 149
5.18.19Enter_normal_mode (13h) 150
5.18.20Exit_inversion_mode (20h) 151
5.18.21Enter_inversion_mode (21h) 152
5.18.22All_Pixel_Off (22h) 153
5.18.23All_Pixel_On (23h) 154
5.18.24Set_gamma_curve (26h) 155
5.18.25Set_display_off (28h) 156
5.18.26Set_display_on (29h) 157
5.18.27Write_memory_start (2Ch) 158
5.18.28Tearing effect line off (34h) 159
5.18.29Set_tear_on (35h) 160
5.18.30Set_address_mode (36h) 161
5.18.31Idle mode off (38h) 163
5.18.32Enter_Idle_mode (39h) 164
5.18.33Set_pixel_format (3Ah) 165

Himax Confidential
Do Not Copy

5.18.34Write_memory_contiune (3Ch) 166
5.18.35Set tear scan lines (44h) 167
5.18.36Get the current scanline(45h) 168
5.18.37Write display brightness (51h) 169
5.18.38Read display brightness value (52h) 170
5.18.39Write CTRL display (53h) 171
5.18.40Read CTRL value display (54h) 172
5.18.41Write content adaptive brightness control (55h) 173
5.18.42Read content adaptive brightness control (56h) 174
5.18.43Write CABC minimum brightness (5Eh) 175
5.18.44Read CABC minimum brightness (5Fh) 176
5.18.45Read automatic brightness control self-diagnostic result (68h) 177
5.18.46Write Idle Mode Color(80h) 178
5.18.47178
5.18.48Read Idle Mode Color(81h) 179
5.18.49Read_DDB_start (A1h) 180
5.18.50Read_DDB_continue (A8h) 182
5.18.51Read ID1 (DAh) 183
5.18.52Read ID2 (DBh) 184
5.18.53Read ID3 (DCh) 185
5.19 User define command description 186
5.19.1SETSEQUENCE: Set sequence (B0h) 186
5.19.2SETPOWER: Set power (B1h) 188
5.19.3SETDISP: Set display related register (B2h) 197
5.19.4SETCYC: Set display waveform cycles (B4h) 200
5.19.5SETVCOM: Set VCOM voltage (B6h) 205
5.19.6SETTE: Set internal TE function (B7h) 206
5.19.7SETSENSOR: Set temperature sensor (B8h) 208
5.19.8SETEXTC: Set extension command (B9h) 209
5.19.9SETMIPI: Set MIPI control (BAh) 210
5.19.10SETOTP: Set OTP (BBh) 212
5.19.11Set register bank (BDh) 214
5.19.12SETDGCLUT: Set DGC LUT (C1h) 215
5.19.13SETID: Set ID (C3h) 219
5.19.14SETDDB: Set DDB (C4h) 220
5.19.15SETCABC: Set CABC control (C9h) 221
5.19.16SETCABCGAIN (CAh) 223
5.19.17SETPANEL (CCh) 224
5.19.18SETOFFSET (D2h) 225
5.19.19SETGIP0: Set GIP Option0 (D3h) 226
5.19.20Set GIP Option1 (D5h) 231
5.19.21Set GIP Option2 (D6h) 234
5.19.22SETGPO (D9h) 238
5.19.23SETSCALING (DDh) 239
5.19.24SETIDLE (DFh) 240
5.19.25SETGAMMA: Set gamma curve related setting (E0h) 243
5.19.26SETCHEMODE_DYN (E4h) 245
5.19.27SETCHE (E5h) 246
5.19.28SETCESEL: Enable color enhance (E6h) 248
5.19.29SET_SP_CMD (E9h) 250
5.19.30SETREADINDEX: Set SPI Read Index (FEh) 251
5.19.31GETSPIREAD: SPI Read Command Data (FFh) 252

6. LAYOUT RECOMMENDATION 253

6.1 Maximum layout resistance 253

Himax Confidential
Do Not Copy

List of Contents

6.2 External components connection 254

7. ELECTRICAL CHARACTERISTICS 260

7.1 Absolute maximum ratings 260
7.2 DC characteristics 261
7.3 AC characteristics 262
7.3.1 DBI Type C interface characteristics 262
7.3.2 DSI D-PHY electrical characteristics 263
7.3.3 Timings for DSI video mode 271

8. ORDERING INFORMATION 273

Himax Confidential
Do Not Copy

List of Figures

Figure 4.1: Serial data stream, write mode 19
Figure 4.2: Serial data stream, write mode 19
Figure 4.3: Type C:Serial interface protocol 3-wire/4-wire read mode 20
Figure 4.4: Display module data transfer recovery 20
Figure 4.5: DSI transmitter and receiver interface 21
Figure 4.6: DSI transmitter and receiver interface 22
Figure 4.7: Clock lane mode state diagram 24
Figure 4.8: Data lane mode state diagram 26
Figure 4.9: Escape mode state machine 27
Figure 4.10: Escape mode timing sequence 28
Figure 4.11: High speed data transmission state machine 29
Figure 4.12: High speed data transmission timing sequence 29
Figure 4.13: Switching the Clock Lane between Clock Transmission and LP Mode 30
Figure 4.14: Turnaround state machine 32
Figure 4.15: Turnaround timing sequence 32
Figure 4.16: Multiple packets transmission 33
Figure 4.17: Structure of the short packet 33
Figure 4.18: Structure of the long packet 34
Figure 4.19: The format of data ID 34
Figure 4.20: Show short- / long-packet transmission command sequence 35
Figure 5.1: Tearing effect output signal mode 1 45
Figure 5.2: TE delay output 45
Figure 5.3: Tearing Effect Output signal mode 2 46
Figure 5.4: TE Output for TELINE setting 46
Figure 5.5: Tearing Effect Output signal 46
Figure 5.6: Tearing effect output line –tearing effect line timing 47
Figure 5.7: Tearing effect output line–definition of tf, tr 47
Figure 5.8: OSC architecture 48
Figure 5.9: Zig-Zag inversion mode(ZZ_2PL=0) 50
Figure 5.10: Zig-Zag2 Inversion mode(ZZ_2PL=1) 51
Figure 5.11: LCD power generation scheme for HX5186 and PFM mode 52
Figure 5.12: LCD power generation scheme for external power mode 52
Figure 5.17: DC/DC converter circuit of external VSP/VSN/VGH/VGL 54
Figure 5.13: DC/DC converter circuit (PFM type D) 55
Figure 5.14: DC/DC converter circuit (PFM type A) 56
Figure 5.15: DC/DC converter circuit (PFM type C) 57
Figure 5.16: DC/DC converter circuit of HX5186-A/B/C 58
Figure 5.17: DC/DC converter circuit of external VSP/VSN 59
Figure 5.18: Idle mode grayscale control 60
Figure 5.19: Idle Mode On/Off Sequence 61
Figure 5.20: Gamma adjustments different of source driver with digital gamma correction 62
Figure 5.21: Grayscale control 63
Figure 5.22: Gamma resister stream and gamma reference voltage 65
Figure 5.23: Block diagram of digital gamma correction 80
Figure 5.24: Sleep out flow chart–command and self-diagnostic functions 84
Figure 5.25: Sleep out flow chart internal function detection 85
Figure 5.26: Power on/off sequence 86
Figure 5.27: VDD1/VDD3 input power on sequence 87
Figure 5.28: VDD1/VDD3 input power off sequence 87
Figure 5.29: VDD1/ VSP/VSN input power on sequence 88
Figure 5.30: VDD1/VSP/VSN input power off sequence 88
Figure 5.31: CABC block diagram 90
Figure 5.32: Module architecture 91
Figure 5.33: CABC gain / CABC duty generation 92
Figure 5.34: CABC_PWM_OUT output duty 93

Himax Confidential
Himax Confidential
Do Not Copy

List of Figures

Figure 5.35: Tempeture sensor diagram 95
Figure 5.36: OTP programming sequence 108
Figure 5.37: OTP programming sequence example 1 110
Figure 5.38: OTP programming sequence example 2 111
Figure 5.39: OTP read sequence flow of index 0x00h 112
Figure 5.40: FRM power on sequence 113
Figure 5.41: Display pattern sequence example in FRM 113
Figure 7.1: DBI Type C interface characteristics 262
Figure 7.2: Electrical functions of a fully D-PHY transceiver 263
Figure 7.3: Shows both the HS and LP signal levels 263
Figure 7.4: Differential HS signals for HS receiver 265
Figure 7.5: Input glitch rejections of low-power receivers 266
Figure 7.6: DDR clock definition 268
Figure 7.7: Data to clock timing definitions 269
Figure 7.8: Skew window of transmittor and receiver 270
Figure 7.9: Vertical timings for DSI I/F 271
Figure 7.10: Horizontal timing for DSI video mode I/F 272

Himax Confidential
Do Not Copy

This information contained herein is the exclusive property of Himax and shall not be distributed, reproduced, or disclosed in whole or in part without prior written permission of Himax.

List of Tables

Table 4.1: Interface selection 18
Table 4.2: Global Operation Timing Parameters for Data Lane 30
Table 4.3: Global Operation Timing Parameters for Clock Lane 31
Table 4.4: Data types for processor-sourced packets 36
Table 4.5: Shows the error report bit definitions 44
Table 4.6: The complete set of peripheral-to-processor data types 44
Table 5.1: AC characteristics of tearing effect signal 47
Table 5.2: Source output for Panel resolution 49
Table 5.3: Power mode setting 53
Table 5.4: Gamma-adjustment registers 64
Table 5.5: Edge adjustment 0~7, 21~28 66
Table 5.6: Center adjustment 8~20 66
Table 5.7: VinP/N0 67
Table 5.8: VinP/N1 67
Table 5.9: VinP/N3 67
Table 5.10: VinP/N5 68
Table 5.11: VinP/N7 68
Table 5.12: VinP/N9 68
Table 5.13: VinP/N12 69
Table 5.14: VinP/N15 69
Table 5.15: VinP/N20 69
Table 5.16: VinP/N28 70
Table 5.17: VinP/N40 70
Table 5.18: VinP/N52 70
Table 5.19: VinP/N76 71
Table 5.20: VinP/N100 71
Table 5.21: VinP/N128 71
Table 5.22: VinP/N156 72
Table 5.23: VinP/N180 72
Table 5.24: VinP/N204 72
Table 5.25: VinP/N216 73
Table 5.26: VinP/N228 73
Table 5.27: VinP/N236 73
Table 5.28: VinP/N240 74
Table 5.29: VinP/N243 74
Table 5.30: VinP/N246 74
Table 5.31: VinP/N248 75
Table 5.32: VinP/N250 75
Table 5.33: VinP/N252 75
Table 5.34: VinP/N254 76
Table 5.35: VinP/N255 76
Table 5.36: Voltage calculation formula of 256-grayscale voltage (positive/negative polarity) 79
Table 5.37: DGC Look-up table 81
Table 5.38: Characteristics of output or bi-directional (I/O) pins 82
Table 5.39: Characteristics of input pins 82
Table 5.40: CABC timing table 93
Table 5.41: OTP table 107
Table 5.42: OTP programming sequence 109
Table 6.1: Maximum layout resistance 253
Table 6.2: HX5186-A/B/C mode external components 254
Table 6.3: PFM type A external components 255
Table 6.4: PFM type C external components 256
Table 6.5: PFM type D external components 257
Table 6.6: External VDD1 / VSP / VSN / VGH / VGL external components 258
Table 6.7: External VDD1 / VSP / VSN external components 259

Himax Confidential
Do Not Copy

This information contained herein is the exclusive property of Himax and shall not be distributed, reproduced, or disclosed in whole or in part without prior written permission of Himax.

List of Tables

Table 7.1: Absolute maximum rating 260
Table 7.2: DC characteristic 261
Table 7.3: DBI Type C interface characteristics 262
Table 7.4: LP transmitter DC specifications 264
Table 7.5: LP transmitter AC specifications 264
Table 7.6: HS receiver DC Specifications 265
Table 7.7: HS receiver AC Specifications 265
Table 7.8: LP receiver DC specifications 266
Table 7.9: LP receiver AC specifications 266
Table 7.10: Contention detector DC specifications 267
Table 7.11: Reverse HS data transmission timing parameters 269
Table 7.12: Data to clock timing specifications 270
Table 7.13: Vertical timings for DSI I/F 271
Table 7.14: Horizontal timings for DSI video mode I/F 272

Himax Confidential
Do Not Copy

This information contained herein is the exclusive property of Himax and shall not be distributed, reproduced, or disclosed in whole or in part without prior written permission of Himax.

1. General Description

This document describes Himax's HX8394-F supports WXGA resolution driving controller. The HX8394-F is designed to provide a single-chip solution that combines source driver control, gate driver control and power supply circuit to drive a TFT dot matrix LCD with 720RGB x 1280 dots at maximum.

The HX8394-F can be operated in low-voltage condition for the interface and integrated internal boosters that produce the liquid crystal voltage, breeder resistance and the voltage follower circuit for liquid crystal driver. In addition, The HX8394-F also supports various functions to reduce the power consumption of a LCD system via software control.

The HX8394-F supports several interface modes, including MIPI DBI Type C interface mode and MIPI DSI (Display Serial Interface) interface mode.

The HX8394-F is suitable for any small portable battery-driven and long-term driving products, such as cellular phones, tablet and other mobile devices.

Himax Confidential
Do Not Copy

This information contained herein is the exclusive property of Himax and shall not be distributed, reproduced, or disclosed in whole or in part without prior written permission of Himax.

2. Features

2.1 Display

- Single chip solution for a WXGA type TFT LCD display
- Resolution:
  - 720RGB x (480 + (8 x NL))
  - 640RGB x (480 + (8 x NL))
  - 600RGB x (480 + (8 x NL))
- Display color modes
  - Full color mode:
    - 16.7M colours **(24-bit 8(R):8(G):8(B))**
  - Reduce color mode:
    - 262k colours **(18-bit 6(R):6(G):6(B))**
    - 65k colours **(16-bit 5(R):6(G):5(B))**
    - 8 colors **(Idle mode on)**: 8 colors **(3-bit binary mode)**

2.2 Display module

- Support 2162 source channel outputs
- Gate driver control signals for GIP
- Support 1-dot / 2-dot / 4-dot / 8-dot / Zig-Zag / Zig-Zag2/Column inversion
- Output voltage level
  - Positive source output voltage level: VSPR to VSSA
  - Negative source output voltage level: VSNR to VSSA
  - Positive gate driver output voltage level: VGH to VSSA
  - Negative gate driver output voltage level: VGL to VSSA
  - VCOM= -4V ~ -0.3V, each step=10mV

Himax Confidential
Do Not Copy

This information contained herein is the exclusive property of Himax and shall not be distributed, reproduced, or disclosed in whole or in part without prior written permission of Himax.

2.3 DC/DC booster

- External charge pump: support HX5186-C mode
- PFM booster: support type A/C/D
- External power mode:
  1. VSP+VSN+VDD1
  2. VSP+VSN+VGH+VGL+VDD1

2.4 Display / Control interface

- Display interface types supported
  - MIPI-DSI **(Display Serial Interface)** interface
    - Support DSI Version 1.1
    - Support D-PHY version 1.1

2.5 Input power

- Logic and interface power supply **(VDD1)**: 1.65V to 3.6V
- Analog power supply **(VDD3)**: 2.5V to 3.6V
  - For PCCS[1:0]=10, VDD3 range is 2.5V ~ 3.6V
  - For PCCS[1:0]=00, 01 and 11, no VDD3 is needed
- Positive source driver power **(External input mode VSP)**: 4.3V to 6.5V
- Negative source driver power **(External input mode VSN)**: -4.3V to -6.5V
- High speed interface power supply **(HS_VCC)**: 1.65V to 3.6V
- OTP programming voltage **(VPP)**: 8.0~8.5V
- VGH,VGL: VGH-VGL<32V

2.6 Miscellaneous

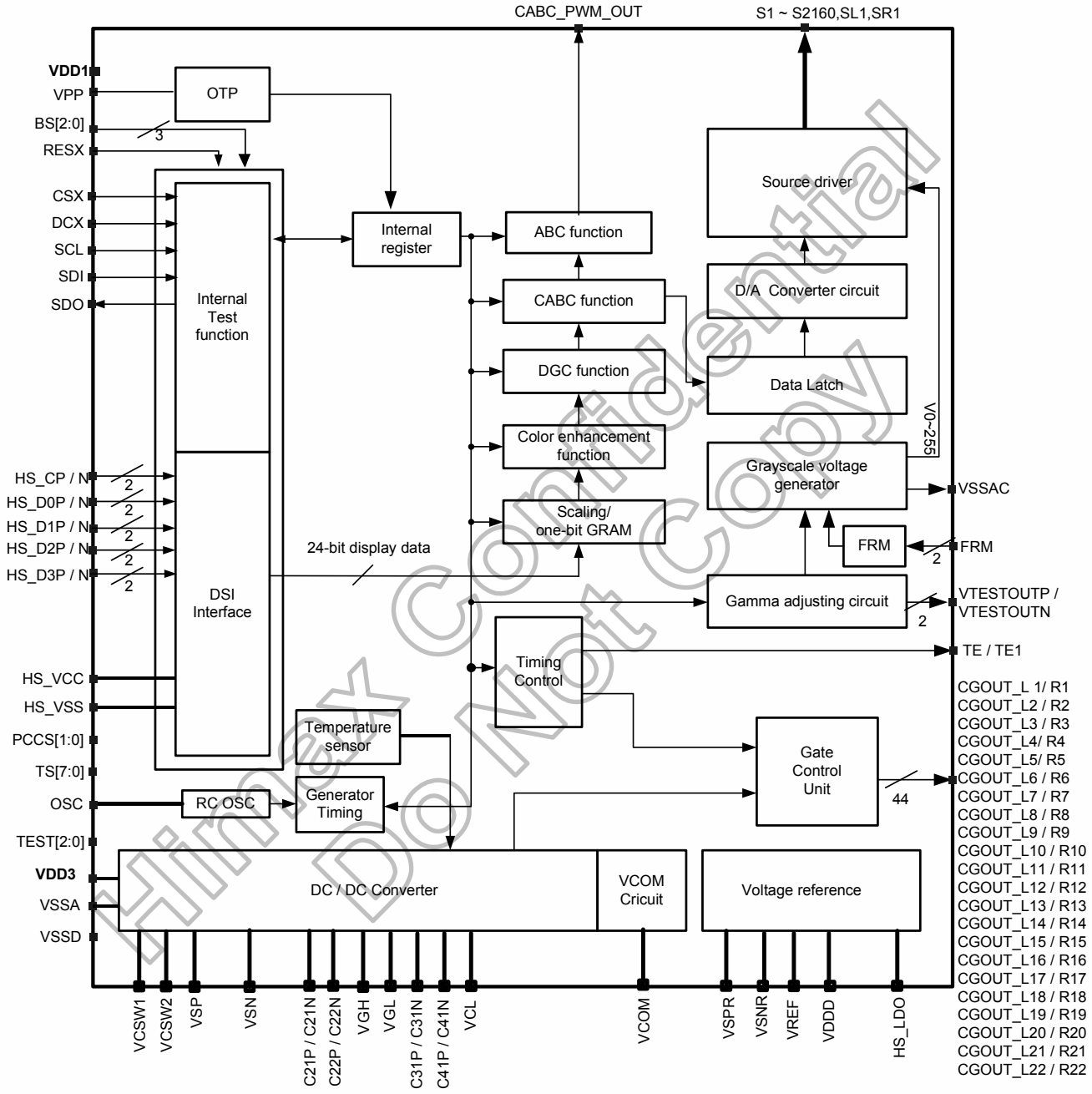
- Software programmable color depth mode
- Oscillator for display clock generation
- Low power consumption, suitable for battery operated systems
- Smart source EQ for power saving
- GAS function for preventing image sticking when abnormal power off
- Optimized layout for COG assembly
- Operation temperature range: -40 to +85 ℃
- Support DC COM driving
- On-chip OTP program voltage generator
- Support CABC (Content Adaptive Brightness Control) function
- Support DGC (Digital Gamma Correction) function
- Support CE (Color Enhancement) function
- Support Scaling function
  - FWVGA(480RGBx854) to HD(720RGBx1280).
  - qHD(540RGBx960) to HD(720RGBx1280).
- Support Temperature sensor function
- Build in Free Funning Mode function(Internal BIST pattern generator)
- Idle mode 1-bit GRAM
- OTP memory to store initialization register settings
- OTP memory to store 2 sets of Gamma and DGC setting
- 3 times MTP for VCOM setting, ID setting

Himax Confidential
Do Not Copy

This information contained herein is the exclusive property of Himax and shall not be distributed, reproduced, or disclosed in whole or in part without prior written permission of Himax.

3. Device Overview

3.1 Block diagram



Himax Confidential
Do Not Copy

3.2 Pin description

<table>
<tr><th colspan="4">Host interface pins</th></tr>
<tr><th>Signals</th><th>I/O</th><th>Pin no.</th><th>Connected</th><th>Description</th></tr>
<tr><td>BS2 ~ BS0</td><td>I</td><td>6</td><td>VSSD / VDD1</td><td>These pins must be connected to VDD1(logic=1) or VSSD(logic=0).
BS2 ~ BS0 are used for the combination of polarity swap and data lane swap of DSI.
<table>
<tr><th>BS[2:0]</th><th>HS_D0N</th><th>HS_D0P</th><th>HS_D1N</th><th>HS_D1P</th><th>HS_CN</th><th>HS_CP</th><th>HS_D2N</th><th>HS_D2P</th><th>HS_D3N</th><th>HS_D3P</th></tr>
<tr><td>“000”</td><td>D3N</td><td>D3P</td><td>D2N</td><td>D2P</td><td>CN</td><td>CP</td><td>D1N</td><td>D1P</td><td>D0N</td><td>D0P</td></tr>
<tr><td>“001”</td><td>D3P</td><td>D3N</td><td>D2P</td><td>D2N</td><td>CP</td><td>CN</td><td>D1P</td><td>D1N</td><td>D0P</td><td>D0N</td></tr>
<tr><td>”010”</td><td>D0N</td><td>D0P</td><td>D1N</td><td>D1P</td><td>CN</td><td>CP</td><td>D2N</td><td>D2P</td><td>D3N</td><td>D3P</td></tr>
<tr><td>”011”</td><td>D0P</td><td>D0N</td><td>D1P</td><td>D1N</td><td>CP</td><td>CN</td><td>D2P</td><td>D2N</td><td>D3P</td><td>D3N</td></tr>
<tr><td>”100”</td><td>D2N</td><td>D2P</td><td>D1N</td><td>D1P</td><td>CN</td><td>CP</td><td>D0N</td><td>D0P</td><td>D3N</td><td>D3P</td></tr>
<tr><td>”101”</td><td>D2P</td><td>D2N</td><td>D1P</td><td>D1N</td><td>CP</td><td>CN</td><td>D0P</td><td>D0N</td><td>D3P</td><td>D3N</td></tr>
<tr><td>”110”</td><td>D3N</td><td>D3P</td><td>D0N</td><td>D0P</td><td>CN</td><td>CP</td><td>D1N</td><td>D1P</td><td>D2N</td><td>D2P</td></tr>
<tr><td>”111”</td><td>D3P</td><td>D3N</td><td>D0P</td><td>D0N</td><td>CP</td><td>CN</td><td>D1P</td><td>D1N</td><td>D2P</td><td>D2N</td></tr>
</table></td></tr>
<tr><td>CSX</td><td>I</td><td>2</td><td>MPU</td><td>Chip select signal. For test only.
Low: Chip can be accessed;
High: Chip cannot be accessed.
If this pin is not used, connect it to VDD1 or VSSD .</td></tr>
<tr><td>DCX</td><td>I</td><td>2</td><td>MPU</td><td>Data / Command Selection pin. For test only.
If this pin is not used, connect it to VDD1 or VSSD .</td></tr>
<tr><td>RESX</td><td>I</td><td>4</td><td>MPU or reset circuit</td><td>Reset pin. Setting either pin low initializes the LSI. It must be reset after power is supplied.</td></tr>
<tr><td>SCL</td><td>I</td><td>2</td><td>MPU</td><td>Serial Clock. For test only.
If this pin is not used, connect it to VDD1 or VSSD .</td></tr>
<tr><td>SDI</td><td>I</td><td>2</td><td>MPU</td><td>Serial data input pin. If not used, let it to open. For test only.</td></tr>
<tr><td>SDO</td><td>O</td><td>2</td><td>MPU</td><td>Serial data output pin. If not used, let it to open. For test only.</td></tr>
<tr><td>HSYNC</td><td>I</td><td>2</td><td>MPU</td><td>Line synchronizing signal. For test only.
If this pin is not used, connect it to VDD1 or VSSD .</td></tr>
<tr><td>DE</td><td>I</td><td>2</td><td>MPU</td><td>Data enable signal. For test only.
If this pin is not used, connect it to VDD1 or VSSD.</td></tr>
<tr><td>VSYNC</td><td>I</td><td>2</td><td>MPU</td><td>Frame synchronizing signal. For test only.
If this pin is not used, connect it to VDD1 or VSSD .</td></tr>
<tr><td>PCLK</td><td>I</td><td>2</td><td>MPU</td><td>Dot clock signal. For test only.
If this pin is not used, connect it to VDD1 or VSSD .</td></tr>
<tr><td>DB23~0</td><td>I</td><td>24</td><td>MPU</td><td>Let unused pins open for each mode.</td></tr>
<tr><th colspan="4">Source driver output pins</th></tr>
<tr><td>S1 to S2160,
SL1, SR1</td><td>O</td><td>2162</td><td>LCD</td><td>Output voltages applied to the liquid crystal.
<table>
<tr><th rowspan="2">Panel Resolution</th><th colspan="2">Source channels</th></tr>
<tr><th>1 / 2 / 4 / 8 dot,
Column inversion</th><th>Zig-Zag inversion</th></tr>
<tr><td>720RGB</td><td>S1~S2160</td><td>S1~S2160,SL1,SR1</td></tr>
<tr><td>640RGB</td><td>S1~S960, S1201~S2160</td><td>S1~S960,
S1201~S2160,SL1,SR1</td></tr>
<tr><td>600RGB</td><td>S1~S900, S1261~S2160</td><td>S1~S900,
S1261~S2160,SL1,SR1</td></tr>
</table></td></tr>
<tr><td>TE</td><td>O</td><td>6</td><td>MPU</td><td>Tearing Effect pin.</td></tr>
<tr><td>TE1</td><td>O</td><td>6</td><td>MPU</td><td>Tearing Effect pin of each scan line.</td></tr>
<tr><th colspan="4">GIP control singal and bias voltage</th></tr>
<tr><td>CGOUT_L1~22</td><td>O</td><td>22</td><td>LCD</td><td>Signals for left side GIP on panel view. (Left side in IC bump view)
Unused pins should be left open.</td></tr>
<tr><td>CGOUT_R1~22</td><td>O</td><td>22</td><td>LCD</td><td>Signals for right side GIP on panel view. (Right side in IC bump view)
Unused pins should be left open.</td></tr>
<tr><th colspan="4">Power supply pins</th></tr>
<tr><td>VDD1</td><td>I</td><td>31</td><td>Power supply</td><td>A power supply for the logic power and I/O circuit. VDD1 range is 1.65 to 3.6V. It must be connected to a stabilizing capacitor 2.2uF to VSSA.</td></tr>
<tr><td>VDD3</td><td>I</td><td>19</td><td>Power supply</td><td>A power supply for the analog power.
It must be connected to a stabilizing capacitor 2.2uF to VSSA.
PCCS[1:0]=10, VDD3 range is 2.5V ~ 3.6V.</td></tr>
</table>

Himax Confidential Do Not Copy

				PCCS[1:0]=00, 01 and 11, VDD3 range is 2.5V ~6.5V.
VSSA	P	16	Power supply	Analoge ground. VSSA=0V. When using the COG method, connect to VSSD on the FPC to prevent noise.
VSSAC	P	4	Power supply	Analoge ground. It must be connected to VSSA on the FPC.
VSSD	P	46	Power supply	Ground for the internal logic. VSSD=0V. When using the COG method, connect to VSSA on the FPC to prevent noise.
VPP	I	6	Power supply	External high voltage pin used in OTP mode and operates at 8.25V. If not used, let it open.
VSP	I	7	Stabilizing capacitor	Input voltage generated from HX5186-C, PFM or external power IC. **VSP range is 4.3V to 6.5V.** It must be connected to a stabilizing capactor between VSP and VSSA.
VSN	I	7	Stabilizing capacitor	Input voltage generated from HX5186-C, PFM or external power IC. **VSP range is -4.3V to -6.5V.** It must be connected to a stabilizing capactor between VSN and VSSA.
Output pins of power and reference voltage				
PCCS1 ~ PCCS0	I	4	VSSD / VDD1	Select the VSP/VSN pumping method as listed below: PCCS[1:0] | Power mode | Input Power | VSP | VSN | VGH/VGL 00 | External VSP+VSN+VGH+VGL | VDD1 | External | External | External 01 | PFM-D (External VSP) | VDD1 | External | Internal PFM | Internal DC/DC 10 | HX5186 or PFM-A or PFM-C | VDD3 VDD1 | Internal PFM or HX5186 | Internal PFM or HX5186 | Internal DC/DC 11 | External VSP+VSN | VDD1 | External | External | Internal DC/DC PCCS[1:0] must be connected to VSSD (logic=0) or VDD1(logic=1). **Note:** For External Power Mode(PCCS[1:0]=00,01 and 11), VDD3 is not needed. Please let VDD3 open or connect it to VSP.
VCL	I	5	Stabilizing capacitor	Output voltage from the set-up circuit. **(-VDD3 or -2.5~-3.1V)** It is generated from VDD3. It must be connected a stabilizing capacitor 1.0uF to VSSA.
VSPR	O	3	Stabilizing capacitor	Positive regulated voltage output for Gamma. **(3.1V to VSP - 0.3)**
VSNR	O	3	Stabilizing capacitor	Negative regulated voltage output for Gamma. **(-3.1V to VSN + 0.3)**
VDDD	O	25	Stabilizing capacitor	Internal logic voltage output. It must be connected a stabilizing capacitor 2.2uF to VSSA.
VGH	O	6	Stabilizing capacitor	Output voltage from the step-up circuit, it is generated from VSP and VSN. Connect to a stabilizing capacitor 1uF between VSSA and VGH.
VGL	O	13	Stabilizing capacitor	Output voltage from the step-up circuit, it is generated from VSP and VSN. Connect to a stabilizing capacitor 1uF between VSSA and VGL. Place a schottkey barrier diode between VSSA and VGL.
VCOM	O	6	Stabilizing capacitor	The power supply of common voltage in DC com driving. The voltage range is -4V to -0.3V. It must be connected to a stabilizing capacitor 2.2uF to VSSA.
VREF	O	3	Stabilizing capacitor	Internal reference voltage.
DC/DC pumping				
C21P, C21N C22P, C22N	I/O	14 14	Step-up Capacitor	Connect to the step-up capacitors according to the DC/DC pumping factor for pumping VGH voltage.
C31P, C31N	I/O	14	Step-up Capacitor	Connect to the step-up capacitors according to the DC/DC pumping factor for pumping tVGL voltage.
C41P, C41N	I/O	14	Step-up Capacitor	Connect to the step-up capacitors according to the DC/DC pumping factor for pumping the VCL voltage.
VCSW1, VCSW2	I	16	-	VCSW1 and VCSW2 should be connected to HX5186-C for HX5186-C mode.
CABC & ABC				
CABC_PWM_OUT	O	4	Led driver	Backlight on/off control pin. If CABC function is used, the pin can be

Himax Confidential
This information contained herein is the exclusive property of Himax and shall not be distributed, reproduced, or disclosed in whole or in part without prior written permission of Himax.

				connected to external LED driver IC. The output voltage level is 0 or VDD1.
High speed interface parts				
HS_D0P, HS_D0N	I/O	6/6	DSI Host	MIPI-DSI Data differential signal input pins. **(Data lane 0)** if not used , Please connected to VSSD or open.
HS_CP, HS_CN	I	6/6	DSI Host	MIPI-DSI CLOCK differential signal input pins. If not used , Please connected to VSSD or open.
HS_D1P, HS_D1N	I	6/6	DSI Host	MIPI-DSI Data differential signal input pins. **(Data lane 1)** if not used , Please connected to VSSD or open.
HS_D2P, HS_D2N	I	6/6	DSI Host	MIPI-DSI Data differential signal input pins. **(Data lane 2)** if not used , Please connected to VSSD or open.
HS_D3P, HS_D3N	I	6/6	DSI Host	MIPI-DSI Data differential signal input pins. **(Data lane 3)** if not used , Please connected to VSSD or open.
HS_VCC	P	12	Power Supply	Power supply for the MIPI DSI analog power. (1.65V to 3.6V)
HS_VSS	P	17	Ground	MIPI DSI analogy ground. VSS=0V. When using the COG method, connect to VSSA on the FPC to prevent noise.
HS_LDO	O	12	Capacitor	DSI regulator output pin. **(1.5V)** Connect to a stabilizing capacitor 1uF between VSS and LDO. If not used, please open these pins.
Test pins				
OSC	I	2	Open	Oscillator input for test purpose. If not used, please let it open or connected to VDD1. **(Weak pull high)**.
TEST[2:0]	I	6	Open	A test pin. This pin is for internal logic function test. It can be output on FPC. If not used, connect it to VSSD. **(Weak pull low)**.
TS0~7	O	8	Open	A test pin. This pin is for internal logic function test. It can be output on FPC.
VTESTOUTP	O	2	Open	A test pin. This pin will output Gamma voltage. It can be output on FPC.
VTESTOUTN	O	2	Open	A test pin. This pin will output Gamma voltage. It can be output on FPC.
FRM	I	2	Open	A test pin. This pin for by internal logic function test. It can be output on FPC. If not used, please connect it to VSSD. **(Weak pull low)**. This pin is used for Himax engineering mode.
DUMMYR1 DUMMYR2 DUMMYR3 DUMMYR4	-	16	Open	Dummy pads. Available for measuring the COG contact resistance. They are short-circuited within the chip.
DUMMY	-	397	Open	Dummy pads(Not used). Let it open.

FRM	Free Running Mode
Low	Disable
High	Enable

Himax Confidential
Do Not Copy

4. Interface

4.1 System interface

The HX8394-F supports MIPI DSI **(Display Serial Interface)** interface. BS2~BS0 are used for the combination of polarity swap and data lane swap of DSI.

BS[2:0]	HS_D0N	HS_D0P	HS_D1N	HS_D1P	HS_CN	HS_CP	HS_D2N	HS_D2P	HS_D3N	HS_D3P
“000”	D3N	D3P	D2N	D2P	CN	CP	D1N	D1P	D0N	D0P
“001”	D3P	D3N	D2P	D2N	CP	CN	C1P	C1N	D0P	D0N
”010”	D0N	D0P	D1N	D1P	CN	CP	D2N	D2P	D3N	D3P
”011”	D0P	D0N	D1P	D1N	CP	CN	D2P	D2N	D3P	D3N
”100”	D2N	D2P	D1N	D1P	CN	CP	D0N	D0P	D3N	D3P
”101”	D2P	D2N	D1P	D1N	CP	CN	D0P	D0N	D3P	D3N
”110”	D3N	D3P	D0N	D0P	CN	CP	D1N	D1P	D2N	D2P
”111”	D3P	D3N	D0P	D0N	CP	CN	D1P	D1N	D2P	D2N

Table 4.1: Interface selection

4.1.1 Serial data transfer interface(MIPI DBI type C)

The HX8394-F supports DBI Type C option 1, 3-wire serial bus to access command set under DSI I/F. The 3-wire serial bus is use: chip select line **(CSX)**, serial input data **(SDI)**,serial output data **(SDO)** and serial transfer clock line **(SCL)**.

Himax Confidential
Do Not Copy

4.1.2 Serial data write mode

The 3-wire serial data packet contains a control bit D/CX and a transmission byte. If DCX is low, the transmission byte is command byte. If D/CX is high, the transmission byte is stored in to command register. The MSB is transmitted first. The serial interface is initialized when CSX is high. In this state, SCL clock pulse or serial input/output data **(SDI and SDO)** have no effect. A falling edge on CSX enables the serial interface and indicates the start of data transmission.

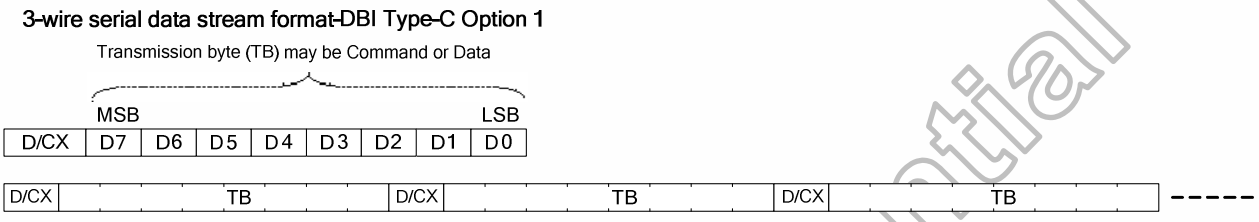


Figure 4.1: Serial data stream, write mode

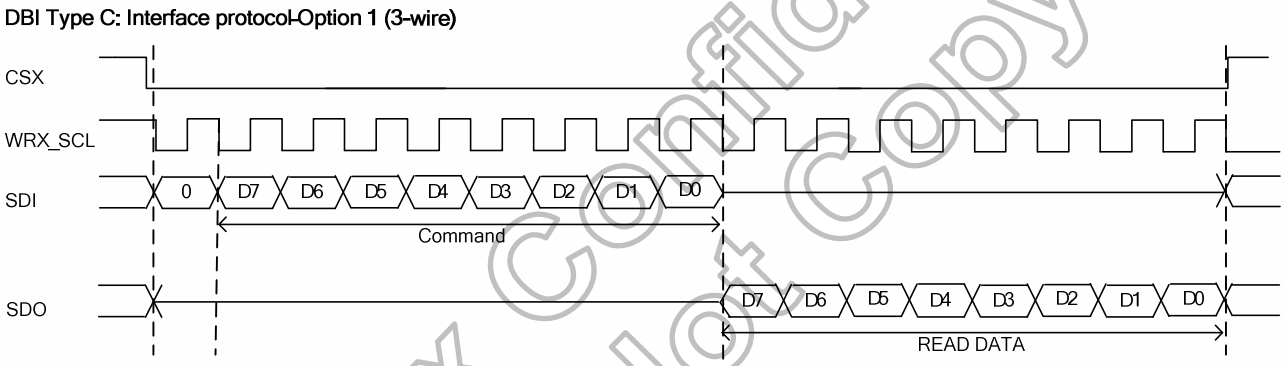


Figure 4.2: Serial data stream, write mode

Himax Confidential
Do Not Copy

This information contained herein is the exclusive property of Himax and shall not be distributed, reproduced, or disclosed in whole or in part without prior written permission of Himax.

4.1.3 Serial data read mode

The micro-controller first has to send a command and then the following byte is transmitted in the opposite direction. The 3-wire serial read data format which just needs 8-bit.

DBI Type C: Interface protocol-Option 1 (3-wire)

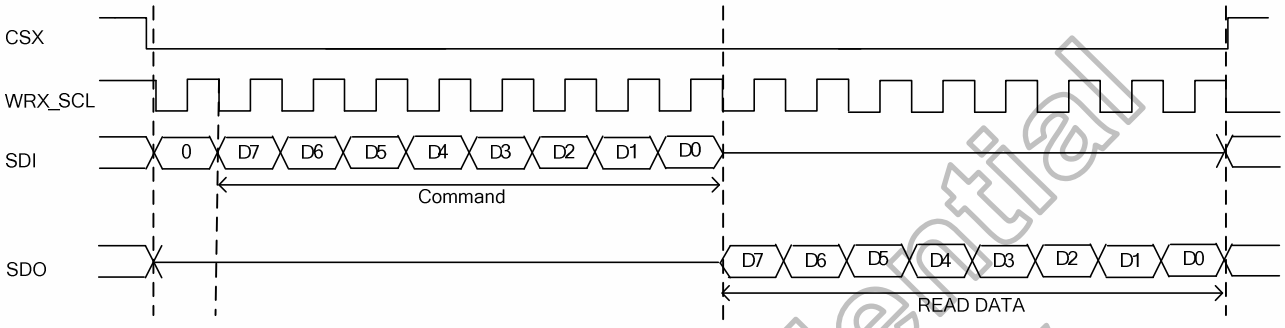


Figure 4.3: Type C:Serial interface protocol 3-wire/4-wire read mode

If there is a break on data transmission when transmit a command before a whole byte has been completed, then the display module will have reset the interface such that it will be ready to receive the same byte re-transmitted when the chip select line **(CSX)** is next activated. See the following figure.

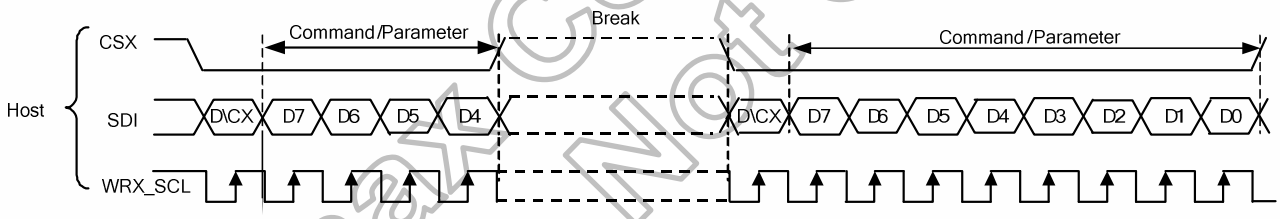


Figure 4.4: Display module data transfer recovery

Himax Confidential
Do Not Copy

4.2 DSI system interface

The selection of interface is by BS[2:0]=”000”~”111“ the DSI specifies the interface between a host processor and a peripheral such as a display module. Figure 4.5 shows a simplified DSI interface. From a conceptual viewpoint, a DSI-compliant interface also sends pixels or commands to the peripheral, and can read back status or pixel information from the peripheral. The main difference is that DSI serializes all pixel data, commands, and events.

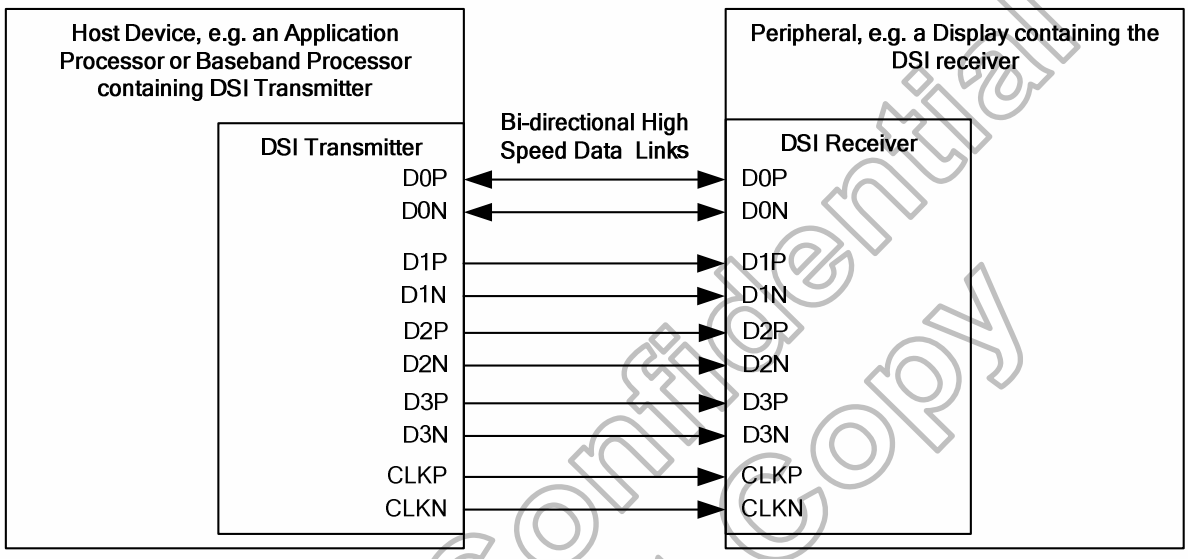


Figure 4.5: DSI transmitter and receiver interface

Please refer to “**DRAFT MIPI Alliance Standard for DSI**” for DSI detail specifications.

The data lane number is selected by register RBAh.

Himax Confidential
Do Not Copy

4.2.1 DSI layer definitions

According Figure 4.6 DSI transmitter and receiver interface to understand simple interface block diagram. Then under diagram is internal block for DSI which include four layers: PHY Layer, Lane Management Layer, Low level protocol and Application Layer.

The PHY Layer specifies the characteristics of transmission medium and electrical parameters for signaling the timing relationship between clock and Data Lanes.

The Lane Management Layer specifies DSI is Lane-scalable for increased performance. The data signals maybe transmission through one or more channel depending on the bandwidth requirements of the application.

The Protocol Layer specifies at the lowest level, DSI protocol specifies the sequence and value of bits and bytes traversing the interface. It specifies how bytes are organized into defined groups called packets.

The Application Layer describes higher-level encoding and interpretation of data contained in the data stream. The DSI specification describes the mapping of pixel values, commands and command's parameters to bytes in the packet assembly.

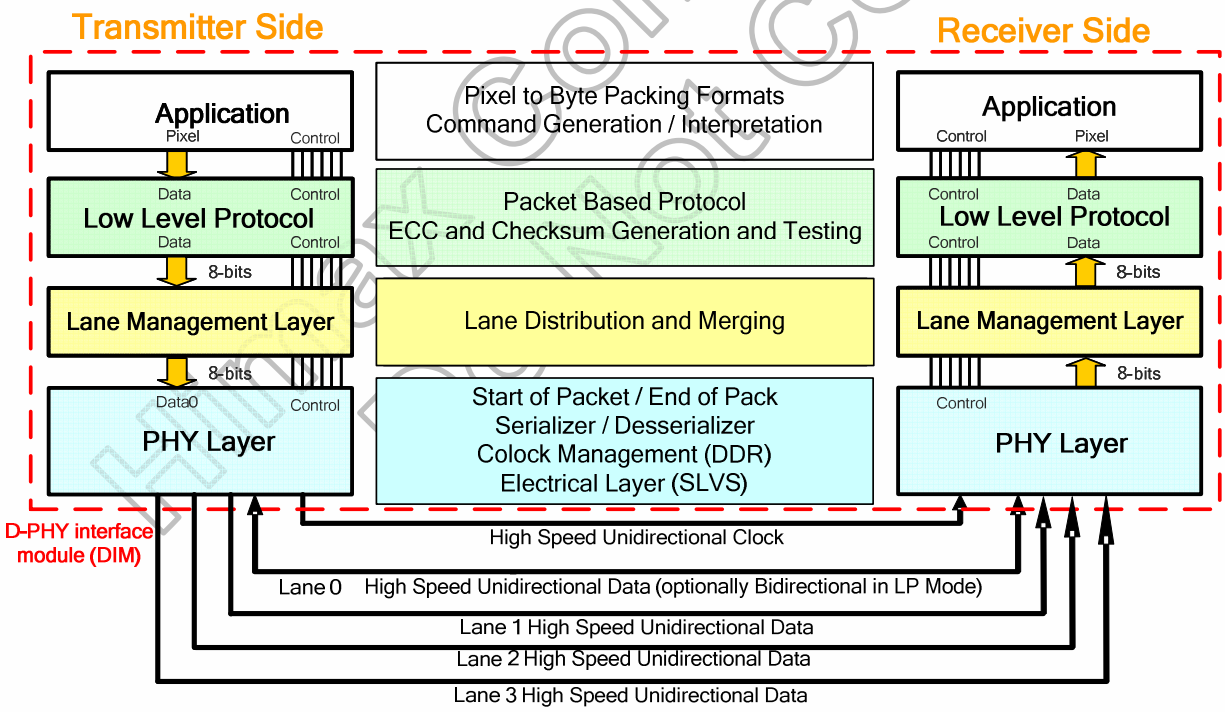


Figure 4.6: DSI transmitter and receiver interface

Himax Confidential Do Not Copy

This information contained herein is the exclusive property of Himax and shall not be distributed, reproduced, or disclosed in whole or in part without prior written permission of Himax.

4.2.2 Lane states

The HX8394-F uses Data Lane and Clock Lane differential pairs for DSI. Both differential lane pairs can be driven LP **(Low Power)** or HS **(High Speed)** mode.

LP mode means each line of the differential pairs are used in independently and single-ended. In LP mode differential receiver is disable **(Termination resistor of the receiver is disable)**. In LP mode there are four possible Low-Power Lane states **(LP-00, LP-01, LP-10, LP-11)**.

HS mode means the differential pairs are not used in single-end and termination resistor of the receiver is enable. There are different modes and protocol in each mode when transfer display data frim MCU to the display module.

The state code of HS and LP Lane pair are defined as below:

State Code	Line Voltage Levels		High-Speed	Low-Power	
	Dp-Line	Dn-Line	Burst Mode	Control Mode	Escape Mode
HS-0	HS Low	HS High	Differential-0	Note1	Note1
HS-1	HS High	HS Low	Differential-1	Note1	Note1
LP-00	LP Low	LP Low	N/A	Bridge	Space
LP-01	LP Low	LP High	N/A	HS-Rqst	Mark-0
LP-10	LP High	LP Low	N/A	LP-Rqst	Mark-1
LP-11	LP High	LP High	N/A	Stop	Note2

Note: (1) During High-Speed transmission the Low-Power Receivers observe LP-00 on the Lines.
(2) If LP-11 occurs during Escape mode the Lane returns to Stop state (Control Mode LP-11)

Himax Confidential
Do Not Copy

4.2.3 Clock lane mode

Figure 4.7 shows the state diagram for Clock Lane Mode. The Clock Lane has three different power modes: Low Power Stop State, Ultra Low Power State **(ULPS)** and High Speed clock transmission.

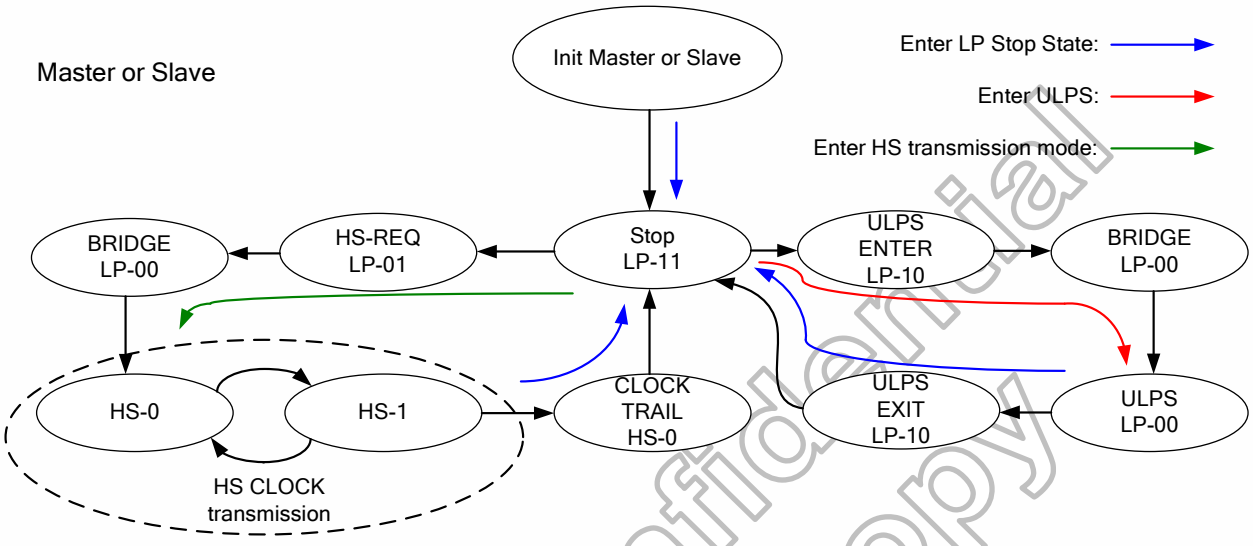
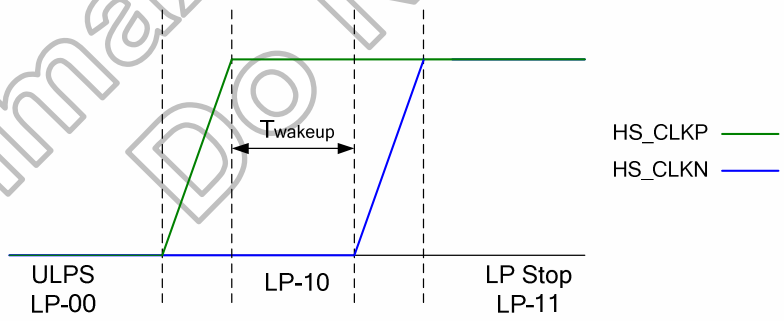


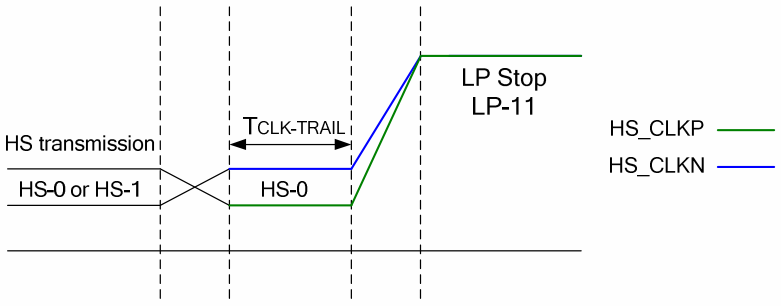
Figure 4.7: Clock lane mode state diagram

Clock Lane can be driven LP-11 to enter Low Power Stop State. There are three ways to enter Lower Power Stop State:

(1) After Initial state. **(HW reset, SW reset, Power on sequence)**
(2) Leaving ULPS: ULPS LP-00 -> LP-10 -> Low Power Stop State LP-11.



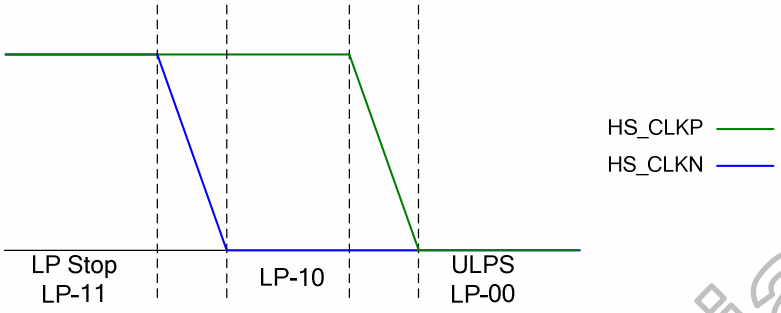
(3) Leaving HS clock transmission mode: HS mode **(HS-0 or HS-1)** -> HS-0 -> Low Power Stop State LP-11.



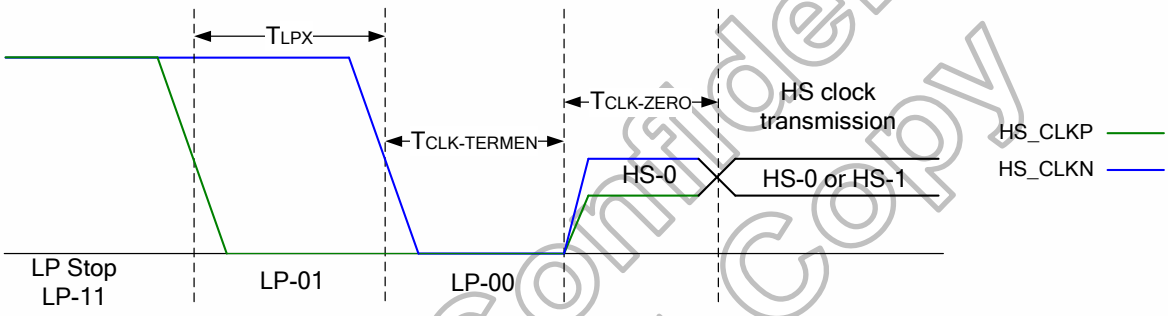
Himax Confidential Do Not Copy

This information contained herein is the exclusive property of Himax and shall not be distributed, reproduced, or disclosed in whole or in part without prior written permission of Himax.

Clock Lane can be driven LP-00 to enter Ultra Low Power State from Low Power Stop State. The flow is Low Power Stop State LP-11 -> LP-10 -> ULPS LP-00.



Clock Lane can be High Speed Clock transmission State from Low Power Stop State. The flow is Low Power Stop State LP-11 -> LP-01 -> LP-00 -> HS-0/1.



Himax Confidential
Do Not Copy

This information contained herein is the exclusive property of Himax and shall not be distributed, reproduced, or disclosed in whole or in part without prior written permission of Himax.

4.2.4 Data lane mode

Figure 4.8 shows the operational flow diagram for Data Lane Mode. There are three operating modes in Data Lane: Escape mode, High-Speed transmission mode and Turnaround.

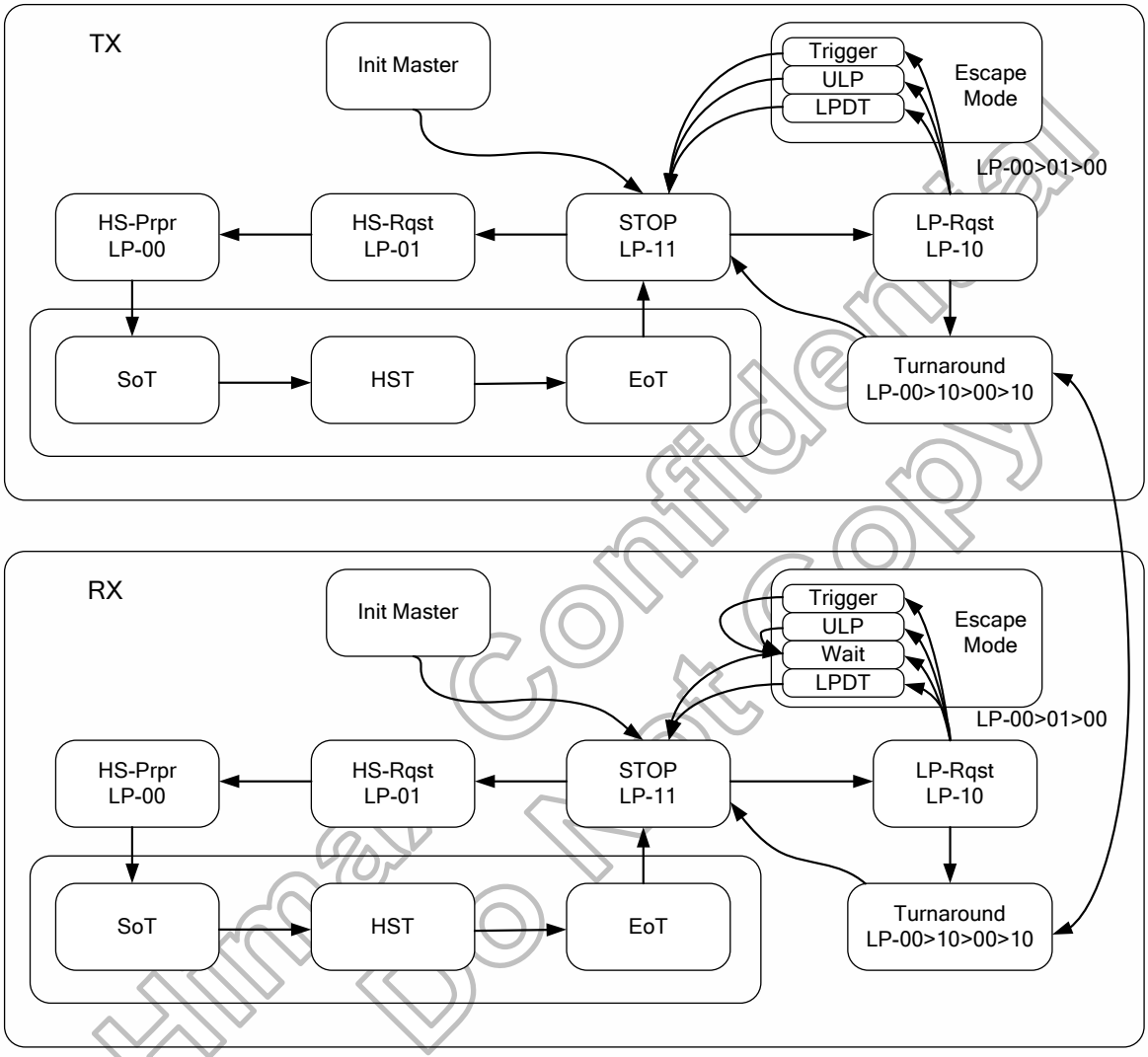


Figure 4.8: Data lane mode state diagram

Himax Confidential Do Not Copy

This information contained herein is the exclusive property of Himax and shall not be distributed, reproduced, or disclosed in whole or in part without prior written permission of Himax.

4.2.5 Escape mode

Data Lane0 is used in Escape Mode when data lane in LP mode. Data Lane shall enter Escape mode via LP-11 -> LP-10 -> LP-00 -> LP-01 -> LP-00 and exit Escape mode via LP-10 -> LP-11.

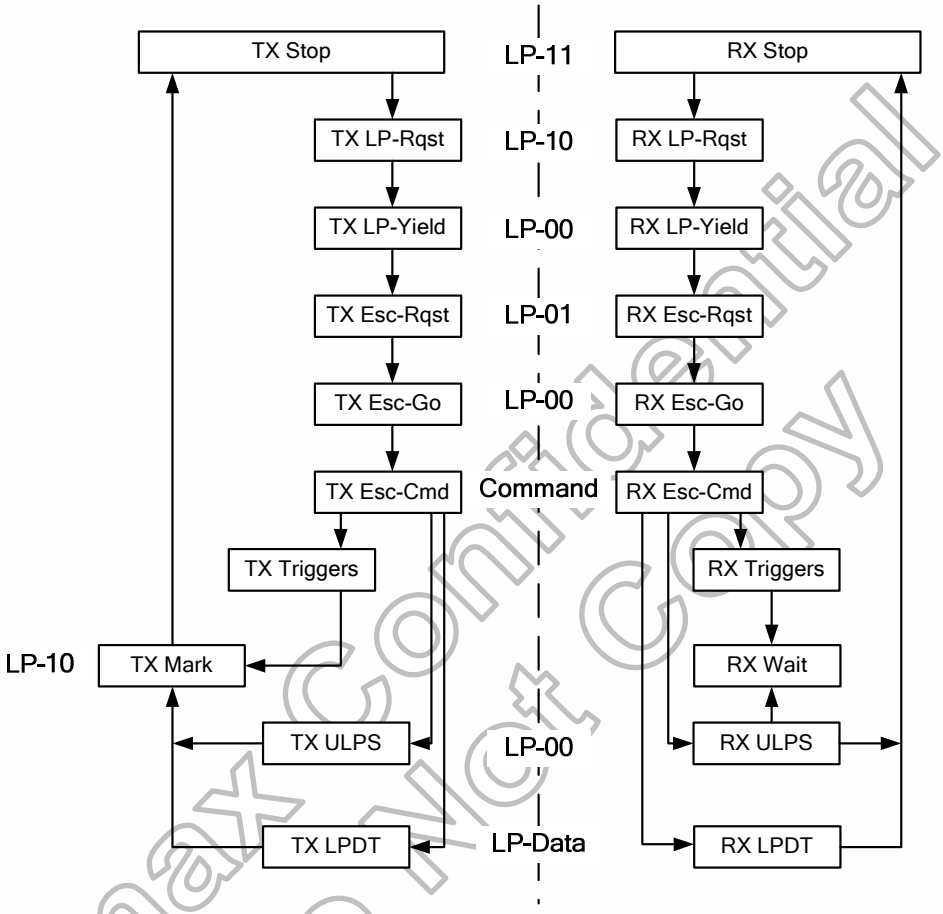


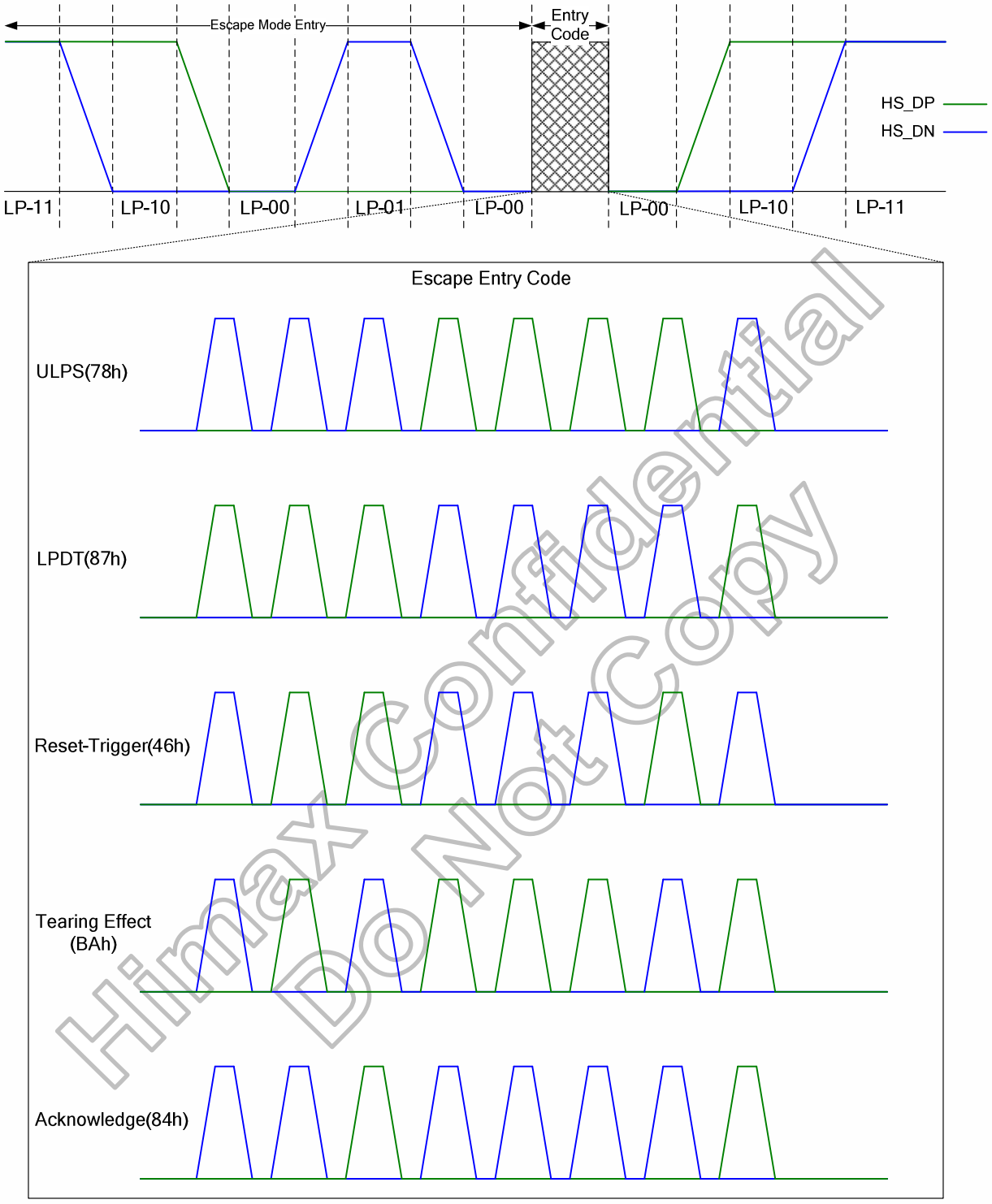
Figure 4.9: Escape mode state machine

Once Escape mode is entered, the transmitter shall send an 8-bit entry code to indicate the requested action. The Entry Code as follows:

(1) Trigger **(Reset-Trigger(46h), Tearing effect(BAh), Acknowledge(84h))**
(2) Drive Data Lane to Ultra Low Power State **(78h)**
(3) Send Low Power Data Transmission **(87h)**

Himax Confidential
Do Not Copy

This information contained herein is the exclusive property of Himax and shall not be distributed, reproduced, or disclosed in whole or in part without prior written permission of Himax.



Himax Confidential
Do Not Copy

Figure 4.10: Escape mode timing sequence

4.2.6 High speed data transmission

The display module can enter High Speed Data Transimission when Clock Lane in the High Speed Clock Mode. All Data Lane enter High Speed Data Tranmission synchronously but may end at different time. Data Lane enter High Speed Data Transmission flow: LP-11 -> LP-01 -> LP-00 -> SoT **(0001_1101)**. And exit High Speed Data Transmission flow: Toggles differential state immediately after last payload data bit and keeps that state for a time $T_{HS-TRAIL}$.

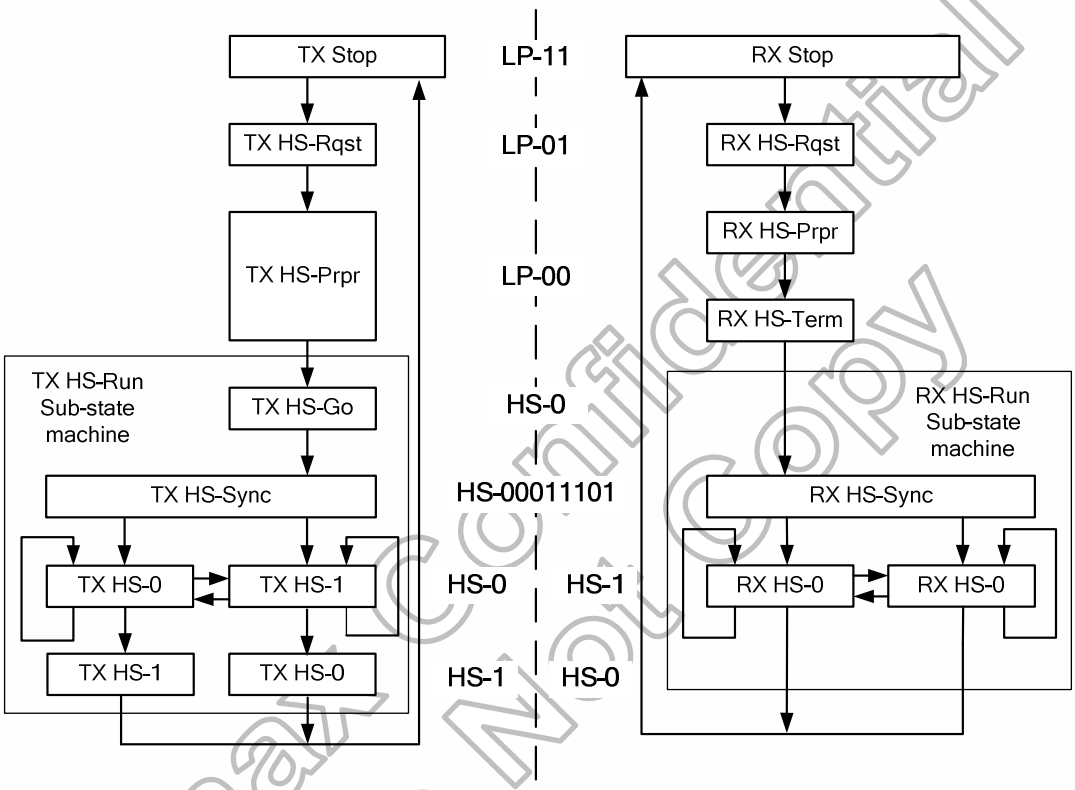


Figure 4.11: High speed data transmission state machine

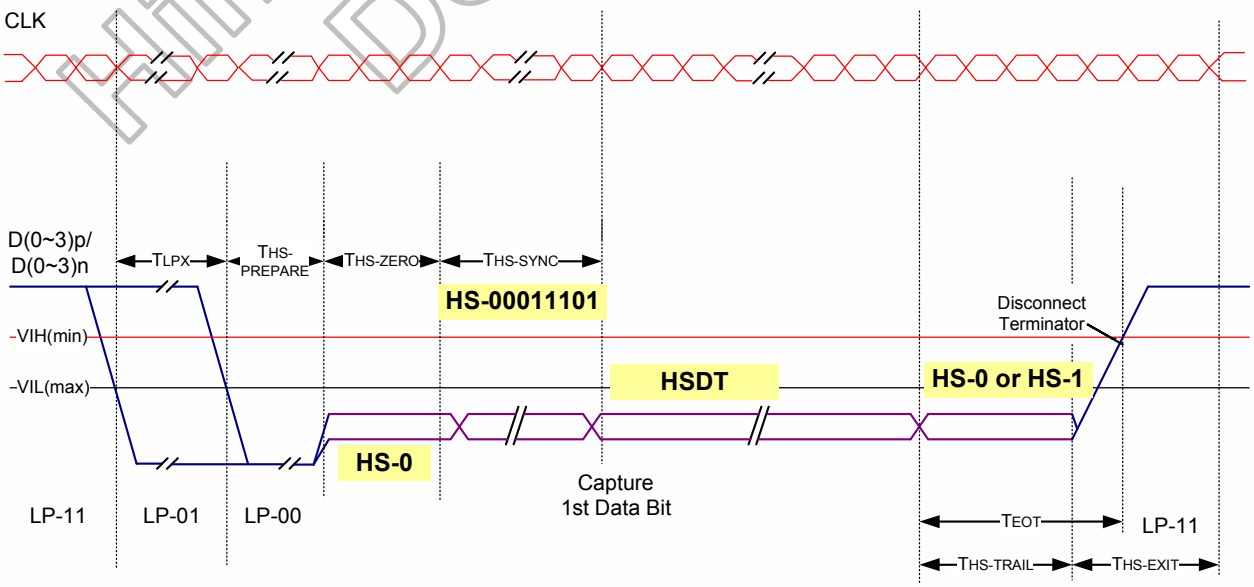


Figure 4.12: High speed data transmission timing sequence

Himax Confidential
Do Not Copy

Parameter	Description	Min.	Typ.	Max.	Unit
$T_{HS-PREPARE}$	Time that the transmitter drives the Data Lane LP-00 Line state immediately before the HS-0 Line state starting the HS transmission.	40+4*UI	-	85+6*UI	ns
$T_{HS-PREPARE}$+ $T_{HS-ZERO}$	THS-PREPARE + time that the transmitter drives the HS-0 state prior to transmitting the Sync sequence.	145+10*UI	-	-	ns
T_{EOT}	Transmitted time interval from the start of THS-TRAIL or TCLK-TRAIL, to the start of the LP-11 state following a HS burst.	-	-	105ns+12*UI	ns
$T_{HS-TRAIL}$	Time that the transmitter drives the flipped differential state after last payload data bit of a HS transmission burst.	60ns+4*UI	-	-	ns
$T_{HS-EXIT}$	Time that the transmitter drives LP-11 following a HS burst.	100	-	-	ns

Table 4.2: Global Operation Timing Parameters for Data Lane

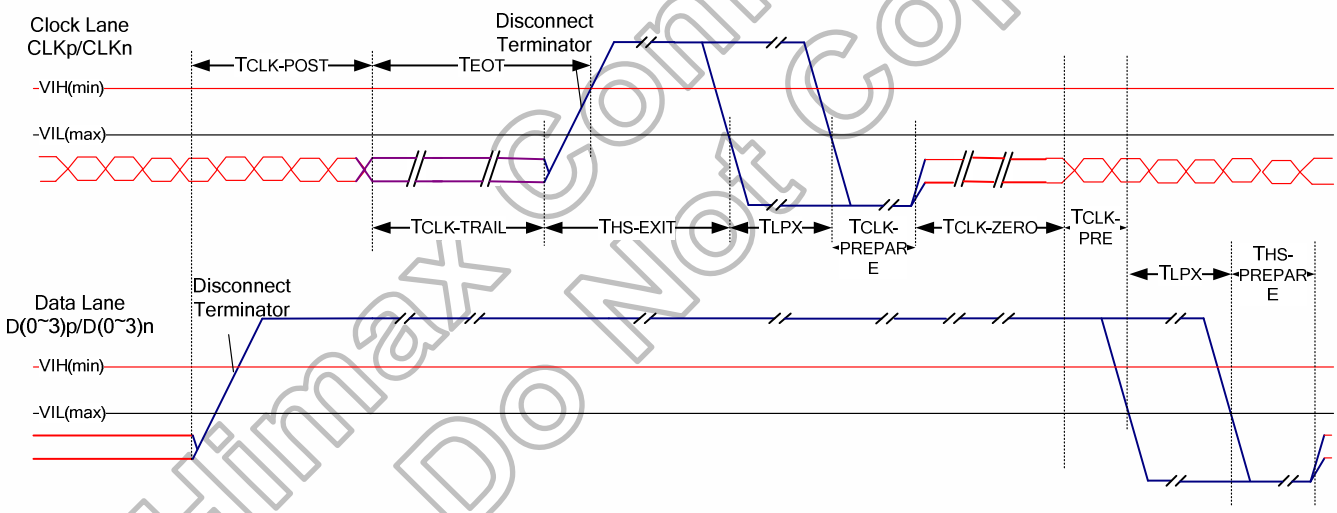


Figure 4.13: Switching the Clock Lane between Clock Transmission and LP Mode

Parameter	Description	Min.	Typ.	Max.	Unit
$T_{CLK-POST}$	Time that the transmitter continues to send HS clock after the last associated Data Lane has transitioned to LP Mode.	60+52*UI	-	-	ns
$T_{CLK-TRAIL}$	Time that the transmitter drives the flipped differential state after last payload data bit of a HS transmission burst.	60	-	-	ns
$T_{CLK-PREPARE}$	Time that the HS clock shall be driven by the transmitter prior to any associated Data Lane beginning the transition from LP to HS mode.	38	-	95	ns

Himax Confidential
Do Not Copy

$T_{CLK-PREPARE+}$ $T_{CLK-ZERO}$	TCLK-PREPARE + time that the transmitter drives the HS-0 state prior to starting the Clock.	300	-	-	ns
$T_{CLK-PRE}$	Time that the HS clock shall be driven by the transmitter prior to any associated Data Lane beginning the transition from LP to HS mode.	8*UI	-	-	ns

Table 4.3: Global Operation Timing Parameters for Clock Lane

Himax Confidential
Do Not Copy

This information contained herein is the exclusive property of Himax and shall not be distributed, reproduced, or disclosed in whole or in part without prior written permission of Himax.

4.2.7 Bi-directional data lane turnaround

The transmission direction od a bi-directional Data Lane can be swapped by means of a Link Turnaround procedure. This procedure enables information transfer in the opposite direction of the current direction. The procedure is the same for either a change from Forward-to-Reverse direction or Reverse-to-Forward direction.

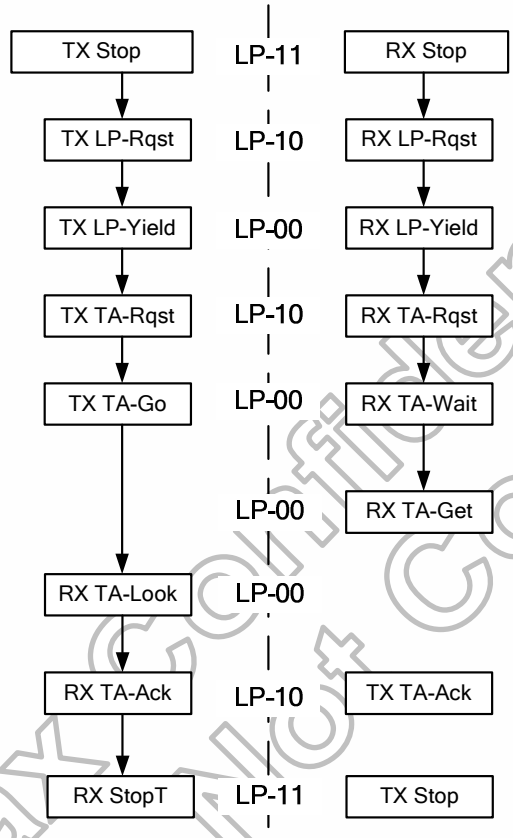


Figure 4.14: Turnaround state machine

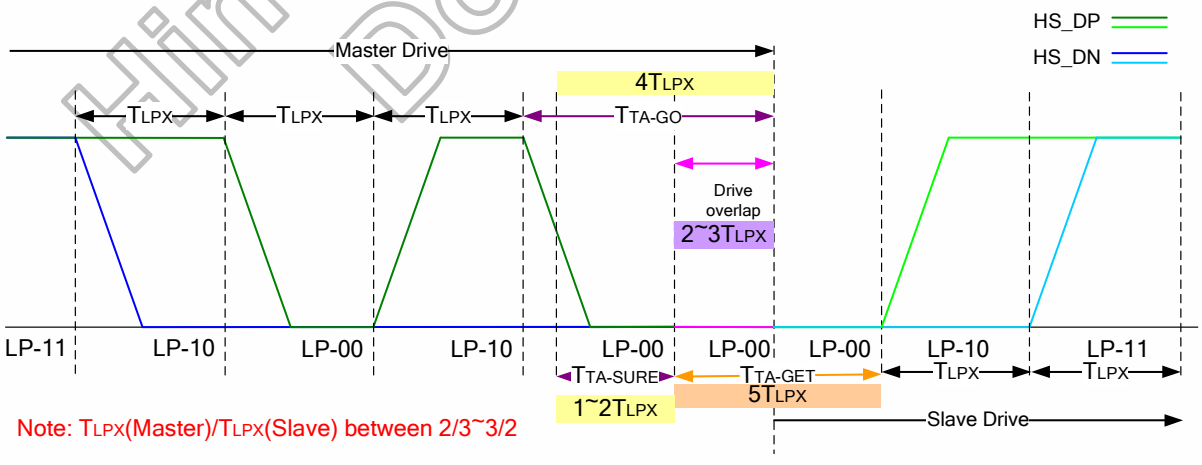
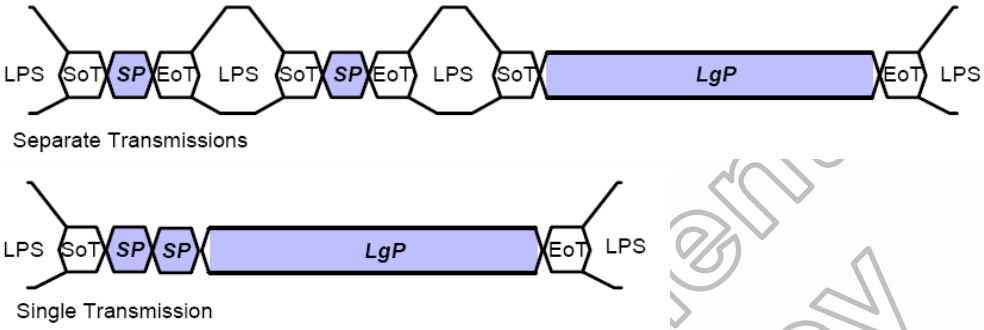


Figure 4.15: Turnaround timing sequence

Himax Confidential
Do Not Copy

4.2.8 DSI protocol

The protocol layer appends packet-protocol information and headers. The receiver side of a DSI Link performs the converse of the transmitter side, decomposing the packet into parallel data, signal events and commands. The DSI protocol permits multiple packets which is useful for events such as peripheral initialization, where many registers may be loaded separate write commands at system startup. Figure 4.15 illustrates multiple HS Transmission packets.

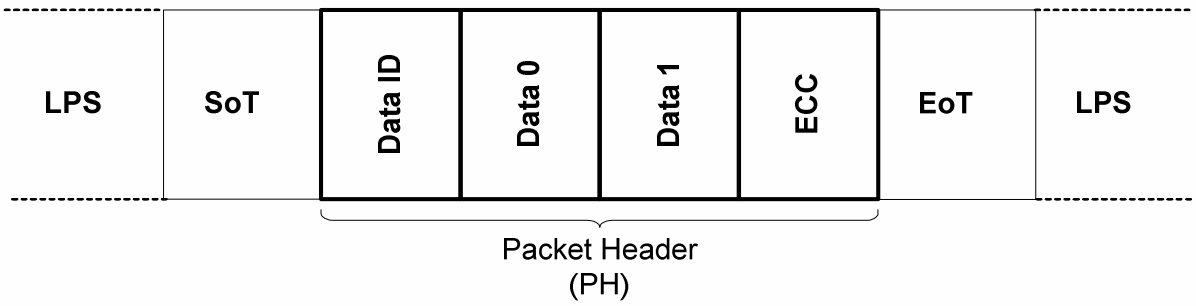


LPS：Low power state
SoT：Start of Transmission
SP：Short Packet
LgP：Long Packet
EoT：End of Transmission

Figure 4.16: Multiple packets transmission

The packet includes two types which are Long packet and Short packet. The first byte of the packet, the Data Identifier **(DI)**, includes information specifying the type of the packet. Command Mode systems send commands and an associated set of parameters, with the number of parameters depending on the command type.

Short packets are four bytes in length including the ECC. Short packet is used for most Command Mode commands and associated parameters. Where Short packets format include an 8-bit Data ID followed by two command or data and an 8-bit ECC. Figure 4.16 shows the structure of the Short packet.



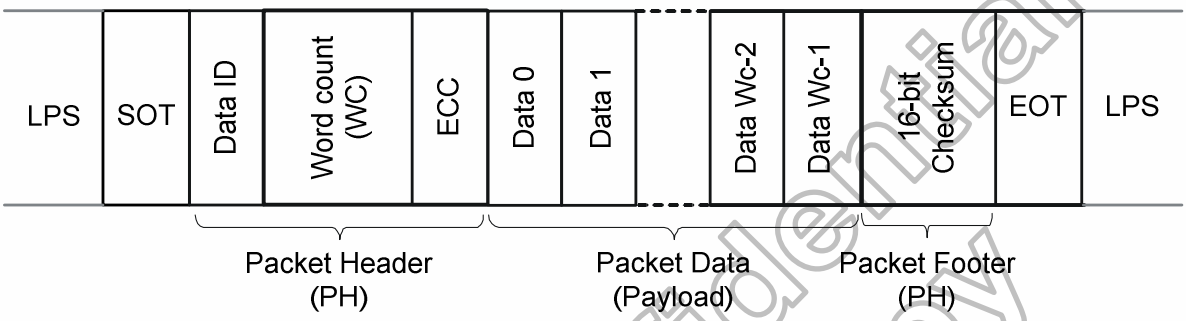
DI(Data ID) **:** Contain Virtual Channel Identifier and Data Type.
ECC(Error Correction Code) **:** The Error Correction Code allows single-bit errors to be corrected and 2-bit errors to be detected in the Short packet.

Figure 4.17: Structure of the short packet

Himax Confidential
Do Not Copy

Long packets specify the payload length using a two-byte Word Count field and then the payload maybe from 0 to 65,535 bytes in length. Long packets permit transmission of large blocks of pixel or other data. Figure 4.17 shows the structure of the Long packet. Long Packet Header composed of three elements: an 8-bit Data Identifier, a 16-bit Word Count, and 8-bit ECC. An application-specific Data Payload has Word Count * bytes following the Packet Header. The Packet Footer has one element, a 16-bit checksum. Long packets can be from 6 to 65,541 bytes in length.

Where 65,541 bytes = 4 bytes PH + (2^{16}-1)bytes Payload + 2 bytes PF



DI (Data ID) **:** Contain Virtual Channel Identifier and Data Type.
WC (Word Count) **:** The receiver use WC to determine the packet end.
ECC (Error Correction Code) **:** The Error Correction Code allows single-bit errors to be corrected and 2-bit errors to be detected in the Packet Header.
PF (Packet Footer) **:** Mean 16-bit Checksum.

Figure 4.18: Structure of the long packet

According to packet form, basic elements include DI and ECC. Figure 4.18 the shows format of Data ID.

DI7	DI6	DI5	DI4	DI3	DI2	DI1	DI0
VC **(Virtual Channel)**		DT **(Data Type)**					

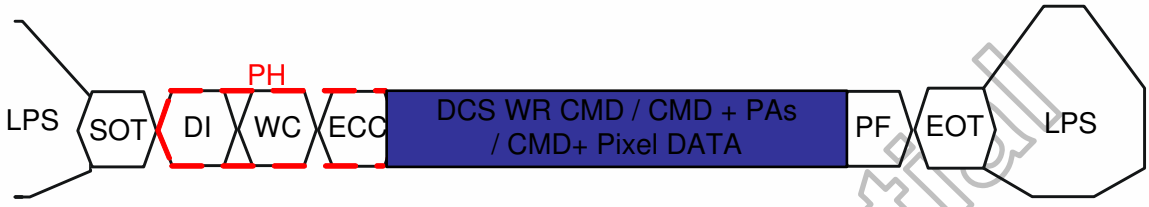
DI[7:6] ➔ These two bits identify the data as directed to one of four virtual channels.
DI[5:0]: These six bits specify the Data Type.

Figure 4.19: The format of data ID

Himax Confidential Do Not Copy

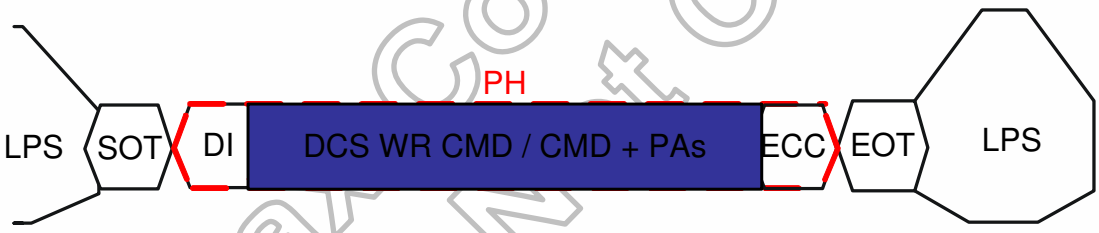
Due to Data Type **(DT)** mean format of transmission type, Figure 4.19 show Short- / Long-packet transmission command sequence.

Long packet write Command / Parameters / Pixel Datas



DI Write suitable Data type.
WC Write number of Payload Data.
Ex: One CMD write, WC setting as 1.
CMD + PAs write, WC setting as number of (CMD+PAs).
CMD + DATA write, WC setting as number of (CMD + Pixel DATA).

Short packet write Command / Parameters



DI Write suitable Data type.
Ex: One CMD write, DI + DCS WR CMD
CMD + PAs write, DI + DCS WR CMD + PAs

Figure 4.20: Show short- / long-packet transmission command sequence

Himax Confidential
Do Not Copy

4.2.9 Processor to peripheral (forward direction) packets data types

The set of transaction types sent from the host processor to a peripheral, such as a display module, are shown in Table 4.2 Data Types for Processor-sourced Packets.

Data type, hex	Data type, binary	Description packet	Size
01h	00 0001	Sync Event, V Sync Start	Short
11h	01 0001	Sync Event, V Sync End	Short
21h	10 0001	Sync Event, H Sync Start	Short
31h	11 0001	Sync Event, H Sync End	Short
08h	00 1000	End of Transmission packet **(EoTp)**	Short
05h	000101	DCS WRITE, no parameter	Short
15h	010101	DCS WRITE, 1 parameter	Short
06h	00 0110	DCS READ, no parameters	Short
37h	11 0111	Set Maximum Return Packet Size	Short
09h	00 1001	Null Packet, no data	Long
19h	01 1001	Blanking Packet, no data	Long
39h	11 1001	DCS Long Write/write_LUT Command Packet	Long
0Eh	00 1110	Packed Pixel Stream, 16-bit RGB, 5-6-5 Format	Long
1Eh	01 1110	Packed Pixel Stream, 18-bit RGB, 6-6-6 Format	Long
2Eh	10 1110	Loosely Packed Pixel Stream, 18-bit RGB, 6-6-6 Format	Long
3Eh	11 1110	Packed Pixel Stream, 24-bit RGB, 8-8-8 Format	Long
X0h and XFh, unspecified	xx 0000 xx 1111	DO NOT USE All unspecified codes are reserved	-

Table 4.4: Data types for processor-sourced packets

Himax Confidential
Do Not Copy

Under tables list all detail function of all data types

Sync event (H start, H end, V start, V end), data type=xx 0001 (x1h)		
Data type, hex	**Function description**	**Number of bytes**
01h	V Sync start, Start of VSA pulse.	4 bytes **(DI+00h+00h+ECC)**
11h	V Sync End, End of VSA pulse.	
21h	H Sync Start, Start of I pulse.	
31h	H Sync End, End IHSA pulse.	
Note: V Sync Start and V Sync End event represents the start and end of the VSA, respectively. Similarly H Sync Start and H Sync End event represents the start and end olhe I, respectively.		

EoT Packet		
Data type, hex	**Function description**	**Number of bytes**
08h	End of Transmission Packet (EoTp) (08,0F,0F,01)	4 bytes (DI+Data0+Data1+ECC)
Note: The main objective of the EoTp is to enhance overall robustness of the system during HS transmission mode. Therefore, DSI transmitters should not generate an EoTp when transmitting in LP mode.		

Color Mode Off /On Command		
Data type, hex	**Function description**	**Number of bytes**
02h	Color Mode Off Packet (02,00,00,0B)	4 bytes (DI+Data0+Data1+ECC)
12h	Color Mode On Packet (12,00,00,18)	
Note: Color Mode Off is a Short packet command that returns a Video Mode display module from low-color mode to normal display operation. Color Mode On is a Short packet command that switches a Video Mode display module to a low-color mode for power saving.		

Display status (shutdown command, turn-on command)		
Data type, hex	**Function description**	**Number of bytes**
22h	Shutdown Peripheral command that turns off the display in a Video Mode display for power saving.	4 bytes **(DI+00h+00h +ECC)**
32h	Turn On Peripheral command that turns on the display in Video Mode display for normal display.	
Note: When use shutdown command, interface shall remain powered in order to receive the turn-on, or wake-up, command.		

DCS read/write function		
Data type, hex	**Function description**	**Number of bytes**
06h	DCS Read command, the returned data **shall be Long packet format.**	4 bytes **(DI+Data0+Data1+ECC)**
05h and 15h	DCS Short Write command, 0 or 1 parameter, Data Types = 00 0101**(05h)**, 01 0101 **(15h)**, Respectively.	4 bytes **(DI+Data0+Data1+ECC)**
39h	DCS Long Write/ Write _ LUT Command is used to send larger blocks of data to a display module that implements the Display Command Set.	Up to 65541 bytes **(DI + WC + ECC + DCS CMD. + Payload DATA + PF)**

Note: (1) For write part, If DCS Short Write command, followed by BTA, the peripheral shall respond with ACK when without error was detected in the transmission **(Host → Slave)**. Unless an error was detected, the peripheral shall respond with Acknowledge with Error Report.

For example: 05h DCS WRITE for no parameter command set.

05h	CMD	0	ECC

Ex. 05h, 29h, 00, 1Ch Display On(29h)

For example: 15h DCS WRITE for only one parameter command set.

15h	CMD	Par	ECC

Ex. 15h, 36h, 08h, 11h — MADCTL(36h)-BGR bit=1

(2) When use DCS Read Command, the Set Max Return Packet Size command will limit the size of returning

Himax Confidential Do Not Copy

packets.

(3) The peripheral shall respond to DCS Read Command Request in one of the following ways:

- If an error was detected by the peripheral, it shall send *Acknowledge with Error Report*. So the peripheral shall transmit the requested READ data packet with suitable ECC in the same transmission.
- If no error was detected by the peripheral, it shall send the requested READ packet **(Short or Long)** with appropriate ECC and Checksum, if either or both features are enabled.

(4) One byte <= Length of payload DATA <= 2^{16}-1

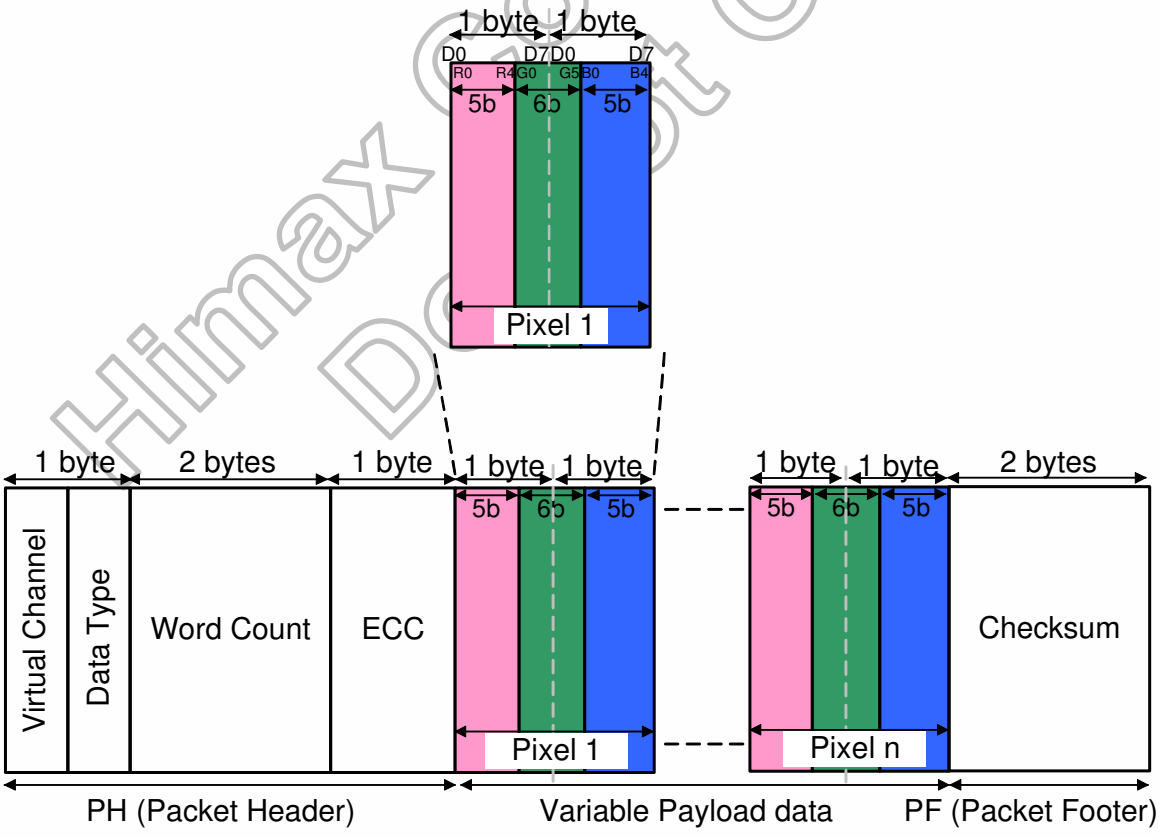
Himax Confidential
Do Not Copy

This information contained herein is the exclusive property of Himax and shall not be distributed, reproduced, or disclosed in whole or in part without prior written permission of Himax.

Maximum Return packet size setting		
Data type, hex	**Function description**	**Number of bytes**
37h	Set Maximum Return Packet Size that specifies the maximum size of the payload in a Long packet transmitted from peripheral back to the host processor.	4 bytes **(DI + Maximum Return Packet Size + ECC)**
Note: The two-byte value is transmitted with LS byte first. And during a power-on or Reset sequence, the Maximum Return Packet Size shall be set by the peripheral to a default value of one.		

Variable data packet		
Data type, hex	**Function description**	**Number of bytes**
09h	Null Packet is a mechanism for keeping the serial Data Lane**(s)** in High-Speed mode while sending dummy data.	Up to 65541 bytes **(DI + WC + ECC + DCS CMD. + Payload DATA + PF)**
19h	Blanking packet is used to convey blanking timing information in a Long packet.	
Note: (1) When Null Packet, the Payload Data belong "null" Data, actual data values sent are irrelevant because the peripheral does not capture or store the data. (2) When Blanking packet, the packet represents a period between active scan lines of a Video Mode display,		

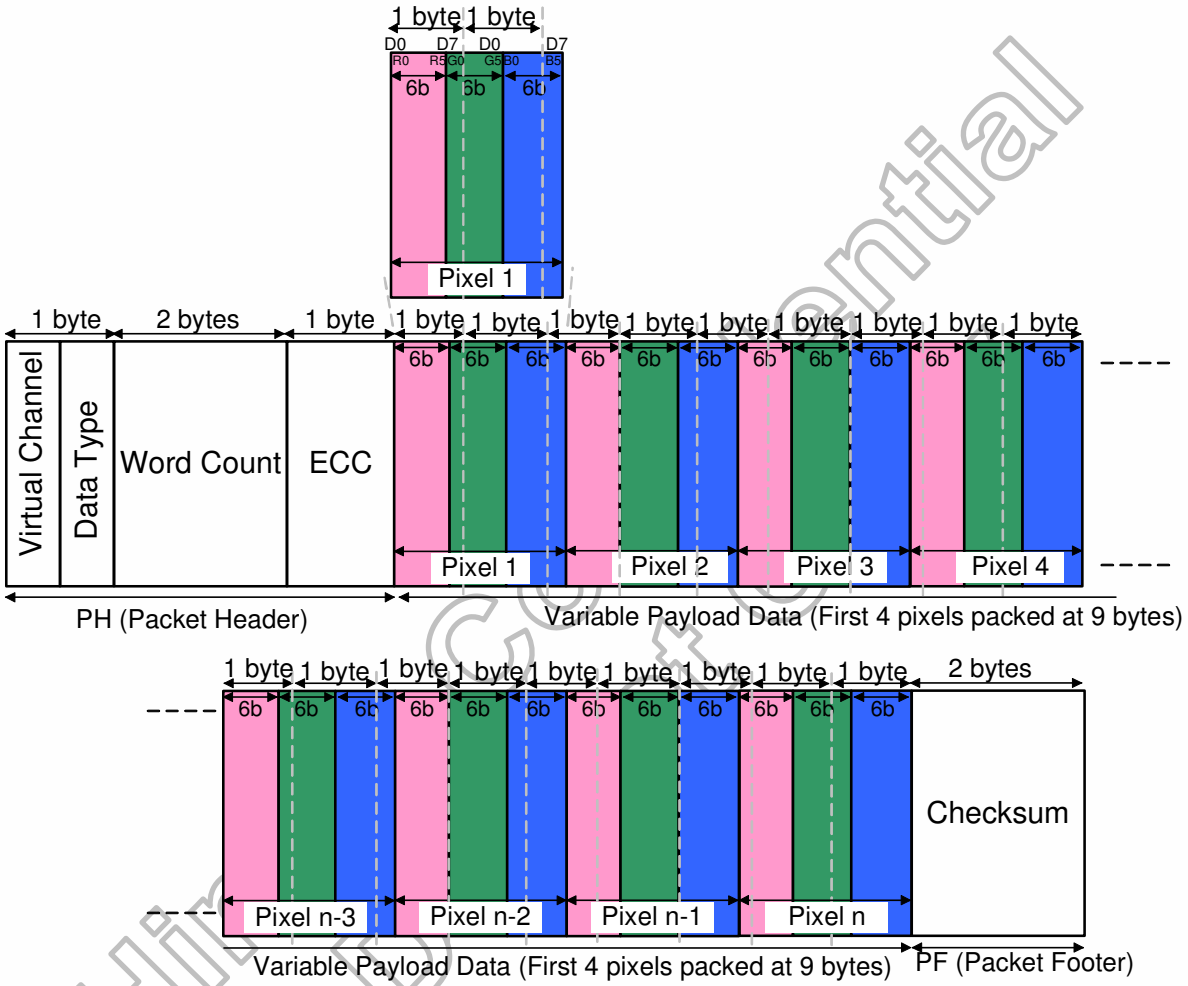
Data stream format		
Data type, hex	**Function description**	**Number of bytes**
0Eh	Packed Pixel Stream 16-Bit Format is used to transmit image data formatted as 16-bit pixels to a Video Mode display module. Pixel format is "**(5 bits)** red, **(6 bits)** green and **(5 bits)** blue".	Up to 65541 bytes **(DI + WC + ECC + DCS CMD. + Payload DATA + PF)**



Note: Within a color component, the "LSB is sent first, the MSB last ".

Himax Confidential
Do not Copy

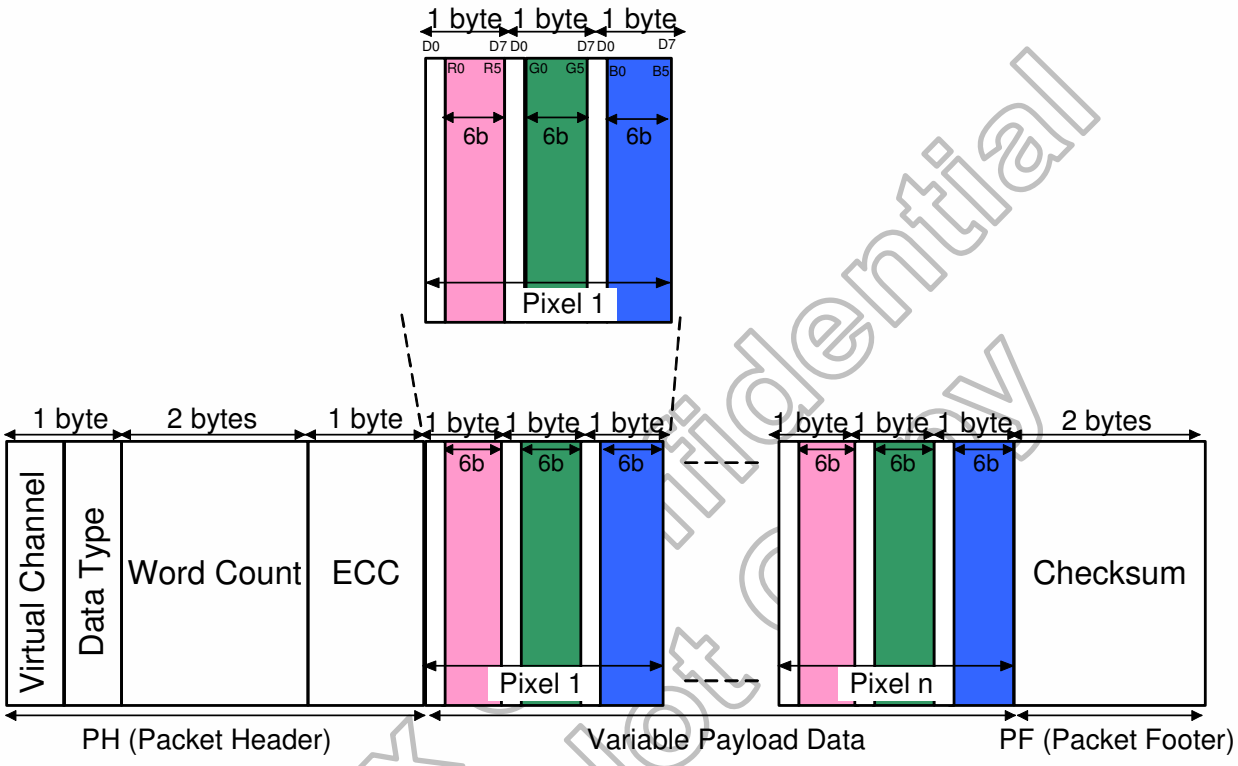
Data stream format		
Data type, hex	**Function description**	**Number of bytes**
1Eh	Packed Pixel Stream 18-Bit Format is used to transmit image data formatted as 18-bit pixels to a Video Mode display module. Pixel format is "**(6 bits)** red, **(6 bits)** green and **(6 bits)** blue".	Up to 65541 bytes **(DI + WC + ECC + DCS CMD. + Payload DATA + PF)**



Note: Within a color component, the LSB is sent first and the MSB last and pixel boundaries only line up with byte boundaries every four pixels **(nine bytes)**. Preferably, display modules employing this format have a horizontal extent **(width in pixels)** evenly divisible by four, so no partial bytes remain at the end of the display line data. It is possible to send pixel data that represent a line width that is not a multiple of four pixels, but display logic on the receiver end shall dispose of the extra bits of the partial byte at the end of active display and ensure a "clean start" for the next line.

Himax Confidential

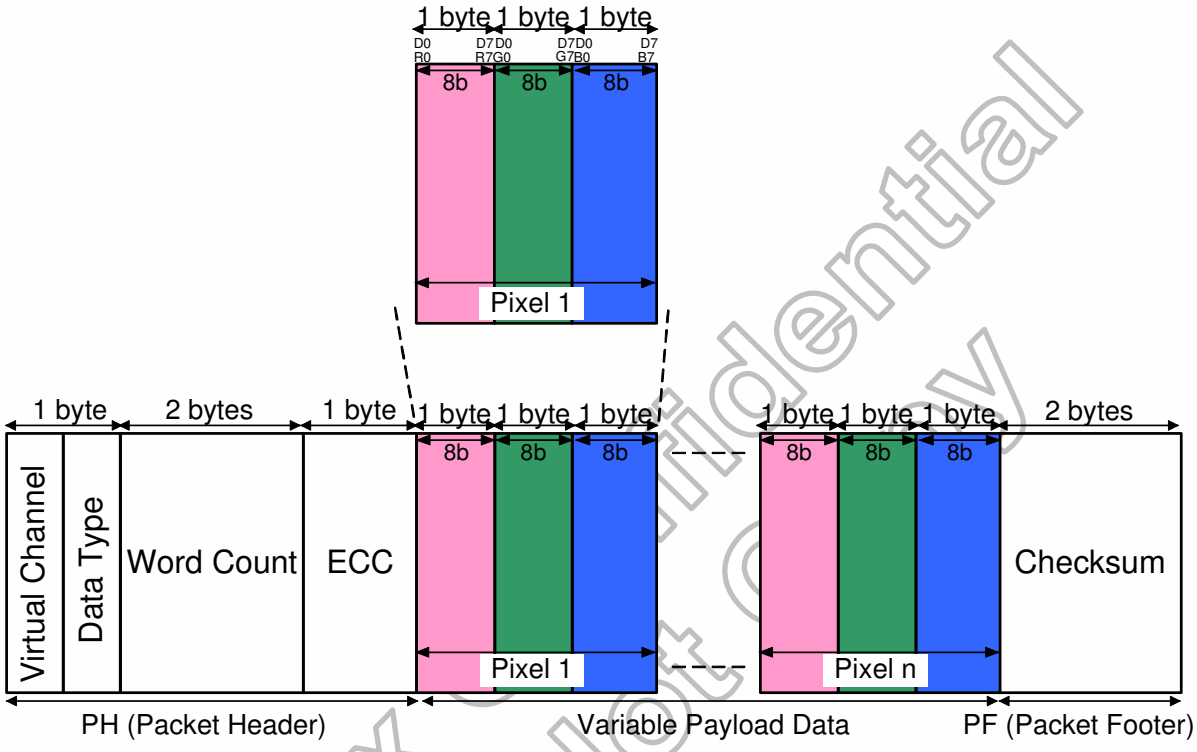
Data stream format		
Data type, hex	**Function description**	**Number of bytes**
2Eh	Packed Pixel Stream 18-Bit Format, each R, G, or B color component is one byte form, but the valid pixel bits occupy bits [7:2] and bits [1:0] of are ignored. Pixel format is "**(6 bits)** red, **(6 bits)** green and **(6 bits)** blue".	Up to 65541 bytes **(DI + WC + ECC** **+ DCS CMD.** **+ Payload DATA + PF)**



Note: Within a color component, the LSB is sent first, the MSB last and With this format, pixel boundaries line up with byte boundaries every three bytes.

Himax Confidential
Do Not Copy

Packed pixel stream, 24-bit format		
Data type, hex	**Function description**	**Number of bytes**
3Eh	Packed Pixel Stream 24-Bit Format is used to transmit image data formatted as 24-bit pixels to a Video Mode display module. Pixel format is **(8 bits)** red, **(8 bits)** green and **(8 bits)** blue.	Up to 65541 bytes **(DI + WC + ECC** **+ DCS CMD.** **+ Payload DATA + PF)**



Note: Within a color component, the LSB is sent first, the MSB last and With this format, pixel boundaries line up with byte boundaries every three bytes.

Himax Confidential
Do Not Copy

4.2.10 Peripheral to processor (reverse direction) packet data type

HX8394-F has the bidirectional capability for returning READ data, ACK or error information to the host processor. The packet structure for peripheral-to-processor transactions is the same as that for the processor-to-peripheral direction.

Peripheral-to-processor transactions are of four basic types:

A. *Tearing Effect* is a Trigger message sent to convey display timing information to the host processor. Trigger messages are single byte packets sent by a peripheral's PHY layer in response to a signal form the DSI protocol layer.
B. *Acknowledge* is a Trigger Message sent when the current transmission, as well as all preceding transmissions since the last peripheral to host communication is received by the peripheral with no errors.
C. *Acknowledge and Error Report* is a Short packet sent if any errors were detected in preceding transmission from the host processor. Once reported, accumulated errors in the error register are cleared.
D. *Response to Read Request* may be Short or Long packet that returns data requested by the preceding READ command from the processor.

In general, if the host processor completes a transmission to the peripheral with BTA asserted, the peripheral shall respond with one or more appropriate packet(s), and then return bus ownership to the host processor. If BTA is not asserted following a transmission from the host processor, the peripheral shall not communicate an Acknowledge or other error information back to the host processor.

The processor-to-peripheral transactions with BTA asserted, can contain under form.

A. Following a non-Read command in which no error was detected, the peripheral shall respond with Acknowledge.
B. Following a Read request in which no error was detected, the peripheral shall send the requested READ data.
C. Following a Read request in which the ECC error was detected and corrected, the Peripheral shall send the requested READ data in a Long or Short packet, followed by a 4-byte (Acknowledge with Error Report) packet in the same LP transmission. The Error Report shall have the ECC Error flag set.
D. Following a non-Read command in which the ECC error was detected and corrected, the peripheral shall proceed to execute the command, and shall respond to BTA by sending a 4-byte (Acknowledge with Error Report) packet, the Error Report shall have the ECC Error flag set.
E. Following any command in which SoT Error, SoT Sync Error, EoT Sync Error, LP Transmit Sync Error, checksum error or DSI VC ID Invalid was detected, or the DSI command was not recognized, the peripheral shall send a 4-byte Acknowledge with Error Report response, with the appropriate error flags set in the two-byte error field. Only the ACK/Error Report packet shall be transmitted; no read or write accesses shall take place on the peripheral in response.

Himax Confidential
Do Not Copy

This information contained herein is the exclusive property of Himax and shall not be distributed, reproduced, or disclosed in whole or in part without prior written permission of Himax.

Which,

A. "*Acknowledge*" is sent using a Trigger message which is one byte: 00100001
B. "*Acknowledge with Error Report*" include 4 bytes which are DI, 2 bytes Error report and ECC.
C. "*Response to Read Request*" are Long packet format.

An error report is comprised of two bytes following the DI byte, with an ECC byte following the error report bytes. Table 4.5 shows the Error Report Bit Definitions. And Table 4.6 list complete set of peripheral-to-processor Data Types.

Bit	Description
0	SoT Error
1	SoT Sync Error
2	Reserved
3	Escape Mode Entry Command Error
4	Low-Power Transmit Sync Error
5	Peripheral Timeout Error
6	Reserved
7	Reserved
8	ECC Error, single-bit (Detected and corrected)
9	ECC Error, multi-bit (Detected, not corrected)
10	Checksum Error (Long packet only)
11	DSI Data Type Not Recognized
12	DSI VC ID Invalid
13	Reserved
14	Reserved
15	Reserved

Table 4.5: Shows the error report bit definitions

Data type, hex	Data type, binary	Description packet	Size
02h	00 0010	Acknowledge with Error Report	Short
1Ch	01 1100	DCS Long READ Response	Long
Others **(00h→3Fh)**		Reserved	-

Table 4.6: The complete set of peripheral-to-processor data types

Acknowledge types		
Data type, hex	**Function description**	**Number of bytes**
02	Get Acknowledge with Error report when Error occurs from processor transmission.	4 bytes
Note: When processor transmits complete Payload, following signal by BTA, peripheral must respond to processor. With error ➔ Acknowledge with error report **(Short packet)**, Without error ➔ request READ data or Acknowledge **(Trigger message)**.		

DCS Read types		
Data type, hex	**Function description**	**Number of bytes**
1Ch	This is the Long-packet response to DCS Long Read Request.	Up to 65541 bytes **(DI + WC + ECC + DCS CMD. + Payload DATA + PF)**
Note: (1)If the peripheral is Checksum capable, is shall return a calculated two-byte Checksum appended to the N-byte payload data. If the peripheral does not support Checksum, it shall return 0000h. If the DCS command itself is possibly corrupt, due to an uncorrectable ECC error, SoT or SoT Sync error, the requested READ data packet shall not be sent after the Acknowledge with Error Report packet be sent. (2)There is no dummy read byte in the read response packet.		

Himax Confidential Do not Copy

This information contained herein is the exclusive property of Himax and shall not be distributed, reproduced, or disclosed in whole or in part without prior written permission of Himax.

5. Function Description

5.1 Tearing effect output line

The Tearing Effect output line supplies a panel synchronization signal. This signal can be enabled or disabled by the Tearing Effect Line Off & On commands. The mode of the Tearing Effect signal is defined by the parameter of the Tearing Effect Line On command. The signal can be used by the MPU to synchronize frame memory writing when displaying video images.

Tearing Effect Line Modes

Mode 1, the Tearing Effect Output signal consists of V-Blanking Information only:

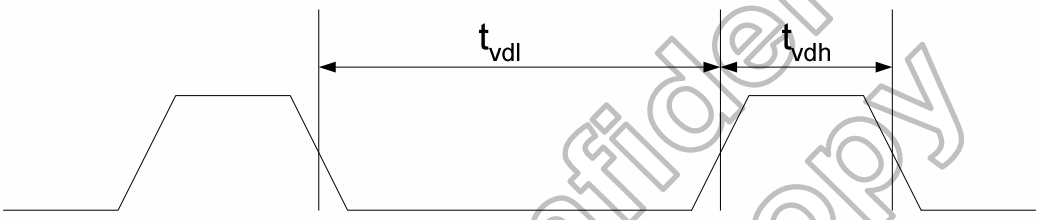


Figure 5.1: Tearing effect output signal mode 1

tvdh= The LCD display is not updated from the Frame Memory
tvdl= The LCD display is updated from the Frame Memory **(Except Invisible Line – see below)**
Under Mode1, the TE output timing will be defined by TEP[10:0] setting.
Ex: 1. FB=BP=0x01h **(3 line)** .
TEP[10:0]=0, then TE signal will output after last line finished.
TEP[10:0]=7, then TE signal will output at second line start.

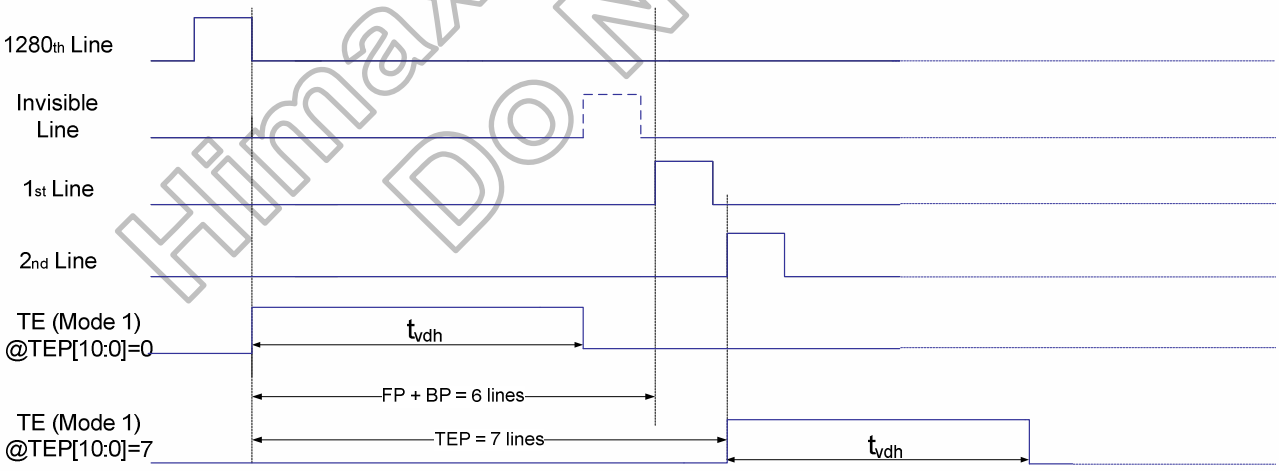
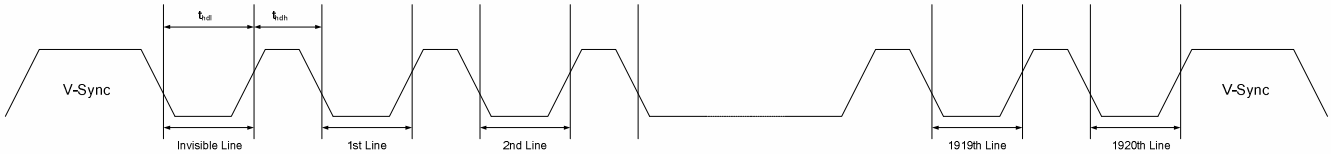


Figure 5.2: TE delay output

Mode 2, the Tearing Effect Output signal consists of V-Blanking and H-Blanking Information, there is one V-sync and N H-sync pulses per field.

N: 480+ 8xNL[7:0].



Himax Confidential Do Not Copy

This information contained herein is the exclusive property of Himax and shall not be distributed, reproduced, or disclosed in whole or in part without prior written permission of Himax.

Figure 5.3: Tearing Effect Output signal mode 2

thdh= The LCD display is not updated

thdl= The LCD display is updated

Under Mode2, the H-sync pulses output amount will be defined by TESL[15:0] setting.

Ex: 1. TESL[15:0]=0, then TE signal will like TE mode 1.

TESL[15:0]=1, then TE signal will output 1280 H-sync.

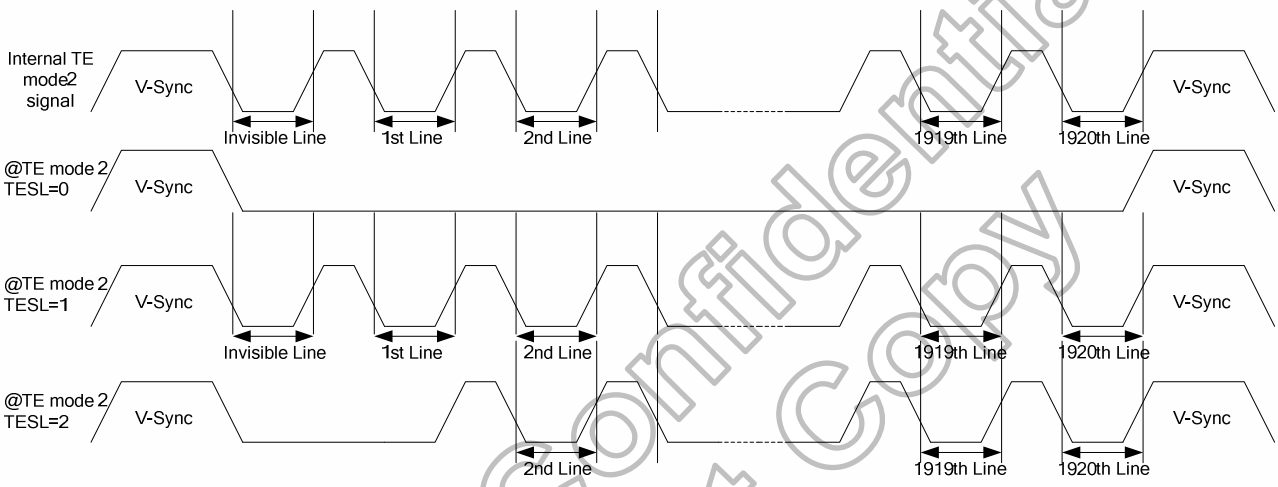


Figure 5.4: TE Output for TELINE setting

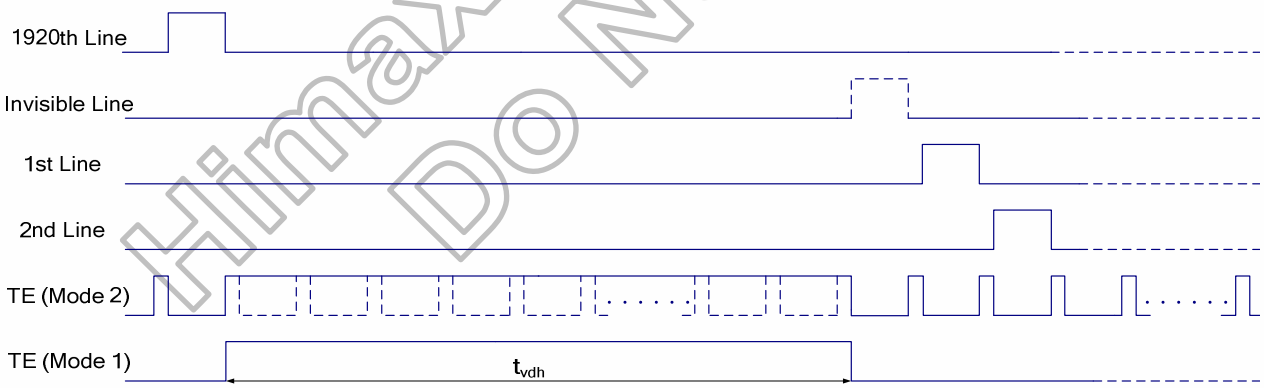


Figure 5.5: Tearing Effect Output signal

Note: During Sleep In Mode, the Tearing Output Pin is active Low

Himax Confidential Do Not Copy

5.1.1 Tearing effect line timing

The Tearing Effect signal is described below:

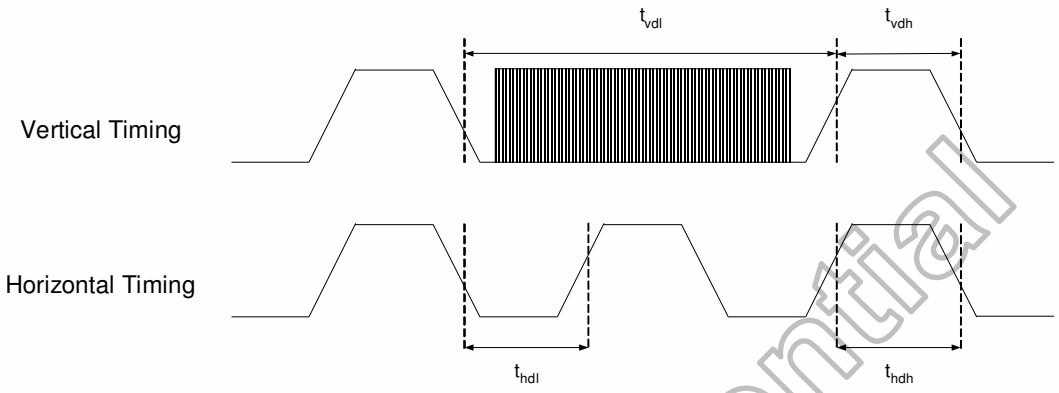


Figure 5.6: Tearing effect output line –tearing effect line timing

Idle Mode Off **(Resolution 720x1280 RGB, Frame Rate=60 Hz)**

Symbol	Parameter	Min.	Max.	Unit
tvdl	Vertical Timing Low Duration	15	-	ms
tvdh	Vertical Timing High Duration	VFP+VHP+VBP	-	us
tr	Rise time	-	15	ns
tf	Fall time	-	15	ns

Note: The timings in Table 5.1 apply when MADCTL ML=0 and ML=1

Table 5.1: AC characteristics of tearing effect signal

The signal's rise and fall times **(tf, tr)** are stipulated to be equal to or less than 15ns.

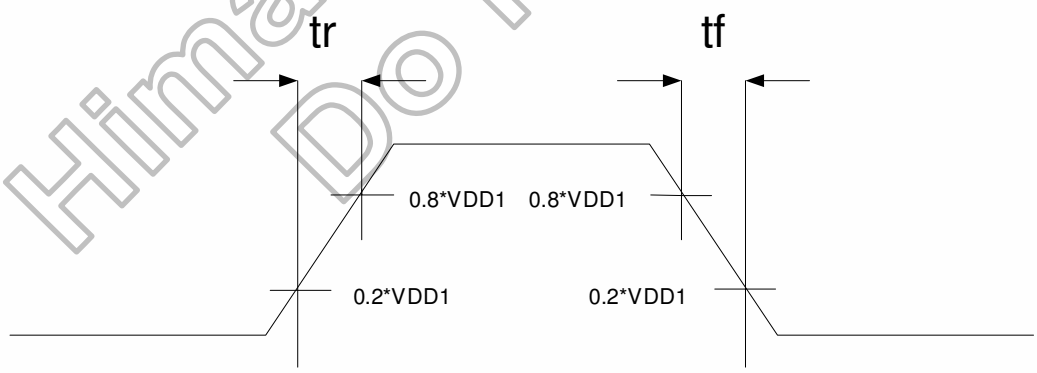


Figure 5.7: Tearing effect output line–definition of tf, tr

Himax Confidential
Do Not Copy

5.2 Oscillator

The HX8394-F can oscillate an internal R-C oscillator with an internal oscillation resistor **(Rf)**. The oscillation frequency is changed according to the internal register UADJ[4:0] if needed. Please refer to OSC control register CBh. The default frequency is 45MHz. The oscillation frequency tolerance is ±5%.

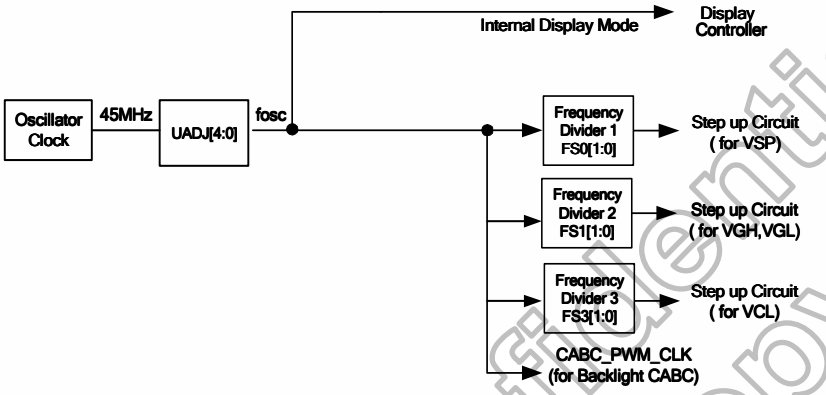


Figure 5.8: OSC architecture

Himax Confidential
Do Not Copy

This information contained herein is the exclusive property of Himax and shall not be distributed, reproduced, or disclosed in whole or in part without prior written permission of Himax.

5.3 Source driver

The HX8394-F contains 2162 channels of source driver (Normal S1~S2160; Zig-zag S1~S2160,SL1,SR1) which is used for driving the source line of TFT LCD panel. The source driver converts the digital data into the analog voltage for 2160 channels and generates corresponding gray scale voltage output, which can realize a 16.7M colors display simultaneously. Since the output circuit of this source driver incorporates an operational amplifier, a positive and a negative voltage can be alternately output from each channel.

H_RES[2:0]	Resolution	Source channels
000	720RGBx(480+ (8xNL))	S1~S2160
001	640RGBx(480+(8xNL))	S1~S960, S1200~S2160
010	600RGBx(480+(8xNL))	S1~S900,S1260~S2160

Table 5.2: Source output for Panel resolution

Himax Confidential
Do Not Copy

This information contained herein is the exclusive property of Himax and shall not be distributed, reproduced, or disclosed in whole or in part without prior written permission of Himax.

5.3.1 Zig-Zag inversion

The HX8394-F supports Zig-Zag inversion which can reduce power consumption. This inversion uses the same polarity as column inversion for data line and has almost the same display quality as 1-dot inversion.

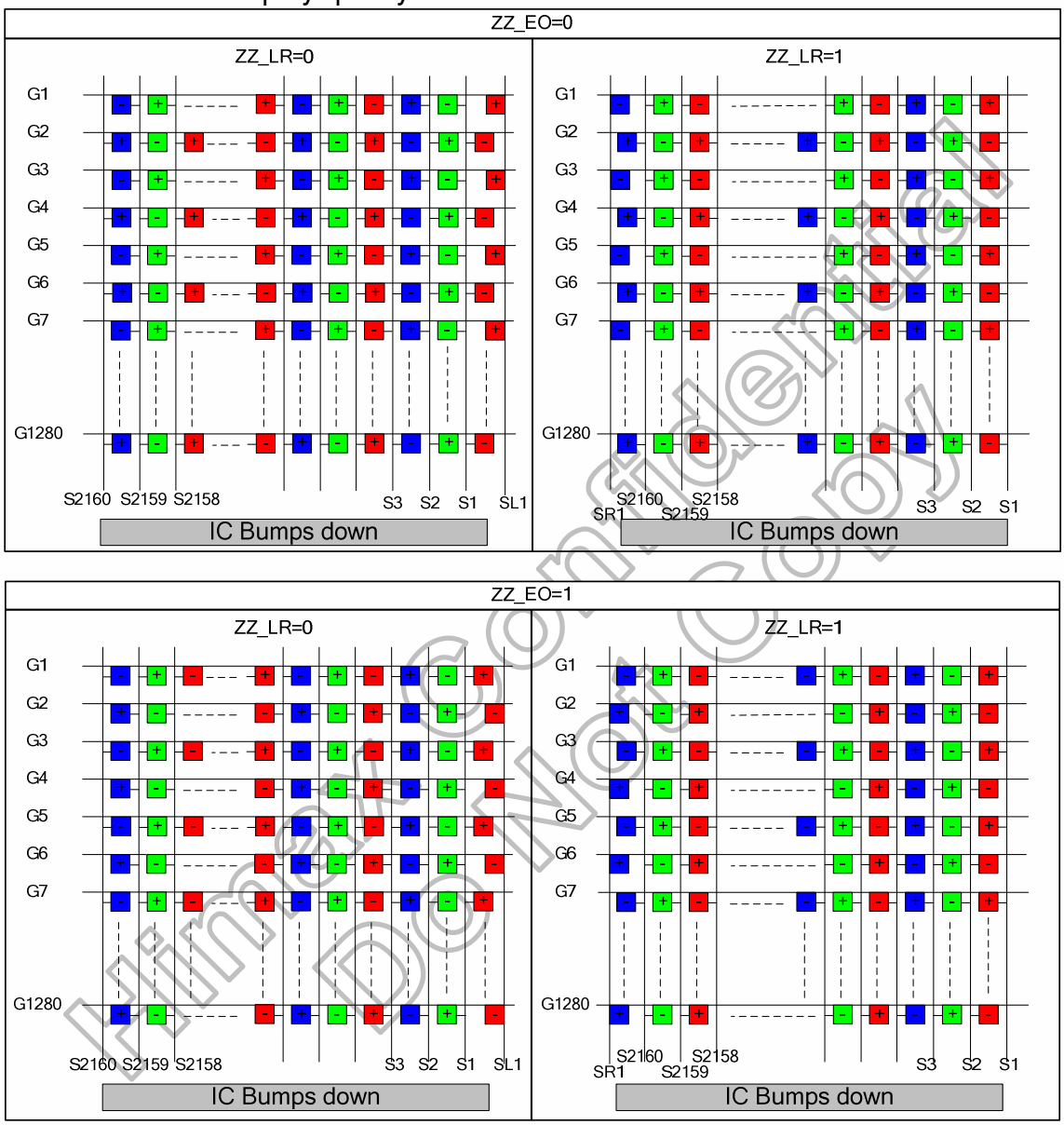


Figure 5.9: Zig-Zag inversion mode(ZZ_2PL=0)

Himax Confidential DO Not Copy

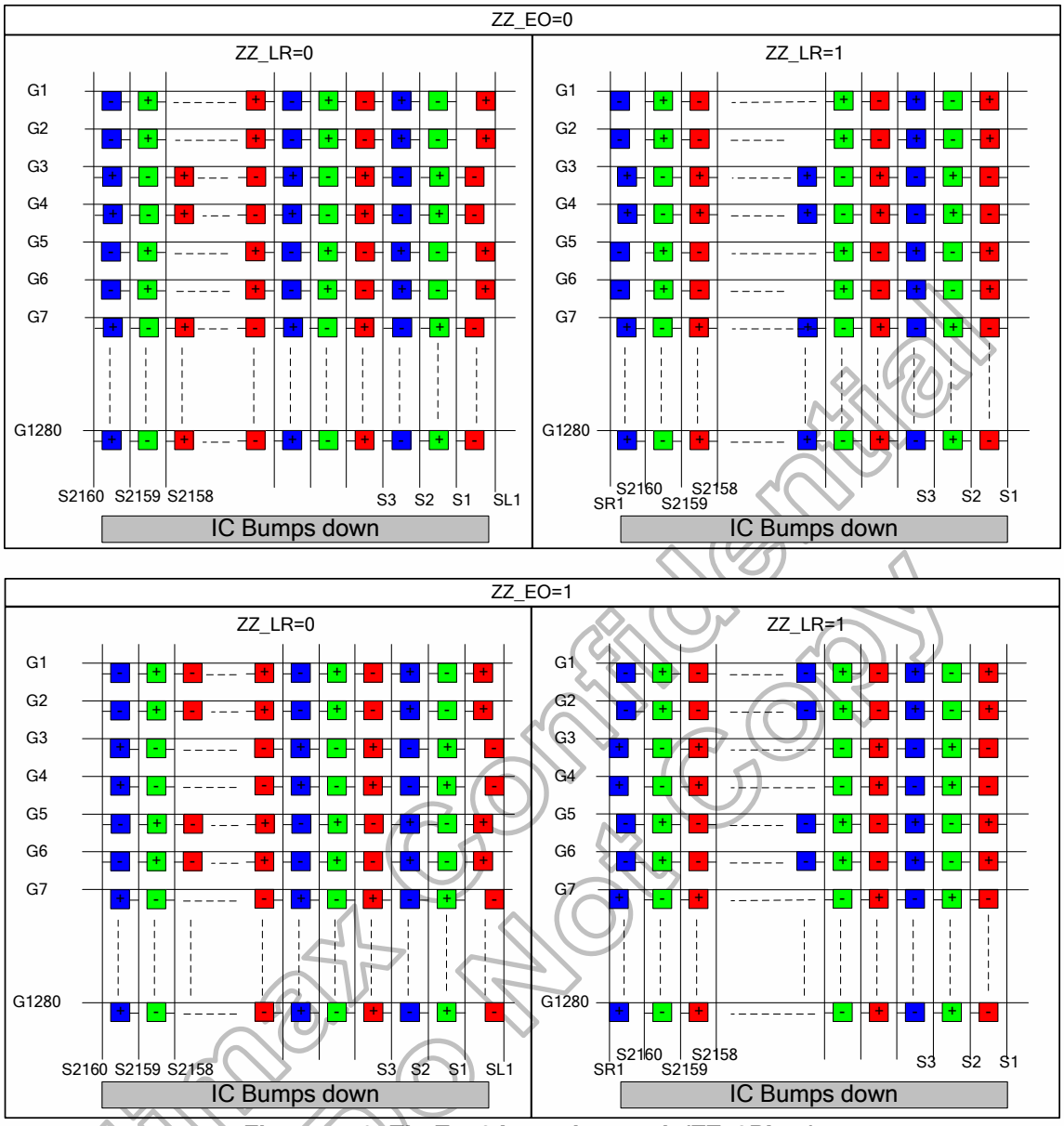


Figure 5.10: Zig-Zag2 Inversion mode(ZZ_2PL=1)

Himax Confidential
Do Not Copy

5.4 LCD power generation scheme

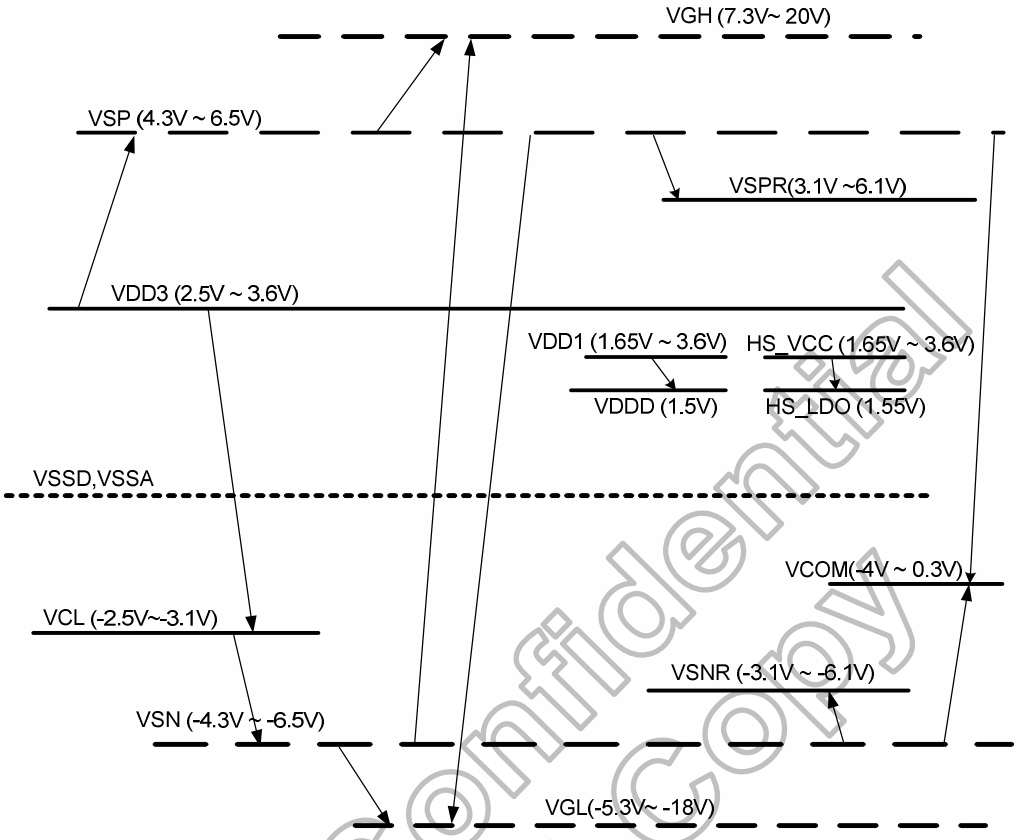


Figure 5.11: LCD power generation scheme for HX5186 and PFM mode

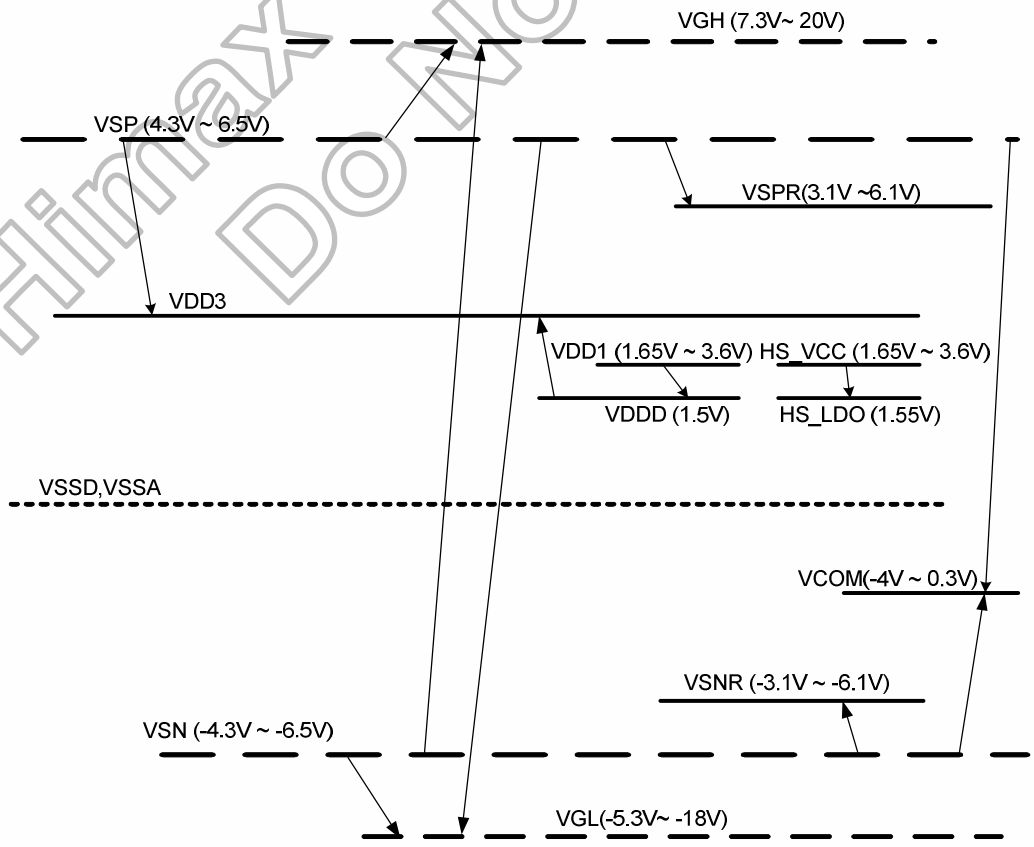


Figure 5.12: LCD power generation scheme for external power mode

Himax Confidential
Do Not Copy

This information contained herein is the exclusive property of Himax and shall not be distributed, reproduced, or disclosed in whole or in part without prior written permission of Himax.

5.5 DC/DC converter circuit

5.5.1 Charge pump and step up circuit mode

HX8394-F supports various kinds of power generation mode, including PFM Type A, PFM Type C, PFM Type D and external HX5186-A/B/C and external VSP & VSN power mode. All power power mode can be set by hardware pins PCCS[1:0] as below:

PCCS[1:0]	Power mode	Input Power	VSP	VSN	VGH/VGL
00	External VSP+VSN+VGH+VGL	VDD1	External	External	External
01	PFM-D (External VSP)	VDD1	External	Internal PFM	Internal DC/DC
10	HX5186 or PFM-A or PFM-C	VDD3, VDD1	Internal PFM or HX5186	Internal PFM or HX5186	Internal DC/DC
11	External VSP+VSN	VDD1	External	External	Internal DC/DC

Note: When use PCCS=[00,11] ,VDD3 is not needed.

Table 5.3: Power mode setting

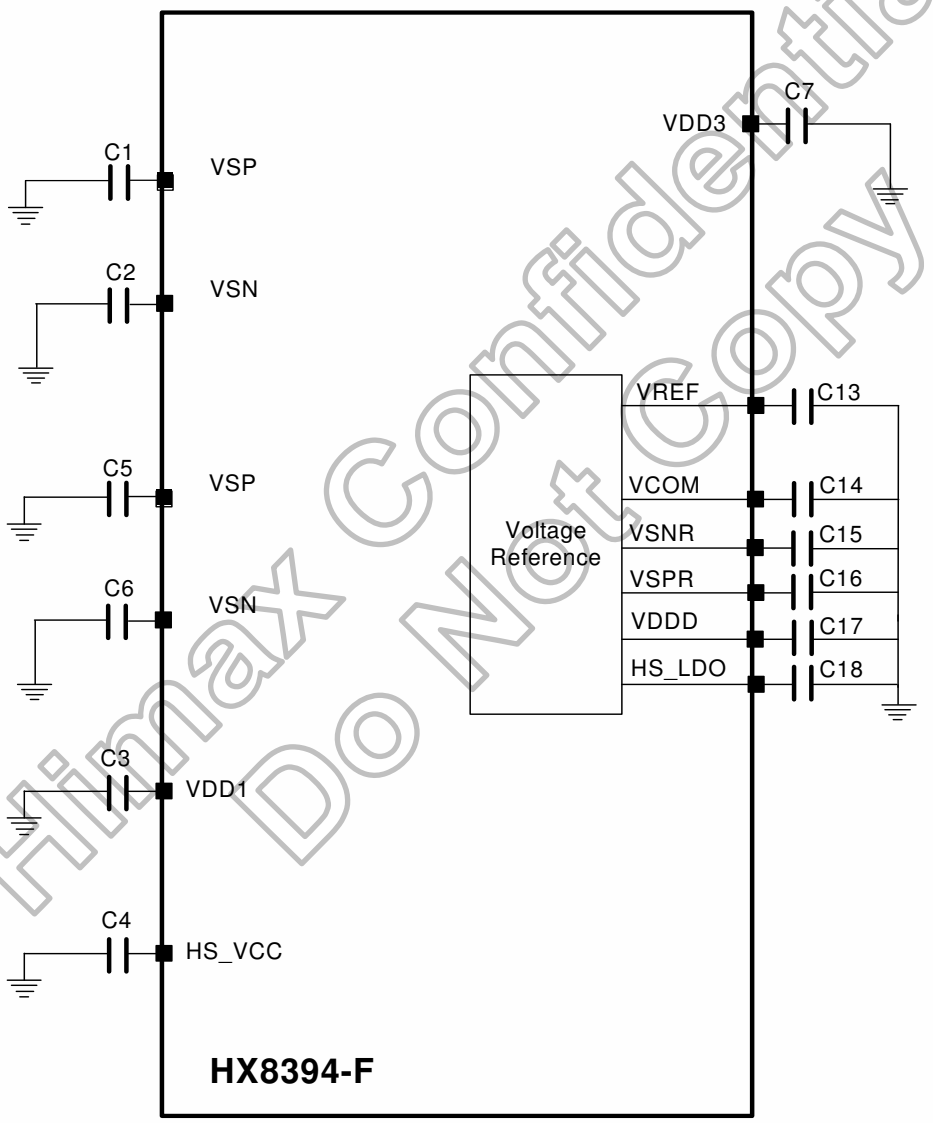
Himax Confidential
Do Not Copy

This information contained herein is the exclusive property of Himax and shall not be distributed, reproduced, or disclosed in whole or in part without prior written permission of Himax.

5.5.2 Use external VSP,VSN,VGH,VGL circuit

The input voltage range of VSP is from 4.3V ~ 6.5V. The input voltage range of VSN is from -4.3V ~ -6.5V. The input voltage range of VGH is from 7.3V ~ 20V. The input voltage range of VGL is from -5.3V ~ -18V. Please set VGH-VGL<32V, and VGL<VSN, VSP<VGH.

Use external VSP,VSN,VGH,VGL circuit. (PCCS[1:0]="00")



Himax Confidential
Do Not Copy

Figure 5.13: DC/DC converter circuit of external VSP/VSN/VGH/VGL

5.5.3 Use PFM DC/DC converter

The PFM DC-DC converter generates the high voltage level VSP/VSN required for source drivers. HX8394-F contains sub-circuits of the PFM boost converter, including a precision 1.8V reference voltage, comparator, PFM controlling logic, and the output buffer. The boost converter uses a external power transistor to provide maximum efficiency and to minimize the number of external components. The output voltage of the boost converter can be set from +4.3V to +6.5V (VSP) and -4.3V to -6.5V (VSN).

Use PFM DC/DC converter (PCCS[1:0]="01")

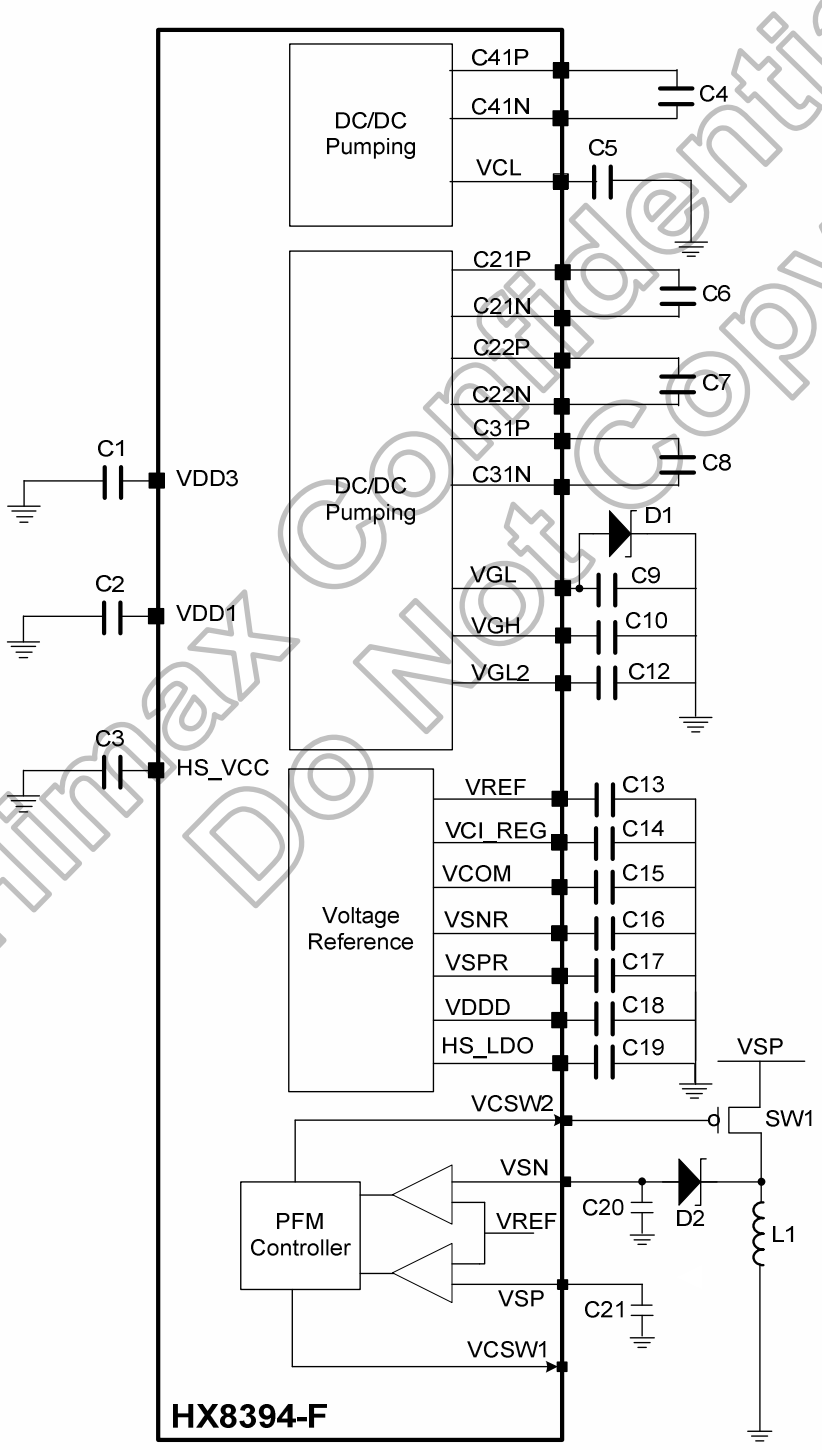


Figure 5.14: DC/DC converter circuit (PFM type D)

Himax Confidential
Do Not Copy

Use PFM DC/DC converter (PCCS[1:0]="10") and (POWMOD[1:0]="00")

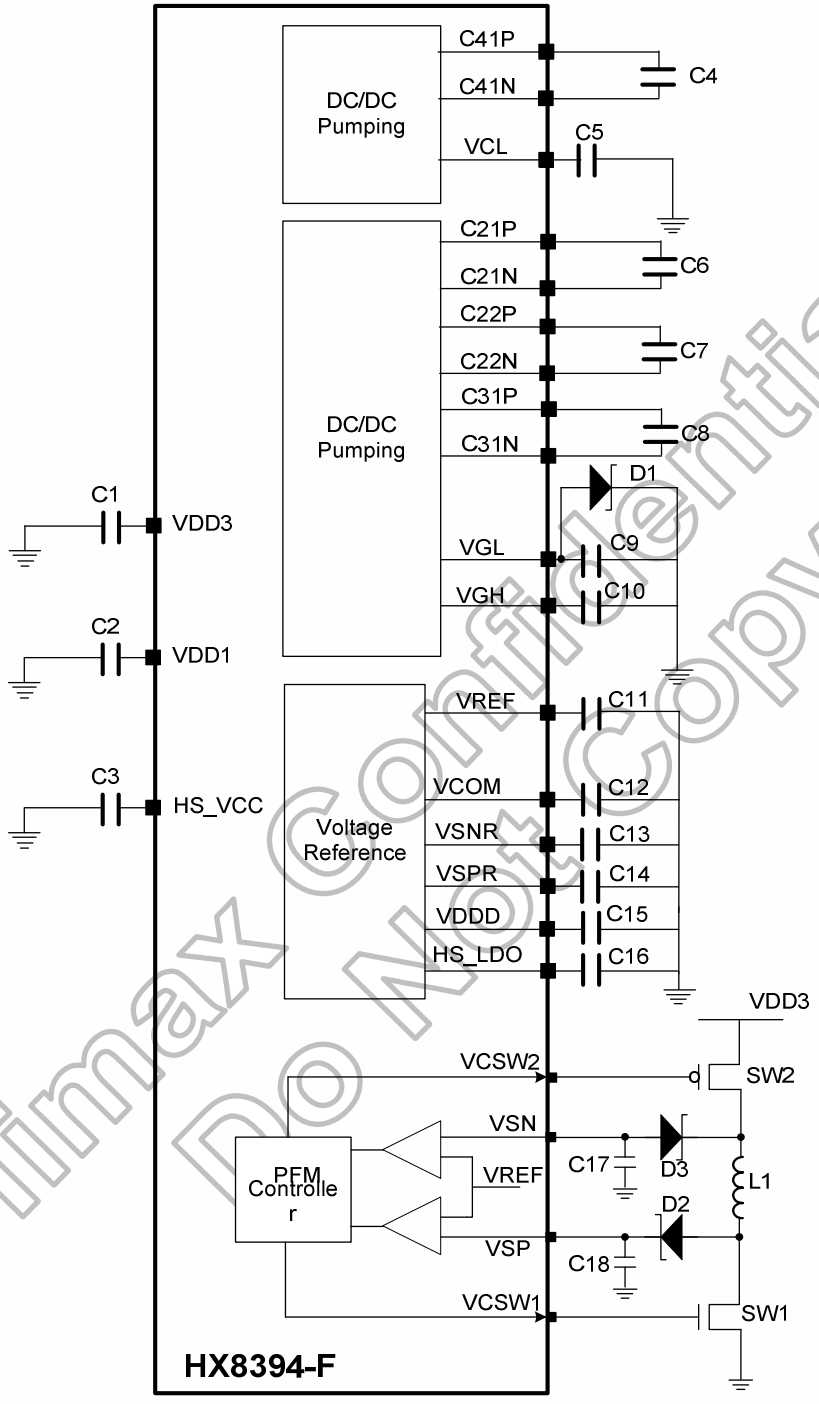


Figure 5.15: DC/DC converter circuit (PFM type A)

Himax Confidential
Do Not Copy

Use PFM DC/DC converter (PCCS[1:0]="10") and (POWMOD[1:0]="01")

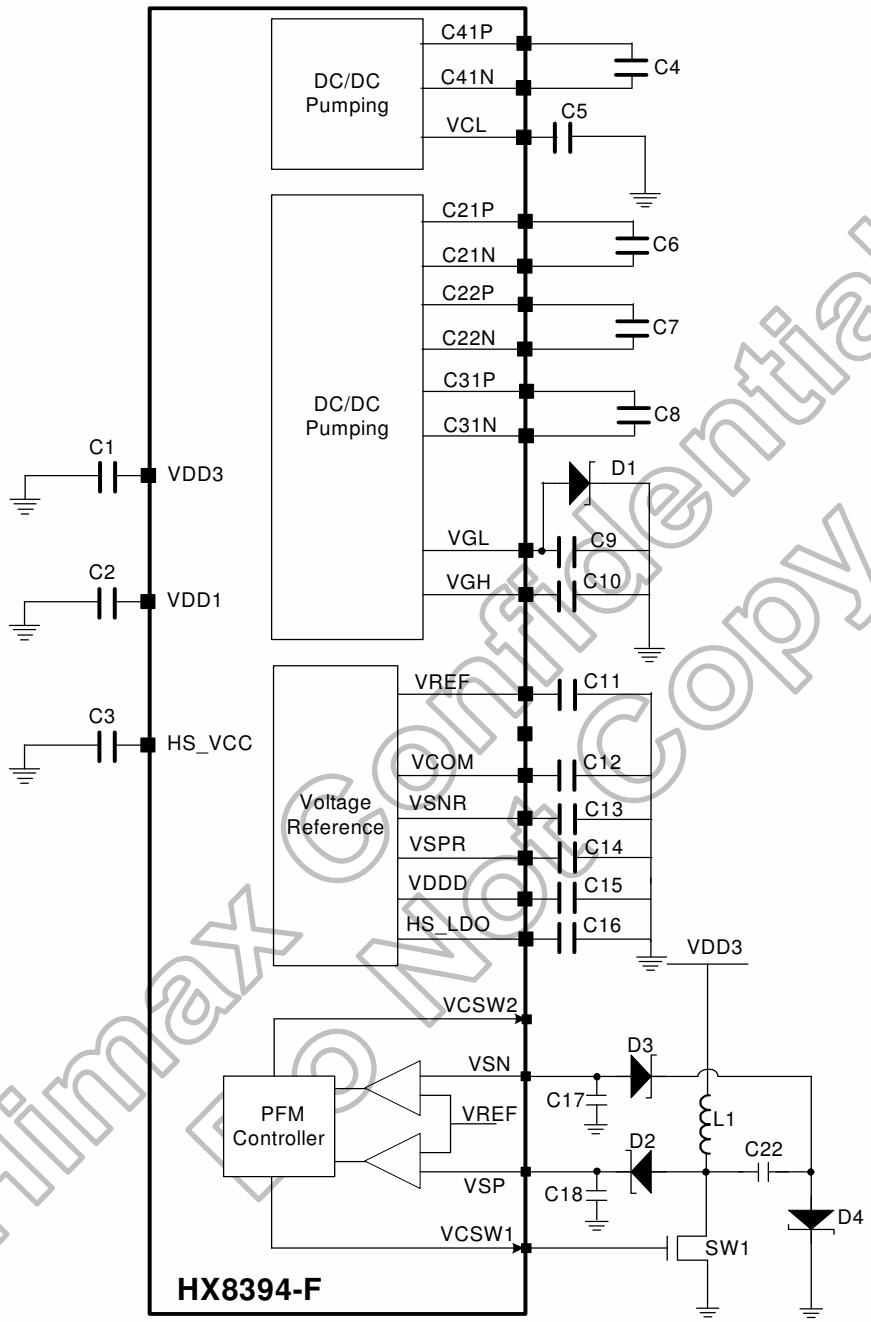


Figure 5.16: DC/DC converter circuit (PFM type C)

Himax Confidential
Do Not Copy

5.5.4 Use HX5186-A/B/C

The HX5186-A/B/C is highly efficient switching voltage generator circuits that generate the high voltage level VSP/VSN required for source drivers. HX8394-F contains Charge Pump Controller for HX5186, including a comparator for VSP/VSN feedback control. HX5186-A/B/C can provide maximum efficiency and use minimum number of external components. The output voltage of the boost converter can be set from 4.3V to 6.5V (VSP) and -4.3V to -6.5V (VSN).

(PCCS[1:0]="10") and (POWMOD[1:0]="10" ,"11")

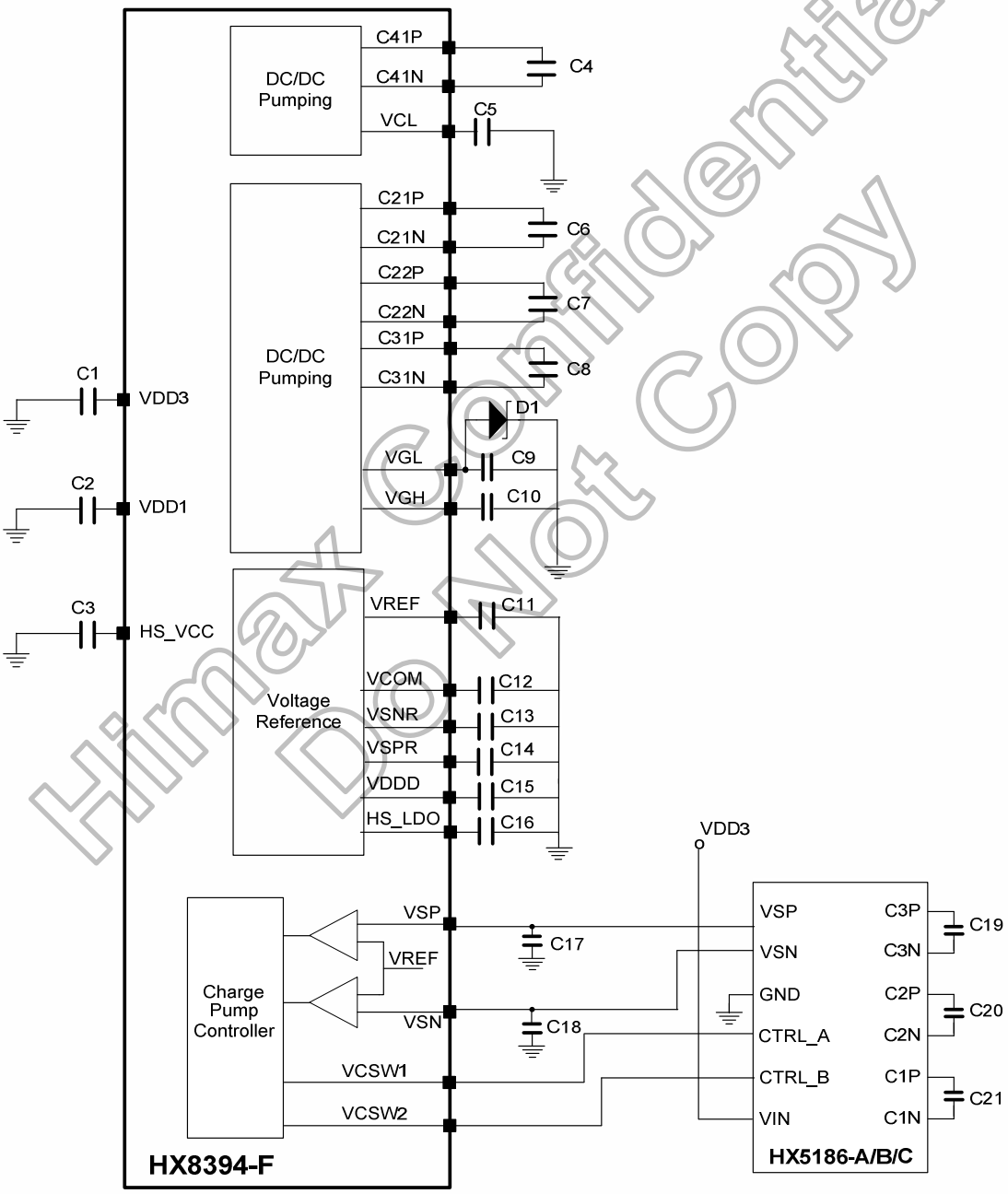


Figure 5.17: DC/DC converter circuit of HX5186-A/B/C

Himax Confidential
Do Not Copy

5.5.5 Use external VSP,VSN circuit

The input voltage range of VSP is from 4.3V ~ 6.5V. The input voltage range of VSN is from -4.3V ~ -6.5V.

Use external VSP,VSN circuit. (PCCS[1:0]="11")

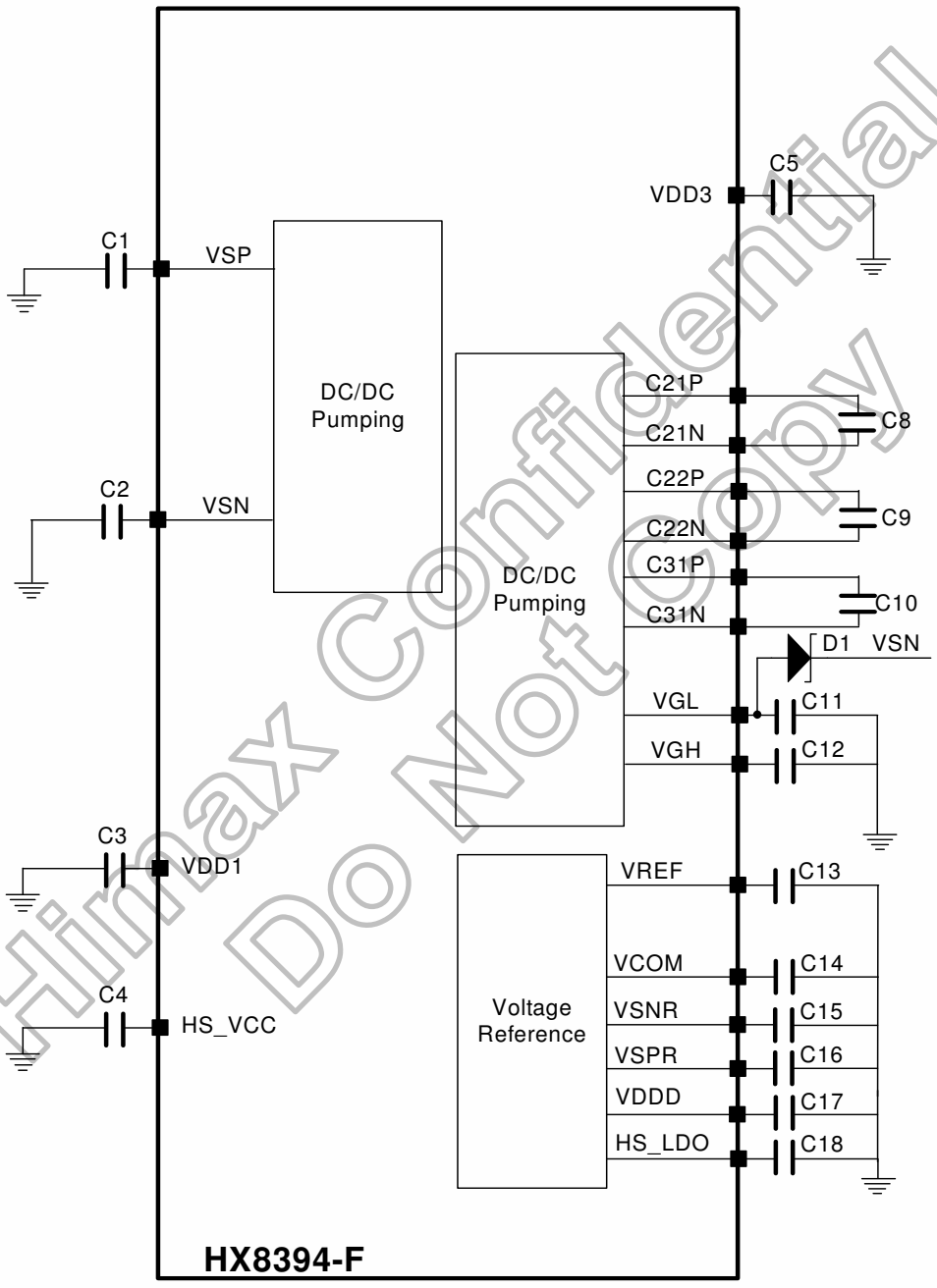
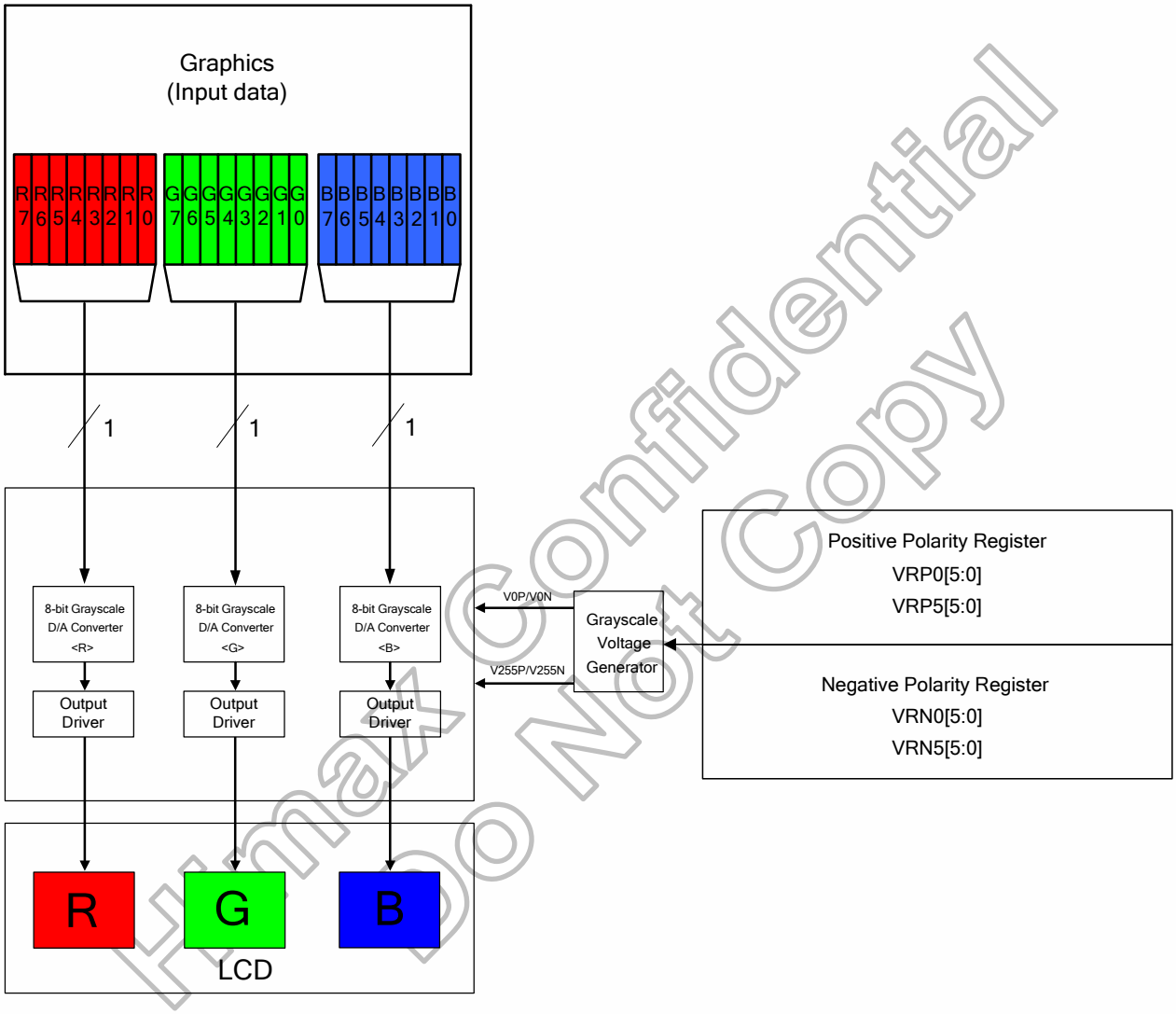


Figure 5.18: DC/DC converter circuit of external VSP/VSN

Himax Confidential
Do Not Copy

5.6 8-color Idle display mode

The HX8394-F supports an 8-color idle display mode. The grayscale level to be used is V0 and V255 with R7, G7, B7 decoding, and the other levels (V1-V254) are halted to reduce power consumption. In idle display mode, the Gamma-micro-adjustment registers are invalid and only the upper bits of RGB are used for display.



Himax Confidential
Do Not Copy

Figure 5.19: Idle mode grayscale control

This information contained herein is the exclusive property of Himax and shall not be distributed, reproduced, or disclosed in whole or in part without prior written permission of Himax.

5.7 1-bit GRAM Idle Display Mode

HX8394-F support display data from GRAM in Idle mode. User can use 2Ch/3Ch command to write image data into GRAM. R/G/B MSB bit data stored in GRAM. GRAM write direction not support MX/MY/MV function.

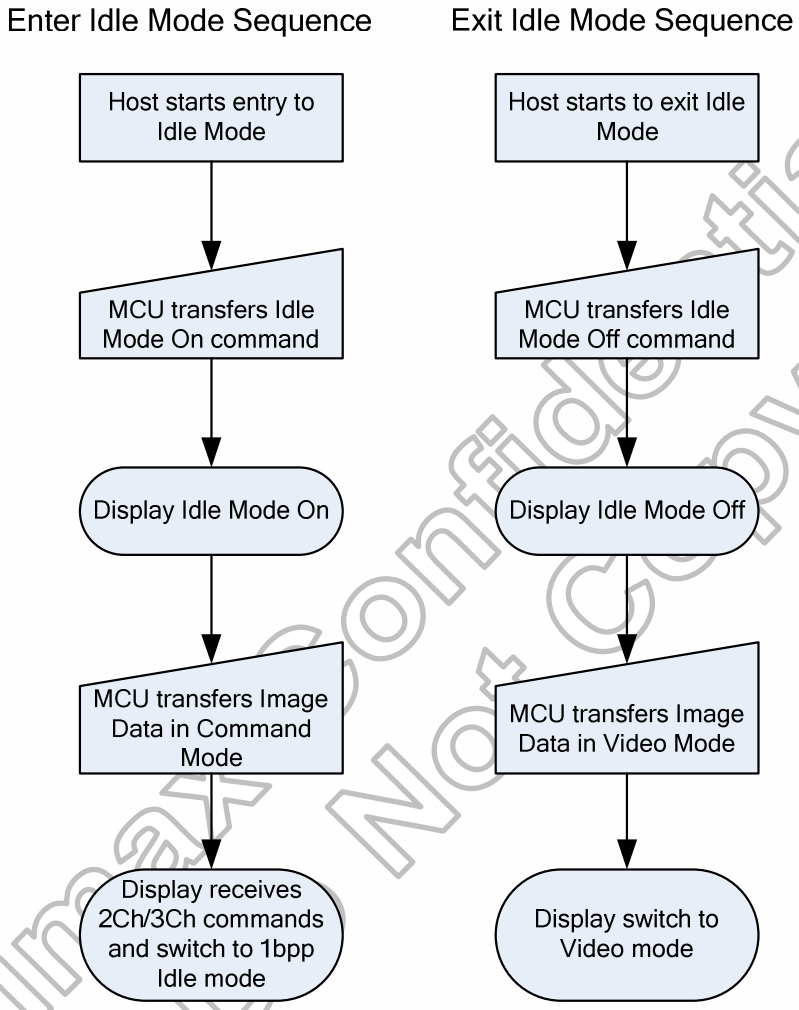


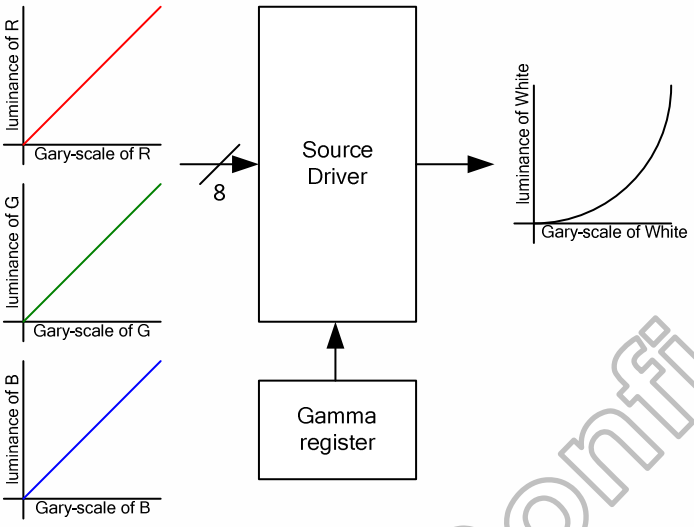
Figure 5.20: Idle Mode On/Off Sequence

Himax Confidential Do Not Copy

5.8 Gamma characteristic correction function

The HX8394-F offers two kinds of Gamma adjustment ways to come to accord with LC characteristic, one kind is through Source Driver directly, another one is adjusted by the digital gamma correction. The Gamma adjustment way is selected by internal register DGC_EN bit.

A) Gamma adjustment of Source Driver



B) Gamma adjustment of Digital Gamma Correction

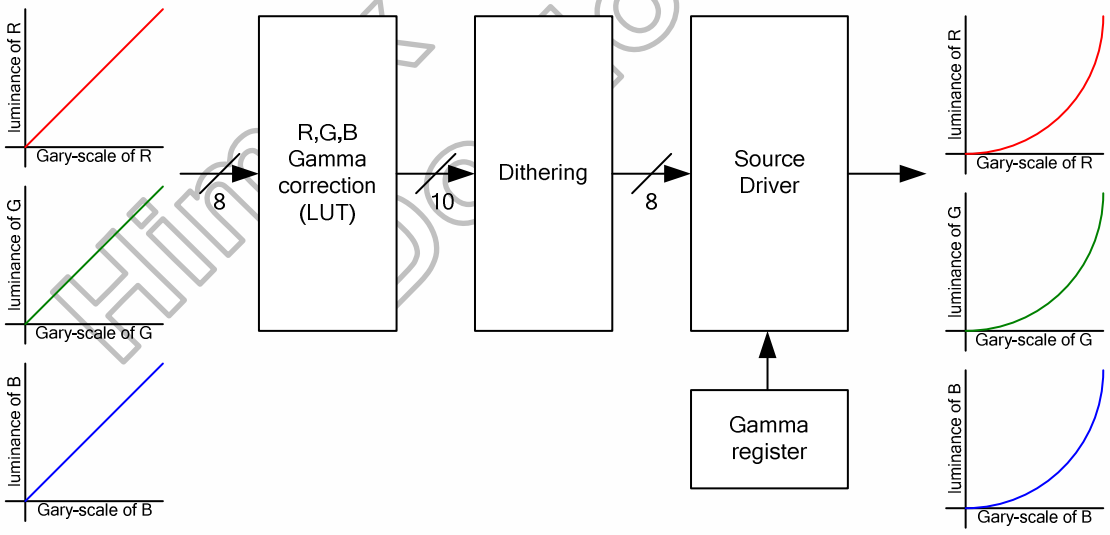
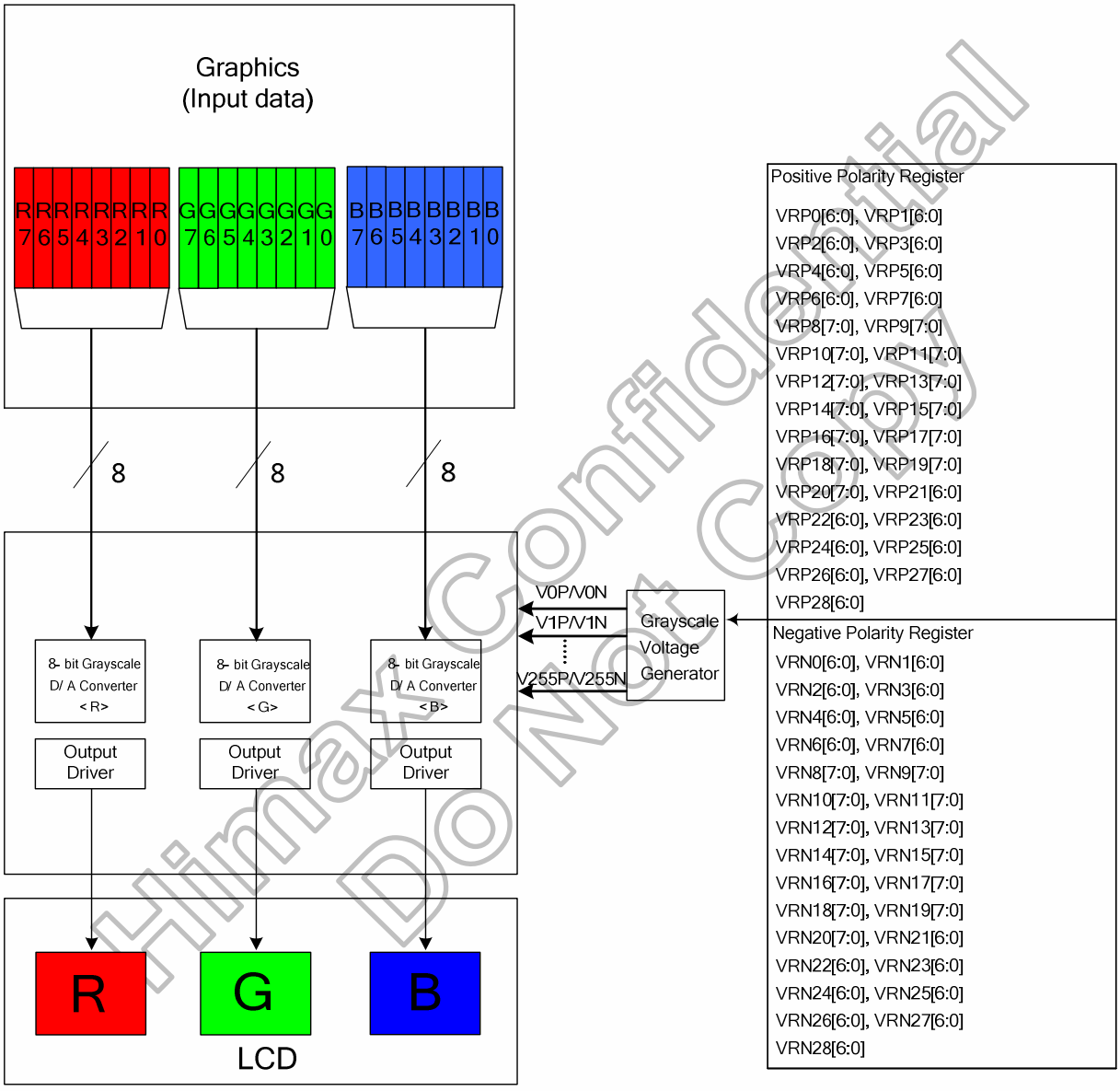


Figure 5.21: Gamma adjustments different of source driver with digital gamma correction

Himax Confidential Do not Copy

The HX8394-F incorporates gamma adjustment function for the 16,777,216-color display (256 grayscale for each R, G, B color). Gamma adjustment operation is implemented by deciding the 29 grayscale levels firstly in gamma adjustment control registers to match the LCD panel. Then total 256 grayscale levels are generated in grayscale voltage generator. These registers are available for both polarities.



Himax Confidential
Do Not Copy

Figure 5.22: Grayscale control

5.8.1 Gamma-characteristics adjustment register

The HX8394-F has register groups for specifying a series grayscale voltage that meets the Gamma-characteristics for the LCD panel used. These registers are divided into two groups, which correspond to the gradient, amplitude, and macro adjustment of the voltage for the grayscale characteristics. The polarity of each register can be specified independently.

(1) Center adjustment registers

This gamma adjustment registers are used to adjust the reference gamma voltage for center grayscale level. This function is implemented by controlling the 256-to-1 selector in the gamma resister stream for reference gamma voltage generation. These registers are available for both positive and negative polarities.

(2) Edge adjustment registers

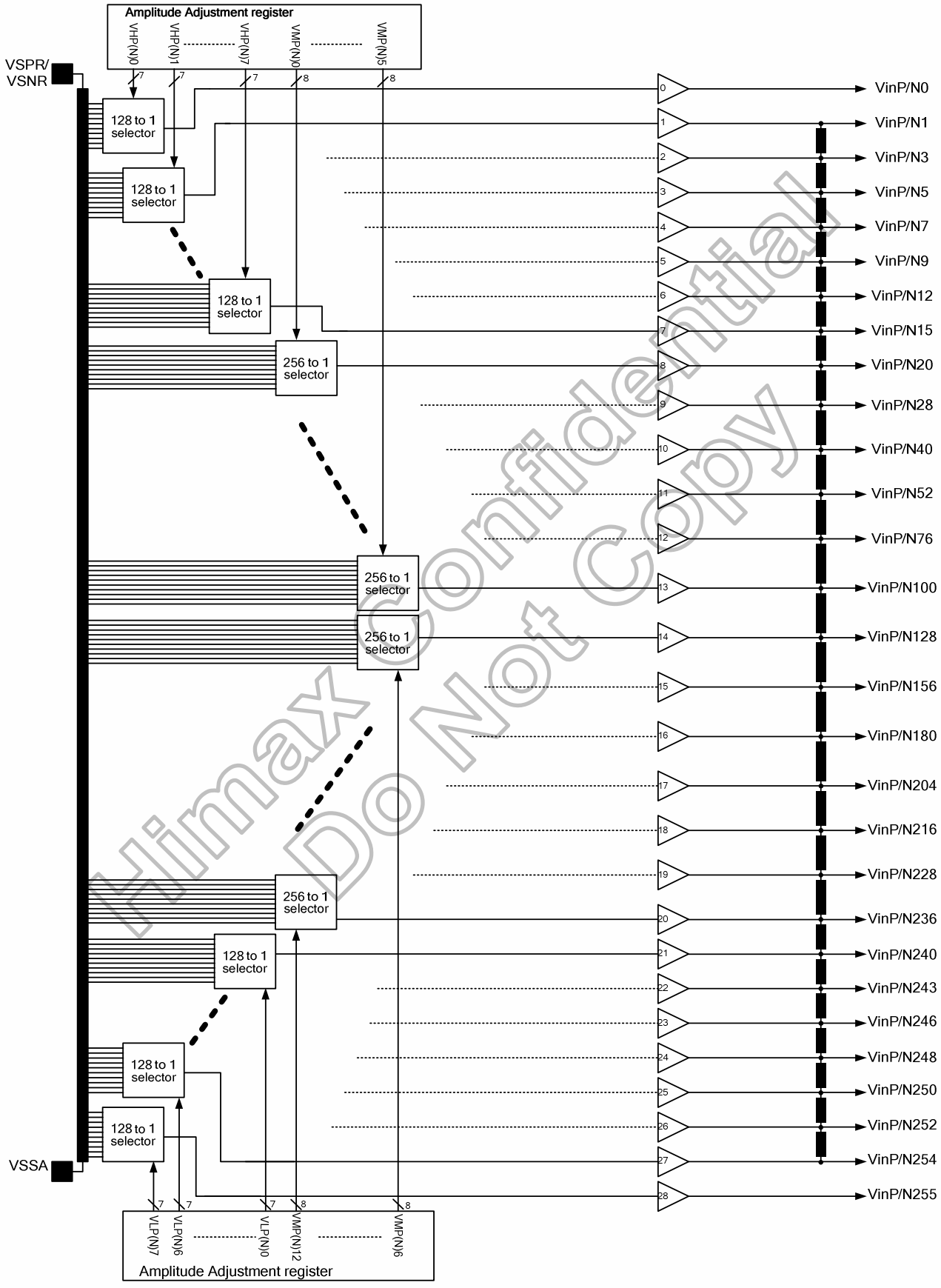
This gamma adjustment registers are used to adjust the reference gamma voltage for both edge grayscale level. This function is implemented by controlling the 128-to-1 selector in the gamma resister stream for reference gamma voltage generation. These registers are available for both positive and negative polarities.

Register Groups	Positive Polarity	Negative Polarity	Description
Up Edge adjustment	VHP0 6-0	VHN0 6-0	128-to-1 selector (voltage level of grayscale 0)
	VHP1 6-0	VHN1 6-0	128-to-1 selector (voltage level of grayscale 1)
	VHP2 6-0	VHN2 6-0	128-to-1 selector (voltage level of grayscale 3)
	VHP3 6-0	VHN3 6-0	128-to-1 selector (voltage level of grayscale 5)
	VHP4 6-0	VHN4 6-0	128-to-1 selector (voltage level of grayscale 7)
	VHP5 6-0	VHN5 6-0	128-to-1 selector (voltage level of grayscale 9)
	VHP6 6-0	VHN6 6-0	128-to-1 selector (voltage level of grayscale 12)
	VHP7 6-0	VHN7 6-0	128-to-1 selector (voltage level of grayscale 15)
Center adjustment	VMP0 7-0	VMN0 7-0	255-to-1 selector (voltage level of grayscale 20)
	VMP1 7-0	VMN1 7-0	255-to-1 selector (voltage level of grayscale 28)
	VMP2 7-0	VMN2 7-0	255-to-1 selector (voltage level of grayscale 40)
	VMP3 7-0	VMN3 7-0	255-to-1 selector (voltage level of grayscale 52)
	VMP4 7-0	VMN4 7-0	255-to-1 selector (voltage level of grayscale 76)
	VMP5 7-0	VMN5 7-0	255-to-1 selector (voltage level of grayscale 100)
	VMP6 7-0	VMN6 7-0	255-to-1 selector (voltage level of grayscale 128)
	VMP7 7-0	VMN7 7-0	255-to-1 selector (voltage level of grayscale 156)
	VMP8 7-0	VMN8 7-0	255-to-1 selector (voltage level of grayscale 180)
	VMP9 7-0	VMN9 7-0	255-to-1 selector (voltage level of grayscale 204)
	VMP10 7-0	VMN10 7-0	255-to-1 selector (voltage level of grayscale 216)
	VMP11 7-0	VMN11 7-0	255-to-1 selector (voltage level of grayscale 228)
	VMP12 7-0	VMN12 7-0	255-to-1 selector (voltage level of grayscale 236)
Down Edge adjustment	VLP0 6-0	VLN0 6-0	128-to-1 selector (voltage level of grayscale 240)
	VLP1 6-0	VLN1 6-0	128-to-1 selector (voltage level of grayscale 243)
	VLP2 6-0	VLN2 6-0	128-to-1 selector (voltage level of grayscale 246)
	VLP3 6-0	VLN3 6-0	128-to-1 selector (voltage level of grayscale 248)
	VLP4 6-0	VLN4 6-0	128-to-1 selector (voltage level of grayscale 250)
	VLP5 6-0	VLN5 6-0	128-to-1 selector (voltage level of grayscale 252)
	VLP6 6-0	VLN6 6-0	128-to-1 selector (voltage level of grayscale 254)
	VLP7 6-0	VLN7 6-0	128-to-1 selector (voltage level of grayscale 255)

Table 5.4: Gamma-adjustment registers

Himax Confidential
Do Not Copy

Gamma resister stream and 8 to 1 selector



Himax Confidential
Do Not Copy

Figure 5.23: Gamma resister stream and gamma reference voltage

Variable resister

There are two types of variable resistors, one is for center adjustment and the other is for edge adjustment. The resistances are decided by setting values in the center adjustment and edge adjustment registers. Their relationships are shown below.

Value in Register VH(P/N)0~7 [6:0]	Resistance VR(P/N) 0~7	Value in Register VL(P/N)0~7 [6:0]	Resistance VL(P/N) 0~7
000_0000	0R	000_0000	0R
000_0001	2R	000_0001	2R
000_0010	4R	000_0010	4R
000_0011	6R	000_0011	6R
•	•	•	•
•	•	•	•
011_1101	122R	011_1101	122R
011_1110	124R	011_1110	124R
011_1111	126R	011_1111	126R
100_0000	128R	100_0000	128R
100_0001	130R	100_0001	130R
100_0010	132R	100_0010	132R
•	•	•	•
•	•	•	•
111_1101	250R	111_1101	250R
111_1110	252R	111_1110	252R
111_1111	254R	111_1111	254R

Table 5.5: Edge adjustment 0~7, 21~28

Value in Register VM(P/N)0~12 [7:0]	Resistance VM(P/N)0~12
0000_0000	0R
0000_0001	1R
0000_0010	2R
•	•
•	•
0101_0101	85R
0101_0110	86R
0101_0111	87R
•	•
•	•
1101_0101	213R
1101_0110	214R
1101_0111	215R
•	•
•	•
1111_1101	253R
1111_1110	254R
1111_1111	255R

Table 5.6: Center adjustment 8~20

Himax Confidential Do Not Copy

The grayscale levels are determined by the following formulas:

Reference voltage	Macro adjustment value	VinP/N0 formula
VinP/N0	VHP0 [6:0] = 000_0000	VSP/NR
	VHP0 [6:0] = 000_0001	((600R – 2R) / 600R) * VSP/NR
	VHP0 [6:0] = 000_0010	((600R – 4R) / 600R) * VSP/NR
	VHP0 [6:0] = 000_0011	((600R – 6R) / 600R) * VSP/NR
	VHP0 [6:0] = 000_0100	((600R – 8R) / 600R) * VSP/NR
	VHP0 [6:0] = 000_0101	((600R – 10R) / 600R) * VSP/NR
	:	:
	VHP0 [6:0] = 100_0001	((600R – 130R) / 600R) * VSP/NR
	VHP0 [6:0] = 100_0010	((600R – 132R) / 600R) * VSP/NR
	VHP0 [6:0] = 100_0011	((600R – 134R) / 600R) * VSP/NR
	VHP0 [6:0] = 100_0100	((600R – 136R) / 600R) * VSP/NR
	VHP0 [6:0] = 100_0101	((600R – 138R) / 600R) * VSP/NR
	:	:
	VHP0 [6:0] = 111_1011	((600R – 246R) / 600R) * VSP/NR
	VHP0 [6:0] = 111_1100	((600R – 248R) / 600R) * VSP/NR
	VHP0 [6:0] = 111_1101	((600R – 250R) / 600R) * VSP/NR
	VHP0 [6:0] = 111_1110	((600R – 252R) / 600R) * VSP/NR
	VHP0 [6:0] = 111_1111	((600R – 254R) / 600R) * VSP/NR

Table 5.7: VinP/N0

Reference voltage	Macro adjustment value	VinP/N1 formula
VinP/N1	VHP1 [6:0] = 000_0000	(596R / 600R) * VSP/NR
	VHP1 [6:0] = 000_0001	((596R – 2R) / 600R) * VSP/NR
	VHP1 [6:0] = 000_0010	((596R – 4R) / 600R) * VSP/NR
	VHP1 [6:0] = 000_0011	((596R – 6R) / 600R) * VSP/NR
	VHP1 [6:0] = 000_0100	((596R – 8R) / 600R) * VSP/NR
	VHP1 [6:0] = 000_0101	((596R – 10R) / 600R) * VSP/NR
	:	:
	VHP1 [6:0] = 100_0001	((596R – 130R) / 600R) * VSP/NR
	VHP1 [6:0] = 100_0010	((596R – 132R) / 600R) * VSP/NR
	VHP1 [6:0] = 100_0011	((596R – 134R) / 600R) * VSP/NR
	VHP1 [6:0] = 100_0100	((596R – 136R) / 600R) * VSP/NR
	VHP1 [6:0] = 100_0101	((596R – 138R) / 600R) * VSP/NR
	:	:
	VHP1 [6:0] = 111_1011	((596R – 246R) / 600R) * VSP/NR
	VHP1 [6:0] = 111_1100	((596R – 248R) / 600R) * VSP/NR
	VHP1 [6:0] = 111_1101	((596R – 250R) / 600R) * VSP/NR
	VHP1 [6:0] = 111_1110	((596R – 252R) / 600R) * VSP/NR
	VHP1 [6:0] = 111_1111	((596R – 254R) / 600R) * VSP/NR

Table 5.8: VinP/N1

Reference voltage	Macro adjustment value	VinP/N3 formula
VinP/N3	VHP2 [6:0] = 000_0000	(592R / 600R) * VSP/NR
	VHP2 [6:0] = 000_0001	((592R – 2R) / 600R) * VSP/NR
	VHP2 [6:0] = 000_0010	((592R – 4R) / 600R) * VSP/NR
	VHP2 [6:0] = 000_0011	((592R – 6R) / 600R) * VSP/NR
	VHP2 [6:0] = 000_0100	((592R – 8R) / 600R) * VSP/NR
	VHP2 [6:0] = 000_0101	((592R – 10R) / 600R) * VSP/NR
	:	:
	VHP2 [6:0] = 100_0001	((592R – 130R) / 600R) * VSP/NR
	VHP2 [6:0] = 100_0010	((592R – 132R) / 600R) * VSP/NR
	VHP2 [6:0] = 100_0011	((592R – 134R) / 600R) * VSP/NR
	VHP2 [6:0] = 100_0100	((592R – 136R) / 600R) * VSP/NR
	VHP2 [6:0] = 100_0101	((592R – 138R) / 600R) * VSP/NR
	:	:
	VHP2 [6:0] = 111_1011	((592R – 246R) / 600R) * VSP/NR
	VHP2 [6:0] = 111_1100	((592R – 248R) / 600R) * VSP/NR
	VHP2 [6:0] = 111_1101	((592R – 250R) / 600R) * VSP/NR
	VHP2 [6:0] = 111_1110	((592R – 252R) / 600R) * VSP/NR
	VHP2 [6:0] = 111_1111	((592R – 254R) / 600R) * VSP/NR

Table 5.9: VinP/N3

Himax Confidential Do Not Copy

Reference voltage	Macro adjustment value	VinP/N5 formula
VinP/N5	VHP3 [6:0] = 000_0000	(588R / 600R) * VSP/NR
	VHP3 [6:0] = 000_0001	((588R – 2R) / 600R) * VSP/NR
	VHP3 [6:0] = 000_0010	((588R – 4R) / 600R) * VSP/NR
	VHP3 [6:0] = 000_0011	((588R – 6R) / 600R) * VSP/NR
	VHP3 [6:0] = 000_0100	((588R – 8R) / 600R) * VSP/NR
	VHP3 [6:0] = 000_0101	((588R – 10R) / 600R) * VSP/NR
	:	:
	VHP3 [6:0] = 100_0001	((588R – 130R) / 600R) * VSP/NR
	VHP3 [6:0] = 100_0010	((588R – 132R) / 600R) * VSP/NR
	VHP3 [6:0] = 100_0011	((588R – 134R) / 600R) * VSP/NR
	VHP3 [6:0] = 100_0100	((588R – 136R) / 600R) * VSP/NR
	VHP3 [6:0] = 100_0101	((588R – 138R) / 600R) * VSP/NR
	:	:
	VHP3 [6:0] = 111_1011	((588R – 246R) / 600R) * VSP/NR
	VHP3 [6:0] = 111_1100	((588R – 248R) / 600R) * VSP/NR
	VHP3 [6:0] = 111_1101	((588R – 250R) / 600R) * VSP/NR
	VHP3 [6:0] = 111_1110	((588R – 252R) / 600R) * VSP/NR
	VHP3 [6:0] = 111_1111	((588R – 254R) / 600R) * VSP/NR

Table 5.10: VinP/N5

Reference voltage	Macro adjustment value	VinP/N7 formula
VinP/N7	VHP4 [6:0] = 000_0000	(580R / 600R) * VSP/NR
	VHP4 [6:0] = 000_0001	((580R – 2R) / 600R) * VSP/NR
	VHP4 [6:0] = 000_0010	((580R – 4R) / 600R) * VSP/NR
	VHP4 [6:0] = 000_0011	((580R – 6R) / 600R) * VSP/NR
	VHP4 [6:0] = 000_0100	((580R – 8R) / 600R) * VSP/NR
	VHP4 [6:0] = 000_0101	((580R – 10R) / 600R) * VSP/NR
	:	:
	VHP4 [6:0] = 100_0001	((580R – 130R) / 600R) * VSP/NR
	VHP4 [6:0] = 100_0010	((580R – 132R) / 600R) * VSP/NR
	VHP4 [6:0] = 100_0011	((580R – 134R) / 600R) * VSP/NR
	VHP4 [6:0] = 100_0100	((580R – 136R) / 600R) * VSP/NR
	VHP4 [6:0] = 100_0101	((580R – 138R) / 600R) * VSP/NR
	:	:
	VHP4 [6:0] = 111_1011	((580R – 246R) / 600R) * VSP/NR
	VHP4 [6:0] = 111_1100	((580R – 248R) / 600R) * VSP/NR
	VHP4 [6:0] = 111_1101	((580R – 250R) / 600R) * VSP/NR
	VHP4 [6:0] = 111_1110	((580R – 252R) / 600R) * VSP/NR
	VHP4 [6:0] = 111_1111	((580R – 254R) / 600R) * VSP/NR

Table 5.11: VinP/N7

Reference voltage	Macro adjustment value	VinP/N9 formula
VinP/N9	VHP5 [6:0] = 000_0000	(576R / 600R) * VSP/NR
	VHP5 [6:0] = 000_0001	((576R – 2R) / 600R) * VSP/NR
	VHP5 [6:0] = 000_0010	((576R – 4R) / 600R) * VSP/NR
	VHP5 [6:0] = 000_0011	((576R – 6R) / 600R) * VSP/NR
	VHP5 [6:0] = 000_0100	((576R – 8R) / 600R) * VSP/NR
	VHP5 [6:0] = 000_0101	((576R – 10R) / 600R) * VSP/NR
	:	:
	VHP5 [6:0] = 100_0001	((576R – 130R) / 600R) * VSP/NR
	VHP5 [6:0] = 100_0010	((576R – 132R) / 600R) * VSP/NR
	VHP5 [6:0] = 100_0011	((576R – 134R) / 600R) * VSP/NR
	VHP5 [6:0] = 100_0100	((576R – 136R) / 600R) * VSP/NR
	VHP5 [6:0] = 100_0101	((576R – 138R) / 600R) * VSP/NR
	:	:
	VHP5 [6:0] = 111_1011	((576R – 246R) / 600R) * VSP/NR
	VHP5 [6:0] = 111_1100	((576R – 248R) / 600R) * VSP/NR
	VHP5 [6:0] = 111_1101	((576R – 250R) / 600R) * VSP/NR
	VHP5 [6:0] = 111_1110	((576R – 252R) / 600R) * VSP/NR
	VHP5 [6:0] = 111_1111	((576R – 254R) / 600R) * VSP/NR

Table 5.12: VinP/N9

Himax Confidential
Do Not Distribute

Reference voltage	Macro adjustment value	VinP/N12 formula
VinP/N12	VHP6 [6:0] = 000_0000	(568R / 600R) * VSP/NR
	VHP6 [6:0] = 000_0001	((568R – 2R) / 600R) * VSP/NR
	VHP6 [6:0] = 000_0010	((568R – 4R) / 600R) * VSP/NR
	VHP6 [6:0] = 000_0011	((568R – 6R) / 600R) * VSP/NR
	VHP6 [6:0] = 000_0100	((568R – 8R) / 600R) * VSP/NR
	VHP6 [6:0] = 000_0101	((568R – 10R) / 600R) * VSP/NR
	:	:
	VHP6 [6:0] = 100_0001	((568R – 130R) / 600R) * VSP/NR
	VHP6 [6:0] = 100_0010	((568R – 132R) / 600R) * VSP/NR
	VHP6 [6:0] = 100_0011	((568R – 134R) / 600R) * VSP/NR
	VHP6 [6:0] = 100_0100	((568R – 136R) / 600R) * VSP/NR
	VHP6 [6:0] = 100_0101	((568R – 138R) / 600R) * VSP/NR
	:	:
	VHP6 [6:0] = 111_1011	((568R – 246R) / 600R) * VSP/NR
	VHP6 [6:0] = 111_1100	((568R – 248R) / 600R) * VSP/NR
	VHP6 [6:0] = 111_1101	((568R – 250R) / 600R) * VSP/NR
	VHP6 [6:0] = 111_1110	((568R – 252R) / 600R) * VSP/NR
	VHP6 [6:0] = 111_1111	((568R – 254R) / 600R) * VSP/NR

Table 5.13: VinP/N12

Reference voltage	Macro adjustment value	VinP/N15 formula
VinP/N15	VHP7 [6:0] = 000_0000	(552R / 600R) * VSP/NR
	VHP7 [6:0] = 000_0001	((552R – 2R) / 600R) * VSP/NR
	VHP7 [6:0] = 000_0010	((552R – 4R) / 600R) * VSP/NR
	VHP7 [6:0] = 000_0011	((552R – 6R) / 600R) * VSP/NR
	VHP7 [6:0] = 000_0100	((552R – 8R) / 600R) * VSP/NR
	VHP7 [6:0] = 000_0101	((552R – 10R) / 600R) * VSP/NR
	:	:
	VHP7 [6:0] = 100_0001	((552R – 130R) / 600R) * VSP/NR
	VHP7 [6:0] = 100_0010	((552R – 132R) / 600R) * VSP/NR
	VHP7 [6:0] = 100_0011	((552R – 134R) / 600R) * VSP/NR
	VHP7 [6:0] = 100_0100	((552R – 136R) / 600R) * VSP/NR
	VHP7 [6:0] = 100_0101	((552R – 138R) / 600R) * VSP/NR
	:	:
	VHP7 [6:0] = 111_1011	((552R – 246R) / 600R) * VSP/NR
	VHP7 [6:0] = 111_1100	((552R – 248R) / 600R) * VSP/NR
	VHP7 [6:0] = 111_1101	((552R – 250R) / 600R) * VSP/NR
	VHP7 [6:0] = 111_1110	((552R – 252R) / 600R) * VSP/NR
	VHP7 [6:0] = 111_1111	((552R – 254R) / 600R) * VSP/NR

Table 5.14: VinP/N15

Reference voltage	Macro adjustment value	VinP/N20 formula
VinP/N20	VMP0 [7:0] = 0000_0000	(536R / 600R) * VSP/NR
	VMP0 [7:0] = 0000_0001	((536R – 1R) / 600R) * VSP/NR
	VMP0 [7:0] = 0000_0010	((536R – 2R) / 600R) * VSP/NR
	VMP0 [7:0] = 0000_0011	((536R – 3R) / 600R) * VSP/NR
	VMP0 [7:0] = 0000_0100	((536R – 4R) / 600R) * VSP/NR
	VMP0 [7:0] = 0000_0101	((536R – 5R) / 600R) * VSP/NR
	:	:
	VMP0 [7:0] = 1000_0001	((536R – 129R) / 600R) * VSP/NR
	VMP0 [7:0] = 1000_0010	((536R – 130R) / 600R) * VSP/NR
	VMP0 [7:0] = 1000_0011	((536R – 131R) / 600R) * VSP/NR
	VMP0 [7:0] = 1000_0100	((536R – 132R) / 600R) * VSP/NR
	VMP0 [7:0] = 1000_0101	((536R – 133R) / 600R) * VSP/NR
	:	:
	VMP0 [7:0] = 1111_1011	((536R – 251R) / 600R) * VSP/NR
	VMP0 [7:0] = 1111_1100	((536R – 252R) / 600R) * VSP/NR
	VMP0 [7:0] = 1111_1101	((536R – 253R) / 600R) * VSP/NR
	VMP0 [7:0] = 1111_1110	((536R – 254R) / 600R) * VSP/NR
	VMP0 [7:0] = 1111_1111	((536R – 255R) / 600R) * VSP/NR

Table 5.15: VinP/N20

Himax Confidential Do Not Copy

Reference voltage	Macro adjustment value	VinP/N28 formula
VinP/N28	VMP1 [7:0] = 0000_0000	(528R / 600R) * VSP/NR
	VMP1 [7:0] = 0000_0001	((528R – 1R) / 600R) * VSP/NR
	VMP1 [7:0] = 0000_0010	((528R – 2R) / 600R) * VSP/NR
	VMP1 [7:0] = 0000_0011	((528R – 3R) / 600R) * VSP/NR
	VMP1 [7:0] = 0000_0100	((528R – 4R) / 600R) * VSP/NR
	VMP1 [7:0] = 0000_0101	((528R – 5R) / 600R) * VSP/NR
	:	:
	VMP1 [7:0] = 1000_0001	((528R – 129R) / 600R) * VSP/NR
	VMP1 [7:0] = 1000_0010	((528R – 130R) / 600R) * VSP/NR
	VMP1 [7:0] = 1000_0011	((528R – 131R) / 600R) * VSP/NR
	VMP1 [7:0] = 1000_0100	((528R – 132R) / 600R) * VSP/NR
	VMP1 [7:0] = 1000_0101	((528R – 133R) / 600R) * VSP/NR
	:	:
	VMP1 [7:0] = 1111_1011	((528R – 251R) / 600R) * VSP/NR
	VMP1 [7:0] = 1111_1100	((528R – 252R) / 600R) * VSP/NR
	VMP1 [7:0] = 1111_1101	((528R – 253R) / 600R) * VSP/NR
	VMP1 [7:0] = 1111_1110	((528R – 254R) / 600R) * VSP/NR
	VMP1 [7:0] = 1111_1111	((528R – 255R) / 600R) * VSP/NR

Table 5.16: VinP/N28

Reference voltage	Macro adjustment value	VinP/N40 formula
VinP/N40	VMP2 [7:0] = 0000_0000	(516R / 600R) * VSP/NR
	VMP2 [7:0] = 0000_0001	((516R – 1R) / 600R) * VSP/NR
	VMP2 [7:0] = 0000_0010	((516R – 2R) / 600R) * VSP/NR
	VMP2 [7:0] = 0000_0011	((516R – 3R) / 600R) * VSP/NR
	VMP2 [7:0] = 0000_0100	((516R – 4R) / 600R) * VSP/NR
	VMP2 [7:0] = 0000_0101	((516R – 5R) / 600R) * VSP/NR
	:	:
	VMP2 [7:0] = 1000_0001	((516R – 129R) / 600R) * VSP/NR
	VMP2 [7:0] = 1000_0010	((516R – 130R) / 600R) * VSP/NR
	VMP2 [7:0] = 1000_0011	((516R – 131R) / 600R) * VSP/NR
	VMP2 [7:0] = 1000_0100	((516R – 132R) / 600R) * VSP/NR
	VMP2 [7:0] = 1000_0101	((516R – 133R) / 600R) * VSP/NR
	:	:
	VMP2 [7:0] = 1111_1011	((516R – 251R) / 600R) * VSP/NR
	VMP2 [7:0] = 1111_1100	((516R – 252R) / 600R) * VSP/NR
	VMP2 [7:0] = 1111_1101	((516R – 253R) / 600R) * VSP/NR
	VMP2 [7:0] = 1111_1110	((516R – 254R) / 600R) * VSP/NR
	VMP2 [7:0] = 1111_1111	((516R – 255R) / 600R) * VSP/NR

Table 5.17: VinP/N40

Reference voltage	Macro adjustment value	VinP/N52 formula
VinP/N52	VMP3 [7:0] = 0000_0000	(492R / 600R) * VSP/NR
	VMP3 [7:0] = 0000_0001	((492R – 1R) / 600R) * VSP/NR
	VMP3 [7:0] = 0000_0010	((492R – 2R) / 600R) * VSP/NR
	VMP3 [7:0] = 0000_0011	((492R – 3R) / 600R) * VSP/NR
	VMP3 [7:0] = 0000_0100	((492R – 4R) / 600R) * VSP/NR
	VMP3 [7:0] = 0000_0101	((492R – 5R) / 600R) * VSP/NR
	:	:
	VMP3 [7:0] = 1000_0001	((492R – 129R) / 600R) * VSP/NR
	VMP3 [7:0] = 1000_0010	((492R – 130R) / 600R) * VSP/NR
	VMP3 [7:0] = 1000_0011	((492R – 131R) / 600R) * VSP/NR
	VMP3 [7:0] = 1000_0100	((492R – 132R) / 600R) * VSP/NR
	VMP3 [7:0] = 1000_0101	((492R – 133R) / 600R) * VSP/NR
	:	:
	VMP3 [7:0] = 1111_1011	((492R – 251R) / 600R) * VSP/NR
	VMP3 [7:0] = 1111_1100	((492R – 252R) / 600R) * VSP/NR
	VMP3 [7:0] = 1111_1101	((492R – 253R) / 600R) * VSP/NR
	VMP3 [7:0] = 1111_1110	((492R – 254R) / 600R) * VSP/NR
	VMP3 [7:0] = 1111_1111	((492R – 255R) / 600R) * VSP/NR

Table 5.18: VinP/N52

Himax Confidential Do Not Copy

Reference voltage	Macro adjustment value	VinP/N76 formula
VinP/N76	VMP4 [7:0] = 0000_0000	(464R / 600R) * VSP/NR
	VMP4 [7:0] = 0000_0001	((464R – 1R) / 600R) * VSP/NR
	VMP4 [7:0] = 0000_0010	((464R – 2R) / 600R) * VSP/NR
	VMP4 [7:0] = 0000_0011	((464R – 3R) / 600R) * VSP/NR
	VMP4 [7:0] = 0000_0100	((464R – 4R) / 600R) * VSP/NR
	VMP4 [7:0] = 0000_0101	((464R – 5R) / 600R) * VSP/NR
	:	:
	VMP4 [7:0] = 1000_0001	((464R – 129R) / 600R) * VSP/NR
	VMP4 [7:0] = 1000_0010	((464R – 130R) / 600R) * VSP/NR
	VMP4 [7:0] = 1000_0011	((464R – 131R) / 600R) * VSP/NR
	VMP4 [7:0] = 1000_0100	((464R – 132R) / 600R) * VSP/NR
	VMP4 [7:0] = 1000_0101	((464R – 133R) / 600R) * VSP/NR
	:	:
	VMP4 [7:0] = 1111_1011	((464R – 251R) / 600R) * VSP/NR
	VMP4 [7:0] = 1111_1100	((464R – 252R) / 600R) * VSP/NR
	VMP4 [7:0] = 1111_1101	((464R – 253R) / 600R) * VSP/NR
	VMP4 [7:0] = 1111_1110	((464R – 254R) / 600R) * VSP/NR
	VMP4 [7:0] = 1111_1111	((464R – 255R) / 600R) * VSP/NR

Table 5.19: VinP/N76

Reference voltage	Macro adjustment value	VinP/N100 formula
VinP/N100	VMP5 [7:0] = 0000_0000	(452R / 600R) * VSP/NR
	VMP5 [7:0] = 0000_0001	((452R – 1R) / 600R) * VSP/NR
	VMP5 [7:0] = 0000_0010	((452R – 2R) / 600R) * VSP/NR
	VMP5 [7:0] = 0000_0011	((452R – 3R) / 600R) * VSP/NR
	VMP5 [7:0] = 0000_0100	((452R – 4R) / 600R) * VSP/NR
	VMP5 [7:0] = 0000_0101	((452R – 5R) / 600R) * VSP/NR
	:	:
	VMP5 [7:0] = 1000_0001	((452R – 129R) / 600R) * VSP/NR
	VMP5 [7:0] = 1000_0010	((452R – 130R) / 600R) * VSP/NR
	VMP5 [7:0] = 1000_0011	((452R – 131R) / 600R) * VSP/NR
	VMP5 [7:0] = 1000_0100	((452R – 132R) / 600R) * VSP/NR
	VMP5 [7:0] = 1000_0101	((452R – 133R) / 600R) * VSP/NR
	:	:
	VMP5 [7:0] = 1111_1011	((452R – 251R) / 600R) * VSP/NR
	VMP5 [7:0] = 1111_1100	((452R – 252R) / 600R) * VSP/NR
	VMP5 [7:0] = 1111_1101	((452R – 253R) / 600R) * VSP/NR
	VMP5 [7:0] = 1111_1110	((452R – 254R) / 600R) * VSP/NR
	VMP5 [7:0] = 1111_1111	((452R – 255R) / 600R) * VSP/NR

Table 5.20: VinP/N100

Reference voltage	Macro adjustment value	VinP/N128 formula
VinP/N128	VMP6 [7:0] = 0000_0000	(427R / 600R) * VSP/NR
	VMP6 [7:0] = 0000_0001	((427R – 1R) / 600R) * VSP/NR
	VMP6 [7:0] = 0000_0010	((427R – 2R) / 600R) * VSP/NR
	VMP6 [7:0] = 0000_0011	((427R – 3R) / 600R) * VSP/NR
	VMP6 [7:0] = 0000_0100	((427R – 4R) / 600R) * VSP/NR
	VMP6 [7:0] = 0000_0101	((427R – 5R) / 600R) * VSP/NR
	:	:
	VMP6 [7:0] = 1000_0001	((427R – 129R) / 600R) * VSP/NR
	VMP6 [7:0] = 1000_0010	((427R – 130R) / 600R) * VSP/NR
	VMP6 [7:0] = 1000_0011	((427R – 131R) / 600R) * VSP/NR
	VMP6 [7:0] = 1000_0100	((427R – 132R) / 600R) * VSP/NR
	VMP6 [7:0] = 1000_0101	((427R – 133R) / 600R) * VSP/NR
	:	:
	VMP6 [7:0] = 1111_1011	((427R – 251R) / 600R) * VSP/NR
	VMP6 [7:0] = 1111_1100	((427R – 252R) / 600R) * VSP/NR
	VMP6 [7:0] = 1111_1101	((427R – 253R) / 600R) * VSP/NR
	VMP6 [7:0] = 1111_1110	((427R – 254R) / 600R) * VSP/NR
	VMP6 [7:0] = 1111_1111	((427R – 255R) / 600R) * VSP/NR

Table 5.21: VinP/N128

Himax Confidential
Do Not Copy!

Reference voltage	Macro adjustment value	VinP/N156 formula
VinP/N156	VMP7 [7:0] = 0000_0000	(403R / 600R) * VSP/NR
	VMP7 [7:0] = 0000_0001	((403R – 1R) / 600R) * VSP/NR
	VMP7 [7:0] = 0000_0010	((403R – 2R) / 600R) * VSP/NR
	VMP7 [7:0] = 0000_0011	((403R – 3R) / 600R) * VSP/NR
	VMP7 [7:0] = 0000_0100	((403R – 4R) / 600R) * VSP/NR
	VMP7 [7:0] = 0000_0101	((403R – 5R) / 600R) * VSP/NR
	:	:
	VMP7 [7:0] = 1000_0001	((403R – 129R) / 600R) * VSP/NR
	VMP7 [7:0] = 1000_0010	((403R – 130R) / 600R) * VSP/NR
	VMP7 [7:0] = 1000_0011	((403R – 131R) / 600R) * VSP/NR
	VMP7 [7:0] = 1000_0100	((403R – 132R) / 600R) * VSP/NR
	VMP7 [7:0] = 1000_0101	((403R – 133R) / 600R) * VSP/NR
	:	:
	VMP7 [7:0] = 1111_1011	((403R – 251R) / 600R) * VSP/NR
	VMP7 [7:0] = 1111_1100	((403R – 252R) / 600R) * VSP/NR
	VMP7 [7:0] = 1111_1101	((403R – 253R) / 600R) * VSP/NR
	VMP7 [7:0] = 1111_1110	((403R – 254R) / 600R) * VSP/NR
	VMP7 [7:0] = 1111_1111	((403R – 255R) / 600R) * VSP/NR

Table 5.22: VinP/N156

Reference voltage	Macro adjustment value	VinP/N180 formula
VinP/N180	VMP8 [7:0] = 0000_0000	(391R / 600R) * VSP/NR
	VMP8 [7:0] = 0000_0001	((391R – 1R) / 600R) * VSP/NR
	VMP8 [7:0] = 0000_0010	((391R – 2R) / 600R) * VSP/NR
	VMP8 [7:0] = 0000_0011	((391R – 3R) / 600R) * VSP/NR
	VMP8 [7:0] = 0000_0100	((391R – 4R) / 600R) * VSP/NR
	VMP8 [7:0] = 0000_0101	((391R – 5R) / 600R) * VSP/NR
	:	:
	VMP8 [7:0] = 1000_0001	((391R – 129R) / 600R) * VSP/NR
	VMP8 [7:0] = 1000_0010	((391R – 130R) / 600R) * VSP/NR
	VMP8 [7:0] = 1000_0011	((391R – 131R) / 600R) * VSP/NR
	VMP8 [7:0] = 1000_0100	((391R – 132R) / 600R) * VSP/NR
	VMP8 [7:0] = 1000_0101	((391R – 133R) / 600R) * VSP/NR
	:	:
	VMP8 [7:0] = 1111_1011	((391R – 251R) / 600R) * VSP/NR
	VMP8 [7:0] = 1111_1100	((391R – 252R) / 600R) * VSP/NR
	VMP8 [7:0] = 1111_1101	((391R – 253R) / 600R) * VSP/NR
	VMP8 [7:0] = 1111_1110	((391R – 254R) / 600R) * VSP/NR
	VMP8 [7:0] = 1111_1111	((391R – 255R) / 600R) * VSP/NR

Table 5.23: VinP/N180

Reference voltage	Macro adjustment value	VinP/N204 formula
VinP/N204	VMP9 [7:0] = 0000_0000	(363R / 600R) * VSP/NR
	VMP9 [7:0] = 0000_0001	((363R – 1R) / 600R) * VSP/NR
	VMP9 [7:0] = 0000_0010	((363R – 2R) / 600R) * VSP/NR
	VMP9 [7:0] = 0000_0011	((363R – 3R) / 600R) * VSP/NR
	VMP9 [7:0] = 0000_0100	((363R – 4R) / 600R) * VSP/NR
	VMP9 [7:0] = 0000_0101	((363R – 5R) / 600R) * VSP/NR
	:	:
	VMP9 [7:0] = 1000_0001	((363R – 129R) / 600R) * VSP/NR
	VMP9 [7:0] = 1000_0010	((363R – 130R) / 600R) * VSP/NR
	VMP9 [7:0] = 1000_0011	((363R – 131R) / 600R) * VSP/NR
	VMP9 [7:0] = 1000_0100	((363R – 132R) / 600R) * VSP/NR
	VMP9 [7:0] = 1000_0101	((363R – 133R) / 600R) * VSP/NR
	:	:
	VMP9 [7:0] = 1111_1011	((363R – 251R) / 600R) * VSP/NR
	VMP9 [7:0] = 1111_1100	((363R – 252R) / 600R) * VSP/NR
	VMP9 [7:0] = 1111_1101	((363R – 253R) / 600R) * VSP/NR
	VMP9 [7:0] = 1111_1110	((363R – 254R) / 600R) * VSP/NR
	VMP9 [7:0] = 1111_1111	((363R – 255R) / 600R) * VSP/NR

Table 5.24: VinP/N204

Himax Confidential
Do Not Copy!

Reference voltage	Macro adjustment value	VinP/N216 formula
VinP/N216	VMP10 [7:0] = 0000_0000	(339R / 600R) * VSP/NR
	VMP10 [7:0] = 0000_0001	((339R – 1R) / 600R) * VSP/NR
	VMP10 [7:0] = 0000_0010	((339R – 2R) / 600R) * VSP/NR
	VMP10 [7:0] = 0000_0011	((339R – 3R) / 600R) * VSP/NR
	VMP10 [7:0] = 0000_0100	((339R – 4R) / 600R) * VSP/NR
	VMP10 [7:0] = 0000_0101	((339R – 5R) / 600R) * VSP/NR
	:	:
	VMP10 [7:0] = 1000_0001	((339R – 129R) / 600R) * VSP/NR
	VMP10 [7:0] = 1000_0010	((339R – 130R) / 600R) * VSP/NR
	VMP10 [7:0] = 1000_0011	((339R – 131R) / 600R) * VSP/NR
	VMP10 [7:0] = 1000_0100	((339R – 132R) / 600R) * VSP/NR
	VMP10 [7:0] = 1000_0101	((339R – 133R) / 600R) * VSP/NR
	:	:
	VMP10 [7:0] = 1111_1011	((339R – 251R) / 600R) * VSP/NR
	VMP10 [7:0] = 1111_1100	((339R – 252R) / 600R) * VSP/NR
	VMP10 [7:0] = 1111_1101	((339R – 253R) / 600R) * VSP/NR
	VMP10 [7:0] = 1111_1110	((339R – 254R) / 600R) * VSP/NR
	VMP10 [7:0] = 1111_1111	((339R – 255R) / 600R) * VSP/NR

Table 5.25: VinP/N216

Reference voltage	Macro adjustment value	VinP/N228 formula
VinP/N228	VMP11 [7:0] = 0000_0000	(327R / 600R) * VSP/NR
	VMP11 [7:0] = 0000_0001	((327R – 1R) / 600R) * VSP/NR
	VMP11 [7:0] = 0000_0010	((327R – 2R) / 600R) * VSP/NR
	VMP11 [7:0] = 0000_0011	((327R – 3R) / 600R) * VSP/NR
	VMP11 [7:0] = 0000_0100	((327R – 4R) / 600R) * VSP/NR
	VMP11 [7:0] = 0000_0101	((327R – 5R) / 600R) * VSP/NR
	:	:
	VMP11 [7:0] = 1000_0001	((327R – 129R) / 600R) * VSP/NR
	VMP11 [7:0] = 1000_0010	((327R – 130R) / 600R) * VSP/NR
	VMP11 [7:0] = 1000_0011	((327R – 131R) / 600R) * VSP/NR
	VMP11 [7:0] = 1000_0100	((327R – 132R) / 600R) * VSP/NR
	VMP11 [7:0] = 1000_0101	((327R – 133R) / 600R) * VSP/NR
	:	:
	VMP11 [7:0] = 1111_1011	((327R – 251R) / 600R) * VSP/NR
	VMP11 [7:0] = 1111_1100	((327R – 252R) / 600R) * VSP/NR
	VMP11 [7:0] = 1111_1101	((327R – 253R) / 600R) * VSP/NR
	VMP11 [7:0] = 1111_1110	((327R – 254R) / 600R) * VSP/NR
	VMP11 [7:0] = 1111_1111	((327R – 255R) / 600R) * VSP/NR

Table 5.26: VinP/N228

Reference voltage	Macro adjustment value	VinP/N236 formula
VinP/N236	VMP12 [7:0] = 0000_0000	(319R / 600R) * VSP/NR
	VMP12 [7:0] = 0000_0001	((319R – 1R) / 600R) * VSP/NR
	VMP12 [7:0] = 0000_0010	((319R – 2R) / 600R) * VSP/NR
	VMP12 [7:0] = 0000_0011	((319R – 3R) / 600R) * VSP/NR
	VMP12 [7:0] = 0000_0100	((319R – 4R) / 600R) * VSP/NR
	VMP12 [7:0] = 0000_0101	((319R – 5R) / 600R) * VSP/NR
	:	:
	VMP12 [7:0] = 1000_0001	((319R – 129R) / 600R) * VSP/NR
	VMP12 [7:0] = 1000_0010	((319R – 130R) / 600R) * VSP/NR
	VMP12 [7:0] = 1000_0011	((319R – 131R) / 600R) * VSP/NR
	VMP12 [7:0] = 1000_0100	((319R – 132R) / 600R) * VSP/NR
	VMP12 [7:0] = 1000_0101	((319R – 133R) / 600R) * VSP/NR
	:	:
	VMP12 [7:0] = 1111_1011	((319R – 251R) / 600R) * VSP/NR
	VMP12 [7:0] = 1111_1100	((319R – 252R) / 600R) * VSP/NR
	VMP12 [7:0] = 1111_1101	((319R – 253R) / 600R) * VSP/NR
	VMP12 [7:0] = 1111_1110	((319R – 254R) / 600R) * VSP/NR
	VMP12 [7:0] = 1111_1111	((319R – 255R) / 600R) * VSP/NR

Table 5.27: VinP/N236

Himax Confidential
Do Not Copy!

Reference voltage	Macro adjustment value	VinP/N240 formula
VinP/N240	VLP0 [6:0] = 000_0000	(302R / 600R) * VSP/NR
	VLP0 [6:0] = 000_0001	((302R – 2R) / 600R) * VSP/NR
	VLP0 [6:0] = 000_0010	((302R – 4R) / 600R) * VSP/NR
	VLP0 [6:0] = 000_0011	((302R – 6R) / 600R) * VSP/NR
	VLP0 [6:0] = 000_0100	((302R – 8R) / 600R) * VSP/NR
	VLP0 [6:0] = 000_0101	((302R – 10R) / 600R) * VSP/NR
	:	:
	VLP0 [6:0] = 100_0001	((302R – 130R) / 600R) * VSP/NR
	VLP0 [6:0] = 100_0010	((302R – 132R) / 600R) * VSP/NR
	VLP0 [6:0] = 100_0011	((302R – 134R) / 600R) * VSP/NR
	VLP0 [6:0] = 100_0100	((302R – 136R) / 600R) * VSP/NR
	VLP0 [6:0] = 100_0101	((302R – 138R) / 600R) * VSP/NR
	:	:
	VLP0 [6:0] = 111_1011	((302R – 246R) / 600R) * VSP/NR
	VLP0 [6:0] = 111_1100	((302R – 248R) / 600R) * VSP/NR
	VLP0 [6:0] = 111_1101	((302R – 250R) / 600R) * VSP/NR
	VLP0 [6:0] = 111_1110	((302R – 252R) / 600R) * VSP/NR
	VLP0 [6:0] = 111_1111	((302R – 254R) / 600R) * VSP/NR

Table 5.28: VinP/N240

Reference voltage	Macro adjustment value	VinP/N243 formula
VinP/N243	VLP1 [6:0] = 000_0000	(286R / 600R) * VSP/NR
	VLP1 [6:0] = 000_0001	((286R – 2R) / 600R) * VSP/NR
	VLP1 [6:0] = 000_0010	((286R – 4R) / 600R) * VSP/NR
	VLP1 [6:0] = 000_0011	((286R – 6R) / 600R) * VSP/NR
	VLP1 [6:0] = 000_0100	((286R – 8R) / 600R) * VSP/NR
	VLP1 [6:0] = 000_0101	((286R – 10R) / 600R) * VSP/NR
	:	:
	VLP1 [6:0] = 100_0001	((286R – 130R) / 600R) * VSP/NR
	VLP1 [6:0] = 100_0010	((286R – 132R) / 600R) * VSP/NR
	VLP1 [6:0] = 100_0011	((286R – 134R) / 600R) * VSP/NR
	VLP1 [6:0] = 100_0100	((286R – 136R) / 600R) * VSP/NR
	VLP1 [6:0] = 100_0101	((286R – 138R) / 600R) * VSP/NR
	:	:
	VLP1 [6:0] = 111_1011	((286R – 246R) / 600R) * VSP/NR
	VLP1 [6:0] = 111_1100	((286R – 248R) / 600R) * VSP/NR
	VLP1 [6:0] = 111_1101	((286R – 250R) / 600R) * VSP/NR
	VLP1 [6:0] = 111_1110	((286R – 252R) / 600R) * VSP/NR
	VLP1 [6:0] = 111_1111	((286R – 254R) / 600R) * VSP/NR

Table 5.29: VinP/N243

Reference voltage	Macro adjustment value	VinP/N246 formula
VinP/N246	VLP2 [6:0] = 000_0000	(278R / 600R) * VSP/NR
	VLP2 [6:0] = 000_0001	((278R – 2R) / 600R) * VSP/NR
	VLP2 [6:0] = 000_0010	((278R – 4R) / 600R) * VSP/NR
	VLP2 [6:0] = 000_0011	((278R – 6R) / 600R) * VSP/NR
	VLP2 [6:0] = 000_0100	((278R – 8R) / 600R) * VSP/NR
	VLP2 [6:0] = 000_0101	((278R – 10R) / 600R) * VSP/NR
	:	:
	VLP2 [6:0] = 100_0001	((278R – 130R) / 600R) * VSP/NR
	VLP2 [6:0] = 100_0010	((278R – 132R) / 600R) * VSP/NR
	VLP2 [6:0] = 100_0011	((278R – 134R) / 600R) * VSP/NR
	VLP2 [6:0] = 100_0100	((278R – 136R) / 600R) * VSP/NR
	VLP2 [6:0] = 100_0101	((278R – 138R) / 600R) * VSP/NR
	:	:
	VLP2 [6:0] = 111_1011	((278R – 246R) / 600R) * VSP/NR
	VLP2 [6:0] = 111_1100	((278R – 248R) / 600R) * VSP/NR
	VLP2 [6:0] = 111_1101	((278R – 250R) / 600R) * VSP/NR
	VLP2 [6:0] = 111_1110	((278R – 252R) / 600R) * VSP/NR
	VLP2 [6:0] = 111_1111	((278R – 254R) / 600R) * VSP/NR

Table 5.30: VinP/N246

Himax Confidential
Do Not Copy!

Reference voltage	Macro adjustment value	VinP/N248 formula
VinP/N248	VLP3 [6:0] = 000_0000	(274R / 600R) * VSP/NR
	VLP3 [6:0] = 000_0001	((274R – 2R) / 600R) * VSP/NR
	VLP3 [6:0] = 000_0010	((274R – 4R) / 600R) * VSP/NR
	VLP3 [6:0] = 000_0011	((274R – 6R) / 600R) * VSP/NR
	VLP3 [6:0] = 000_0100	((274R – 8R) / 600R) * VSP/NR
	VLP3 [6:0] = 000_0101	((274R – 10R) / 600R) * VSP/NR
	:	:
	VLP3 [6:0] = 100_0001	((274R – 130R) / 600R) * VSP/NR
	VLP3 [6:0] = 100_0010	((274R – 132R) / 600R) * VSP/NR
	VLP3 [6:0] = 100_0011	((274R – 134R) / 600R) * VSP/NR
	VLP3 [6:0] = 100_0100	((274R – 136R) / 600R) * VSP/NR
	VLP3 [6:0] = 100_0101	((274R – 138R) / 600R) * VSP/NR
	:	:
	VLP3 [6:0] = 111_1011	((274R – 246R) / 600R) * VSP/NR
	VLP3 [6:0] = 111_1100	((274R – 248R) / 600R) * VSP/NR
	VLP3 [6:0] = 111_1101	((274R – 250R) / 600R) * VSP/NR
	VLP3 [6:0] = 111_1110	((274R – 252R) / 600R) * VSP/NR
	VLP3 [6:0] = 111_1111	((274R – 254R) / 600R) * VSP/NR

Table 5.31: VinP/N248

Reference voltage	Macro adjustment value	VinP/N250 formula
VinP/N250	VLP4 [6:0] = 000_0000	(266R / 600R) * VSP/NR
	VLP4 [6:0] = 000_0001	((266R – 2R) / 600R) * VSP/NR
	VLP4 [6:0] = 000_0010	((266R – 4R) / 600R) * VSP/NR
	VLP4 [6:0] = 000_0011	((266R – 6R) / 600R) * VSP/NR
	VLP4 [6:0] = 000_0100	((266R – 8R) / 600R) * VSP/NR
	VLP4 [6:0] = 000_0101	((266R – 10R) / 600R) * VSP/NR
	:	:
	VLP4 [6:0] = 100_0001	((266R – 130R) / 600R) * VSP/NR
	VLP4 [6:0] = 100_0010	((266R – 132R) / 600R) * VSP/NR
	VLP4 [6:0] = 100_0011	((266R – 134R) / 600R) * VSP/NR
	VLP4 [6:0] = 100_0100	((266R – 136R) / 600R) * VSP/NR
	VLP4 [6:0] = 100_0101	((266R – 138R) / 600R) * VSP/NR
	:	:
	VLP4 [6:0] = 111_1011	((266R – 246R) / 600R) * VSP/NR
	VLP4 [6:0] = 111_1100	((266R – 248R) / 600R) * VSP/NR
	VLP4 [6:0] = 111_1101	((266R – 250R) / 600R) * VSP/NR
	VLP4 [6:0] = 111_1110	((266R – 252R) / 600R) * VSP/NR
	VLP4 [6:0] = 111_1111	((266R – 254R) / 600R) * VSP/NR

Table 5.32: VinP/N250

Reference voltage	Macro adjustment value	VinP/N252 formula
VinP/N252	VLP5 [6:0] = 000_0000	(262R / 600R) * VSP/NR
	VLP5 [6:0] = 000_0001	((262R – 2R) / 600R) * VSP/NR
	VLP5 [6:0] = 000_0010	((262R – 4R) / 600R) * VSP/NR
	VLP5 [6:0] = 000_0011	((262R – 6R) / 600R) * VSP/NR
	VLP5 [6:0] = 000_0100	((262R – 8R) / 600R) * VSP/NR
	VLP5 [6:0] = 000_0101	((262R – 10R) / 600R) * VSP/NR
	:	:
	VLP5 [6:0] = 100_0001	((262R – 130R) / 600R) * VSP/NR
	VLP5 [6:0] = 100_0010	((262R – 132R) / 600R) * VSP/NR
	VLP5 [6:0] = 100_0011	((262R – 134R) / 600R) * VSP/NR
	VLP5 [6:0] = 100_0100	((262R – 136R) / 600R) * VSP/NR
	VLP5 [6:0] = 100_0101	((262R – 138R) / 600R) * VSP/NR
	:	:
	VLP5 [6:0] = 111_1011	((262R – 246R) / 600R) * VSP/NR
	VLP5 [6:0] = 111_1100	((262R – 248R) / 600R) * VSP/NR
	VLP5 [6:0] = 111_1101	((262R – 250R) / 600R) * VSP/NR
	VLP5 [6:0] = 111_1110	((262R – 252R) / 600R) * VSP/NR
	VLP5 [6:0] = 111_1111	((262R – 254R) / 600R) * VSP/NR

Table 5.33: VinP/N252

Himax Confidential
Do Not Copy!

Reference voltage	Macro adjustment value	VinP/N254 formula
VinP/N254	VLP6 [6:0] = 000_0000	(258R / 600R) * VSP/NR
	VLP6 [6:0] = 000_0001	((258R – 2R) / 600R) * VSP/NR
	VLP6 [6:0] = 000_0010	((258R – 4R) / 600R) * VSP/NR
	VLP6 [6:0] = 000_0011	((258R – 6R) / 600R) * VSP/NR
	VLP6 [6:0] = 000_0100	((258R – 8R) / 600R) * VSP/NR
	VLP6 [6:0] = 000_0101	((258R – 10R) / 600R) * VSP/NR
	:	:
	VLP6 [6:0] = 100_0001	((258R – 130R) / 600R) * VSP/NR
	VLP6 [6:0] = 100_0010	((258R – 132R) / 600R) * VSP/NR
	VLP6 [6:0] = 100_0011	((258R – 134R) / 600R) * VSP/NR
	VLP6 [6:0] = 100_0100	((258R – 136R) / 600R) * VSP/NR
	VLP6 [6:0] = 100_0101	((258R – 138R) / 600R) * VSP/NR
	:	:
	VLP6 [6:0] = 111_1011	((258R – 246R) / 600R) * VSP/NR
	VLP6 [6:0] = 111_1100	((258R – 248R) / 600R) * VSP/NR
	VLP6 [6:0] = 111_1101	((258R – 250R) / 600R) * VSP/NR
	VLP6 [6:0] = 111_1110	((258R – 252R) / 600R) * VSP/NR
	VLP6 [6:0] = 111_1111	((258R – 254R) / 600R) * VSP/NR

Table 5.34: VinP/N254

Reference voltage	Macro adjustment value	VinP/N255 formula
VinP/N255	VLP7 [6:0] = 000_0000	(254R / 600R) * VSP/NR
	VLP7 [6:0] = 000_0001	((254R – 2R) / 600R) * VSP/NR
	VLP7 [6:0] = 000_0010	((254R – 4R) / 600R) * VSP/NR
	VLP7 [6:0] = 000_0011	((254R – 6R) / 600R) * VSP/NR
	VLP7 [6:0] = 000_0100	((254R – 8R) / 600R) * VSP/NR
	VLP7 [6:0] = 000_0101	((254R – 10R) / 600R) * VSP/NR
	:	:
	VLP7 [6:0] = 100_0001	((254R – 130R) / 600R) * VSP/NR
	VLP7 [6:0] = 100_0010	((254R – 132R) / 600R) * VSP/NR
	VLP7 [6:0] = 100_0011	((254R – 134R) / 600R) * VSP/NR
	VLP7 [6:0] = 100_0100	((254R – 136R) / 600R) * VSP/NR
	VLP7 [6:0] = 100_0101	((254R – 138R) / 600R) * VSP/NR
	:	:
	VLP7 [6:0] = 111_1011	((254R – 246R) / 600R) * VSP/NR
	VLP7 [6:0] = 111_1100	((254R – 248R) / 600R) * VSP/NR
	VLP7 [6:0] = 111_1101	((254R – 250R) / 600R) * VSP/NR
	VLP7 [6:0] = 111_1110	((254R – 252R) / 600R) * VSP/NR
	VLP7 [6:0] = 111_1111	((254R – 254R) / 600R) * VSP/NR

Table 5.35: VinP/N255

Himax Confidential
Do Not Copy!

Grayscale voltage (NW/NB)	Formula
V0/V255	VinP/N0
V1/V254	VinP/N1
V2/V253	VinP/N0 - (VinP/N0 - VinP/N1)*(R/2R)
V3/V252	VinP/N3
V4/V251	VinP/N3 - (VinP/N3 - VinP/N5)*(1R/2R)
V5/V250	VinP/N5
V6/V249	VinP/N5 - (VinP/N5 - VinP/N7)*(1R/2R)
V7/V248	VinP/N7
V8/V247	VinP/N7 - (VinP/N7 - VinP/N9)*(1R/2R)
V9/V246	VinP/N9
V10/V245	VinP/N9 - (VinP/N9 - VinP/N12)*(1R/3R)
V11/V244	VinP/N9 - (VinP/N9 - VinP/N12)*(2R/3R)
V12/V243	VinP/N12
V13/V242	VinP/N12 - (VinP/N12 - VinP/N15)*(1R/3R)
V14/V241	VinP/N12 - (VinP/N12 - VinP/N15)*(2R/3R)
V15/V240	VinP/N15
V16/V239	VinP/N15 - (VinP/N15 - VinP/N20)*(1R/5R)
V17/V238	VinP/N15 - (VinP/N15 - VinP/N20)*(2R/5R)
V18/V237	VinP/N15 - (VinP/N15 - VinP/N20)*(3R/5R)
V19/V236	VinP/N15 - (VinP/N15 - VinP/N20)*(4R/5R)
V20/V235	VinP/N20
V21/V234	VinP/N20 - (VinP/N20 - VinP/N28)*(1R/8R)
V22/V233	VinP/N20 - (VinP/N20 - VinP/N28)*(2R/8R)
V23/V232	VinP/N20 - (VinP/N20 - VinP/N28)*(3R/8R)
V24/V231	VinP/N20 - (VinP/N20 - VinP/N28)*(4R/8R)
V25/V230	VinP/N20 - (VinP/N20 - VinP/N28)*(5R/8R)
V26/V229	VinP/N20 - (VinP/N20 - VinP/N28)*(6R/8R)
V27/V228	VinP/N20 - (VinP/N20 - VinP/N28)*(7R/8R)
V28/V227	VinP/N28
V29/V226	VinP/N28 - (VinP/N28 - VinP/N40)*(1R/12R)
V30/V225	VinP/N28 - (VinP/N28 - VinP/N40)*(2R/12R)
V31/V224	VinP/N28 - (VinP/N28 - VinP/N40)*(3R/12R)
V32/V223	VinP/N28 - (VinP/N28 - VinP/N40)*(4R/12R)
V33/V222	VinP/N28 - (VinP/N28 - VinP/N40)*(5R/12R)
V34/V221	VinP/N28 - (VinP/N28 - VinP/N40)*(6R/12R)
V35/V220	VinP/N28 - (VinP/N28 - VinP/N40)*(7R/12R)
V36/V219	VinP/N28 - (VinP/N28 - VinP/N40)*(8R/12R)
V37/V218	VinP/N28 - (VinP/N28 - VinP/N40)*(9R/12R)
V38/V217	VinP/N28 - (VinP/N28 - VinP/N40)*(10R/12R)
V39/V216	VinP/N28 - (VinP/N28 - VinP/N40)*(11R/12R)
V40/V215	VinP/N40
V41/V214	VinP/N40 - (VinP/N40 - VinP/N52)*(1R/12R)
V42/V213	VinP/N40 - (VinP/N40 - VinP/N52)*(2R/12R)
V43/V212	VinP/N40 - (VinP/N40 - VinP/N52)*(3R/12R)

Grayscale voltage (NW/NB)	Formula
V44/V211	VinP/N40 - (VinP/N40 - VinP/N52)*(4R/12R)
V45/V210	VinP/N40 - (VinP/N40 - VinP/N52)*(5R/12R)
V46/V209	VinP/N40 - (VinP/N40 - VinP/N52)*(6R/12R)
V47/V208	VinP/N40 - (VinP/N40 - VinP/N52)*(7R/12R)
V48/V207	VinP/N40 - (VinP/N40 - VinP/N52)*(8R/12R)
V49/V206	VinP/N40 - (VinP/N40 - VinP/N52)*(9R/12R)
V50/V205	VinP/N40 - (VinP/N40 - VinP/N52)*(10R/12R)
V51/V204	VinP/N40 - (VinP/N40 - VinP/N52)*(11R/12R)
V52/V203	VinP/N52
V53/V202	VinP/N52 - (VinP/N52 - VinP/N76)*(1R/24R)
V54/V201	VinP/N52 - (VinP/N52 - VinP/N76)*(2R/24R)
V55/V200	VinP/N52 - (VinP/N52 - VinP/N76)*(3R/24R)
V56/V199	VinP/N52 - (VinP/N52 - VinP/N76)*(4R/24R)
V57/V198	VinP/N52 - (VinP/N52 - VinP/N76)*(5R/24R)
V58/V197	VinP/N52 - (VinP/N52 - VinP/N76)*(6R/24R)
V59/V196	VinP/N52 - (VinP/N52 - VinP/N76)*(7R/24R)
V60/V195	VinP/N52 - (VinP/N52 - VinP/N76)*(8R/24R)
V61/V194	VinP/N52 - (VinP/N52 - VinP/N76)*(9R/24R)
V62/V193	VinP/N52 - (VinP/N52 - VinP/N76)*(10R/24R)
V63/V192	VinP/N52 - (VinP/N52 - VinP/N76)*(11R/24R)
V64/V191	VinP/N52 - (VinP/N52 - VinP/N76)*(12R/24R)
V65/V190	VinP/N52 - (VinP/N52 - VinP/N76)*(13R/24R)
V66/V189	VinP/N52 - (VinP/N52 - VinP/N76)*(14R/24R)
V67/V188	VinP/N52 - (VinP/N52 - VinP/N76)*(15R/24R)
V68/V187	VinP/N52 - (VinP/N52 - VinP/N76)*(16R/24R)
V69/V186	VinP/N52 - (VinP/N52 - VinP/N76)*(17R/24R)
V70/V185	VinP/N52 - (VinP/N52 - VinP/N76)*(18R/24R)
V71/V184	VinP/N52 - (VinP/N52 - VinP/N76)*(19R/24R)
V72/V183	VinP/N52 - (VinP/N52 - VinP/N76)*(20R/24R)
V73/V182	VinP/N52 - (VinP/N52 - VinP/N76)*(21R/24R)
V74/V181	VinP/N52 - (VinP/N52 - VinP/N76)*(22R/24R)
V75/V180	VinP/N52 - (VinP/N52 - VinP/N76)*(23R/24R)
V76/V179	VinP/N76
V77/V178	VinP/N76 - (VinP/N76 - VinP/N100)*(1R/24R)
V78/V177	VinP/N76 - (VinP/N76 - VinP/N100)*(2R/24R)
V79/V176	VinP/N76 - (VinP/N76 - VinP/N100)*(3R/24R)
V80/V175	VinP/N76 - (VinP/N76 - VinP/N100)*(4R/24R)
V81/V174	VinP/N76 - (VinP/N76 - VinP/N100)*(5R/24R)
V82/V173	VinP/N76 - (VinP/N76 - VinP/N100)*(6R/24R)
V83/V172	VinP/N76 - (VinP/N76 - VinP/N100)*(7R/24R)
V84/V171	VinP/N76 - (VinP/N76 - VinP/N100)*(8R/24R)
V85/V170	VinP/N76 - (VinP/N76 - VinP/N100)*(9R/24R)
V86/V169	VinP/N76 - (VinP/N76 - VinP/N100)*(10R/24R)
V87/V168	VinP/N76 - (VinP/N76 - VinP/N100)*(11R/24R)

Himax Confidential
Do Not Copy

Grayscale voltage (NW/NB)	Formula
V88/V167	VinP/N76 - (VinP/N76 - VinP/N100)*(12R/24R)
V89/V166	VinP/N76 - (VinP/N76 - VinP/N100)*(13R/24R)
V90/V165	VinP/N76 - (VinP/N76 - VinP/N100)*(14R/24R)
V91/V164	VinP/N76 - (VinP/N76 - VinP/N100)*(15R/24R)
V92/V163	VinP/N76 - (VinP/N76 - VinP/N100)*(16R/24R)
V93/V162	VinP/N76 - (VinP/N76 - VinP/N100)*(17R/24R)
V94/V161	VinP/N76 - (VinP/N76 - VinP/N100)*(18R/24R)
V95/V160	VinP/N76 - (VinP/N76 - VinP/N100)*(19R/24R)
V96/V159	VinP/N76 - (VinP/N76 - VinP/N100)*(20R/24R)
V97/V158	VinP/N76 - (VinP/N76 - VinP/N100)*(21R/24R)
V98/V157	VinP/N76 - (VinP/N76 - VinP/N100)*(22R/24R)
V99/V156	VinP/N76 - (VinP/N76 - VinP/N100)*(23R/24R)
V100/V155	VinP/N100
V101/V154	VinP/N100 - (VinP/N100 - VinP/N128)*(1R/28R)
V102/V153	VinP/N100 - (VinP/N100 - VinP/N128)*(2R/28R)
V103/V152	VinP/N100 - (VinP/N100 - VinP/N128)*(3R/28R)
V104/V151	VinP/N100 - (VinP/N100 - VinP/N128)*(4R/28R)
V105/V150	VinP/N100 - (VinP/N100 - VinP/N128)*(5R/28R)
V106/V149	VinP/N100 - (VinP/N100 - VinP/N128)*(6R/28R)
V107/V148	VinP/N100 - (VinP/N100 - VinP/N128)*(7R/28R)
V108/V147	VinP/N100 - (VinP/N100 - VinP/N128)*(8R/28R)
V109/V146	VinP/N100 - (VinP/N100 - VinP/N128)*(9R/28R)
V110/V145	VinP/N100 - (VinP/N100 - VinP/N128)*(10R/28R)
V111/V144	VinP/N100 - (VinP/N100 - VinP/N128)*(11R/28R)
V112/V143	VinP/N100 - (VinP/N100 - VinP/N128)*(12R/28R)
V113/V142	VinP/N100 - (VinP/N100 - VinP/N128)*(13R/28R)
V114/V141	VinP/N100 - (VinP/N100 - VinP/N128)*(14R/28R)
V115/V140	VinP/N100 - (VinP/N100 - VinP/N128)*(15R/28R)
V116/V139	VinP/N100 - (VinP/N100 - VinP/N128)*(16R/28R)
V117/V138	VinP/N100 - (VinP/N100 - VinP/N128)*(17R/28R)
V118/V137	VinP/N100 - (VinP/N100 - VinP/N128)*(18R/28R)
V119/V136	VinP/N100 - (VinP/N100 - VinP/N128)*(19R/28R)
V120/V135	VinP/N100 - (VinP/N100 - VinP/N128)*(20R/28R)
V121/V134	VinP/N100 - (VinP/N100 - VinP/N128)*(21R/28R)
V122/V133	VinP/N100 - (VinP/N100 - VinP/N128)*(22R/28R)
V123/V132	VinP/N100 - (VinP/N100 - VinP/N128)*(23R/28R)
V124/V131	VinP/N100 - (VinP/N100 - VinP/N128)*(24R/28R)
V125/V130	VinP/N100 - (VinP/N100 - VinP/N128)*(25R/28R)
V12V129	VinP/N100 - (VinP/N100 - VinP/N128)*(26R/28R)
V127/V128	VinP/N100 - (VinP/N100 - VinP/N128)*(27R/28R)
V128/V127	VinP/N128
V129/V126	VinP/N128 - (VinP/N128 - VinP/N156)*(1R/28R)
V130/V125	VinP/N128 - (VinP/N128 - VinP/N156)*(2R/28R)
V131/V124	VinP/N128 - (VinP/N128 - VinP/N156)*(3R/28R)

Grayscale voltage (NW/NB)	Formula
V132/V123	VinP/N128 - (VinP/N128 - VinP/N156)*(4R/28R)
V133/V122	VinP/N128 - (VinP/N128 - VinP/N156)*(5R/28R)
V134/V121	VinP/N128 - (VinP/N128 - VinP/N156)*(6R/28R)
V135/V120	VinP/N128 - (VinP/N128 - VinP/N156)*(7R/28R)
V136/V119	VinP/N128 - (VinP/N128 - VinP/N156)*(8R/28R)
V137/V118	VinP/N128 - (VinP/N128 - VinP/N156)*(9R/28R)
V138/V117	VinP/N128 - (VinP/N128 - VinP/N156)*(10R/28R)
V139/V116	VinP/N128 - (VinP/N128 - VinP/N156)*(11R/28R)
V140/V115	VinP/N128 - (VinP/N128 - VinP/N156)*(12R/28R)
V141/V114	VinP/N128 - (VinP/N128 - VinP/N156)*(13R/28R)
V142/V113	VinP/N128 - (VinP/N128 - VinP/N156)*(14R/28R)
V143/V112	VinP/N128 - (VinP/N128 - VinP/N156)*(15R/28R)
V144/V111	VinP/N128 - (VinP/N128 - VinP/N156)*(16R/28R)
V145/V110	VinP/N128 - (VinP/N128 - VinP/N156)*(17R/28R)
V146/V109	VinP/N128 - (VinP/N128 - VinP/N156)*(18R/28R)
V147/V108	VinP/N128 - (VinP/N128 - VinP/N156)*(19R/28R)
V148/V107	VinP/N128 - (VinP/N128 - VinP/N156)*(20R/28R)
V149/V106	VinP/N128 - (VinP/N128 - VinP/N156)*(21R/28R)
V150/V105	VinP/N128 - (VinP/N128 - VinP/N156)*(22R/28R)
V151/V104	VinP/N128 - (VinP/N128 - VinP/N156)*(23R/28R)
V152/V103	VinP/N128 - (VinP/N128 - VinP/N156)*(24R/28R)
V153/V102	VinP/N128 - (VinP/N128 - VinP/N156)*(25R/28R)
V154/V101	VinP/N128 - (VinP/N128 - VinP/N156)*(26R/28R)
V155/V100	VinP/N128 - (VinP/N128 - VinP/N156)*(27R/28R)
V156/V99	VinP/N156
V157/V98	VinP/N156 - (VinP/N156 - VinP/N180)*(1R/24R)
V158/V97	VinP/N156 - (VinP/N156 - VinP/N180)*(2R/24R)
V159/V96	VinP/N156 - (VinP/N156 - VinP/N180)*(3R/24R)
V160/V95	VinP/N156 - (VinP/N156 - VinP/N180)*(4R/24R)
V161/V94	VinP/N156 - (VinP/N156 - VinP/N180)*(5R/24R)
V162/V93	VinP/N156 - (VinP/N156 - VinP/N180)*(6R/24R)
V163/V92	VinP/N156 - (VinP/N156 - VinP/N180)*(7R/24R)
V164/V91	VinP/N156 - (VinP/N156 - VinP/N180)*(8R/24R)
V165/V90	VinP/N156 - (VinP/N156 - VinP/N180)*(9R/24R)
V166/V89	VinP/N156 - (VinP/N156 - VinP/N180)*(10R/24R)
V167/V88	VinP/N156 - (VinP/N156 - VinP/N180)*(11R/24R)
V168/V87	VinP/N156 - (VinP/N156 - VinP/N180)*(12R/24R)
V169/V86	VinP/N156 - (VinP/N156 - VinP/N180)*(13R/24R)
V170/V85	VinP/N156 - (VinP/N156 - VinP/N180)*(14R/24R)
V171/V84	VinP/N156 - (VinP/N156 - VinP/N180)*(15R/24R)
V172/V83	VinP/N156 - (VinP/N156 - VinP/N180)*(16R/24R)
V173/V82	VinP/N156 - (VinP/N156 - VinP/N180)*(17R/24R)
V174/V81	VinP/N156 - (VinP/N156 - VinP/N180)*(18R/24R)
V175/V80	VinP/N156 - (VinP/N156 - VinP/N180)*(19R/24R)

Himax Confidential
Do Not Copy

Grayscale voltage (NW/NB)	Formula
V176/V79	VinP/N156 - (VinP/N156 - VinP/N180)*(20R/24R)
V177/V78	VinP/N156 - (VinP/N156 - VinP/N180)*(21R/24R)
V178/V77	VinP/N156 - (VinP/N156 - VinP/N180)*(22R/24R)
V179/V76	VinP/N156 - (VinP/N156 - VinP/N180)*(23R/24R)
V180/V75	VinP/N180
V181/V74	VinP/N180 - (VinP/N180 - VinP/N204)*(1R/24R)
V182/V73	VinP/N180 - (VinP/N180 - VinP/N204)*(2R/24R)
V183/V72	VinP/N180 - (VinP/N180 - VinP/N204)*(3R/24R)
V184/V71	VinP/N180 - (VinP/N180 - VinP/N204)*(4R/24R)
V185/V70	VinP/N180 - (VinP/N180 - VinP/N204)*(5R/24R)
V186/V69	VinP/N180 - (VinP/N180 - VinP/N204)*(6R/24R)
V187/V68	VinP/N180 - (VinP/N180 - VinP/N204)*(7R/24R)
V188/V67	VinP/N180 - (VinP/N180 - VinP/N204)*(8R/24R)
V189/V66	VinP/N180 - (VinP/N180 - VinP/N204)*(9R/24R)
V190/V65	VinP/N180 - (VinP/N180 - VinP/N204)*(10R/24R)
V191/V64	VinP/N180 - (VinP/N180 - VinP/N204)*(11R/24R)
V192/V63	VinP/N180 - (VinP/N180 - VinP/N204)*(12R/24R)
V193/V62	VinP/N180 - (VinP/N180 - VinP/N204)*(13R/24R)
V194/V61	VinP/N180 - (VinP/N180 - VinP/N204)*(14R/24R)
V195/V60	VinP/N180 - (VinP/N180 - VinP/N204)*(15R/24R)
V196/V59	VinP/N180 - (VinP/N180 - VinP/N204)*(16R/24R)
V197/V58	VinP/N180 - (VinP/N180 - VinP/N204)*(17R/24R)
V198/V57	VinP/N180 - (VinP/N180 - VinP/N204)*(18R/24R)
V199/V56	VinP/N180 - (VinP/N180 - VinP/N204)*(19R/24R)
V200/V55	VinP/N180 - (VinP/N180 - VinP/N204)*(20R/24R)
V201/V54	VinP/N180 - (VinP/N180 - VinP/N204)*(21R/24R)
V202/V53	VinP/N180 - (VinP/N180 - VinP/N204)*(22R/24R)
V203/V52	VinP/N180 - (VinP/N180 - VinP/N204)*(23R/24R)
V204/V51	VinP/N204
V205/V50	VinP/N204 - (VinP/N204 - VinP/N216)*(1R/12R)
V206/V49	VinP/N204 - (VinP/N204 - VinP/N216)*(2R/12R)
V207/V48	VinP/N204 - (VinP/N204 - VinP/N216)*(3R/12R)
V208/V47	VinP/N204 - (VinP/N204 - VinP/N216)*(4R/12R)
V209/V46	VinP/N204 - (VinP/N204 - VinP/N216)*(5R/12R)
V210/V45	VinP/N204 - (VinP/N204 - VinP/N216)*(6R/12R)
V211/V44	VinP/N204 - (VinP/N204 - VinP/N216)*(7R/12R)
V212/V43	VinP/N204 - (VinP/N204 - VinP/N216)*(8R/12R)
V213/V42	VinP/N204 - (VinP/N204 - VinP/N216)*(9R/12R)
V214/V41	VinP/N204 - (VinP/N204 - VinP/N216)*(10R/12R)
V215/V40	VinP/N204 - (VinP/N204 - VinP/N216)*(11R/12R)

Grayscale voltage (NW/NB)	Formula
V216/V39	VinP/N216
V217/V38	VinP/N216 - (VinP/N216 - VinP/N228)*(1R/12R)
V218/V37	VinP/N216 - (VinP/N216 - VinP/N228)*(2R/12R)
V219/V36	VinP/N216 - (VinP/N216 - VinP/N228)*(3R/12R)
V220/V35	VinP/N216 - (VinP/N216 - VinP/N228)*(4R/12R)
V221/V34	VinP/N216 - (VinP/N216 - VinP/N228)*(5R/12R)
V222/V33	VinP/N216 - (VinP/N216 - VinP/N228)*(6R/12R)
V223/V32	VinP/N216 - (VinP/N216 - VinP/N228)*(7R/12R)
V224/V31	VinP/N216 - (VinP/N216 - VinP/N228)*(8R/12R)
V225/V30	VinP/N216 - (VinP/N216 - VinP/N228)*(9R/12R)
V226/V29	VinP/N216 - (VinP/N216 - VinP/N228)*(10R/12R)
V227/V28	VinP/N216 - (VinP/N216 - VinP/N228)*(11R/12R)
V228/V27	VinP/N228
V229/V26	VinP/N228 - (VinP/N228 - VinP/N236)*(1R/8R)
V230/V25	VinP/N228 - (VinP/N228 - VinP/N236)*(2R/8R)
V231/V24	VinP/N228 - (VinP/N228 - VinP/N236)*(3R/8R)
V232/V23	VinP/N228 - (VinP/N228 - VinP/N236)*(4R/8R)
V233/V22	VinP/N228 - (VinP/N228 - VinP/N236)*(5R/8R)
V234/V21	VinP/N228 - (VinP/N228 - VinP/N236)*(6R/8R)
V235/V20	VinP/N228 - (VinP/N228 - VinP/N236)*(7R/8R)
V236/V19	VinP/N236
V237/V18	VinP/N236 - (VinP/N236 - VinP/N240)*(1R/4R)
V238/V17	VinP/N236 - (VinP/N236 - VinP/N240)*(2R/4R)
V239/V16	VinP/N236 - (VinP/N236 - VinP/N240)*(3R/4R)
V240/V15	VinP/N240
V241/V14	VinP/N240 - (VinP/N240 - VinP/N243)*(1R/3R)
V242/V13	VinP/N240 - (VinP/N240 - VinP/N243)*(2R/3R)
V243/V12	VinP/N243
V244/V11	VinP/N243 - (VinP/N243 - VinP/N246)*(1R/3R)
V245/V10	VinP/N243 - (VinP/N243 - VinP/N246)*(2R/3R)
V246/V9	VinP/N246
V247/V8	VinP/N246 - (VinP/N246 - VinP/N248)*(1R/2R)
V248/V7	VinP/N248
V249/V6	VinP/N248 - (VinP/N248 - VinP/N250)*(1R/2R)
V250/V5	VinP/N250
V251/V4	VinP/N250 - (VinP/N250 - VinP/N252)*(1R/2R)
V252/V3	VinP/N252
V253/V2	VinP/N252 - (VinP/N252 - VinP/N254)*(1R/2R)
V254/V1	VinP/N254
V255/V0	VinP/N255

Table 5.36: Voltage calculation formula of 256-grayscale voltage (positive/negative polarity)

Himax Confidential

5.8.2 Gray voltage generator for digital gamma correction

The HX8394-F digital gamma correction can reach the independent GAMMA curve of RGB. HX8394-F utilizes DGC_LUT (Digital Gamma Correction Look Up Table) to change input data from 8-bit into 10-bit and sends 10-bit data to Dithering circuit, and then drive Source Driver via Dithering circuit.

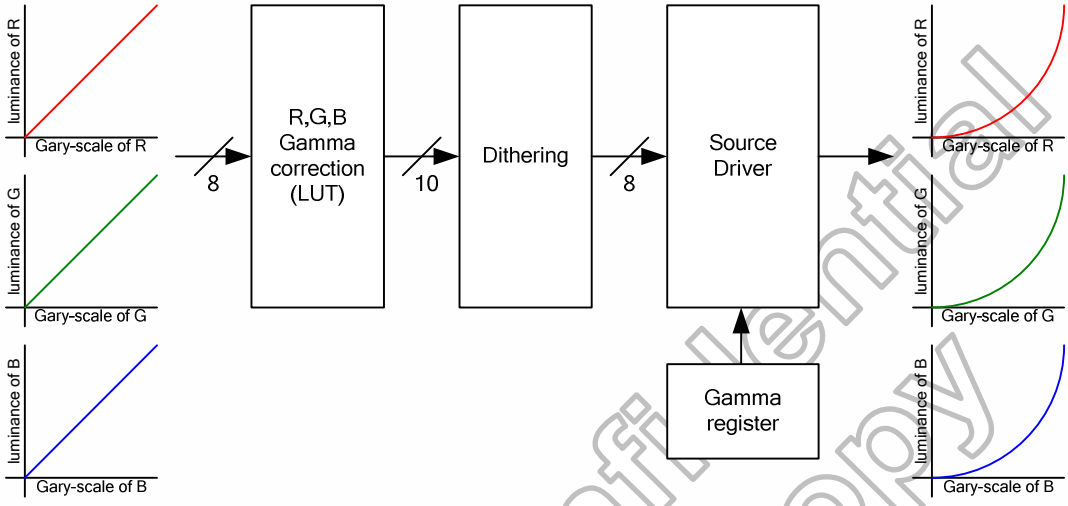


Figure 5.24: Block diagram of digital gamma correction

There are 126 bytes DGC LUT to set R, G, B gamma independently. When DGC_EN=1, R, G, B gamma will mapping V0, V8, V16, ..., V240, V248, V255 voltage to the LUT register setting gray level voltage.

LUT	D7	D6	D5	D4	D3	D2	D1	D0	Default
1st	R009	R008	R007	R006	R005	R004	R003	R002	00h
2nd	R019	R018	R017	R016	R015	R014	R013	R012	08h
3rd	R029	R028	R027	R026	R025	R024	R023	R022	10h
:	:	:	:	:	:	:	:	:	:
32th	R319	R318	R317	R316	R315	R314	R313	R312	F8h
33th	R329	R328	R327	R326	R325	R324	R323	R322	FFh
34th	R001	R000	R011	R010	R021	R020	R031	R030	00h
35th	R041	R040	R051	R050	R061	R060	R071	R070	00h
:	:	:	:	:	:	:	:	:	:
41th	R281	R280	R291	R290	R301	R300	R311	R310	00h
42th	R321	R320	0	0	0	0	0	0	00h
43th	G009	G008	G007	G006	G005	G004	G003	G002	00h
44th	G019	G018	G017	G016	G015	G014	G013	G012	08h
45th	G029	G028	G027	G026	G025	G024	G023	G022	10h
:	:	:	:	:	:	:	:	:	:
74th	G319	G318	G317	G316	G315	G314	G313	G312	F8h
75th	G329	G328	G327	G326	G325	G324	G323	G322	FFh
76th	G001	G000	G011	G010	G021	G020	G031	G030	00h
77th	G041	G040	G051	G050	G061	G060	G071	G070	00h
:	:	:	:	:	:	:	:	:	:
83th	G281	G280	G291	G290	G301	G300	G311	G310	00h
84th	G321	G320	0	0	0	0	0	0	00h
85th	B009	B008	B007	B006	B005	B004	B003	B002	00h
86th	B019	B018	B017	B016	B015	B014	B013	B012	08h
87th	B029	B028	B027	B026	B025	B024	B023	B022	10h
:	:	:	:	:	:	:	:	:	:
116th	B319	B318	B317	B316	B315	B314	B313	B312	F8h

Himax Confidential Do not Copy

117th	B329	B328	B327	B326	B325	B324	B323	B322	FFh
118th	B001	B000	B011	B010	B021	B020	B031	B030	00h
119th	B041	B040	B051	B050	B061	B060	B071	B070	00h
:	:	:	:	:	:	:	:	:	:
:	:	:	:	:	:	:	:	:	:
125th	B281	B280	B291	B290	B301	B300	B311	B310	00h
126th	B321	B320	0	0	0	0	0	0	00h

Table 5.37: DGC Look-up table

Himax Confidential
Do Not Copy

5.9 Characteristics of I/O

5.9.1 Output or bi-directional (I/O) pins

Output or bi-directional pins	After power on	After hardware reset	After software reset
TE	Low	Low	Low
TE1	Low	Low	Low
SDO	High-Z (Inactive)	High-Z (Inactive)	High-Z (Inactive)
CABC_PWM_OUT	Low	Low	Low

Note: Please follow D9h GPO output setting.

Table 5.38: Characteristics of output or bi-directional (I/O) pins

5.9.2 Input pins

Input pins	During power on process	After power on	After hardware reset	After software reset	During power off process
RESX	Input invalid	Input valid	Input valid	Input valid	Input invalid
CSX	Input invalid	Input valid	Input valid	Input valid	Input invalid
DCX	Input invalid	Input valid	Input valid	Input valid	Input invalid
SCL	Input invalid	Input valid	Input valid	Input valid	Input invalid
DB[23:0], SDI	Input invalid	Input valid	Input valid	Input valid	Input invalid
HSYNC	Input invalid	Input valid	Input valid	Input valid	Input invalid
VSYNC	Input invalid	Input valid	Input valid	Input valid	Input invalid
PCLK	Input invalid	Input valid	Input valid	Input valid	Input invalid
DE	Input invalid	Input valid	Input valid	Input valid	Input invalid
OSC	Input invalid	Input valid	Input valid	Input valid	Input invalid
TEST[2:0]	Low	Low	Low	Low	Low
FRM	Low	Low	Low	Low	Low
PCCS[1:0]	Input invalid	Input valid	Input valid	Input valid	Input invalid

Table 5.39: Characteristics of input pins

Himax Confidential
Do Not Copy

5.10 GIP control singal

HX8394-F is a single chip solution for a WXGA GIP (Gate In Panel) type TFT LCD display. There are many GIP/ASG type TFT panels that correspond to different GIP timing. Therefore, the GIP setting must be setup to the correct GIP/ASG timing for the normal display. The GIP timing adjustment is related to register 0xD5h SETGIP.

The GIP control signals (CGOUT1_L/R ~ CGOUT22_L/R) are for panel used. The assignment of each panel type is specified on the application note. Regarding the GIP/ASG timing, please refer to HX8394-F application note.

Himax Confidential
Do Not Copy

This information contained herein is the exclusive property of Himax and shall not be distributed, reproduced, or disclosed in whole or in part without prior written permission of Himax.

5.11 Sleep Out –command and self-diagnostic functions of the display module

5.11.1 Register loading detection

Sleep Out-command (See “Sleep Out (11h)”) is a trigger for an internal function of the display module, which indicates, if the display module loading function of factory default values from OTP (or similar device) to registers of the display controller is working properly. There are compared factory values of the OTP and register values of the display controller by the display controller. If those both values (OTP and register values) are same, there is inverted (=increased by 1) a bit, which is defined in command “Read Display Self-Diagnostic Result (0Fh)” (=RDDSDR) (The used bit of this command is D7). If those both values are not same, this bit (D7) is not inverted (=increased by 1).

The flow chart for this internal function is following:

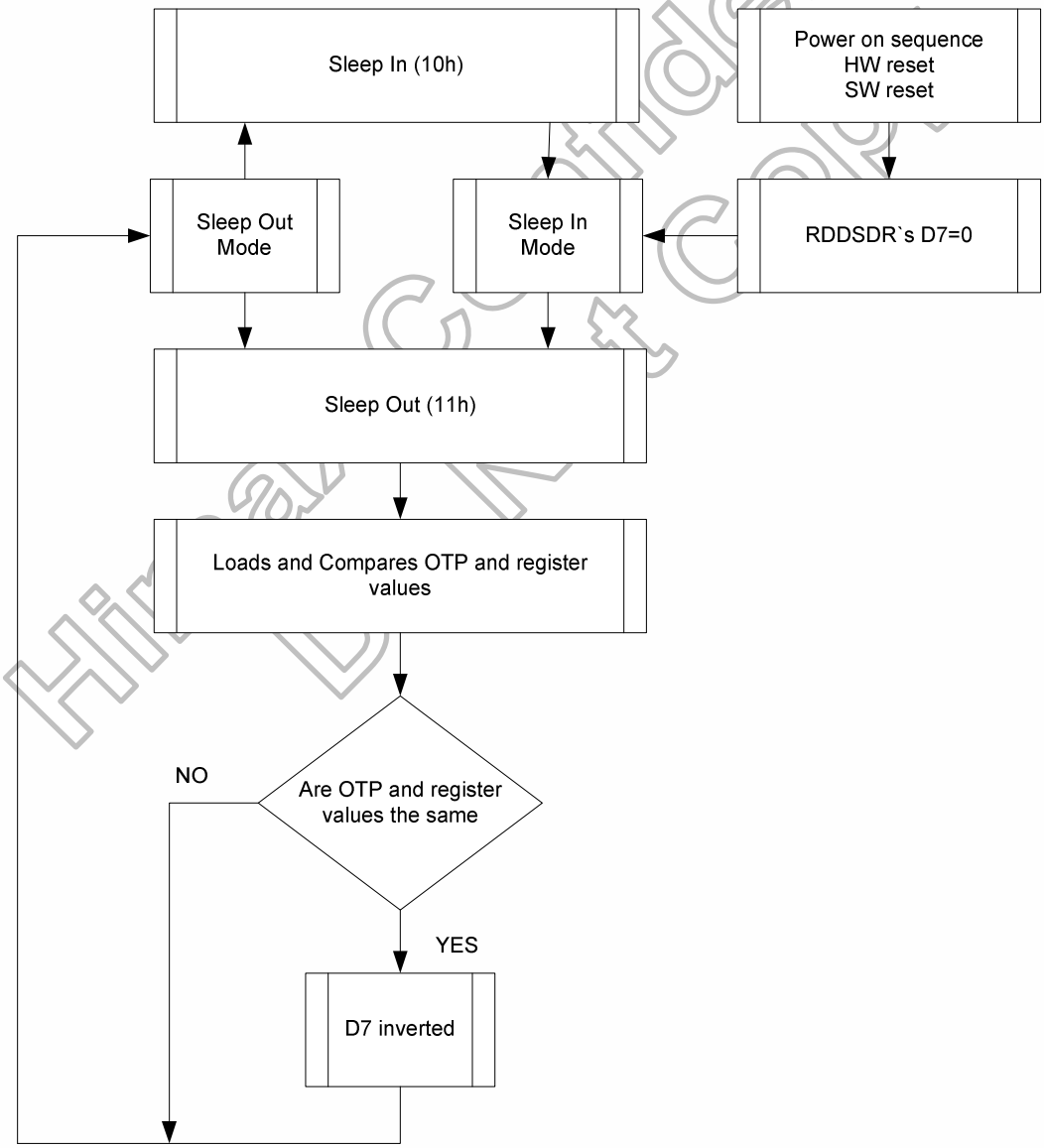


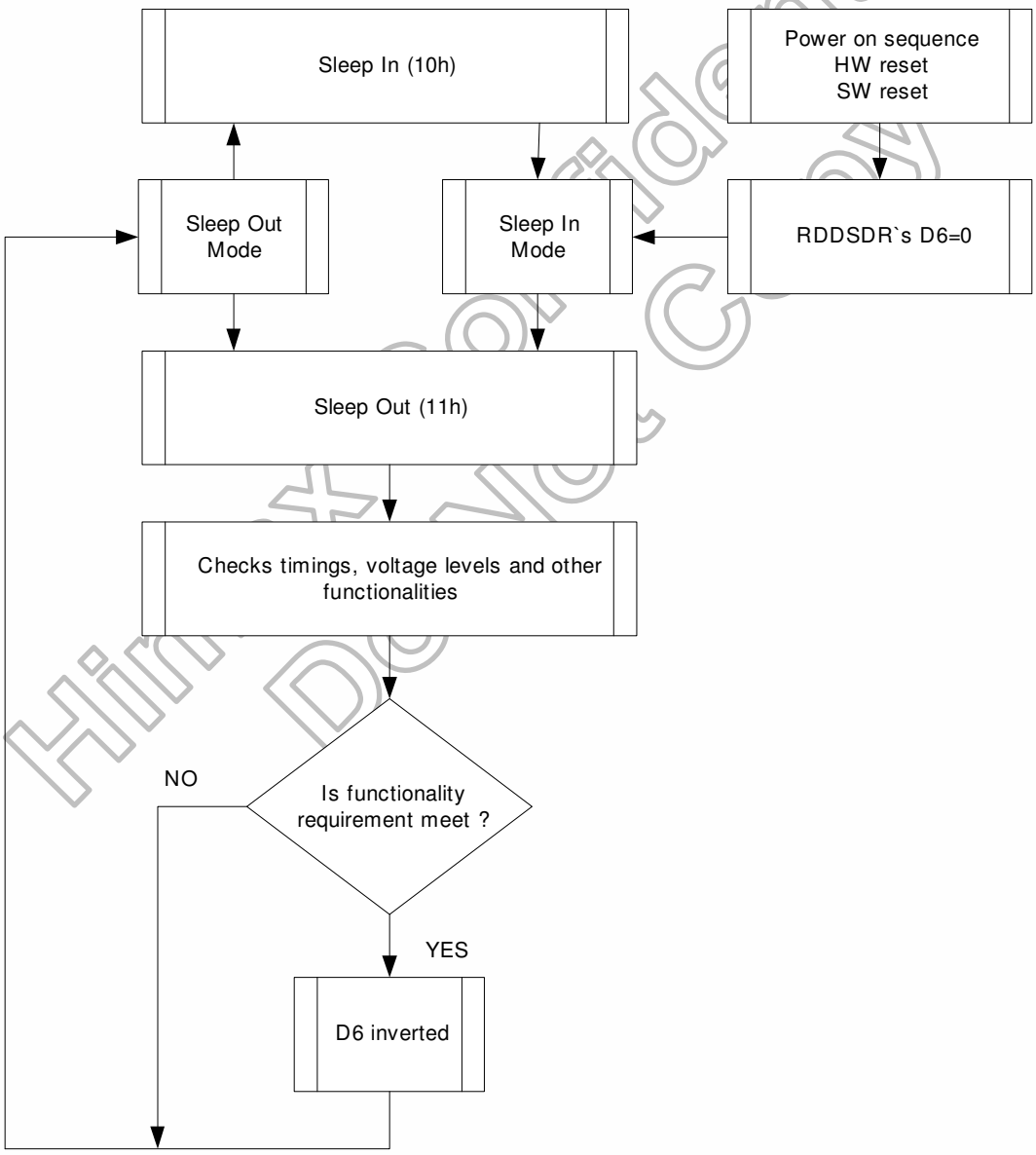
Figure 5.25: Sleep out flow chart–command and self-diagnostic functions

Himax Confidential Do Not Copy

5.11.2 Functionality detection

Sleep Out-command (See “Sleep Out (11h)”) is a trigger for an internal function of the display module, which indicates, if the display module is still running and meets functionality requirements.

The internal function (=the display controller) is comparing, if the display module still meets functionality requirements (e.g. booster voltage levels, timings, etc.). If functionality requirement is met, 1 bit will be inverted (=increased by 1), which is defined in command “Read Display Self- Diagnostic Result (0Fh)” (=RDDSDR) (The used bit of this command is D6). If functionality requirement is not the same, this bit (D6) is not inverted (=increased by 1). The flow chart for this internal function is shown as below.



Note: There is needed 120m sec. after Sleep Out –command, when there is changing from Sleep In–mode toSleep Out –mode, before there is possible to check if Customer’s functionality requirements are met and a value of RDDSDR’s D6 is valid. Otherwise, there is 5msec delay for D6’s value, when Sleep Out –command is sent in Sleep Out –mode.

Figure 5.26: Sleep out flow chart internal function detection

Himax Confidential
This information contained herein is the exclusive property of Himax and shall not be distributed, reproduced, or disclosed in whole or in part without prior written permission of Himax.

5.12 Power on/off sequence

The Power supply On/Off, Sleep In/Out and Display On/Off sequence is illustrated below.

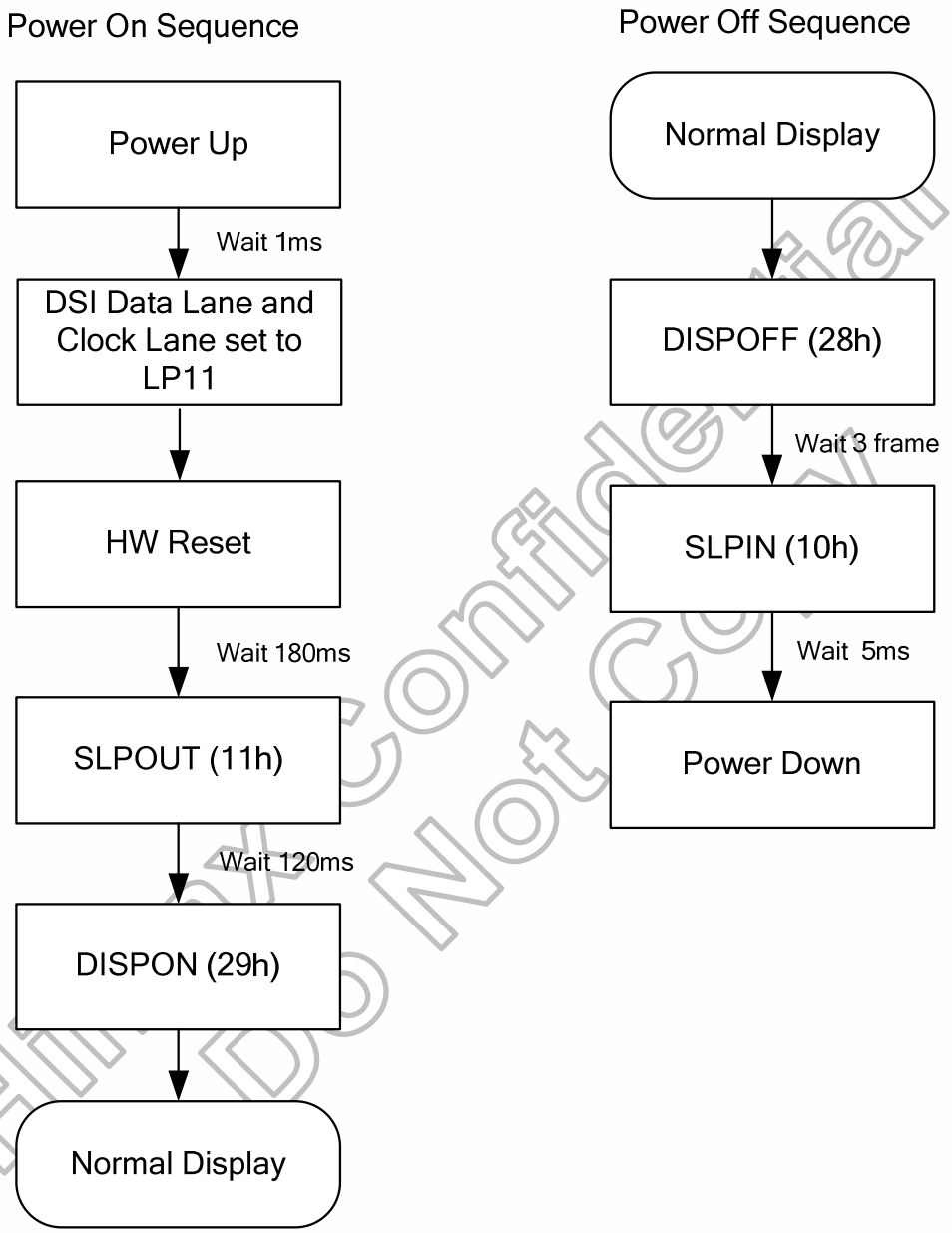


Figure 5.27: Power on/off sequence

Himax Confidential
Do Not Copy

5.12.1 VDD1/VDD3 input power sequence (PCCS[1:0]=”01”, “10”)

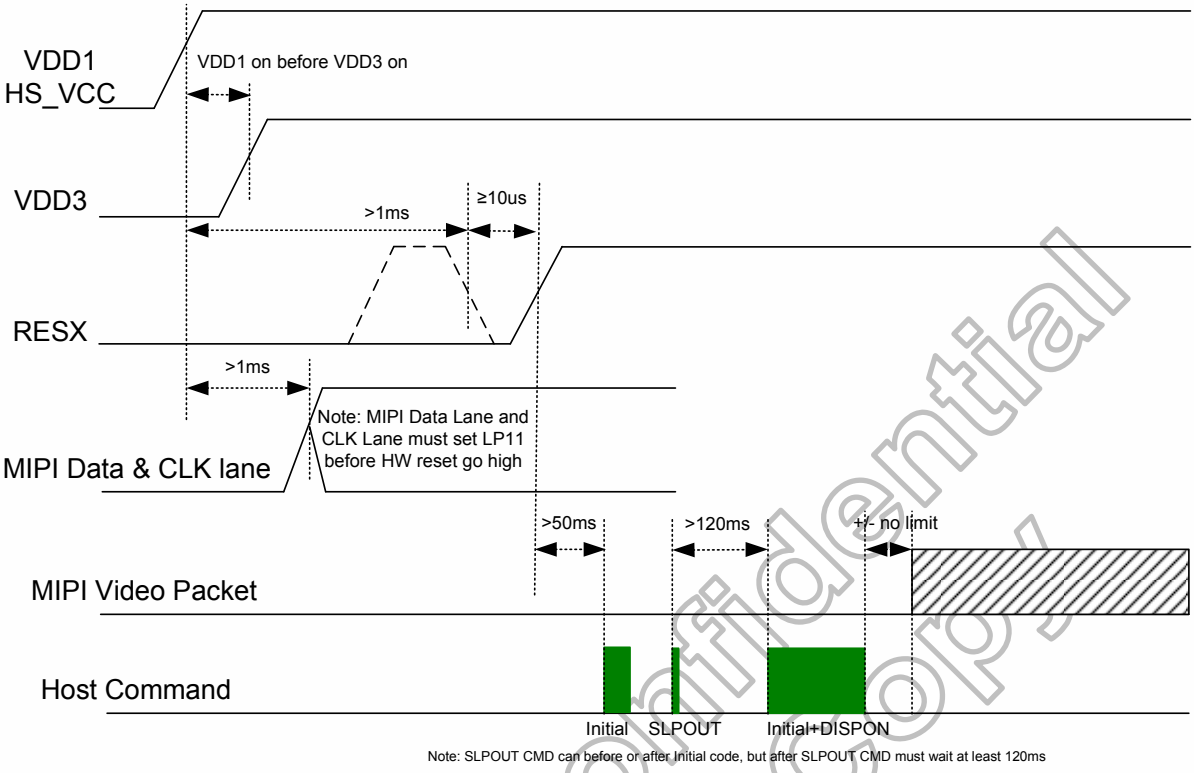


Figure 5.28: VDD1/VDD3 input power on sequence

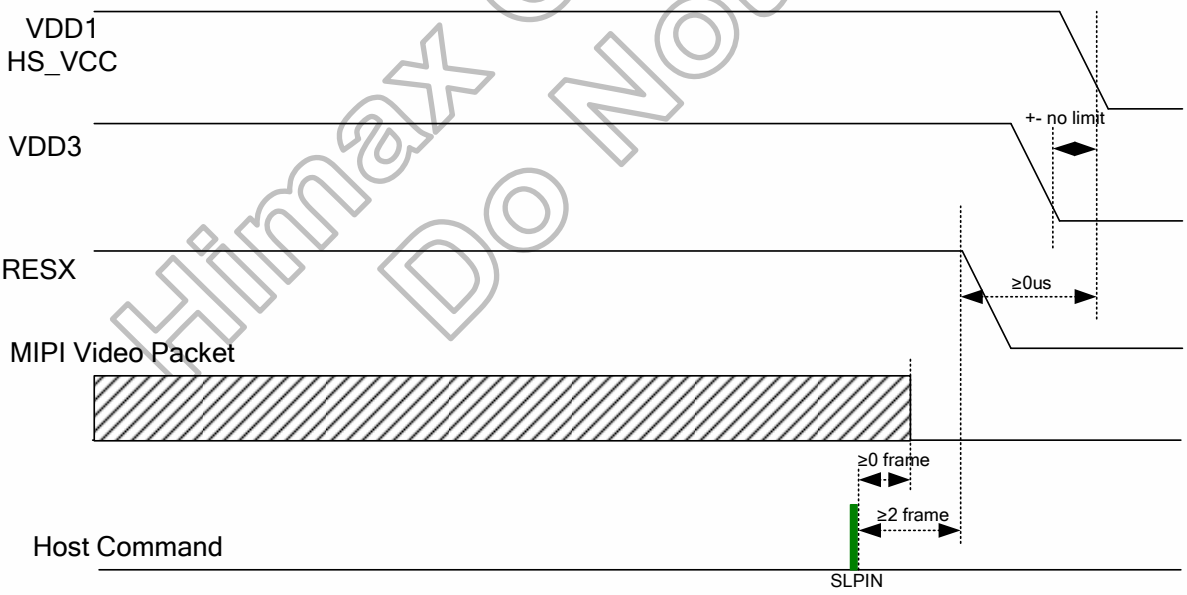


Figure 5.29: VDD1/VDD3 input power off sequence

Himax Confidential
Do Not Copy

5.12.2 VDD1/VSP/VSN input power sequence (PCCS[1:0]=”11”)

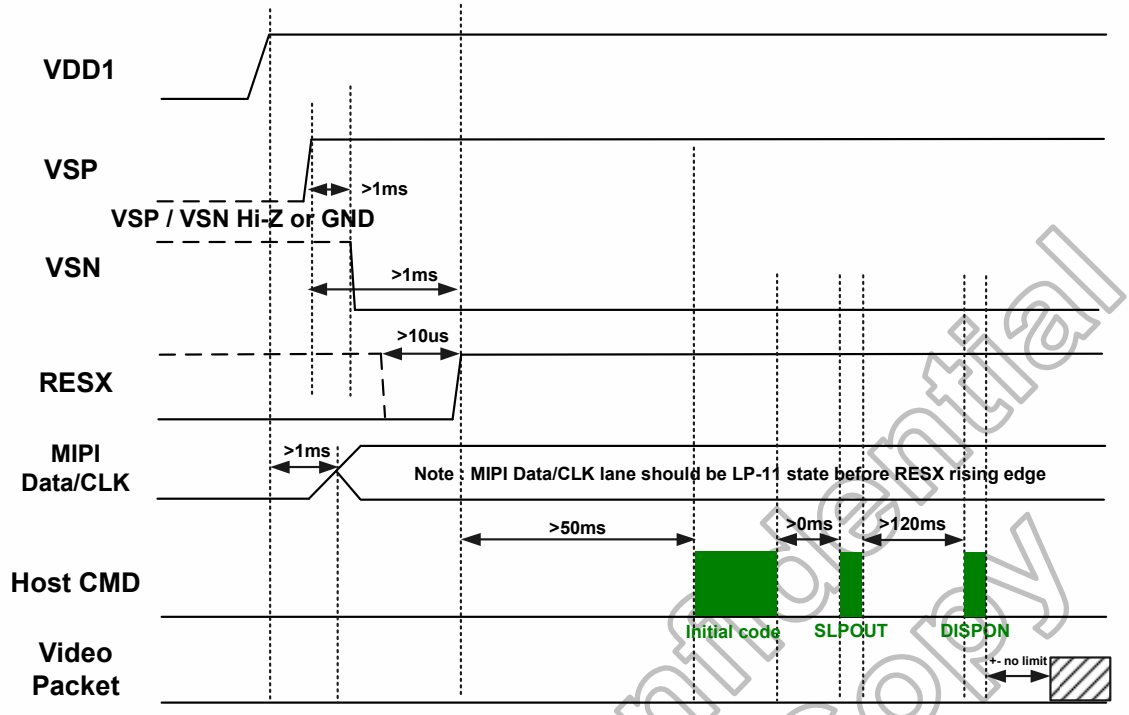


Figure 5.30: VDD1/ VSP/VSN input power on sequence

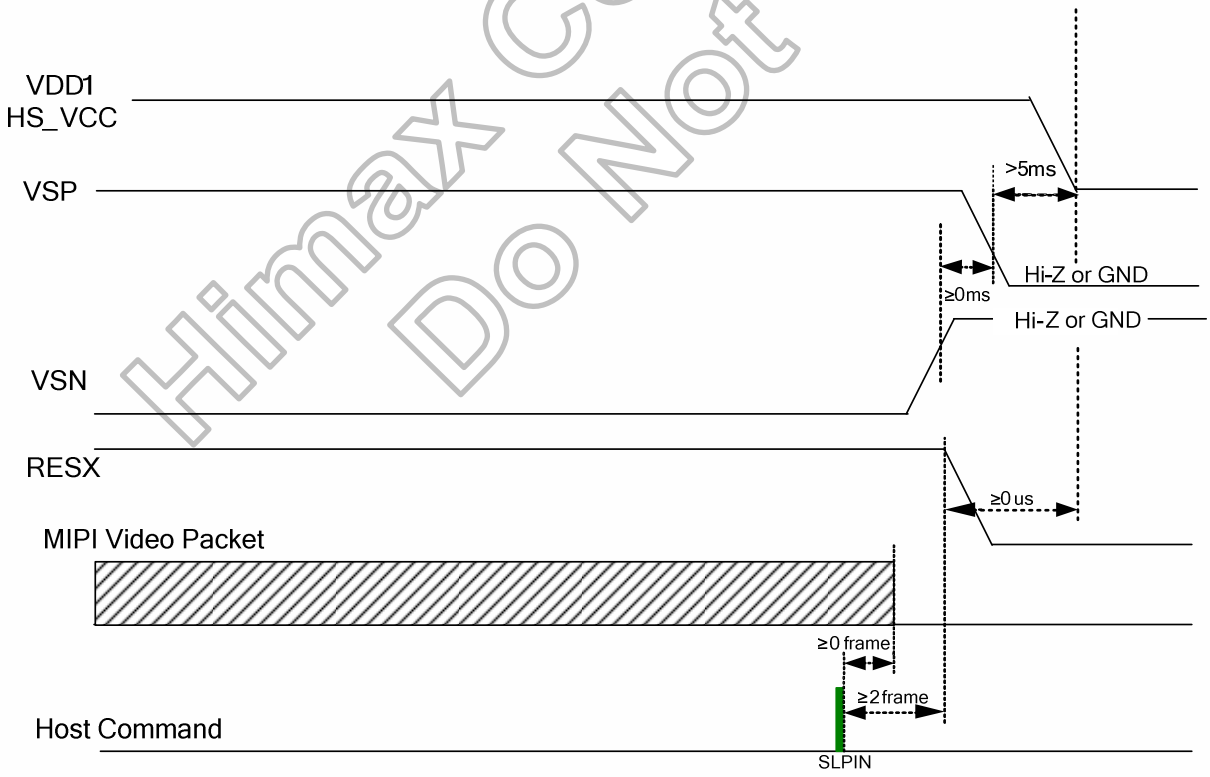
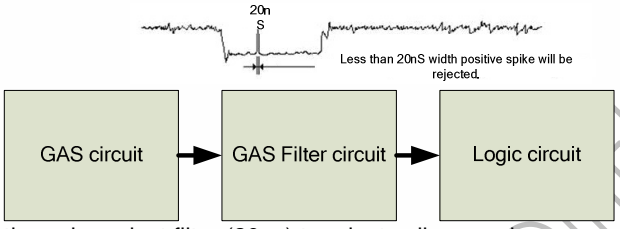


Figure 5.31: VDD1/VSP/VSN input power off sequence

Himax Confidential
Do Not Copy

5.13 Uncontrolled power off

The uncontrolled power off means a situation when e.g. there is removed a battery without the controlled power off sequence. There will not be any damages for the display module or the display module will not cause any damages for the host or lines of the interface. At an uncontrolled power off the display will go blank and there will not be any visible effects within 1 second on the display (blank display) and remains blank until “Power On Sequence” powers it up.



Note: HX8394-F is support the noise reject filter (20ns) to reject spike or noise.

Himax Confidential
Do Not Copy

5.14Content adaptive brightness control (CABC) function

The general block diagram of the CABC and the brightness control is illustrated below:

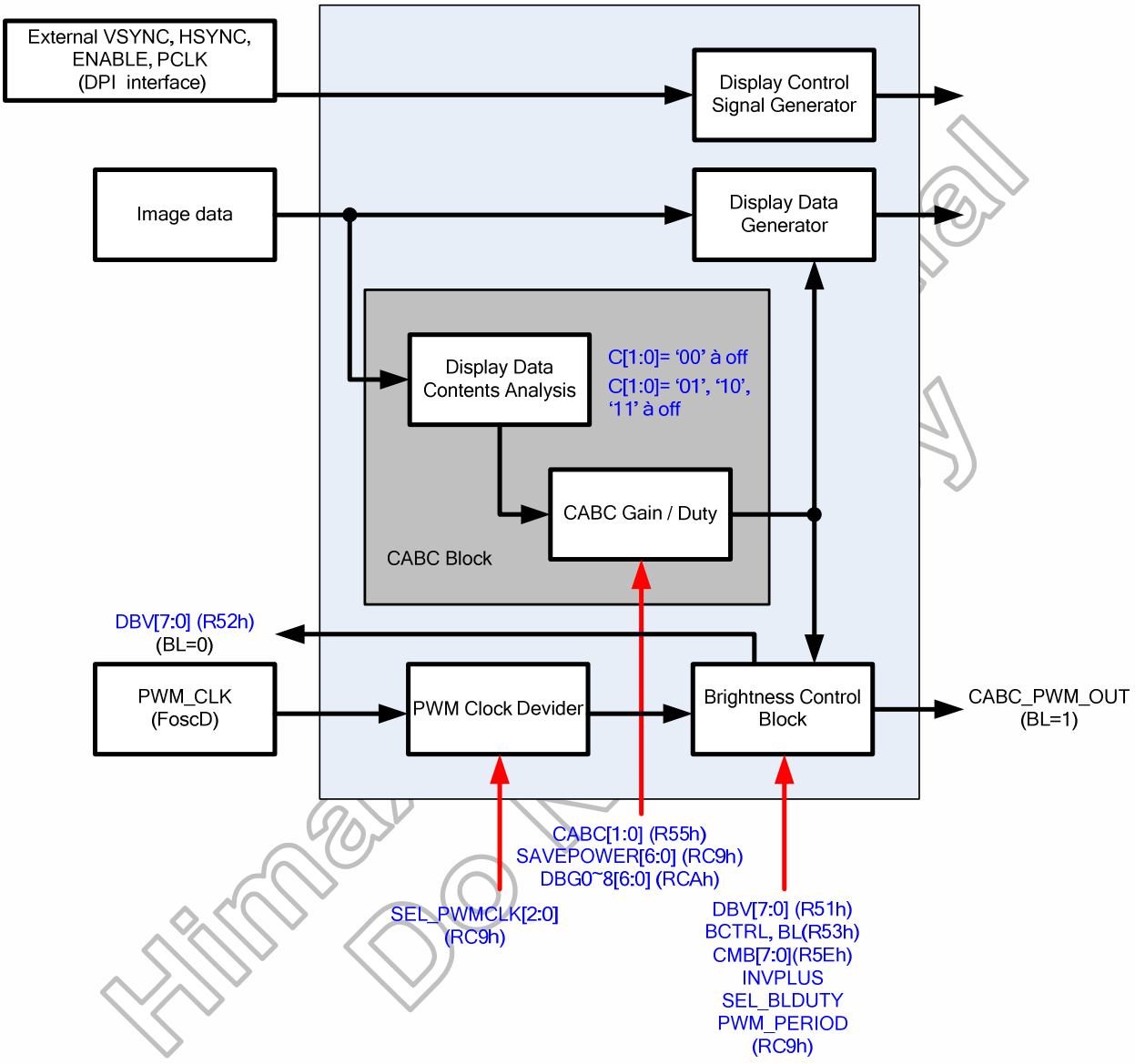


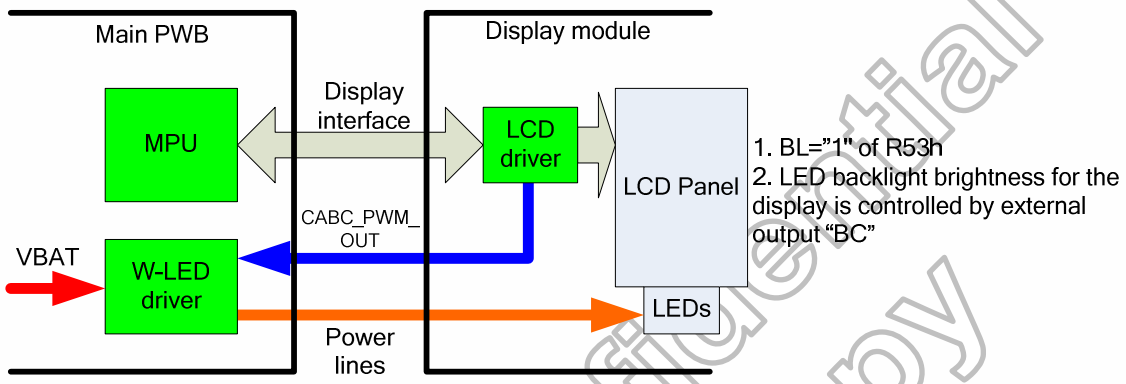
Figure 5.32: CABC block diagram

This information contained herein is the exclusive property of Himax and shall not be distributed, reproduced, or disclosed in whole or in part without prior written permission of Himax.

5.14.1 Module architectures

HX8394-F can support two module architectures for CABC operation. The BL bit setting of R53h can be used to select used display module architecture. White LED driver circuit for display backlight is located on the main PWB, not in the display module both in architecture I and II.

- Architecture I



- Architecture II

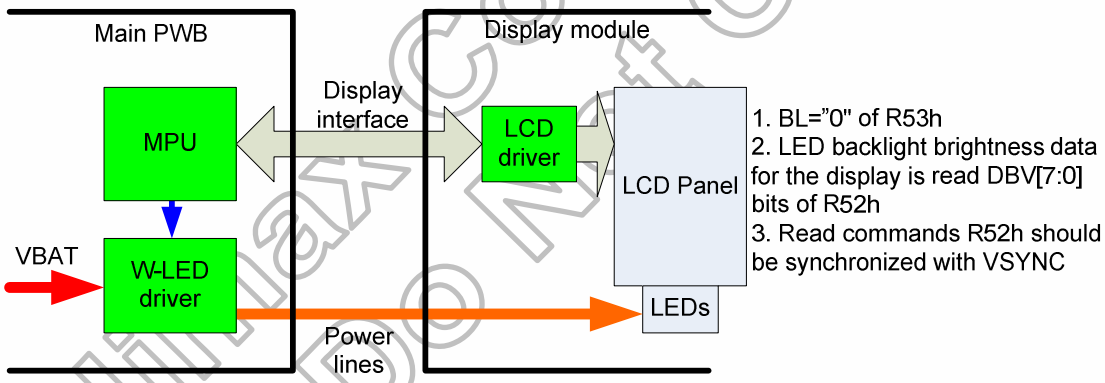


Figure 5.33: Module architecture

Himax Confidential
Do Not Copy

5.14.2 CABC block

There are DBG0~8[6:0] register bits in CABC block to define the “CABC gain”/ “CABC duty” table. Every DBGx[6:0] has 33 gain/duty value setting.

After one-frame display data content analysis, LSI will generate one CABC gain / CABC duty value calculated from DBG0~8[6:0] register bits setting (by using interpolated method) for display data generating and for backlight PWM pulse generating.

Please note that the CABC gain / CABC duty value calculated by the LSI is one of the 33 gain/duty value setting in DBGxx[6:0].

Please note that : Duty (valid level period (LED on) / one complete period)=1/ gain.

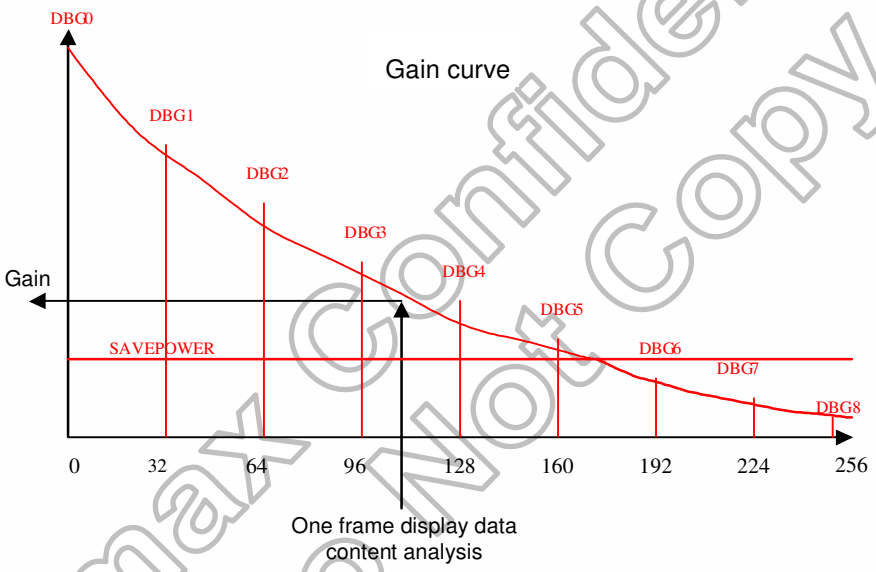


Figure 5.34: CABC gain / CABC duty generation

For power saving of backlight module, there are SAVEPOWER[6:0] bits to define the “minimum gain”/ “maximum duty” of CABC block output. If the CABC gain / duty after one-frame display data contents analysis is smaller **(gain)** / larger **(duty)** than SAVEPOWER[6:0] bits setting, the CABC block will output CABC gain / duty equal to SAVEPOWER[6:0] and ignore the result of display data contents analysis.

Himax Confidential
Do Not Copy

5.14.3 Brightness control block

There is an external output signal from brightness block, CABC_PWM_OUT, to control the LED driver IC in order to control display brightness.

There are resister bits, DBV[7:0] of R51h, for display brightness of manual brightness setting. The CABC_PWM_OUT duty is calculated as (DBV[7:0]) / 255 x CABC duty (generated after one-frame display data content analysis).

For ex: CABC_PWM_OUT period=2.95 ms, and DBV[7:0] (R51h)='228_{DEC}' and CABC duty is 74%. Then CABC_PWM_OUT duty=(228) / 255 x 74.42%≡66.54%. Correspond to the CABC_PWM_OUT period=2.95 ms, the high-level of CABC_PWM_OUT (high effective)=1.96ms, and the low-level of CABC_PWM_OUT =0.99ms.

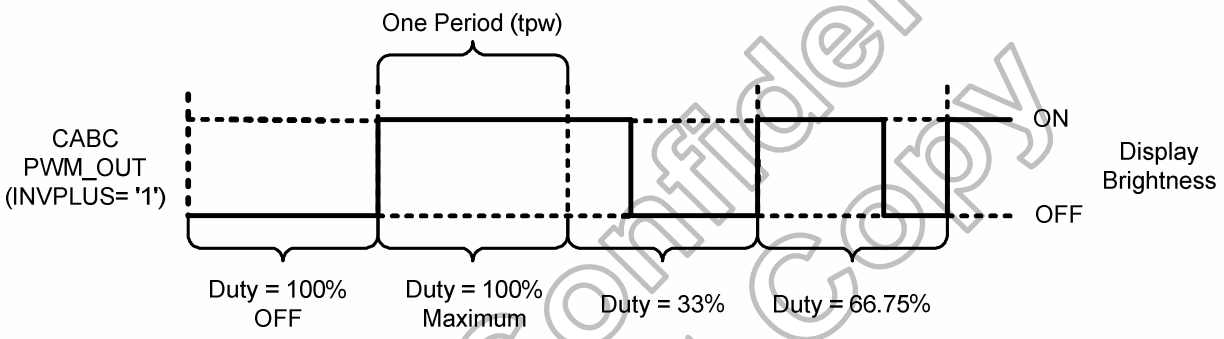


Figure 5.35: CABC_PWM_OUT output duty

Symbol	Parameter	Min.	Max.	Unit	Description
tpw	Pulse width	0.0333	8.33	ms	-

Table 5.40: CABC timing table

Note: (1) The signal rise and fall times (tf, tr) are stipulated to be equal to or less than 15ns.
(2) The pulse width range by setting CABC related registers is locate between 0.0333ms to 8.33ms.

When Architecture II module is used (BL='0') with the example below, the CABC_PWM_OUT is always output low and the DBV[7:0] (R51h) will be read a value as 169_{DEC} ((169) / 255≡66.27%).

Himax Confidential
Do Not Copy

5.14.4 Minimum brightness setting of CABC function

CABC function is automatically reduced backlight brightness based on image contents. In the case of the combination with the CABC or manual brightness setting, display brightness is too dark. It must affect to image quality degradation. CABC minimum brightness setting (CMB[7:0] bits of R5Eh) is to avoid too much brightness reduction.

When CABC is active, CABC can not reduce the display brightness to less than CABC minimum brightness setting. Image processing function is worked as normal, even if the brightness can not be changed.

This function does not affect to the other function, manual brightness setting. Manual brightness can be set the display brightness to less than CABC minimum brightness. Smooth transition and dimming function can be worked as normal.

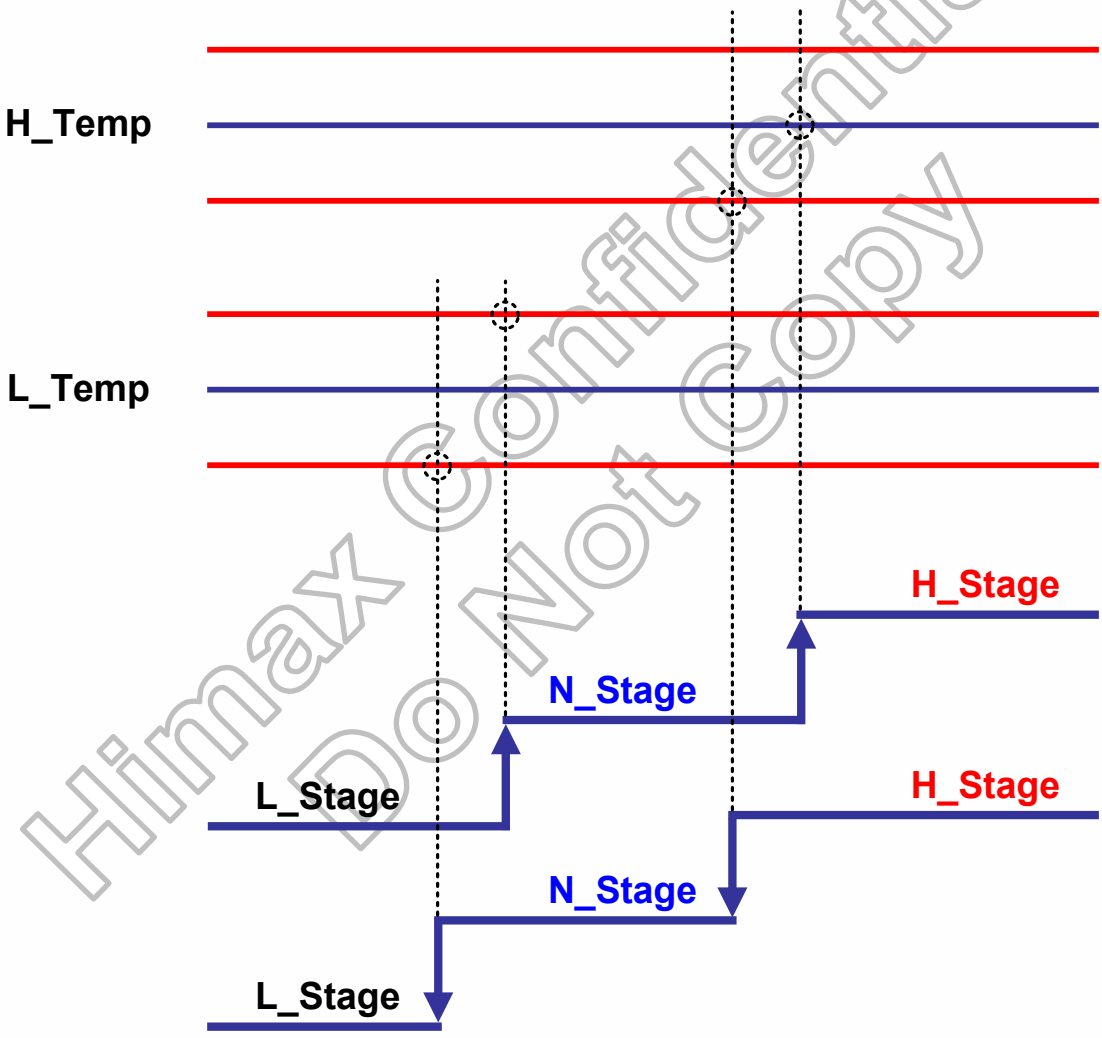
When display brightness is turned off (BCTRL='0' of R53h), CABC minimum brightness setting is ignored. "CMB[7:0], Read CABC minimum brightness (R5Fh)" always read the setting value of "CMB[7:0], Write CABC minimum brightness (R5Eh)"

Himax Confidential
Do Not Copy

5.15Temperature sensor

HX8394-F supports temperature sensor function that changes to H_Temp or L_Temp mode in different environment temperature and it also achieves low power consumption to save power dissipation. Besides, this function can cover panel GIP timing deviation when environment in high or low temperature.

At HX8394-F temperature sensor setting, user can set the temperature upper (H_Temp) and lower boundary (L_Temp). Following those setting, voltage and source output will automatically change to suitable value when environment temperature reaches upper or lower boundary setting.



Himax Confidential
Do Not Copy

Figure 5.36: Tempeture sensor diagram

5.16OTP programming

5.16.1 OTP table

OTP_INDEX (HEX)	B7	B6	B5	B4	B3	B2	B1	B0
0	ID1_1[7:0]							
1	ID2_1[7:0]							
2	ID3_1[7:0]							
3	ID4_1[7:0]							
4	ID1_2[7:0]							
5	ID2_2[7:0]							
6	ID3_2[7:0]							
7	ID4_2[7:0]							
8	ID1_3[7:0]							
9	ID2_3[7:0]							
A	ID3_3[7:0]							
B	ID4_3[7:0]							
C	NVALID_ID1	NVALID_ID2	NVALID_ID3	-	-	-	-	-
D	VCMC_F1[7:0]							
E	VCMC_B1[7:0]							
F	VCMC_F2[7:0]							
10	VCMC_B2[7:0]							
11	VCMC_F3[7:0]							
12	VCMC_B3[7:0]							
13	-	-	VCMC_B38	VCMC_B38	VCMC_B28	VCMC_B28	VCMC_B18	VCMC_B18
14	NVALID_VCMC1	NVALID_VCMC2	NVALID_VCMC3	-	-	-	-	-
15	NVALID_PANEL	-	-	-	SS_PANEL	GS_PANEL	REV_PANE	BGR_PANEL
45	NVALID_GAMMA	VHP_0[6:0]						
46	-	VHP_1[6:0]						
47	-	VHP_2[6:0]						
48	-	VHP_3[6:0]						
49	-	VHP_4[6:0]						
4A	-	VHP_5[6:0]						
4B	-	VHP_6[6:0]						
4C	-	VHP_7[6:0]						
4D	VMP_0[7:0]							
4E	VMP_1[7:0]							
4F	VMP_2[7:0]							
50	VMP_3[7:0]							
51	VMP_4[7:0]							
52	VMP_5[7:0]							
53	VMP_6[7:0]							
54	VMP_7[7:0]							
55	VMP_8[7:0]							
56	VMP_9[7:0]							
57	VMP_10[7:0]							
58	VMP_11[7:0]							
59	VMP_12[7:0]							
5A	-	VLP_0[6:0]						
5B	-	VLP_1[6:0]						
5C	-	VLP_2[6:0]						
5D	-	VLP_3[6:0]						
5E	-	VLP_4[6:0]						
5F	-	VLP_5[6:0]						
60	-	VLP_6[6:0]						
61	-	VLP_7[6:0]						
62	-	VHN_0[6:0]						
63	-	VHN_1[6:0]						
64	-	VHN_2[6:0]						
65	-	VHN_3[6:0]						

This information contained herein is the exclusive property of Himax and shall not be distributed, reproduced, or disclosed in whole or in part without prior written permission of Himax.

66	-	VHN_4[6:0]						
67	-	VHN_5[6:0]						
68	-	VHN_6[6:0]						
69	-	VHN_7[6:0]						
6A	VMN_0[7:0]							
6B	VMN_1[7:0]							
6C	VMN_21[7:0]							
6D	VMN_3[7:0]							
6E	VMN_4[7:0]							
6F	VMN_5[7:0]							
70	VMN_6[7:0]							
71	VMN_7[7:0]							
72	VMN_8[7:0]							
73	VMN_9[7:0]							
74	VMN_10[7:0]							
75	VMN_11[7:0]							
76	VMN_12[7:0]							
77	-	VLN_0[6:0]						
78	-	VLN_1[6:0]						
79	-	VLN_2[6:0]						
7A	-	VLN_3[6:0]						
7B	-	VLN_4[6:0]						
7C	-	VLN_5[6:0]						
7D	-	VLN_6[6:0]						
7E	-	VLN_7[6:0]						
7F	NVALID_DGC	-	-	-	-	-	-	DGC_EN
80	R_GAMMA0[9:2]							
81	R_GAMMA1[9:2]							
82	R_GAMMA2[9:2]							
83	R_GAMMA3[9:2]							
84	R_GAMMA4[9:2]							
85	R_GAMMA5[9:2]							
86	R_GAMMA6[9:2]							
87	R_GAMMA7[9:2]							
88	R_GAMMA8[9:2]							
89	R_GAMMA9[9:2]							
8A	R_GAMMA10[9:2]							
8B	R_GAMMA11[9:2]							
8C	R_GAMMA12[9:2]							
8D	R_GAMMA13[9:2]							
8E	R_GAMMA14[9:2]							
8F	R_GAMMA15[9:2]							
90	R_GAMMA16[9:2]							
91	R_GAMMA17[9:2]							
92	R_GAMMA18[9:2]							
93	R_GAMMA19[9:2]							
94	R_GAMMA20[9:2]							
95	R_GAMMA21[9:2]							
96	R_GAMMA22[9:2]							
97	R_GAMMA23[9:2]							
98	R_GAMMA24[9:2]							
99	R_GAMMA25[9:2]							
9A	R_GAMMA26[9:2]							
9B	R_GAMMA27[9:2]							
9C	R_GAMMA28[9:2]							
9D	R_GAMMA29[9:2]							
9E	R_GAMMA30[9:2]							
9F	R_GAMMA31[9:2]							
A0	R_GAMMA32[9:2]							
A1	R_GAMMA0[1:0]		R_GAMMA1[1:0]		R_GAMMA2[1:0]		R_GAMMA3[1:0]	
A2	R_GAMMA4[1:0]		R_GAMMA5[1:0]		R_GAMMA6[1:0]		R_GAMMA7[1:0]	
A3	R_GAMMA8[1:0]		R_GAMMA9[1:0]		R_GAMMA10[1:0]		R_GAMMA11[1:0]	
A4	R_GAMMA12[1:0]		R_GAMMA13[1:0]		R_GAMMA14[1:0]		R_GAMMA15[1:0]	
A5	R_GAMMA16[1:0]		R_GAMMA17[1:0]		R_GAMMA18[1:0]		R_GAMMA19[1:0]	

This information contained herein is the exclusive property of Himax and shall not be distributed, reproduced, or disclosed in whole or in part without prior written permission of Himax.

A6	R_GAMMA20[1:0]		R_GAMMA21[1:0]		R_GAMMA22[1:0]		R_GAMMA23[1:0]	
A7	R_GAMMA24[1:0]		R_GAMMA25[1:0]		R_GAMMA26[1:0]		R_GAMMA27[1:0]	
A8	R_GAMMA28[1:0]		R_GAMMA29[1:0]		R_GAMMA30[1:0]		R_GAMMA31[1:0]	
A9	R_GAMMA32[1:0]		-	-	-	-	R_GAMMAFF_OPT	R_GAMMA_256
AA	G_GAMMA0[9:2]							
AB	G_GAMMA1[9:2]							
AC	G_GAMMA2[9:2]							
AD	G_GAMMA3[9:2]							
AE	G_GAMMA4[9:2]							
AF	G_GAMMA5[9:2]							
B0	G_GAMMA6[9:2]							
B1	G_GAMMA7[9:2]							
B2	G_GAMMA8[9:2]							
B3	G_GAMMA9[9:2]							
B4	G_GAMMA10[9:2]							
B5	G_GAMMA11[9:2]							
B6	G_GAMMA12[9:2]							
B7	G_GAMMA13[9:2]							
B8	G_GAMMA14[9:2]							
B9	G_GAMMA15[9:2]							
BA	G_GAMMA16[9:2]							
BB	G_GAMMA17[9:2]							
BC	G_GAMMA18[9:2]							
BD	G_GAMMA19[9:2]							
BE	G_GAMMA20[9:2]							
BF	G_GAMMA21[9:2]							
C0	G_GAMMA22[9:2]							
C1	G_GAMMA23[9:2]							
C2	G_GAMMA24[9:2]							
C3	G_GAMMA25[9:2]							
C4	G_GAMMA26[9:2]							
C5	G_GAMMA27[9:2]							
C6	G_GAMMA28[9:2]							
C7	G_GAMMA29[9:2]							
C8	G_GAMMA30[9:2]							
C9	G_GAMMA31[9:2]							
CA	G_GAMMA32[9:2]							
CB	G_GAMMA0[1:0]		G_GAMMA1[1:0]		G_GAMMA2[1:0]		G_GAMMA3[1:0]	
CC	G_GAMMA4[1:0]		G_GAMMA5[1:0]		G_GAMMA6[1:0]		G_GAMMA7[1:0]	
CD	G_GAMMA8[1:0]		G_GAMMA9[1:0]		G_GAMMA10[1:0]		G_GAMMA11[1:0]	
CE	G_GAMMA12[1:0]		G_GAMMA13[1:0]		G_GAMMA14[1:0]		G_GAMMA15[1:0]	
CF	G_GAMMA16[1:0]		G_GAMMA17[1:0]		G_GAMMA18[1:0]		G_GAMMA19[1:0]	
D0	G_GAMMA20[1:0]		G_GAMMA21[1:0]		G_GAMMA22[1:0]		G_GAMMA23[1:0]	
D1	G_GAMMA24[1:0]		G_GAMMA25[1:0]		G_GAMMA26[1:0]		G_GAMMA27[1:0]	
D2	G_GAMMA28[1:0]		G_GAMMA29[1:0]		G_GAMMA30[1:0]		G_GAMMA31[1:0]	
D3	G_GAMMA32[1:0]		-	-	-	-	G_GAMMAFF_OPT	G_GAMMA_256
D4	B_GAMMA0[9:2]							
D5	B_GAMMA1[9:2]							
D6	B_GAMMA2[9:2]							
D7	B_GAMMA3[9:2]							
D8	B_GAMMA4[9:2]							
D9	B_GAMMA5[9:2]							
DA	B_GAMMA6[9:2]							
DB	B_GAMMA7[9:2]							
DC	B_GAMMA8[9:2]							
DD	B_GAMMA9[9:2]							
DE	B_GAMMA10[9:2]							
DF	B_GAMMA11[9:2]							
E0	B_GAMMA12[9:2]							
E1	B_GAMMA13[9:2]							
E2	B_GAMMA14[9:2]							
E3	B_GAMMA15[9:2]							
E4	B_GAMMA16[9:2]							

This information contained herein is the exclusive property of Himax and shall not be distributed, reproduced, or disclosed in whole or in part without prior written permission of Himax.

E5	B_GAMMA17[9:2]							
E6	B_GAMMA18[9:2]							
E7	B_GAMMA19[9:2]							
E8	B_GAMMA20[9:2]							
E9	B_GAMMA21[9:2]							
EA	B_GAMMA22[9:2]							
EB	B_GAMMA23[9:2]							
EC	B_GAMMA24[9:2]							
ED	B_GAMMA25[9:2]							
EE	B_GAMMA26[9:2]							
EF	B_GAMMA27[9:2]							
F0	B_GAMMA28[9:2]							
F1	B_GAMMA29[9:2]							
F2	B_GAMMA30[9:2]							
F3	B_GAMMA31[9:2]							
F4	B_GAMMA32[9:2]							
F5	B_GAMMA0[1:0]		B_GAMMA1[1:0]		B_GAMMA2[1:0]		B_GAMMA3[1:0]	
F6	B_GAMMA4[1:0]		B_GAMMA5[1:0]		B_GAMMA6[1:0]		B_GAMMA7[1:0]	
F7	B_GAMMA8[1:0]		B_GAMMA9[1:0]		B_GAMMA10[1:0]		B_GAMMA11[1:0]	
F8	B_GAMMA12[1:0]		B_GAMMA13[1:0]		B_GAMMA14[1:0]		B_GAMMA15[1:0]	
F9	B_GAMMA16[1:0]		B_GAMMA17[1:0]		B_GAMMA18[1:0]		B_GAMMA19[1:0]	
FA	B_GAMMA20[1:0]		B_GAMMA21[1:0]		B_GAMMA22[1:0]		B_GAMMA23[1:0]	
FB	B_GAMMA24[1:0]		B_GAMMA25[1:0]		B_GAMMA26[1:0]		B_GAMMA27[1:0]	
FC	B_GAMMA28[1:0]		B_GAMMA29[1:0]		B_GAMMA30[1:0]		B_GAMMA31[1:0]	
FD	B_GAMMA32[1:0]		-	-	-	-	B_GAMMAF F_OPT	B_GAMMA_ 256
FE								
FF	VALID_DGC _USEFUL							
100	NVALID_GA MMA	VHP_0[6:0]						
101	-	VHP_1[6:0]						
102	-	VHP_2[6:0]						
103	-	VHP_3[6:0]						
104	-	VHP_4[6:0]						
105	-	VHP_5[6:0]						
106	-	VHP_6[6:0]						
107	-	VHP_7[6:0]						
108	VMP_0[7:0]							
109	VMP_1[7:0]							
10A	VMP_2[7:0]							
10B	VMP_3[7:0]							
10C	VMP_4[7:0]							
10D	VMP_5[7:0]							
10E	VMP_6[7:0]							
10F	VMP_7[7:0]							
110	VMP_8[7:0]							
111	VMP_9[7:0]							
112	VMP_10[7:0]							
113	VMP_11[7:0]							
114	VMP_12[7:0]							
115	-	VLP_0[6:0]						
116	-	VLP_1[6:0]						
117	-	VLP_2[6:0]						
118	-	VLP_3[6:0]						
119	-	VLP_4[6:0]						
11A	-	VLP_5[6:0]						
11B	-	VLP_6[6:0]						
11C	-	VLP_7[6:0]						
11D	-	VHN_0[6:0]						
11E	-	VHN_1[6:0]						
11F	-	VHN_2[6:0]						
120	-	VHN_3[6:0]						
121	-	VHN_4[6:0]						
122	-	VHN_5[6:0]						

This information contained herein is the exclusive property of Himax and shall not be distributed, reproduced, or disclosed in whole or in part without prior written permission of Himax.

123	-	VHN_6[6:0]						
124	-	VHN_7[6:0]						
125	VMN_0[7:0]							
126	VMN_1[7:0]							
127	VMN_21[7:0]							
128	VMN_3[7:0]							
129	VMN_4[7:0]							
12A	VMN_5[7:0]							
12B	VMN_6[7:0]							
12C	VMN_7[7:0]							
12D	VMN_8[7:0]							
12E	VMN_9[7:0]							
12F	VMN_10[7:0]							
130	VMN_11[7:0]							
131	VMN_12[7:0]							
132	-	VLN_0[6:0]						
133	-	VLN_1[6:0]						
134	-	VLN_2[6:0]						
135	-	VLN_3[6:0]						
136	-	VLN_4[6:0]						
137	-	VLN_5[6:0]						
138	-	VLN_6[6:0]						
139	-	VLN_7[6:0]						
13A	NVALID_DGC	-	-	-	-	-	-	DGC_EN
13B	R_GAMMA0[9:2]							
13C	R_GAMMA1[9:2]							
13D	R_GAMMA2[9:2]							
13E	R_GAMMA3[9:2]							
13F	R_GAMMA4[9:2]							
140	R_GAMMA5[9:2]							
141	R_GAMMA6[9:2]							
142	R_GAMMA7[9:2]							
143	R_GAMMA8[9:2]							
144	R_GAMMA9[9:2]							
145	R_GAMMA10[9:2]							
146	R_GAMMA11[9:2]							
147	R_GAMMA12[9:2]							
148	R_GAMMA13[9:2]							
149	R_GAMMA14[9:2]							
14A	R_GAMMA15[9:2]							
14B	R_GAMMA16[9:2]							
14C	R_GAMMA17[9:2]							
14D	R_GAMMA18[9:2]							
14E	R_GAMMA19[9:2]							
14F	R_GAMMA20[9:2]							
150	R_GAMMA21[9:2]							
151	R_GAMMA22[9:2]							
152	R_GAMMA23[9:2]							
153	R_GAMMA24[9:2]							
154	R_GAMMA25[9:2]							
155	R_GAMMA26[9:2]							
156	R_GAMMA27[9:2]							
157	R_GAMMA28[9:2]							
158	R_GAMMA29[9:2]							
159	R_GAMMA30[9:2]							
15A	R_GAMMA31[9:2]							
15B	R_GAMMA32[9:2]							
15C	R_GAMMA0[1:0]	R_GAMMA1[1:0]	R_GAMMA2[1:0]	R_GAMMA3[1:0]				
15D	R_GAMMA4[1:0]	R_GAMMA5[1:0]	R_GAMMA6[1:0]	R_GAMMA7[1:0]				
15E	R_GAMMA8[1:0]	R_GAMMA9[1:0]	R_GAMMA10[1:0]	R_GAMMA11[1:0]				
15F	R_GAMMA12[1:0]	R_GAMMA13[1:0]	R_GAMMA14[1:0]	R_GAMMA15[1:0]				
160	R_GAMMA16[1:0]	R_GAMMA17[1:0]	R_GAMMA18[1:0]	R_GAMMA19[1:0]				
161	R_GAMMA20[1:0]	R_GAMMA21[1:0]	R_GAMMA22[1:0]	R_GAMMA23[1:0]				
162	R_GAMMA24[1:0]	R_GAMMA25[1:0]	R_GAMMA26[1:0]	R_GAMMA27[1:0]				

This information contained herein is the exclusive property of Himax and shall not be distributed, reproduced, or disclosed in whole or in part without prior written permission of Himax.

163	R_GAMMA28[1:0]		R_GAMMA29[1:0]		R_GAMMA30[1:0]		R_GAMMA31[1:0]	
164	R_GAMMA32[1:0]		-	-	-	-	R_GAMMAFF_OPT	R_GAMMA_256
165	G_GAMMA0[9:2]							
166	G_GAMMA1[9:2]							
167	G_GAMMA2[9:2]							
168	G_GAMMA3[9:2]							
169	G_GAMMA4[9:2]							
16A	G_GAMMA5[9:2]							
16B	G_GAMMA6[9:2]							
16C	G_GAMMA7[9:2]							
16D	G_GAMMA8[9:2]							
16E	G_GAMMA9[9:2]							
16F	G_GAMMA10[9:2]							
170	G_GAMMA11[9:2]							
171	G_GAMMA12[9:2]							
172	G_GAMMA13[9:2]							
173	G_GAMMA14[9:2]							
174	G_GAMMA15[9:2]							
175	G_GAMMA16[9:2]							
176	G_GAMMA17[9:2]							
177	G_GAMMA18[9:2]							
178	G_GAMMA19[9:2]							
179	G_GAMMA20[9:2]							
17A	G_GAMMA21[9:2]							
17B	G_GAMMA22[9:2]							
17C	G_GAMMA23[9:2]							
17D	G_GAMMA24[9:2]							
17E	G_GAMMA25[9:2]							
17F	G_GAMMA26[9:2]							
180	G_GAMMA27[9:2]							
181	G_GAMMA28[9:2]							
182	G_GAMMA29[9:2]							
183	G_GAMMA30[9:2]							
184	G_GAMMA31[9:2]							
185	G_GAMMA32[9:2]							
186	G_GAMMA0[1:0]		G_GAMMA1[1:0]		G_GAMMA2[1:0]		G_GAMMA3[1:0]	
187	G_GAMMA4[1:0]		G_GAMMA5[1:0]		G_GAMMA6[1:0]		G_GAMMA7[1:0]	
188	G_GAMMA8[1:0]		G_GAMMA9[1:0]		G_GAMMA10[1:0]		G_GAMMA11[1:0]	
189	G_GAMMA12[1:0]		G_GAMMA13[1:0]		G_GAMMA14[1:0]		G_GAMMA15[1:0]	
18A	G_GAMMA16[1:0]		G_GAMMA17[1:0]		G_GAMMA18[1:0]		G_GAMMA19[1:0]	
18B	G_GAMMA20[1:0]		G_GAMMA21[1:0]		G_GAMMA22[1:0]		G_GAMMA23[1:0]	
18C	G_GAMMA24[1:0]		G_GAMMA25[1:0]		G_GAMMA26[1:0]		G_GAMMA27[1:0]	
18D	G_GAMMA28[1:0]		G_GAMMA29[1:0]		G_GAMMA30[1:0]		G_GAMMA31[1:0]	
18E	G_GAMMA32[1:0]		-	-	-	-	G_GAMMAFF_OPT	G_GAMMA_256
18F	B_GAMMA0[9:2]							
190	B_GAMMA1[9:2]							
191	B_GAMMA2[9:2]							
192	B_GAMMA3[9:2]							
193	B_GAMMA4[9:2]							
194	B_GAMMA5[9:2]							
195	B_GAMMA6[9:2]							
196	B_GAMMA7[9:2]							
197	B_GAMMA8[9:2]							
198	B_GAMMA9[9:2]							
199	B_GAMMA10[9:2]							
19A	B_GAMMA11[9:2]							
19B	B_GAMMA12[9:2]							
19C	B_GAMMA13[9:2]							
19D	B_GAMMA14[9:2]							
19E	B_GAMMA15[9:2]							
19F	B_GAMMA16[9:2]							
1A0	B_GAMMA17[9:2]							
1A1	B_GAMMA18[9:2]							

This information contained herein is the exclusive property of Himax and shall not be distributed, reproduced, or disclosed in whole or in part without prior written permission of Himax.

1A2	B_GAMMA19[9:2]							
1A3	B_GAMMA20[9:2]							
1A4	B_GAMMA21[9:2]							
1A5	B_GAMMA22[9:2]							
1A6	B_GAMMA23[9:2]							
1A7	B_GAMMA24[9:2]							
1A8	B_GAMMA25[9:2]							
1A9	B_GAMMA26[9:2]							
1AA	B_GAMMA27[9:2]							
1AB	B_GAMMA28[9:2]							
1AC	B_GAMMA29[9:2]							
1AD	B_GAMMA30[9:2]							
1AE	B_GAMMA31[9:2]							
1AF	B_GAMMA32[9:2]							
1B0	B_GAMMA0[1:0]		B_GAMMA1[1:0]		B_GAMMA2[1:0]		B_GAMMA3[1:0]	
1B1	B_GAMMA4[1:0]		B_GAMMA5[1:0]		B_GAMMA6[1:0]		B_GAMMA7[1:0]	
1B2	B_GAMMA8[1:0]		B_GAMMA9[1:0]		B_GAMMA10[1:0]		B_GAMMA11[1:0]	
1B3	B_GAMMA12[1:0]		B_GAMMA13[1:0]		B_GAMMA14[1:0]		B_GAMMA15[1:0]	
1B4	B_GAMMA16[1:0]		B_GAMMA17[1:0]		B_GAMMA18[1:0]		B_GAMMA19[1:0]	
1B5	B_GAMMA20[1:0]		B_GAMMA21[1:0]		B_GAMMA22[1:0]		B_GAMMA23[1:0]	
1B6	B_GAMMA24[1:0]		B_GAMMA25[1:0]		B_GAMMA26[1:0]		B_GAMMA27[1:0]	
1B7	B_GAMMA28[1:0]		B_GAMMA29[1:0]		B_GAMMA30[1:0]		B_GAMMA31[1:0]	
1B8	B_GAMMA32[1:0]		-	-	-	-	B_GAMMAFF_OPT	B_GAMMA_256
1B9	NVALID_GIP_D3	-	GIP_EQ_OPT[1:0]		-	-	EQ_DELAY_HSYNC[1:0]	
1BA	-	-	-	-	-	EQ_VSEL	EQ_DISC[1:0]	
1BB	EQ_DELAY_ON1[7:0]							
1BC	EQ_DELAY_OFF1[7:0]							
1BD	GTO[7:0]							
1BE	GNO[7:0]							
1BF	USER_GIP_GATE[7:0]							
1C0	USER_GIP_GATE1[7:0]							
1C1	SHR0_3[3:0]				SHR0_2[3:0]			
1C2	SHR0_1[3:0]				SHR0[11:8]			
1C3	SHR0[7:0]							
1C4	-	-	-	-	SHR0_GS[11:8]			
1C5	SHR0_GS[7:0]							
1C6	SHR1_3[3:0]				SHR1_2[3:0]			
1C7	SHR1_1[3:0]				SHR1[11:8]			
1C8	SHR1[7:0]							
1C9	-	-	-	-	SHR1_GS[11:8]			
1CA	SHR1_GS[7:0]							
1CB	SHR2_3[3:0]				SHR2_2[3:0]			
1CC	SHR2_1[3:0]				SHR2[11:8]			
1CD	SHR2[7:0]							
1CE	-	-	-	-	SHR2_GS[11:8]			
1CF	SHR2_GS[7:0]							
1D0	SHP0[3:0]				SCP[3:0]			
1D1	SHP2[3:0]				SHP1[3:0]			
1D2	CHR0[7:0]							
1D3	CHR0_GS[7:0]							
1D4	CHP0[3:0]				CCP0[3:0]			
1D5	CHR1[7:0]							
1D6	CHR1_GS[7:0]							
1D7	CHP1[3:0]				CCP1[3:0]			
1EE	NVALID_CABC	SEL_PWMCLK[2:0]			PWM_PERIOD[16]	BC_CTRL_EN	INVPULS	SEL_BLDUTY
1EF	PWM_PERIOD[15:8]							
1F0	PWM_PERIOD[7:0]							
1F1	CABC_FSYNC	DIM_FRAME[6:0]						
1F2	CABC_DD	CABC_DD_IDLE	EN_DIM_MIX	ABC_CHECKSUM_EN	CABC_FLM[3:0]			
1F7	NVALID_SE	-	-	-	-	-	-	-

This information contained herein is the exclusive property of Himax and shall not be distributed, reproduced, or disclosed in whole or in part without prior written permission of Himax.

	TDPHY							
1F8	DPHYCMD0[7:0]							
1F9	DPHYCMD1[7:0]							
249	NVALID_GIP_D5	CGTS_L_INV[1]	COS1_L[5:0] (CGOUT1_L)					
24A	-	CGTS_R_INV[1]	COS1_R[5:0] (CGOUT1_R)					
24B	-	CGTS_L_INV[2]	COS2_L[5:0] (CGOUT2_L)					
24C	-	CGTS_R_INV[2]	COS2_R[5:0] (CGOUT2_R)					
24D	-	CGTS_L_INV[3]	COS3_L[5:0] (CGOUT3_L)					
24E	-	CGTS_R_INV[3]	COS3_R[5:0] (CGOUT3_R)					
24F	-	CGTS_L_INV[4]	COS4_L[5:0] (CGOUT4_L)					
250	-	CGTS_R_INV[4]	COS4_R[5:0] (CGOUT4_R)					
251	-	CGTS_L_INV[5]	COS5_L[5:0] (CGOUT5_L)					
252	-	CGTS_R_INV[5]	COS5_R[5:0] (CGOUT5_R)					
253	-	CGTS_L_INV[6]	COS6_L[5:0] (CGOUT6_L)					
254	-	CGTS_R_INV[6]	COS6_R[5:0] (CGOUT6_R)					
255	-	CGTS_L_INV[7]	COS7_L[5:0] (CGOUT7_L)					
256	-	CGTS_R_INV[7]	COS7_R[5:0] (CGOUT7_R)					
257	-	CGTS_L_INV[8]	COS8_L[5:0] (CGOUT8_L)					
258	-	CGTS_R_INV[8]	COS8_R[5:0] (CGOUT8_R)					
259	-	CGTS_L_INV[9]	COS9_L[5:0] (CGOUT9_L)					
25A	-	CGTS_R_INV[9]	COS9_R[5:0] (CGOUT9_R)					
25B	-	CGTS_L_INV[10]	COS10_L[5:0] (CGOUT10_L)					
25C	-	CGTS_R_INV[10]	COS10_R[5:0] (CGOUT10_R)					
25D	-	CGTS_L_INV[11]	COS11_L[5:0] (CGOUT11_L)					
25E	-	CGTS_R_INV[11]	COS11_R[5:0] (CGOUT11_R)					
25F	-	CGTS_L_INV[12]	COS12_L[5:0] (CGOUT12_L)					
260	-	CGTS_R_INV[12]	COS12_R[5:0] (CGOUT12_R)					
261	-	CGTS_L_INV[13]	COS13_L[5:0] (CGOUT13_L)					
262	-	CGTS_R_INV[13]	COS13_R[5:0] (CGOUT13_R)					
263	-	CGTS_L_INV[14]	COS14_L[5:0] (CGOUT14_L)					
264	-	CGTS_R_INV[14]	COS14_R[5:0] (CGOUT14_R)					
265	-	CGTS_L_INV[15]	COS15_L[5:0] (CGOUT15_L)					
266	-	CGTS_R_INV[15]	COS15_R[5:0] (CGOUT15_R)					
267	-	CGTS_L_INV[16]	COS16_L[5:0] (CGOUT16_L)					
268	-	CGTS_R_IN	COS16_R[5:0] (CGOUT16_R)					

This information contained herein is the exclusive property of Himax and shall not be distributed, reproduced, or disclosed in whole or in part without prior written permission of Himax.

		V[16]	
269	-	CGTS_L_INV[17]	COS17_L[5:0] (CGOUT17_L)
26A	-	CGTS_R_INV[17]	COS17_R[5:0] (CGOUT17_R)
26B	-	CGTS_L_INV[18]	COS18_L[5:0] (CGOUT18_L)
26C	-	CGTS_R_INV[18]	COS18_R[5:0] (CGOUT18_R)
26D		CGTS_L_INV[19]	COS19_L[5:0] (CGOUT19_L)
26E		CGTS_R_INV[19]	COS19_R[5:0] (CGOUT19_R)
26F		CGTS_L_INV[20]	COS20_L[5:0] (CGOUT20_L)
270		CGTS_R_INV[20]	COS20_R[5:0] (CGOUT20_R)
271	-	CGTS_L_INV[21]	COS21_L[5:0] (CGOUT21_L)
272	-	CGTS_R_INV[21]	COS21_R[5:0] (CGOUT21_R)
273	-	CGTS_L_INV[22]	COS22_L[5:0] (CGOUT22_L)
274	-	CGTS_R_INV[22]	COS22_R[5:0] (CGOUT22_R)
275	NVALID_GIP_D6	CGOUT_L_HIZ_IN[1]	COS1_L_GS[5:0] (CGOUT1_L)
276	-	CGOUT_R_HIZ_IN[1]	COS1_R_GS[5:0] (CGOUT1_R)
277	-	CGOUT_L_HIZ_IN[2]	COS2_L_GS[5:0] (CGOUT2_L)
278	-	CGOUT_R_HIZ_IN[2]	COS2_R_GS[5:0] (CGOUT2_R)
279	-	CGOUT_L_HIZ_IN[3]	COS3_L_GS[5:0] (CGOUT3_L)
27A	-	CGOUT_R_HIZ_IN[3]	COS3_R_GS[5:0] (CGOUT3_R)
27B	-	CGOUT_L_HIZ_IN[4]	COS4_L_GS[5:0] (CGOUT4_L)
27C	-	CGOUT_R_HIZ_IN[4]	COS4_R_GS[5:0] (CGOUT4_R)
27D	-	CGOUT_L_HIZ_IN[5]	COS5_L_GS[5:0] (CGOUT5_L)
27E	-	CGOUT_R_HIZ_IN[5]	COS5_R_GS[5:0] (CGOUT5_R)
27F	-	CGOUT_L_HIZ_IN[6]	COS6_L_GS[5:0] (CGOUT6_L)
280	-	CGOUT_R_HIZ_IN[6]	COS6_R_GS[5:0] (CGOUT6_R)
281	-	CGOUT_L_HIZ_IN[7]	COS7_L_GS[5:0] (CGOUT7_L)
282	-	CGOUT_R_HIZ_IN[7]	COS7_R_GS[5:0] (CGOUT7_R)
283	-	CGOUT_L_HIZ_IN[8]	COS8_L_GS[5:0] (CGOUT8_L)
284	-	CGOUT_R_HIZ_IN[8]	COS8_R_GS[5:0] (CGOUT8_R)
285	-	CGOUT_L_HIZ_IN[9]	COS9_L_GS[5:0] (CGOUT9_L)
286	-	CGOUT_R_HIZ_IN[9]	COS9_R_GS[5:0] (CGOUT9_R)
287	-	CGOUT_L_HIZ_IN[10]	COS10_L_GS[5:0] (CGOUT10_L)
288	-	CGOUT_R_HIZ_IN[10]	COS10_R_GS[5:0] (CGOUT10_R)
289	-	CGOUT_L_H	COS11_L_GS[5:0] (CGOUT11_L)

This information contained herein is the exclusive property of Himax and shall not be distributed, reproduced, or disclosed in whole or in part without prior written permission of Himax.

		IZ_IN[11]	COS11_R_GS[5:0] (CGOUT11_R)					
28A	-	CGOUT_R_HIZ_IN[11]						
28B	-	CGOUT_L_HIZ_IN[12]	COS12_L_GS[5:0] (CGOUT12_L)					
28C	-	CGOUT_R_HIZ_IN[12]	COS12_R_GS[5:0] (CGOUT12_R)					
28D	-	CGOUT_L_HIZ_IN[13]	COS13_L_GS[5:0] (CGOUT13_L)					
28E	-	CGOUT_R_HIZ_IN[13]	COS13_R_GS[5:0] (CGOUT13_R)					
28F	-	CGOUT_L_HIZ_IN[14]	COS14_L_GS[5:0] (CGOUT14_L)					
290	-	CGOUT_R_HIZ_IN[14]	COS14_R_GS[5:0] (CGOUT14_R)					
291	-	CGOUT_L_HIZ_IN[15]	COS15_L_GS[5:0] (CGOUT15_L)					
292	-	CGOUT_R_HIZ_IN[15]	COS15_R_GS[5:0] (CGOUT15_R)					
293	-	CGOUT_L_HIZ_IN[16]	COS16_L_GS[5:0] (CGOUT16_L)					
294	-	CGOUT_R_HIZ_IN[16]	COS16_R_GS[5:0] (CGOUT16_R)					
295		CGOUT_L_HIZ_IN[17]	COS17_L_GS[5:0] (CGOUT17_L)					
296		CGOUT_R_HIZ_IN[17]	COS17_R_GS[5:0] (CGOUT17_R)					
297		CGOUT_L_HIZ_IN[18]	COS18_L_GS[5:0] (CGOUT18_L)					
298		CGOUT_R_HIZ_IN[18]	COS18_R_GS[5:0] (CGOUT18_R)					
299	-	CGOUT_L_HIZ_IN[19]	COS19_L_GS[5:0] (CGOUT19_L)					
29A	-	CGOUT_R_HIZ_IN[19]	COS19_R_GS[5:0] (CGOUT19_R)					
29B	-	CGOUT_L_HIZ_IN[20]	COS20_L_GS[5:0] (CGOUT20_L)					
29C	-	CGOUT_R_HIZ_IN[20]	COS20_R_GS[5:0] (CGOUT20_R)					
29D	-	CGOUT_L_HIZ_IN[21]	COS21_L_GS[5:0] (CGOUT21_L)					
29E	-	CGOUT_R_HIZ_IN[21]	COS21_R_GS[5:0] (CGOUT21_R)					
29F	-	CGOUT_L_HIZ_IN[22]	COS22_L_GS[5:0] (CGOUT22_L)					
2A0	-	CGOUT_R_HIZ_IN[22]	COS22_R_GS[5:0] (CGOUT22_R)					
2ED	NVALID_1BPP	-	-	-	-	NW_PE[2:0]		
2EE	BP_PE[7:0]							
2EF	FP_PE[7:0]							
2F0	RTN_PE[7:0]							
2F1	VCMC_F_IDLE[7:0]							
2F2	VCMC_B_IDLE[7:0]							
2F3	AP_PE[2:0]			-	-	-	VCMC_B_IDLE[8]	VCMC_F_IDLE[8]
2F8	NVALID_CEMODE	-	-	-	-	-	-	DYN_CEH_EN
2F9	HUE_MODE[1:0]		SE_MODE[1:0]		BE_MODE[1:0]		CE_MODE[1:0]	
300	NVALID_POWER	POWMOD[1:0]		AP[2:0]			DSTBY_OPT	DSTB
301	VCLS[2:0]			VRHP[4:0]				
302	VPPS[2:0]			VRHN[4:0]				
303	EQ_VCLEN	-	-	-	VSP_FBOFF (1 no FB)	XDK[2:0]		
304	-	-	CLK_OPT2	CLK_OPT1(	FS0[3:0](VSP_pmping)			

This information contained herein is the exclusive property of Himax and shall not be distributed, reproduced, or disclosed in whole or in part without prior written permission of Himax.

			(VGL)	VGH)				
305	FS1[3:0](VGH_pmping)				FS2[3:0](VGL_pmping)			
306	VGH_RATIO[2:0]			BTP[4:0]				
307	VGL_RATIO[2:0]			BTN[4:0]				
308	-	VGHS[6:0]						
309	-	VGLS[6:0]						
30A	DT1[1:0]		DT2[1:0]		DCDIV[3:0]			
30B	-	DCS[2:0]			-	DC[2:0]		
30C	-	DTPS[2:0]			-	DTP[2:0]		
30D	-	DTNS[2:0]			-	DTN[2:0]		
30E	NVALID_GAS	APF_EN	GASIOVCC_OPT[1:0]		-	GASVCI_OPT[2:0]		
30F	GAS_CP_VSPVSN	GASVSN_OPT[2:0]			-	GASVSP_OPT[2:0]		
312	NVALID_DISP	ZZ_LR	ZZ_EO	ZZ_2PL	-	NW[2:0]		
313	MESSI_ENB	H_RES[2:0]			-	-	-	-
314	NL[7:0]							
315	BP [7:0]							
316	FP [7:0]							
317	RTN[7:0]							
31C	FRM_PATTERN_CYCLE[3:0]				DISP_BIST_EN	FRM_SCAN_CYCLE[2:0]		
320	NVALID_CYC	-	-	-	SAP_PURECOLOR_EN	SAP_XTALK_EN	SAP_PURECOLOR_SW	SAP_XTALK_SW
321	SPON[7:0]							
322	SPOFF[7:0]							
323	CON[7:0]							
324	COFF[7:0]							
325	CON1[7:0]							
326	COFF1[7:0]							
327	EQON1[7:0]							
328	SON[7:0]							
329	SOFF[7:0]							
32A	SAP1_P[3:0]				-	SAP2[2:0]		
32B	-	DX2_EN	EQ_ON	EQ_MIX	-	-	-	-
32C	DX2OFF[7:0]							
32D	SPON_MPU[7:0]							
32E	SPOFF_MPU[7:0]							
32F	CON_MPU[7:0]							
330	COFF_MPU[7:0]							
331	CON1_MPU[7:0]							
332	COFF1_MPU[7:0]							
333	EQON1_MPU[7:0]							
334	SON_MPU[7:0]							
335	SOFF_MPU[7:0]							
336	DX2OFF_MPU[7:0]							
33F	NVALID_CABCDBG	DBG0[6:0]						
340	-	DBG1[6:0]						
341	-	DBG2[6:0]						
342	-	DBG3[6:0]						
343	-	DBG4[6:0]						
344	-	DBG5[6:0]						
345	-	DBG6[6:0]						
346	-	DBG7[6:0]						
347	-	DBG8[6:0]						
348	NVALID_SETDSI	DSISETUP0[6:0]						
349	DSISETUP1[7:0]							
354	NVALID_SENSOR	TSENSOR_EN	LT_EN	HT_EN	-	TEMP_GAP[2:0]		
367	NVALID_OFFSET3	-	-	-	-	-	-	-

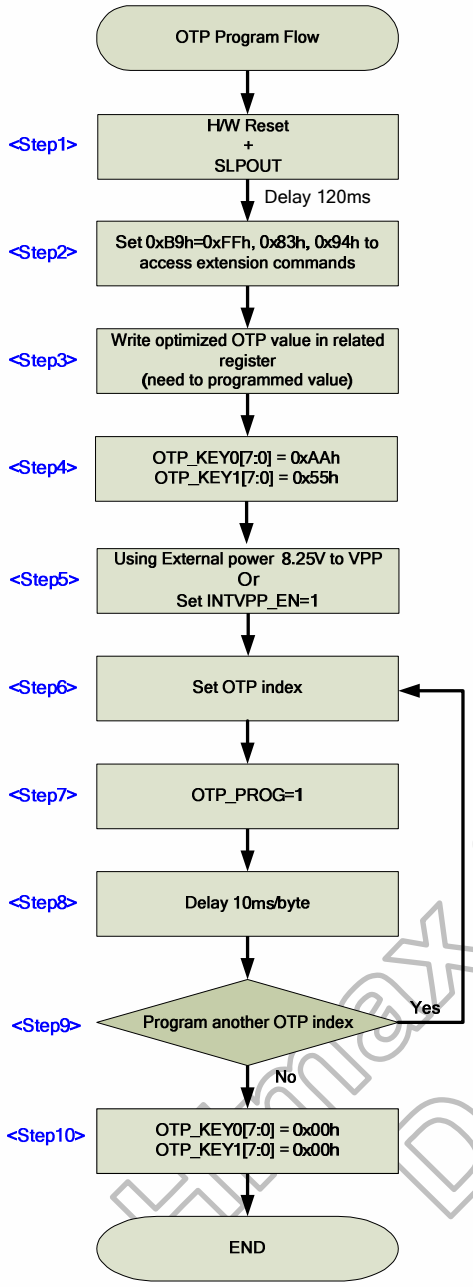
This information contained herein is the exclusive property of Himax and shall not be distributed, reproduced, or disclosed in whole or in part without prior written permission of Himax.

368	VN_REFS[3:0]				VP_REFS[3:0]			
36A	NVALID_SCL	-	-	-	-	-	SCALING_TYPE	SCALING_EN

Table 5.41: OTP table

Himax Confidential
Do Not Copy

5.16.2 OTP programming flow



OTP_KEY0[7:0] OTP_KEY1[7:0]	Description	Note
OTP_KEY0[7:0] = 0xAAh OTP_KEY1[7:0] = 0x55h	Enter OTP program mode	
OTP_KEY0[7:0] = 0x00h OTP_KEY1[7:0] = 0x00h	Leave OTP program mode	
Other value	Invalid	1. If HX8394-F operate on OTP program mode, then keep on OTP program mode. 2. If HX8394-F operate on non-OTP program mode, then keep on non-OTP program mode.

Figure 5.37: OTP programming sequence

Himax Confidential
Do Not Copy

5.16.3 Programming sequence

Step	Operation
1	Power on and reset the module.
2	SLPOUT and set 0xB9h = 0xFFh, 0x83h, 0x94h to access the extension commands.
3	Write optimized values to related registers.
4	Set OTP_KEY0[7:0]=0xAAh and OTP_KEY1[7:0]=0x55h to enter OTP program mode.
5	Set INTVPP_EN=1 for OTP programming state for using internal power mode. Or using the external power 8.25V to VPP.
6	Specify OTP_index, please refer to the OTP table.
7	Set OTP_Mask=0x00h, programming the entire bit of one parameter.
8	Set OTP_PROG=1, Internal register begin write to OTP according to OTP_index.
9	Wait 10ms/byte (**Note 1**)
10	Set OTP_PROG=0, OTP_index programming action done.
11	Complete programming one parameter to OTP. If continue to programming other parameter, return to step (5). Otherwise, set OTP_KEY1[7:0]=0x00h and OTP_KEY1[7:0]=0x00h to leave OTP program mode and power off the module and remove the external power on VPP pin.

Note: (1) When do the OTP programming process, it must be added 5ms delay time after setting OTP_PROG=1.

Table 5.42: OTP programming sequence

Himax Confidential
Do Not Copy

5.16.4 OTP programming example of VCOM setting VCMC

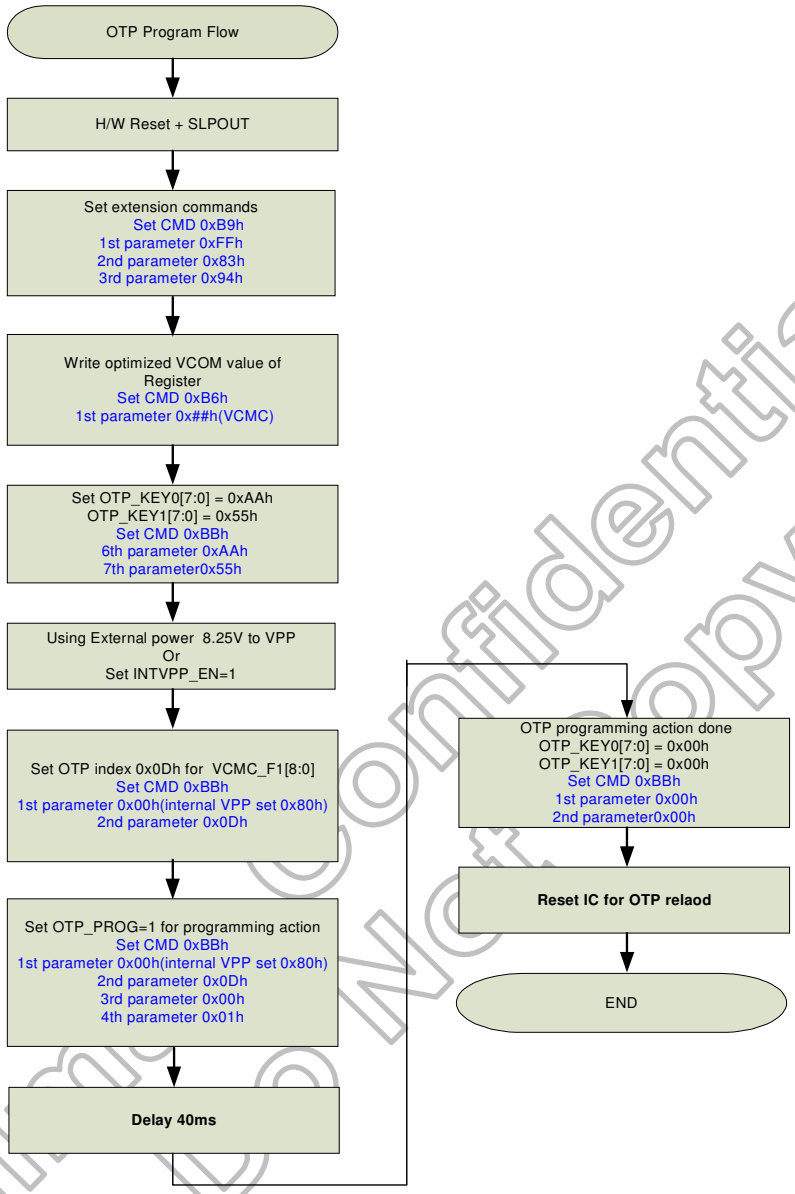
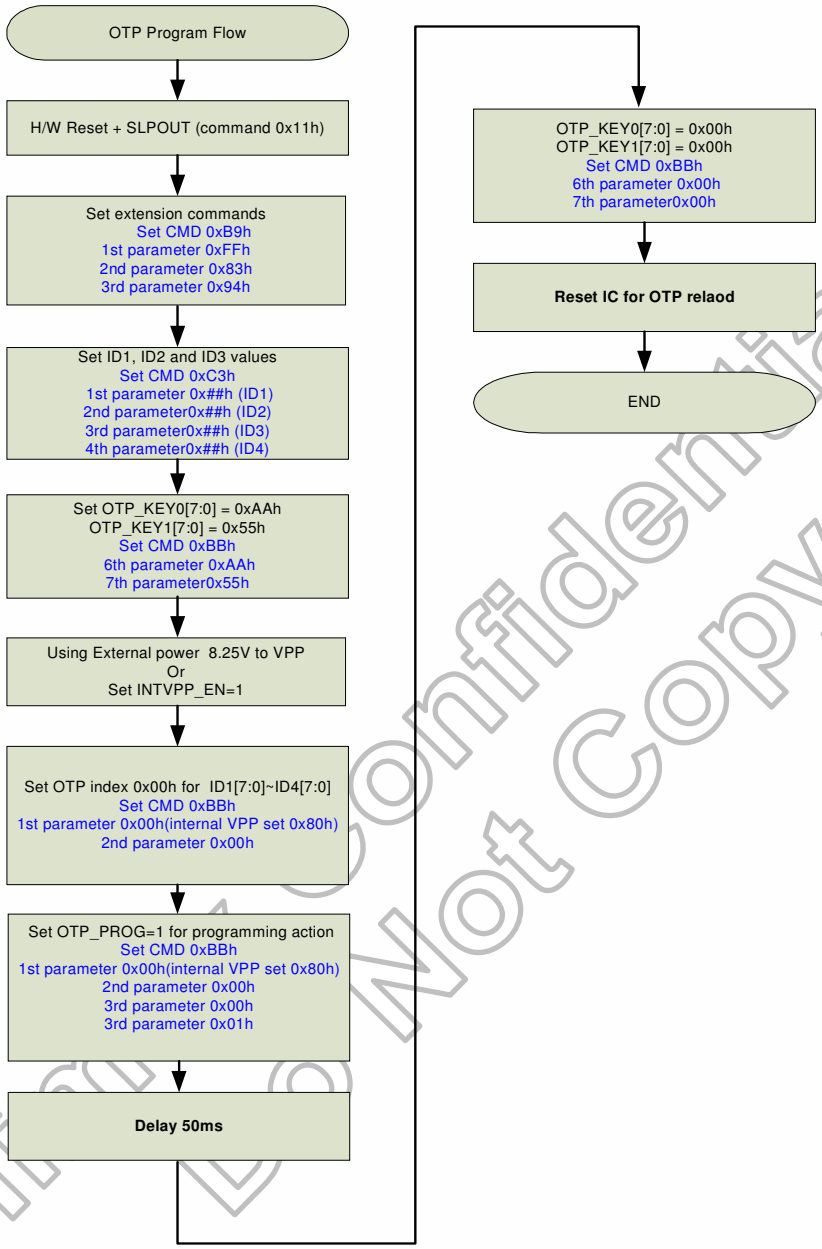


Figure 5.38: OTP programming sequence example 1

Himax Confidential Do Not Copy

5.16.5 OTP programming example of ID1, ID2 and ID3



Himax Confidential
Do Not Copy

Figure 5.39: OTP programming sequence example 2

5.16.6 OTP read example of 0x00h (ID1)

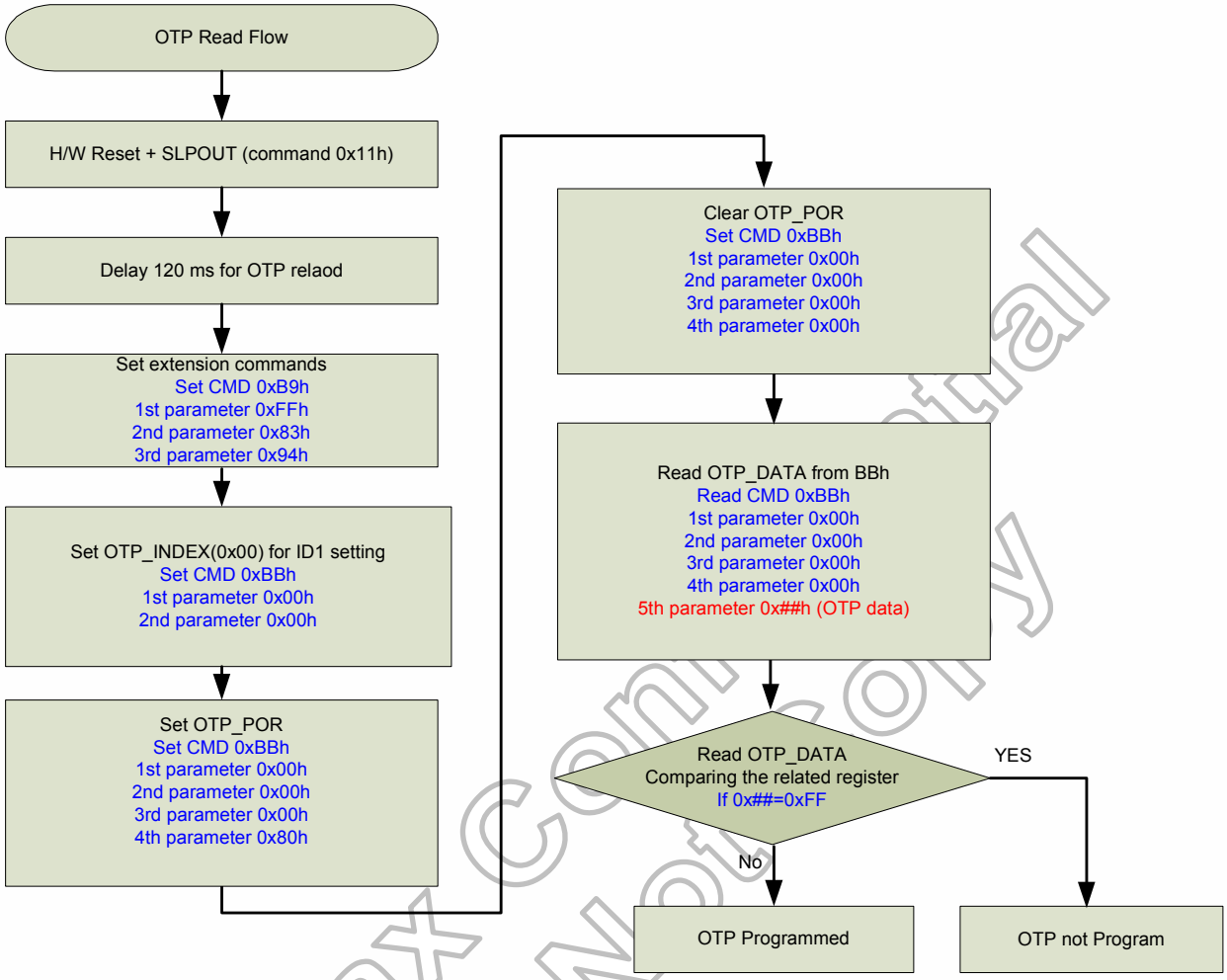


Figure 5.40: OTP read sequence flow of index 0x00h

Himax Confidential
Do Not Copy

5.17FRM (Free running mode)

Burn-in of TFT displays consists of driving each module for 10hr at a temperature of 60°C. In order to drive the modules, it requires extra electronics. To reduce the burn-in cost, it is requested that the driver IC will generate the required display image without requiring extra electronics. We term this a free running mode **(FRM)**. For burn-in, it is sufficient that the display is powered up with a plane saturated black or white pattern. The Display pattern sequence and power on sequence is as below Figure. Free-running mode pattern number and pattern delay time can be further set in register B2h FRM_PATTERN_CYCLE[3:0] and FRM_SCAN_CYCLE[2:0].

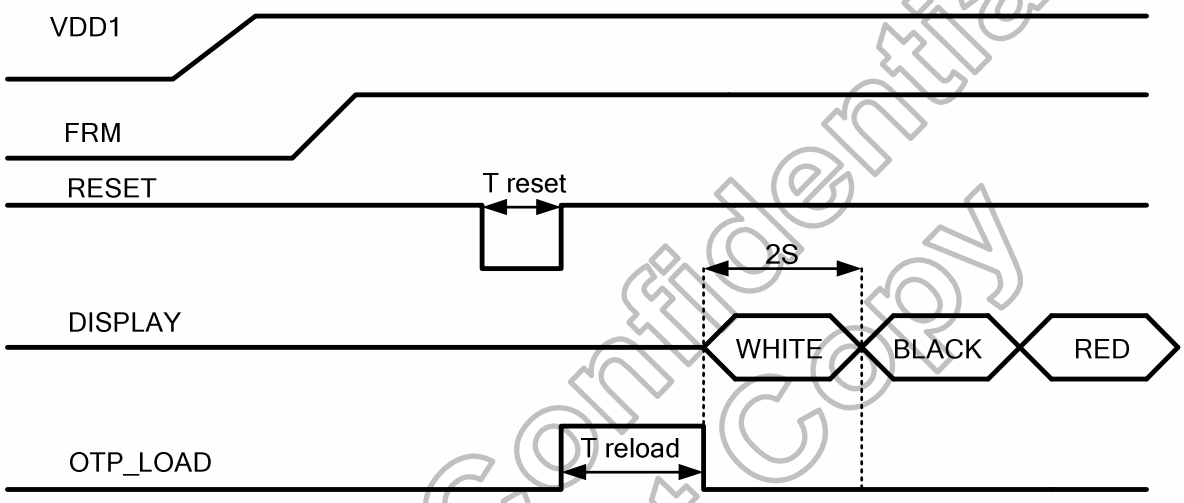


Figure 5.41: FRM power on sequence

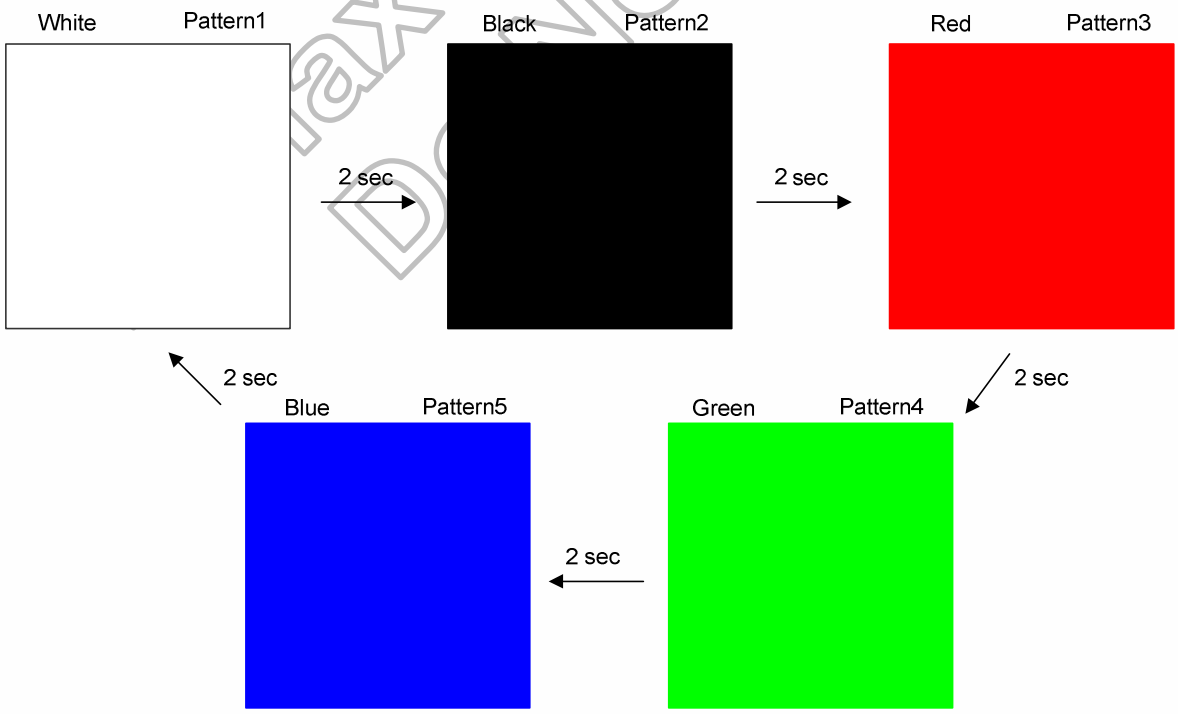


Figure 5.42: Display pattern sequence example in FRM
(FRM_PATTERN_CYCLE[3:0] =4h, FRM_SCAN_CYCLE[2:0]=1h)

Himax Confidential
Do Not Copy

5.18Command list

5.18.1 Standard command

(Hex)	Operation code	DCX	D7	D6	D5	D4	D3	D2	D1	D0	Function	Default (Hex)
00	NOP	0	0	0	0	0	0	0	0	0	No Operation	-
01	SWRESET	0	0	0	0	0	0	0	0	1	Software Reset	-
04	RDDIDIF	0	0	0	0	0	0	1	0	0	Read Display Identification Information	-
		1	ID1[7:0]									-
		1	ID2[7:0]									-
		1	ID3[7:0]									-
05	RDNUMPE	0	0	0	0	0	0	1	0	1	Read Number of DSI Parity Error	-
		1	P[7:0]									-
06	RDRED	0	0	0	0	0	0	1	1	0	Read Red Colour	-
		1	R7	R6	R5	R4	R3	R2	R1	R0	xx	-
07	RDGREEN	0	0	0	0	0	0	1	1	1	Read Green Colour	-
		1	G7	G6	G5	G4	G3	G2	G1	G0	xx	-
08	RDBLUE	0	0	0	0	0	1	0	0	0	Read Blue Colour	-
		1	B7	B6	B5	B4	B3	B2	B1	B0	xx	-
09	RDDST	0	0	0	0	0	1	0	0	1	Read display status	-
		1	D31	D30	D29	0	0	D26	D25	D24		-
		1	D23	D22	D21	D20	D19	0	D17	D16		-
		1	0	0	D13	D12	D11	D10	D9	D8		-
		1	D7	D6	D5	D4	D3	D2	D1	D0		-
0A	RDDPM	0	0	0	0	0	1	0	1	0	Read display power mode	-
		1	D7	D6	0	D4	D3	D2	0	0		-
0B	RDDMADCTL	0	0	0	0	0	1	0	1	1	Read display MADCTL	-
		1	D7	D6	0	0	D3	D2	D1	D0		-
0C	RDDCOLMOD	0	0	0	0	0	1	1	0	0	Read display pixel format	-
		1	0	D6	D5	D4	0	0	0	0		-
0D	RDDIM	0	0	0	0	0	1	1	0	1	Read display image mode	-
		1	0	0	D5	D4	D3	D2	D1	D0		-
0E	RDDSM	0	0	0	0	0	1	1	1	0	Read display signal mode	-
		1	D7	D6	D5	D4	D3	D2	0	D0		-
0F	RDDSDR	0	0	0	0	0	1	1	1	1	Read display self-diagnostic result	-
		1	D7	D6	D5	D4	0	0	0	0		-
10	SLPIN	0	0	0	0	1	0	0	0	0	Sleep In	-
11	SLPOUT	0	0	0	0	1	0	0	0	1	Sleep Out	-
13	NORON	0	0	0	0	1	0	0	1	1	Normal display mode on	-
20	INVOFF	0	0	0	1	0	0	0	0	0	Display inversion off	-
21	INVON	0	0	0	1	0	0	0	0	1	Display inversion on	-
22	ALLPOFF	0	0	0	1	0	0	0	1	0	All pixel off (black)	-
23	ALLPON	0	0	0	1	0	0	0	1	1	All pixel on (white)	-
26	GAMSET	0	0	0	1	0	0	1	1	0	Gamma set	-
		1	GC7	GC6	GC5	GC4	GC3	GC2	GC1	GC0		-
28	DISPOFF	0	0	0	1	0	1	0	0	0	Display off	-
29	DISPON	0	0	0	1	0	1	0	0	1	Display on	-
2C	RAMWR	0	0	0	1	0	1	1	0	0	Memory Write	-
		1	D17	D16	D15	D14	D13	D12	D11	D10	Write data	-
		1	Dx7	Dx6	Dx5	Dx4	Dx3	Dx2	Dx1	Dx0	Write data	-
		1	Dn7	Dn6	Dn5	Dn4	Dn3	Dn2	Dn1	Dn0	Write data	-
34	TEOFF	0	0	0	1	1	0	1	0	0	Tearing Effect Line OFF	-
35	TEON	0	0	0	1	1	0	1	0	1	Tearing Effect Line ON	-
		1	X	X	X	X	X	X	X	M		-
36	MADCTL	0	0	0	1	1	0	1	1	0	Memory Access Control	-
		1	D7	D6	X	X	D3	D2	D1	D0		-
38	IDMOFF	0	0	0	1	1	1	0	0	0	Idle mode off	-
39	IDMON	0	0	0	1	1	1	0	0	1	Idle mode on	-
3A	COLMOD	0	0	0	1	1	1	0	1	0	0	-
		1	X	D6	D5	D4	X	D2	D1	D0		-
3C	RAMWRCON	0	0	0	1	1	1	1	0	0	Memory write	-
		1	D17	D16	D15	D14	D13	D12	D11	D10		-
		1	Dx7	Dx6	Dx5	Dx4	Dx3	Dx2	Dx1	Dx0		-
		1	Dn7	Dn6	Dn5	Dn4	Dn3	Dn2	Dn1	Dn0		-
44	TESL	0	0	1	0	0	0	1	0	0	TESL	-
		1	TELINE[15:8](8'b0)									-

Himax Confidential
Do Not Copy

		1	TELINE[7:0](8'b0)									-
45	GETSCAN	0	0	1	0	0	0	1	0	1	Reture the current scanline SLN[15:0]	-
		1	SLN[15:8]									-
		1	SLN[7:0]									-
51	WRDISBV	0	0	1	0	1	0	0	0	1	Write Display Brightness	-
		1	DBV[7:0]									-
52	RDDISBV	0	0	1	0	1	0	0	1	0	Read Display Brightness Value	-
		1	DBV[7:0]									-
53	WRCTRLD	0	0	1	0	1	0	0	1	1	Write CTRL Display	-
		1	xx	xx	BCTRL	xx	DD	BL	xx	xx		-
54	RDCTRLD	0	0	1	0	1	0	0	1	1	Read Control Value Display	-
		1	0	0	BCTRL	0	DD	BL	0	0		-
55	WRCABC	0	0	1	0	1	0	1	0	1	Write Adaptive Brightness Control	-
		1	xx	xx	xx	xx	xx	xx	CABC[1:0]			-
56	RDCABC	0	0	1	0	1	0	1	1	0	Read Adaptive Brightness Control Content	-
		1	0	0	0	0	0	0	C1	C0		-
5E	WRCABCMB	0	0	1	0	1	1	1	1	0	Write CABC minimum brightness	-
		1	CMB[7:0]									-
5F	RDCABCMB	0	0	1	0	1	1	1	1	1	Read CABC minimum brightness	-
		1	CMB[7:0]									-
68	RDABCSDR	0	0	1	1	0	1	0	0	0	Read Automatic Brightness Control Self-Diagnostic Result	-
		1	D[7:6]		0	0	0	0	0	0		-
80	WRIMCOL	0	1	0	0	0	0	0	0	0	Write Idle Mode Color	-
		1	XX	XX	XX	XX	XX	R	G	B		-
81	RDIMCOL	0	1	0	0	0	0	0	0	0	Read Idle Mode Color	-
		1	0	0	0	0	0	R	G	B		-
A1	Read_DDB_start	0	1	0	1	0	0	0	0	1	Read the DDB from the provided location.	-
		1	x	x	x	x	x	x	x	x		-
		1	x	x	x	x	x	x	x	x		-
		1	x	x	x	x	x	x	x	x		-
A8	Read_DDB_continue	0	1	0	1	0	1	0	0	0	Continue reading the DDB from the last read location.	-
		1	x	x	x	x	x	x	x	x		-
		1	x	x	x	x	x	x	x	x		-
		1	x	x	x	x	x	x	x	x		-
DA	RDID1	0	1	1	0	1	1	0	1	0	Read ID1	-
		1	module's manufacturer[7:0]									-
DB	RDID2	0	1	1	0	1	1	0	1	1	Read ID2	-
		1	LCD module/driver version [7:0]									-
DC	RDID3	0	1	1	0	1	1	1	0	0	Read ID3	-
		1	LCD module/driver ID[7:0]									-

Note: (1) Undefined commands are treated as NOP (00h) command.
(2) B0h to D8h and E0h to FFh are for factory use of display supplier.

Himax Confidential
Do Not Copy

5.18.2 User define command list table

User define command list is available only set “SETEXC” command.
Note: *=Read only

(Hex)	Operation Code	Parameter index	D7	D6	D5	D4	D3	D2	D1	D0	Default (Hex)
B0	SETAUTO	NO_OTP	1	0	1	1	0	0	0	0	
		1	-	-	-	-	-	-	-	-	00
		2	-	-	-		-	-	-	OSC_EN	10
		3	-	VSN_EN	VSP_EN	VGL_EN	VGH_EN	VCL_EN	-	STB	7D
		4	-	-	-	-	GON	DTE	D[1:0]		0C
B1	SETPOWER		1	0	1	1	0	0	0	0	
		Bank_0_1	-	POWMOD[1:0]		AP[2:0]			DSTBY_OPT	DSTB	10
		Bank_0_2	VCLS[2:0]			VRHP[4:0]					10
		Bank_0_3	VPPS[2:0]			VRHN[4:0]					70
		Bank_0_4	EQ_VCLEN	-	-	-	VSP_FBOFF	XDK[2:0]			81
		Bank_0_5	-	-	CLK_OPT2	CLK_OPT1	FS0[3:0]				32
		Bank_0_6	FS1[3:0]				FS2[3:0]				33
		Bank_0_7	VGH_RATIO[2:0]			BTP[4:0]					71
		Bank_0_8	VGL_RATIO[2:0]			BTN[4:0]					71
		Bank_0_9	-	VGHS[6:0]							34
		Bank_0_10	-	VGLS[6:0]							48
		Bank_0_11	DT1[1:0]		DT2[1:0]		DCDIV[3:0]				56
		Bank_0_12	-	DCS[2:0]			-	DC[2:0]			73
		Bank_0_13	-	DTPS[2:0]			-	DTP[2:0]			02
		Bank_0_14	-	DTNS[2:0]			-	DTN[2:0]			02
		Bank_1_1	-	APF_EN	GASIOVCC_OPT[1:0]		-	GASVCI_OPT[2:0]			64
		Bank_1_2	GAS_CP_VSPVSN	GASVSN_OPT[2:0]			-	GASVSP_OPT[2:0]			44
		Bank_1_3	-	-	GASVGL_OPT[1:0]		-		GASVGH_OPT[1:0]		11
B2	SETDISP		1	0	1	1	0	0	1	0	
		Bank_0_1	-	ZZ_LR	ZZ_EO	ZZ_2PL	-	NW[2:0]			40
		Bank_0_2	MESSI_ENB	H_RES[2:0]			-	-	-	-	80
		Bank_0_3	NL[7:0]								64
		Bank_0_4	BP [7:0]								08
		Bank_0_5	FP [7:0]								08
		Bank_0_6	RTN[7:0]								2F
		Bank_0_7	-	-			-	-			00
		Bank_0_8	-	-	-		-	-	-	-	00
		Bank_0_9	-	-	-		-	-	-	-	00
		Bank_0_10	-	-	-	-	-	-	-	-	00
		Bank_0_11	FRM_PATTERN_CYCLE[3:0]				DISP_BIST_EN	FRM_SCAN_CYCLE[2:0]			C0
B4	SETCYC	1	SPON[7:0]								03
		2	SPOFF[7:0]								3F
		3	CON[7:0]								05
		4	COFF[7:0]								3F
		5	CON1[7:0]								05
		6	COFF1[7:0]								3F
		7	EQON1[7:0]								05
		8	SON[7:0]								10
		9	SOFF[7:0]								7E
		10	SAP1_P[3:0]				-	SAP2[2:0]			73
		11	-	DX2_EN	EQ_ON	EQ_MIX	-	-	-	-	20
		12	DX2OFF[7:0]								3F
		13	SPON_MPU[7:0]								03
		14	SPOFF_MPU[7:0]								3F
		15	CON_MPU[7:0]								05
		16	COFF_MPU[7:0]								3F
		17	CON1_MPU[7:0]								05
		18	COFF1_MPU[7:0]								3F

Himax Confidential
Do Not Copy

		19	EQON1_MPU[7:0]								05
		20	SON_MPU[7:0]								10
		21	SOFF_MPU[7:0]								7E
		22	DX2OFF_MPU[7:0]								3F
B6	SETVCOM (OTPx3)		1	0	1	1	0	1	1	0	
		1	VCMC_F[7:0]								46
		2	VCMC_B[7:0]								46
		3	VCOM_TIMES[2:0]			-	-	-	VCMC_B[8]	VCMC_F[8]	E0
B7	SETTE		1	0	1	1	0	1	1	1	
		1	-	TEI[3:0]				TEP[10:8]			00
		2	TEP[7:0]								00
		3	TE_SEL[3:0]				TE1_SEL[3:0]				00
B8	SETSENSOR		1	1	0	1	0	0	0	1	
		bank_0_1	-	TSENSOR_EN	LT_EN	HT_EN	-	-			01
B9	SETEXTC	NO_OTP	1	0	1	1	1	0	0	1	
		1	EXTC1[7:0]								00/FF
		2	EXTC2[7:0]								00/83
		3	EXTC3[7:0]								00/94
BA	SETMIPI		1	0	1	1	1	0	1	0	
		1	-	DSISETUP0[6:0]							63
		2	DSISETUP1[7:0]								03
		3	DPHYCMD0[7:0]								68
		4	DPHYCMD1[7:0]								63
BB	SETOTP	NO_OTP	1	0	1	1	1	0	1	1	
		1	INTVPP_EN	-	OTP_PROG_ALL	-	-	-	OTP_INDEX[9:8]		00
		2	OTP_INDEX[7:0]								00
		3	OTP_DATA[7:0]								00
		4	OTP_POR	OTP_PWE	OTP_PWR_SEL	OTP_PTM[2:0]			OTP_TEST	OTP_PROG	00
		5	OTP_DATA_READ[7:0]								FF
		6	OTP_KEY0[7:0]								00
		7	OTP_KEY1[7:0]								00
		8	OTP_MASK[7:0]								00
		9	-	-	-	-	-	-	OTP_BIT[1:0]		01
BD	SET_BANK	NO_OTP	1	1	1	1	0	0	0	1	
		1	-	-	-	-	-	-	BANK_INDEX[1:0]		00
C1	SETDGCLUT		1	1	0	0	0	0	0	1	
		Bank_0_1	-	-	-	-	-	-	DGC_INV	DGC_EN	00
		Bank_0_2	R_GAMMA0[9:2]								-
		Bank_0_3	R_GAMMA1[9:2]								-
		Bank_0_4	R_GAMMA2[9:2]								-
		Bank_0_5	R_GAMMA3[9:2]								-
		Bank_0_6	R_GAMMA4[9:2]								-
		Bank_0_7	R_GAMMA5[9:2]								-
		Bank_0_8	R_GAMMA6[9:2]								-
		Bank_0_9	R_GAMMA7[9:2]								-
		Bank_0_10	R_GAMMA8[9:2]								-
		Bank_0_11	R_GAMMA9[9:2]								-
		Bank_0_12	R_GAMMA10[9:2]								-
		Bank_0_13	R_GAMMA11[9:2]								-
		Bank_0_14	R_GAMMA12[9:2]								-
		Bank_0_15	R_GAMMA13[9:2]								-
		Bank_0_16	R_GAMMA14[9:2]								-
		Bank_0_17	R_GAMMA15[9:2]								-
		Bank_0_18	R_GAMMA16[9:2]								-
		Bank_0_19	R_GAMMA17[9:2]								-
		Bank_0_20	R_GAMMA18[9:2]								-
		Bank_0_21	R_GAMMA19[9:2]								-

Himax Confidential
Do Not Copy

Bank_0_22	R_GAMMA20[9:2]											-
Bank_0_23	R_GAMMA21[9:2]											-
Bank_0_24	R_GAMMA22[9:2]											-
Bank_0_25	R_GAMMA23[9:2]											-
Bank_0_26	R_GAMMA24[9:2]											-
Bank_0_27	R_GAMMA25[9:2]											-
Bank_0_28	R_GAMMA26[9:2]											-
Bank_0_29	R_GAMMA27[9:2]											-
Bank_0_30	R_GAMMA28[9:2]											-
Bank_0_31	R_GAMMA29[9:2]											-
Bank_0_32	R_GAMMA30[9:2]											-
Bank_0_33	R_GAMMA31[9:2]											-
Bank_0_34	R_GAMMA32[9:2]											-
Bank_0_35	R_GAMMA0[1:0]		R_GAMMA1[1:0]		R_GAMMA2[1:0]		R_GAMMA3[1:0]					-
Bank_0_36	R_GAMMA4[1:0]		R_GAMMA5[1:0]		R_GAMMA6[1:0]		R_GAMMA7[1:0]					-
Bank_0_37	R_GAMMA8[1:0]		R_GAMMA9[1:0]		R_GAMMA10[1:0]		R_GAMMA11[1:0]					-
Bank_0_38	R_GAMMA12[1:0]		R_GAMMA13[1:0]		R_GAMMA14[1:0]		R_GAMMA15[1:0]					-
Bank_0_39	R_GAMMA16[1:0]		R_GAMMA17[1:0]		R_GAMMA18[1:0]		R_GAMMA19[1:0]					-
Bank_0_40	R_GAMMA20[1:0]		R_GAMMA21[1:0]		R_GAMMA22[1:0]		R_GAMMA23[1:0]					-
Bank_0_41	R_GAMMA24[1:0]		R_GAMMA25[1:0]		R_GAMMA26[1:0]		R_GAMMA27[1:0]					-
Bank_0_42	R_GAMMA28[1:0]		R_GAMMA29[1:0]		R_GAMMA30[1:0]		R_GAMMA31[1:0]					-
Bank_0_43	R_GAMMA32[1:0]		-	-	-	-	-	-				01
Bank_1_1	G_GAMMA0[9:2]											-
Bank_1_2	G_GAMMA1[9:2]											-
Bank_1_3	G_GAMMA2[9:2]											-
Bank_1_4	G_GAMMA3[9:2]											-
Bank_1_5	G_GAMMA4[9:2]											-
Bank_1_6	G_GAMMA5[9:2]											-
Bank_1_7	G_GAMMA6[9:2]											-
Bank_1_8	G_GAMMA7[9:2]											-
Bank_1_9	G_GAMMA8[9:2]											-
Bank_1_10	G_GAMMA9[9:2]											-
Bank_1_11	G_GAMMA10[9:2]											-
Bank_1_12	G_GAMMA11[9:2]											-
Bank_1_13	G_GAMMA12[9:2]											-
Bank_1_14	G_GAMMA13[9:2]											-
Bank_1_15	G_GAMMA14[9:2]											-
Bank_1_16	G_GAMMA15[9:2]											-
Bank_1_17	G_GAMMA16[9:2]											-
Bank_1_18	G_GAMMA17[9:2]											-
Bank_1_19	G_GAMMA18[9:2]											-
Bank_1_2	G_GAMMA19[9:2]											

Himax Confidential
Do Not Copy

		0								
		Bank_1_21	G_GAMMA20[9:2]							-
		Bank_1_22	G_GAMMA21[9:2]							-
		Bank_1_23	G_GAMMA22[9:2]							-
		Bank_1_24	G_GAMMA23[9:2]							-
		Bank_1_25	G_GAMMA24[9:2]							-
		Bank_1_26	G_GAMMA25[9:2]							-
		Bank_1_27	G_GAMMA26[9:2]							-
		Bank_1_28	G_GAMMA27[9:2]							-
		Bank_1_29	G_GAMMA28[9:2]							-
		Bank_1_30	G_GAMMA29[9:2]							-
		Bank_1_31	G_GAMMA30[9:2]							-
		Bank_1_32	G_GAMMA31[9:2]							-
		Bank_1_33	G_GAMMA32[9:2]							-
		Bank_1_34	G_GAMMA0[1:0]	G_GAMMA1[1:0]		G_GAMMA2[1:0]		G_GAMMA3[1:0]		-
		Bank_1_35	G_GAMMA4[1:0]	G_GAMMA5[1:0]		G_GAMMA6[1:0]		G_GAMMA7[1:0]		-
		Bank_1_36	G_GAMMA8[1:0]	G_GAMMA9[1:0]		G_GAMMA10[1:0]		G_GAMMA11[1:0]		-
		Bank_1_37	G_GAMMA12[1:0]	G_GAMMA13[1:0]		G_GAMMA14[1:0]		G_GAMMA15[1:0]		-
		Bank_1_38	G_GAMMA16[1:0]	G_GAMMA17[1:0]		G_GAMMA18[1:0]		G_GAMMA19[1:0]		-
		Bank_1_39	G_GAMMA20[1:0]	G_GAMMA21[1:0]		G_GAMMA22[1:0]		G_GAMMA23[1:0]		-
		Bank_1_40	G_GAMMA24[1:0]	G_GAMMA25[1:0]		G_GAMMA26[1:0]		G_GAMMA27[1:0]		-
		Bank_1_41	G_GAMMA28[1:0]	G_GAMMA29[1:0]		G_GAMMA30[1:0]		G_GAMMA31[1:0]		-
		Bank_1_42	G_GAMMA32[1:0]	-	-	-	-	-	-	01
		Bank_2_1	B_GAMMA0[9:2]							-
		Bank_2_2	B_GAMMA1[9:2]							-
		Bank_2_3	B_GAMMA2[9:2]							-
		Bank_2_4	B_GAMMA3[9:2]							-
		Bank_2_5	B_GAMMA4[9:2]							-
		Bank_2_6	B_GAMMA5[9:2]							-
		Bank_2_7	B_GAMMA6[9:2]							-
		Bank_2_8	B_GAMMA7[9:2]							-
		Bank_2_9	B_GAMMA8[9:2]							-
		Bank_2_10	B_GAMMA9[9:2]							-
		Bank_2_11	B_GAMMA10[9:2]							-
		Bank_2_12	B_GAMMA11[9:2]							-
		Bank_2_13	B_GAMMA12[9:2]							-
		Bank_2_14	B_GAMMA13[9:2]							-
		Bank_2_15	B_GAMMA14[9:2]							-
		Bank_2_16	B_GAMMA15[9:2]							-
		Bank_2_17	B_GAMMA16[9:2]							-
		Bank_2_18	B_GAMMA17[9:2]							-
		Bank_2_19	B_GAMMA18[9:2]							-

Himax Confidential
Do Not Copy

		Bank_2_220	B_GAMMA19[9:2]								-
		Bank_2_221	B_GAMMA20[9:2]								-
		Bank_2_222	B_GAMMA21[9:2]								-
		Bank_2_223	B_GAMMA22[9:2]								-
		Bank_2_224	B_GAMMA23[9:2]								-
		Bank_2_225	B_GAMMA24[9:2]								-
		Bank_2_226	B_GAMMA25[9:2]								-
		Bank_2_227	B_GAMMA26[9:2]								-
		Bank_2_228	B_GAMMA27[9:2]								-
		Bank_2_229	B_GAMMA28[9:2]								-
		Bank_2_230	B_GAMMA29[9:2]								-
		Bank_2_231	B_GAMMA30[9:2]								-
		Bank_2_232	B_GAMMA31[9:2]								-
		Bank_2_233	B_GAMMA32[9:2]								-
		Bank_2_234	B_GAMMA0[1:0]		B_GAMMA1[1:0]		B_GAMMA2[1:0]		B_GAMMA3[1:0]		-
		Bank_2_235	B_GAMMA4[1:0]		B_GAMMA5[1:0]		B_GAMMA6[1:0]		B_GAMMA7[1:0]		-
		Bank_2_236	B_GAMMA8[1:0]		B_GAMMA9[1:0]		B_GAMMA10[1:0]		B_GAMMA11[1:0]		-
		Bank_2_237	B_GAMMA12[1:0]		B_GAMMA13[1:0]		B_GAMMA14[1:0]		B_GAMMA15[1:0]		-
		Bank_2_238	B_GAMMA16[1:0]		B_GAMMA17[1:0]		B_GAMMA18[1:0]		B_GAMMA19[1:0]		-
		Bank_2_239	B_GAMMA20[1:0]		B_GAMMA21[1:0]		B_GAMMA22[1:0]		B_GAMMA23[1:0]		-
		Bank_2_240	B_GAMMA24[1:0]		B_GAMMA25[1:0]		B_GAMMA26[1:0]		B_GAMMA27[1:0]		-
		Bank_2_241	B_GAMMA28[1:0]		B_GAMMA29[1:0]		B_GAMMA30[1:0]		B_GAMMA31[1:0]		-
		Bank_2_242	B_GAMMA32[1:0]		-	-	-	-	-	-	-
C3	SETID (OTPx3)		1	1	0	0	0	0	1	1	
		1	ID1[7:0]								83
		2	ID2[7:0]								94
		3	ID3[7:0]								0F
		4	ID4[7:0]								00
		5	ID_TIMES[2:0]*			-	-	-	-	-	E0
C4	SETDDB		1	1	0	0	0	1	0	0	
		1	DDB1[7:0]								00
		2	DDB2[7:0]								00
		3	DDB3[7:0]								00
		4	DDB4[7:0]								00
		5	DDB5[7:0]								00
		6	DDB6[7:0]								00
C9	SETCABC		1	1	0	0	1	0	0	1	
		1	-	SEL_PWMCLK[2:0]			PWM_PERIOD[16]	BC_CTRL_EN	INVPULS	SEL_BLDUTY	13
		2	PWM_PERIOD[15:8]								00
		3	PWM_PERIOD[7:0]								2E
		4	CABC_FSYNC	DIM_FRAME[6:0]							1E
		5	CABC_DD	CABC_DD_IDLE	-	-	CABC_FLM[3:0]				B1
CA	SETCABCGAIN		1	1	0	0	1	0	1	0	
		1	-	DBG0[6:0]							40
		2	-	DBG1[6:0]							3C
		3	-	DBG2[6:0]							38
		4	-	DBG3[6:0]							34
		5	-	DBG4[6:0]							33

Himax Confidential
Do Not Copy

		6	-	DBG5[6:0]							32
		7	-	DBG6[6:0]							2B
		8	-	DBG7[6:0]							24
		9	-	DBG8[6:0]							22
CC	SETPANEL		1	1	0	0	1	1	0	0	
		1	-	-	-	-	SS_PANEL	GS_PANEL	REV_PANEL	BGR_PANEL	00
D2	SETOFFSET		1	1	0	1	0	0	1	0	
		1	VN_REFS[3:0]				VP_REFS[3:0]				55
D3	SETGIP_0		1	1	0	1	0	0	1	1	
		bank_0_1	-	-	GIP_EQ_OPT[1:0]		-	-	EQ_DELAY_HSYNC[1:0]		00
		bank_0_2	-	-	-	-	-	EQ_VSEL	EQ_DISC[1:0]		00
		bank_0_3	EQ_DELAY_ON1[7:0]								00
		bank_0_4	EQ_DELAY_OFF1[7:0]								00
		bank_0_5	GTO[7:0]								00
		bank_0_6	GNO[7:0]								00
		bank_0_7	USER_GIP_GATE[7:0]								08
		bank_0_8	USER_GIP_GATE1[7:0]								08
		bank_0_9	SHR0_3[3:0]				SHR0_2[3:0]				32
		bank_0_10	SHR0_1[3:0]				SHR0[11:8]				10
		bank_0_11	SHR0[7:0]								02
		bank_0_12	-	-	-	-	SHR0_GS[11:8]				00
		bank_0_13	SHR0_GS[7:0]								02
		bank_0_14	SHR1_3[3:0]				SHR1_2[3:0]				32
		bank_0_15	SHR1_1[3:0]				SHR1[11:8]				13
		bank_0_16	SHR1[7:0]								C0
		bank_0_17	-	-	-	-	SHR1_GS[11:8]				00
		bank_0_18	SHR1_GS[7:0]								00
		bank_0_19	SHR2_3[3:0]				SHR2_2[3:0]				32
		bank_0_20	SHR2_1[3:0]				SHR2[11:8]				10
		bank_0_21	SHR2[7:0]								08
		bank_0_22	-	-	-	-	SHR2_GS[11:8]				00
		bank_0_23	SHR2_GS[7:0]								00
		bank_0_24	SHP0[3:0]				SCP[3:0]				4B
		bank_0_25	SHP2[3:0]				SHP1[3:0]				00
		bank_0_26	CHR0[7:0]								06
		bank_0_27	CHR0_GS[7:0]								06
		bank_0_28	CHP0[3:0]				CCP0[3:0]				47
		bank_0_29	CHR1[7:0]								04
		bank_0_30	CHR1_GS[7:0]								00
		bank_0_31	CHP1[3:0]				CCP1[3:0]				27
D5	SETGIP_1		1	1	0	1	0	1	0	1	
		1	-	CGTS_L_INV[1]	COS1_L[5:0] (CGOUT1_L)						18
		2	-	CGTS_R_INV[1]	COS1_R[5:0] (CGOUT1_R)						18
		3	-	CGTS_L_INV[2]	COS2_L[5:0] (CGOUT2_L)						18
		4	-	CGTS_R_INV[2]	COS2_R[5:0] (CGOUT2_R)						18
		5	-	CGTS_L_INV[3]	COS3_L[5:0] (CGOUT3_L)						18
		6	-	CGTS_R_INV[3]	COS3_R[5:0] (CGOUT3_R)						18
		7	-	CGTS_L_INV[4]	COS4_L[5:0] (CGOUT4_L)						18
		8	-	CGTS_R_INV[4]	COS4_R[5:0] (CGOUT4_R)						18
		9	-	CGTS_L_INV[5]	COS5_L[5:0] (CGOUT5_L)						18
		10	-	CGTS_R_INV[5]	COS5_R[5:0] (CGOUT5_R)						18
		11	-	CGTS_L_INV[6]	COS6_L[5:0] (CGOUT6_L)						18
		12	-	CGTS_R_INV[6]	COS6_R[5:0] (CGOUT6_R)						18
		13	-	CGTS_L_INV[7]	COS7_L[5:0] (CGOUT7_L)						18
		14	-	CGTS_R_INV[7]	COS7_R[5:0] (CGOUT7_R)						18
		15	-	CGTS_L_INV[8]	COS8_L[5:0] (CGOUT8_L)						18

Himax Confidential Do Not Copy

		16	-	CGTS_R_INV[8]	COS8_R[5:0] (CGOUT8_R)						18
		17	-	CGTS_L_INV[9]	COS9_L[5:0] (CGOUT9_L)						18
		18	-	CGTS_R_INV[9]	COS9_R[5:0] (CGOUT9_R)						18
		19	-	CGTS_L_INV[10]	COS10_L[5:0] (CGOUT10_L)						18
		20	-	CGTS_R_INV[10]	COS10_R[5:0] (CGOUT10_R)						18
		21	-	CGTS_L_INV[11]	COS11_L[5:0] (CGOUT11_L)						18
		22	-	CGTS_R_INV[11]	COS11_R[5:0] (CGOUT11_R)						18
		23	-	CGTS_L_INV[12]	COS12_L[5:0] (CGOUT12_L)						18
		24	-	CGTS_R_INV[12]	COS12_R[5:0] (CGOUT12_R)						18
		25	-	CGTS_L_INV[13]	COS13_L[5:0] (CGOUT13_L)						18
		26	-	CGTS_R_INV[13]	COS13_R[5:0] (CGOUT13_R)						18
		27	-	CGTS_L_INV[14]	COS14_L[5:0] (CGOUT14_L)						18
		28	-	CGTS_R_INV[14]	COS14_R[5:0] (CGOUT14_R)						18
		29	-	CGTS_L_INV[15]	COS15_L[5:0] (CGOUT15_L)						18
		30	-	CGTS_R_INV[15]	COS15_R[5:0] (CGOUT15_R)						18
		31	-	CGTS_L_INV[16]	COS16_L[5:0] (CGOUT16_L)						18
		32	-	CGTS_R_INV[16]	COS16_R[5:0] (CGOUT16_R)						18
		33	-	CGTS_L_INV[17]	COS17_L[5:0] (CGOUT17_L)						18
		34	-	CGTS_R_INV[17]	COS17_R[5:0] (CGOUT17_R)						18
		35	-	CGTS_L_INV[18]	COS18_L[5:0] (CGOUT18_L)						18
		36	-	CGTS_R_INV[18]	COS18_R[5:0] (CGOUT18_R)						18
		37	-	CGTS_L_INV[19]	COS19_L[5:0] (CGOUT19_L)						18
		38	-	CGTS_R_INV[19]	COS19_R[5:0] (CGOUT19_R)						18
		39	-	CGTS_L_INV[20]	COS20_L[5:0] (CGOUT20_L)						18
		40	-	CGTS_R_INV[20]	COS20_R[5:0] (CGOUT20_R)						18
		41	-	CGTS_L_INV[21]	COS21_L[5:0] (CGOUT21_L)						18
		42	-	CGTS_R_INV[21]	COS21_R[5:0] (CGOUT21_R)						18
		43	-	CGTS_L_INV[22]	COS22_L[5:0] (CGOUT22_L)						18
		44	-	CGTS_R_INV[22]	COS22_R[5:0] (CGOUT22_R)						18
D6	SETGIP_2		1	1	0	1	0	1	1	0	
		1	-	CGOUT_L_HIZ_IN[1]	COS1_L_GS[5:0] (CGOUT1_L)						18
		2	-	CGOUT_R_HIZ_IN[1]	COS1_R_GS[5:0] (CGOUT1_R)						18
		3	-	CGOUT_L_HIZ_IN[2]	COS2_L_GS[5:0] (CGOUT2_L)						18
		4	-	CGOUT_R_HIZ_IN[2]	COS2_R_GS[5:0] (CGOUT2_R)						18
		5	-	CGOUT_L_HIZ_IN[3]	COS3_L_GS[5:0] (CGOUT3_L)						18
		6	-	CGOUT_R_HIZ_IN[3]	COS3_R_GS[5:0] (CGOUT3_R)						18
		7	-	CGOUT_L_HIZ_IN[4]	COS4_L_GS[5:0] (CGOUT4_L)						18
		8	-	CGOUT_R	COS4_R_GS[5:0] (CGOUT4_R)						18

Himax Confidential Do Not Copy

				_HIZ_IN[4]		
		9	-	CGOUT_L_HIZ_IN[5]	COS5_L_GS[5:0] (CGOUT5_L)	18
		10	-	CGOUT_R_HIZ_IN[5]	COS5_R_GS[5:0] (CGOUT5_R)	18
		11	-	CGOUT_L_HIZ_IN[6]	COS6_L_GS[5:0] (CGOUT6_L)	18
		12	-	CGOUT_R_HIZ_IN[6]	COS6_R_GS[5:0] (CGOUT6_R)	18
		13	-	CGOUT_L_HIZ_IN[7]	COS7_L_GS[5:0] (CGOUT7_L)	18
		14	-	CGOUT_R_HIZ_IN[7]	COS7_R_GS[5:0] (CGOUT7_R)	18
		15	-	CGOUT_L_HIZ_IN[8]	COS8_L_GS[5:0] (CGOUT8_L)	18
		16	-	CGOUT_R_HIZ_IN[8]	COS8_R_GS[5:0] (CGOUT8_R)	18
		17	-	CGOUT_L_HIZ_IN[9]	COS9_L_GS[5:0] (CGOUT9_L)	18
		18	-	CGOUT_R_HIZ_IN[9]	COS9_R_GS[5:0] (CGOUT9_R)	18
		19	-	CGOUT_L_HIZ_IN[10]	COS10_L_GS[5:0] (CGOUT10_L)	18
		20	-	CGOUT_R_HIZ_IN[10]	COS10_R_GS[5:0] (CGOUT10_R)	18
		21	-	CGOUT_L_HIZ_IN[11]	COS11_L_GS[5:0] (CGOUT11_L)	18
		22	-	CGOUT_R_HIZ_IN[11]	COS11_R_GS[5:0] (CGOUT11_R)	18
		23	-	CGOUT_L_HIZ_IN[12]	COS12_L_GS[5:0] (CGOUT12_L)	18
		24	-	CGOUT_R_HIZ_IN[12]	COS12_R_GS[5:0] (CGOUT12_R)	18
		25	-	CGOUT_L_HIZ_IN[13]	COS13_L_GS[5:0] (CGOUT13_L)	18
		26	-	CGOUT_R_HIZ_IN[13]	COS13_R_GS[5:0] (CGOUT13_R)	18
		27	-	CGOUT_L_HIZ_IN[14]	COS14_L_GS[5:0] (CGOUT14_L)	18
		28	-	CGOUT_R_HIZ_IN[14]	COS14_R_GS[5:0] (CGOUT14_R)	18
		29	-	CGOUT_L_HIZ_IN[15]	COS15_L_GS[5:0] (CGOUT15_L)	18
		30	-	CGOUT_R_HIZ_IN[15]	COS15_R_GS[5:0] (CGOUT15_R)	18
		31	-	CGOUT_L_HIZ_IN[16]	COS16_L_GS[5:0] (CGOUT16_L)	18
		32	-	CGOUT_R_HIZ_IN[16]	COS16_R_GS[5:0] (CGOUT16_R)	18
		33	-	CGOUT_L_HIZ_IN[17]	COS17_L_GS[5:0] (CGOUT17_L)	18
		34	-	CGOUT_R_HIZ_IN[17]	COS17_R_GS[5:0] (CGOUT17_R)	18
		35	-	CGOUT_L_HIZ_IN[18]	COS18_L_GS[5:0] (CGOUT18_L)	18
		36	-	CGOUT_R_HIZ_IN[18]	COS18_R_GS[5:0] (CGOUT18_R)	18

Himax Confidential
Do Not Copy

		37	-	CGOUT_L_HIZ_IN[19]	COS19_L_GS[5:0] (CGOUT19_L)						18
		38	-	CGOUT_R_HIZ_IN[19]	COS19_R_GS[5:0] (CGOUT19_R)						18
		39	-	CGOUT_L_HIZ_IN[20]	COS20_L_GS[5:0] (CGOUT20_L)						18
		40	-	CGOUT_R_HIZ_IN[20]	COS20_R_GS[5:0] (CGOUT20_R)						18
		41	-	CGOUT_L_HIZ_IN[21]	COS21_L_GS[5:0] (CGOUT21_L)						18
		42	-	CGOUT_R_HIZ_IN[21]	COS21_R_GS[5:0] (CGOUT21_R)						18
		43	-	CGOUT_L_HIZ_IN[22]	COS22_L_GS[5:0] (CGOUT22_L)						18
		44	-	CGOUT_R_HIZ_IN[22]	COS22_R_GS[5:0] (CGOUT22_R)						18
D9	SETGPO		1	1	0	1	1	0	1	0	
		1	ESD_DET	-			TE_GPO[3:0]				00
		2	-	-			TE1_GPO[3:0]				01
		3	-	-			CABC_GPO[3:0]				82
		4	-				SDO_GPO[3:0]				07
DD	SETSCL		1	1	0	1	1	1	0	1	
		2	-	-	-	-	-	-	SCALING_TYPE	SCALING_EN	00
DF	SET1BPP		1	1	0	1	1	1	1	1	
		1	-	-	-	-	-	NW_PE[2:0]			00
		2	BP_PE[7:0]								08
		3	FP_PE[7:0]								08
		4	RTN_PE[7:0]								2F
		5	VCMC_F_IDLE[7:0]								FE
		6	VCMC_B_IDLE[7:0]								FE
		7	AP_PE[2:0]			-	-	-	VCMC_B_IDLE[8]	VCMC_F_IDLE[8]	83
E0	SETGAMMA		1	1	1	0	0	0	0	0	
		1	-	VHP_0[6:0]							00
		2	-	VHP_1[6:0]							04
		3	-	VHP_2[6:0]							08
		4	-	VHP_3[6:0]							0C
		5	-	VHP_4[6:0]							10
		6	-	VHP_5[6:0]							14
		7	-	VHP_6[6:0]							18
		8	-	VHP_7[6:0]							1C
		9	VMP_0[7:0]								20
		10	VMP_1[7:0]								24
		11	VMP_2[7:0]								28
		12	VMP_3[7:0]								2C
		13	VMP_4[7:0]								30
		14	VMP_5[7:0]								34
		15	VMP_6[7:0]								38
		16	VMP_7[7:0]								3C
		17	VMP_8[7:0]								40
		18	VMP_9[7:0]								44
		19	VMP_10[7:0]								48
		20	VMP_11[7:0]								4C
		21	VMP_12[7:0]								50
		22	-	VLP_0[6:0]							54
		23	-	VLP_1[6:0]							58
		24	-	VLP_2[6:0]							5C
		25	-	VLP_3[6:0]							60
		26	-	VLP_4[6:0]							64
		27	-	VLP_5[6:0]							68
		28	-	VLP_6[6:0]							6C
		29	-	VLP_7[6:0]							7F
		30	-	VHN_0[6:0]							00
		31	-	VHN_1[6:0]							04

Himax Confidential Do Not Copy

		32	-	VHN_2[6:0]							08
		33	-	VHN_3[6:0]							0C
		34	-	VHN_4[6:0]							10
		35	-	VHN_5[6:0]							14
		36	-	VHN_6[6:0]							18
		37	-	VHN_7[6:0]							1C
		38	VMN_0[7:0]								20
		39	VMN_1[7:0]								24
		40	VMN_2[7:0]								28
		41	VMN_3[7:0]								2C
		42	VMN_4[7:0]								30
		43	VMN_5[7:0]								34
		44	VMN_6[7:0]								38
		45	VMN_7[7:0]								3C
		46	VMN_8[7:0]								40
		47	VMN_9[7:0]								44
		48	VMN_10[7:0]								48
		49	VMN_11[7:0]								4C
		50	VMN_12[7:0]								50
		51	-	VLN_0[6:0]							54
		52	-	VLN_1[6:0]							58
		53	-	VLN_2[6:0]							5C
		54	-	VLN_3[6:0]							60
		55	-	VLN_4[6:0]							64
		56	-	VLN_5[6:0]							68
		57	-	VLN_6[6:0]							6C
		58	-	VLN_7[6:0]							7F
E4	SETCHEMODE		1	1	1	0	0	1	0	0	
		1	-	-	-	-	-	-	-	DYN_CEH_EN	01
		2	HUE_MODE[1:0]		SE_MODE[1:0]		BE_MODE[1:0]		CE_MODE[1:0]		00
E5	SETCHE		1	1	1	0	0	1	0	1	
		1	-	-	-	-	-	-	-	-	-
		2	-	-	-	SG0_F[4:0]					-
		3	-	-	-	SG64_F[4:0]					-
		4	-	-	-	SG128_F[4:0]					-
		5	-	-	-	SG192_F[4:0]					-
		6	-	-	-	SG255_F[4:0]					-
		7	TH_SAT[7:0]								-
		8	TH_WHITE[7:0]								-
		9	TH_COLOR[7:0]								-
		10	TH_GREY[7:0]								-
		11	-	-	-	-	-	-	-	-	-
		12	-	-	-	CG0_F[4:0]					-
		13	-	-	-	CG64_F[4:0]					-
		14	-	-	-	CG128_F[4:0]					-
		15	-	-	-	CG192_F[4:0]					-
		16	-	-	-	CG255_F[4:0]					-
		17	THC[7:0]								-
		18	-	-	-	-	EG2_F[1:0]		EG1_F[1:0]		-
		19	TH_RFL[7:0]								-
		20	-	-	-	SG16_F[4:0]					-
		21	-	-	-	SG32_F[4:0]					-
		22	-	-	-	SG48_F[4:0]					-
		23	-	-	-	SG80_F[4:0]					-
		24	-	-	-	SG96_F[4:0]					-
		25	-	-	-	SG112_F[4:0]					-
		26	-	-	-	SG144_F[4:0]					-
		27	-	-	-	SG160_F[4:0]					-
		28	-	-	-	SG176_F[4:0]					-
		29	-	-	-	SG208_F[4:0]					-
		30	-	-	-	SG224_F[4:0]					-
		31	-	-	-	SG240_F[4:0]					-
		32	-	-	-	CG16_F[4:0]					-
		33	-	-	-	CG32_F[4:0]					-
		34	-	-	-	CG48_F[4:0]					-
		35	-	-	-	CG80_F[4:0]					-
		36	-	-	-	CG96_F[4:0]					-
		37	-	-	-	CG112_F[4:0]					-
		38	-	-	-	CG144_F[4:0]					-
		39	-	-	-	CG160_F[4:0]					-
		40	-	-	-	CG176_F[4:0]					-
		41	-	-	-	CG208_F[4:0]					-

Himax Confidential Do Not Copy

		42	-	-	-	CG224_F[4:0]					-
		43	-	-	-	CG240_F[4:0]					-
E6	SETCHESEL		1	1	1	0	0	1	1	0	
		1	ENCH00_SEL[6:0]							ENCH00_SAMPLE_SEL	00
		2	ENCH01_SW[4:0]					ENCH01_SE_OFF	ENCH01_BE_OFF	ENCH01_CE_OFF	-
		3	ENCH02_DS[4:0]					ENCH02_SE_DS	ENCH02_BE_DS	ENCH02_CE_DS	-
		4	ENCH03_reg[7:0]								05
		5	ENCH04_reg[7:0]								05
		6	ENCH05_reg[7:0]								-
		7	ENCH06_reg[7:0]								-
		8	ENCH07_reg[7:0]								-
		9	ENCH08_reg[7:0]								-
		10	ENCH09_reg[7:0]								-
		11	ENCH10_reg[7:0]								-
		12	ENCH11_reg[7:0]								-
		13	ENCH12_reg[7:0]								-
		14	ENCH13_reg[7:0]								-
		15	ENCH14_reg[7:0]								-
		16	ENCH15_reg[7:0]								-
		17	ENCH16_reg[7:0]								-
E9	SET_SP_CMD	NO_OTP	1	1	1	0	1	0	0	1	
			FORCE_OPT	RAND_WR_SERIAL_EN	RAND_WR_CNT[5:0]						3F
		1	WR_CMD_CN[7:0]								-
FE	SET READ INDEX		1	1	1	1	1	1	1	0	
		1	CMD_ADD[7:0]								00

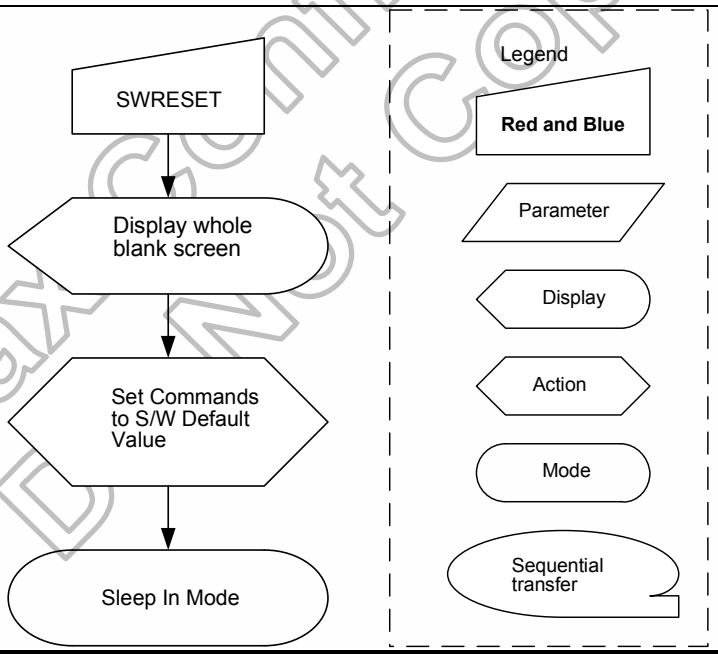
Himax Confidential
Do Not Copy

5.18.3 NOP (00h)

00H	NOP (No Operation)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	0	0	0	0	0	0	0	0	00
Parameter	NO PARAMETER									
Description	This command is an empty command; it does not have any effect on the display module.									
Restriction	-									
Register Availability	Status	Availability								
	Normal Mode On, Idle Mode Off, Sleep Out	Yes								
	Normal Mode On, Idle Mode On, Sleep Out	Yes								
	Sleep In or Booster Off	Yes								
Default	N/A									
Flow Chart	-									

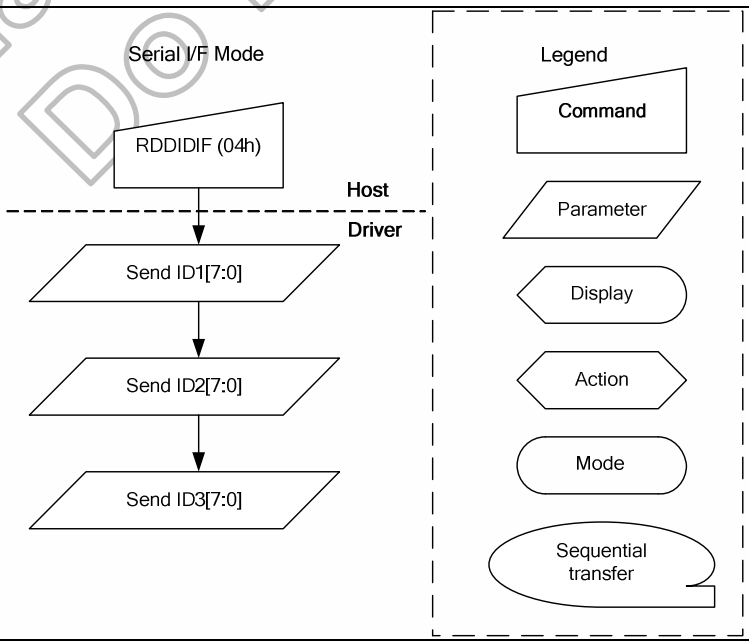
Himax Confidential
Do Not Copy

5.18.4 Software reset (01h)

01H	SWRESET (Software Reset)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	0	0	0	0	0	0	0	1	01
Parameter	NO PARAMETER									
Description	When the Software Reset command is written, it causes a software reset. It resets the commands and parameters to their S/W Reset default values. **(See default tables in each command description.)**									
Restriction	It will be necessary to wait 5msec before sending new command following software reset. The display module loads all display supplier's factory default values to the registers during this 5msec. If Software Reset is applied during Sleep Out mode, it will be necessary to wait 120msec before sending Sleep out command. Software Reset Command cannot be sent during Sleep Out sequence. The host processor needs continuing to send PCLK, HSYNC, VSYNC and DE signals to HX8394-F for two frames after this command is sent.									
Register Availability	Status: Normal Mode On, Idle Mode Off, Sleep Out — Availability: Yes Status: Normal Mode On, Idle Mode On, Sleep Out — Availability: Yes Status: Sleep In or Booster Off — Availability: Yes									
Default	N/A									
Flow Chart										

Himax Confidential Do Not Copy

5.18.5 Read Display Identification Information (04h)

<table>
<tr><th rowspan="2">04H</th><th colspan="10">RDDIDIF (Read Display Identification Information)</th></tr>
<tr><th>DCX</th><th>D7</th><th>D6</th><th>D5</th><th>D4</th><th>D3</th><th>D2</th><th>D1</th><th>D0</th><th>HEX</th></tr>
<tr><td>Command</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td><td>04</td></tr>
<tr><td>1st parameter</td><td>1</td><td colspan="8">ID1[7:0]</td><td>xx</td></tr>
<tr><td>2nd parameter</td><td>1</td><td colspan="8">ID2[7:0]</td><td>xx</td></tr>
<tr><td>3rd parameter</td><td>1</td><td colspan="8">ID3[7:0]</td><td>xx</td></tr>
<tr><td>Description</td><td colspan="11">This read byte returns 24-bit display identification information. The 1st Parameter identifies the LCD module's manufacturer. It is specified by display supplier and for xx is defined as xxHEX. The 2nd Parameter has 2 purposes. Bit7 (MSB) defines the type of panel. 0=Driver (STN B/W), 1=Module (Colour). Bits 6..0 are used to track the LCD module/driver version. It is defined by display supplier and it changes each time a revision is made to the display, material or construction specifications. See Table:
<table>
<tr><th>ID Byte Value V[7:0]</th><th>Version</th><th>Changes</th></tr>
<tr><td>80h</td><td>-</td><td>-</td></tr>
<tr><td>81h</td><td>-</td><td>-</td></tr>
<tr><td>82h</td><td>-</td><td>-</td></tr>
<tr><td>83h</td><td>-</td><td>-</td></tr>
<tr><td>84h</td><td>-</td><td>-</td></tr>
<tr><td>85h</td><td>-</td><td>-</td></tr>
</table>
The 3rd parameter identifies the LCD module/driver. It is specified by display supplier and for this LCD project module is defined as xxHEX.</td></tr>
<tr><td>Restriction</td><td colspan="11">-</td></tr>
<tr><td>Register Availability</td><td colspan="11">
<table>
<tr><th>Status</th><th>Availability</th></tr>
<tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr>
<tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr>
<tr><td>Sleep In or Booster Off</td><td>Yes</td></tr>
</table>
</td></tr>
<tr><td>Default</td><td colspan="11">
<table>
<tr><th>Status</th><th>Default Value</th></tr>
<tr><td>Power On Sequence</td><td>OTP Value</td></tr>
<tr><td>S/W Reset</td><td>OTP Value</td></tr>
<tr><td>H/W Reset</td><td>OTP Value</td></tr>
</table>
</td></tr>
<tr><td>Flow Chart</td><td colspan="11"> 
</td></tr>
</table>

Himax Confidential Do Not Copy

5.18.6 RDNUMPE: Read number of the parity errors (05h)

05H	RDNUMPE (Read Number of the Parity Errors)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	0	0	0	0	0	1	0	1	05
1st parameter	1	P7	P6	P5	P4	P3	P2	P1	P0	xx
Description	The first parameter is telling a number of the errors on DSI. The more detailed description of the bits is below. P[6..0] bits are telling a number of the errors. P[7] is set to '1' if there is overflow with P[6..0] bits. P[7..0] bits are set to '0's **(as well as RDDSM(0Eh)'s D0 is set '0' at the same time)** after there is sent the second parameter information. **(The read function is completed)**									
Restriction	SETEXTC turn on to enable this command									

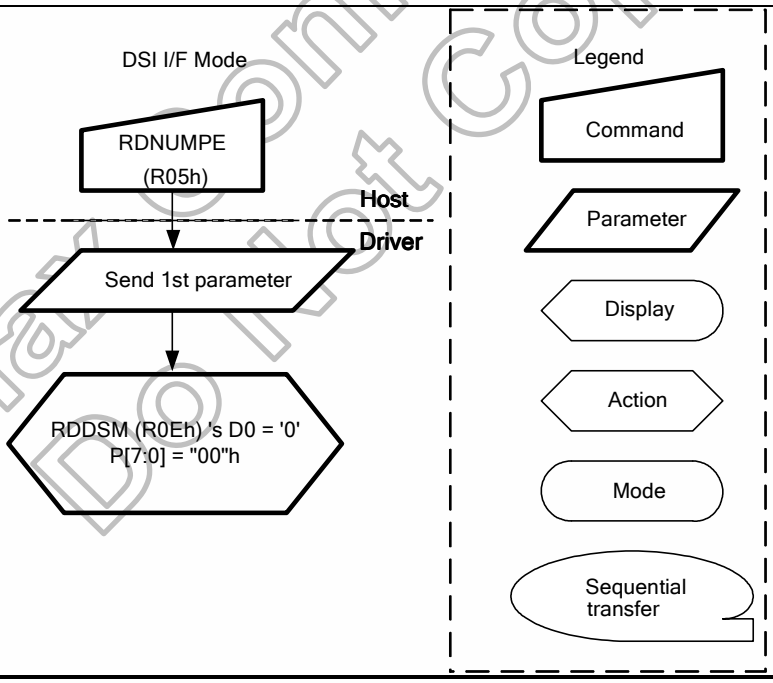
Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Sleep In or Booster Off	Yes

Default

Status	Default Value
Power On Sequence	00h
S/W Reset	00h
H/W Reset	00h

Flow Chart



Himax Confidential Do Not Copy

5.18.7 Get_red_channel (06h)

06H	RDRED (Read Red Colour)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	0	0	0	0	0	1	1	0	06
1st parameter	1	R7	R6	R5	R4	R3	R2	R1	R0	xx
Description	The first parameter is telling red colour value of the first pixel of the frame when there is used DPI I/F. 16 bit format: R5 is MSB and R1 is LSB. R7, R6 and R0 are set to '0'. 18 bit format: R5 is MSB and R0 is LSB. R7 and R6 are set to '0'. 24 bit format: R7 is MSB and R0 is LSB.									
Restriction	-									

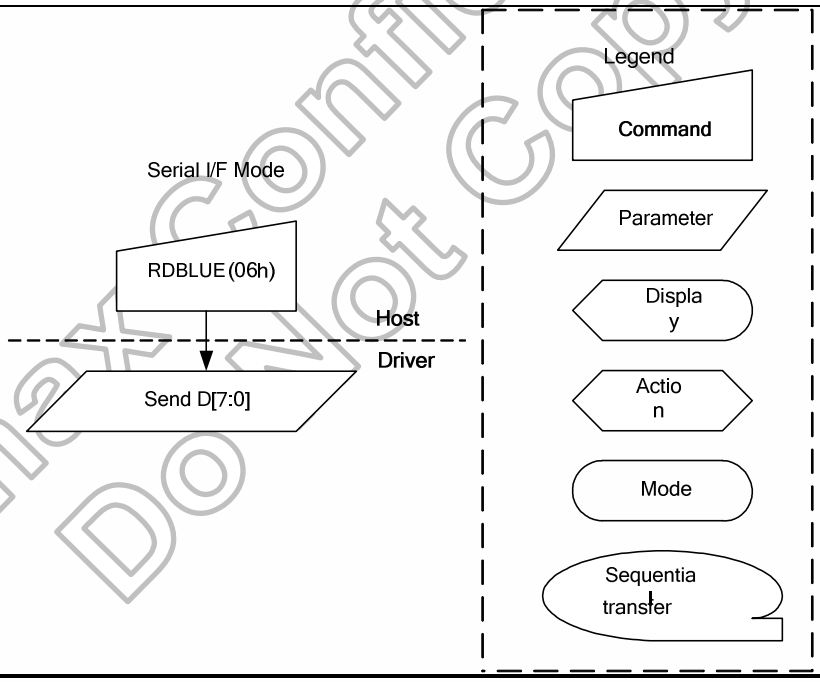
Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Sleep In or Booster Off	Yes

Default

Status	Default Value
Power On Sequence	00h
S/W Reset	00h
H/W Reset	00h

Flow Chart



Himax Confidential Do Not Copy

5.18.8 Get_green_channel (07h)

07H	RDGREEN (Read Green Colour)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	0	0	0	0	0	1	1	1	07
1st parameter	1	G7	G6	G5	G4	G3	G2	G1	G0	xx
Description	The first parameter is telling green colour value of the first pixel of the frame when there is used DPI I/F. 16 and 18 bit formats: G5 is MSB and G0 is LSB. G7 and G6 are set to '0'. 24 bit format: G7 is MSB and G0 is LSB.									
Restriction	-									

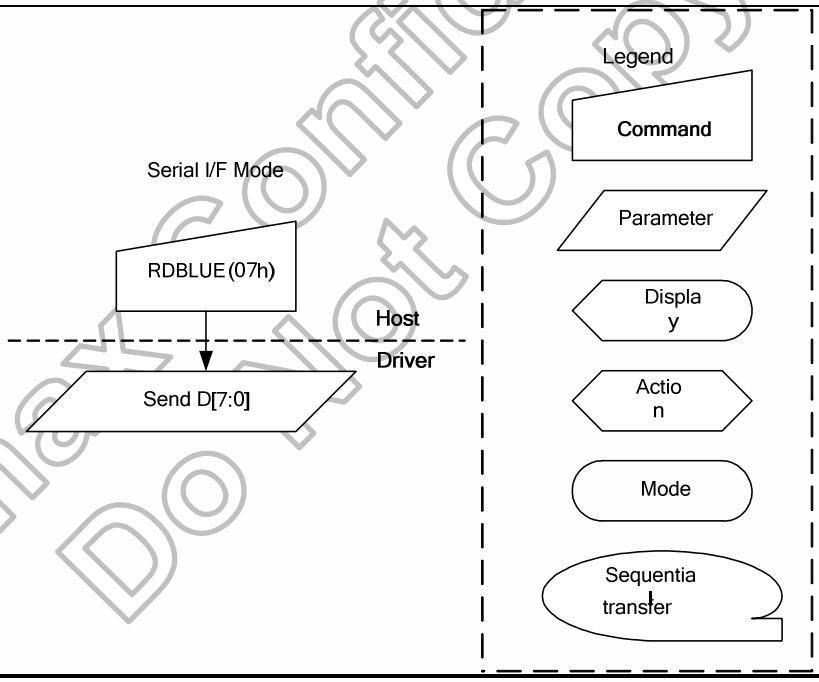
Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Sleep In or Booster Off	Yes

Default

Status	Default Value
Power On Sequence	00h
S/W Reset	00h
H/W Reset	00h

Flow Chart



Himax Confidential Do Not Copy

5.18.9 Get_blue_channel (08h)

08H	RDBLUE (Read Blue Colour)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	0	0	0	0	1	0	0	0	08
1st parameter	1	B7	B6	B5	B4	B3	B2	B1	B0	xx
Description	The first parameter is telling blue colour value of the first pixel of the frame when there is used DPI I/F. 16 bit format: B5 is MSB and B1 is LSB. B7, B6 and B0 are set to '0'. 18 bit format: B5 is MSB and B0 is LSB. B7 and B6 are set to '0'. 24 bit format: B7 is MSB and B0 is LSB.									
Restriction	-									

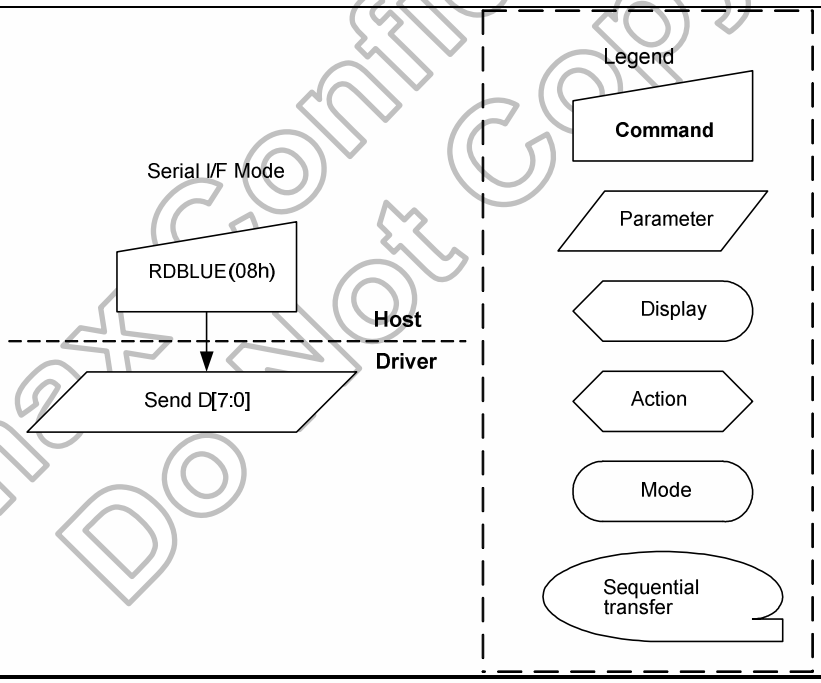
Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Sleep In or Booster Off	Yes

Default

Status	Default Value
Power On Sequence	00h
S/W Reset	00h
H/W Reset	00h

Flow Chart



Himax Confidential Do Not Copy

5.18.10 Read Display Status (09h)

09H	RDDST (Read Display Status)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	0	0	0	0	1	0	0	1	09
1st parameter	1	D[31:24]								xx
2nd parameter	1	D[23:16]								xx
3rd parameter	1	D[15:8]								xx
4th parameter	1	D[7:0]								xx

Description

This command indicates the current status of the display as described in the table below:

Bit	Description	Comment
D31	Booster Voltage Status	-
D30	Page Address Order	Set to '0'
D29	Column Address Order	Set to '0'
D28	Page/Column Order	Set to '0'
D27	Vertical Order	Set to '0'
D26	RGB/BGR Order	-
D25	Horizontal Order	Set to '0'
D24	Source scan sequence	-
D23	Gate scan sequence	-
D22	Interface Colour Pixel Format Definition	-
D21		
D20		
D19	Idle Mode On/Off	-
D18	Partial Mode On/Off	Set to '0'
D17	Sleep In/Out	-
D16	Display Normal Mode On/Off	-
D15	Vertical Scrolling Status	Set to '0'
D14	Horizontal Scrolling Status	Set to '0'
D13	Inversion Status	-
D12	All Pixels On	-
D11	All Pixels Off	-
D10	Display On/Off	-
D9	Tearing Effect Line On/Off	-
D8	Gamma Curve Selection	-
D7		
D6		
D5	Tearing Effect Output Line Mode	-
D4	Horizontal Sync. **(HSYNC, DPI I/F)**	-
D3	Vertical Sync. **(VSYNC, DPI I/F)**	-
D2	Pixel Clock **(DCK, DPI I/F)**	-
D1	Data Enable **(ENABLE, DPI I/F)**	-
D0	For Future Use	Set to '0'

Bit Values are explained overleaf.
Bit D31 – Booster Voltage Status
'0'=Booster Off.
'1'=Booster On.
Bit D30 – Page Address Order
'0'=Top to Bottom **(When MADCTL B7='0')**.
'1'=Bottom to Top **(When MADCTL B7='1')**.
This bit is not applicable for this project, so it is set to '0'
Bit D29 – Column Address Order
'0'=Left to Right **(When MADCTL B6='0')**.
'1'=Right to Left **(When MADCTL B6='1')**.
This bit is not applicable for this project, so it is set to '0'
Bit D28 – Page/Column Order
'0'=Normal **(When MADCTL B5='0').**

Himax Confidential Do not Copy

'1'=Rotation **(When MADCTL B5='1')**.
This bit is not applicable for this project, so it is set to '0'
Bit D27 – Line Address Order
'0'=LCD Refresh Top to Bottom **(When MADCTL B4='0').**
'1'=LCD Refresh Bottom to Top **(When MADCTL B4='1')**.
This bit is not applicable for this project, so it is set to '0'
Bit D26 – RGB/BGR Order
'0'=RGB **(When MADCTL B3='0')**.
'1'=BGR **(When MADCTL B3='1')**.
This bit is not applicable for this project, so it is set to '0'
Bit D24 – Source scan sequence
'0'=Normal **(When MADCTL B1='0')**.
'1'=Flipped **(When MADCTL B1='1')**.
Bit D23 – Gate scan sequence
'0'=Normal **(When MADCTL B0='0')**.
'1'=Flipped **(When MADCTL B0='1')**.
Bits D22, D21, D20 – Interface Colour Pixel Format Definition

Interface Format	D22	D21	D20
Not Defined	0	0	0
Not Defined	0	0	1
Not Defined	0	1	0
Not Defined	0	1	1
Not Defined	1	0	0
16 Bit/Pixel	1	0	1
18 Bit/Pixel	1	1	0
24 Bit/Pixel	1	1	1

Bit D19 – Idle Mode On/Off
'0'=Idle Mode Off.
'1'=Idle Mode On.
Bit D18 – Partial Mode On/Off
'0'=Partial Mode Off.
'1'=Partial Mode On.
This bit is not applicable for this project, so it is set to '0'
Bit D17 – Sleep In/Out
'0'=Sleep In Mode.
'1'=Sleep Out Mode.
Bit D16 – Display Normal Mode On/Off
'0'=Partial or Scrolling Mode.
'1'=Normal Mode.
Bit D15 – Vertical Scrolling On/Off
'0'=Vertical Scrolling is Off.
'1'=Vertical Scrolling is On.
This bit is not applicable for this project, so it is set to '0'
Bit D14 – Horizontal Scrolling Status
This bit is not applicable for this project, so it is set to '0'
Bit D13 – Inversion On/Off
'0'=Inversion is Off.
'1'=Inversion is On.
Bit D12 – All Pixels On.
'0'=Nornal mode.
'1'=All Pixels On.
Bit D11 – All Pixels Off.
0'=Nornal mode.
'1'=All Pixels Off.
Bit D10 – Display On/Off
'0'=Display is Off.
'1'=Display is On.
Bit D9 – Tearing Effect Line On/Off
'0'=Tearing Effect Line Off.
'1'=Tearing Effect On.
Bits D8, D7, D6 – Gamma Curve Selection

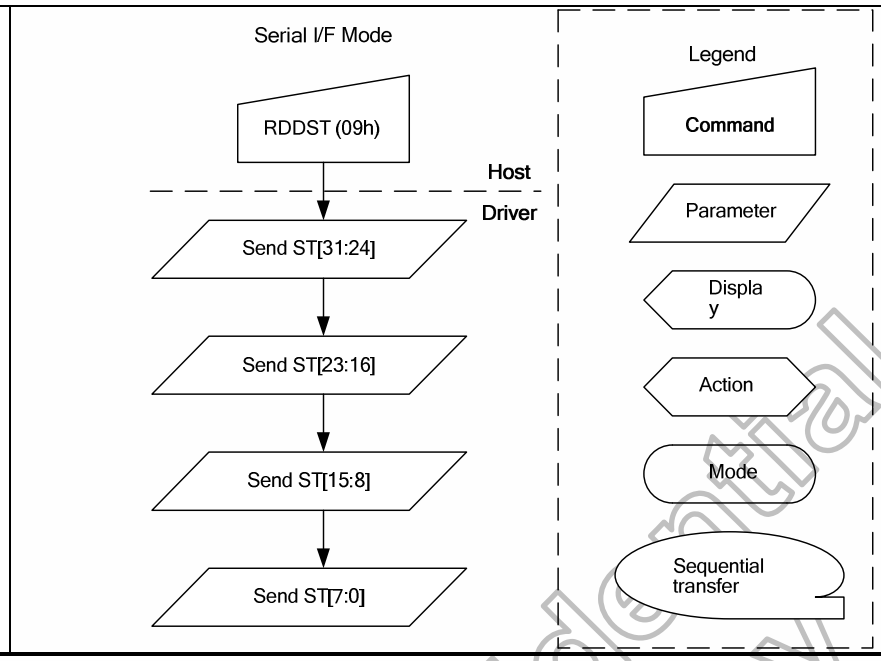
Himax Confidential Do Not Copy

This information contained herein is the exclusive property of Himax and shall not be distributed, reproduced, or disclosed in whole or in part without prior written permission of Himax.

	Gamma Curve Selected / B8 / B7 / B6 / Gamma Set **(26h)** Parameter Gamma Curve 1 / 0 / 0 / 0 / GC0 Gamma Curve 2 / 0 / 0 / 1 / GC1 Gamma Curve 3 / 0 / 1 / 0 / Reload disable Gamma Curve 4 / 0 / 1 / 1 / Setting Inhibit Not Defined / 1 / 0 / 0 / Not Defined Not Defined / 1 / 0 / 1 / Not Defined Not Defined / 1 / 1 / 0 / Not Defined Not Defined / 1 / 1 / 1 / Not Defined Bit D5 – Tearing Effect Line Output Mode. '0'=Mode 1, V-Blanking only. '1'=Mode 2, both H-Blanking and V-Blanking. Bit D4 – Horizontal Sync. **(HSYNC)** On/Off, Note '0'=Horizontal Sync. Line is Off **("Low")**. '1'=Horizontal Sync. Line is On **("High")**. Bit D3 – Vertical Sync. **(VSYNC)** On/Off, Note '0'=Vertical Sync. Line is Off **("Low")**. '1'=Vertical Sync. Line is On **("High")**. Bit D2 – Pixel Clock **(PCLK)** On/Off, Note '0'=Pixel Clock line is Off **("Low")**. '1'=Pixel Clock line is On **("High")**. Bit D1 – Data Enable **(DE)** On/Off, Note '0'=Data Enable line is Off **("Low")**. '1'=Data Enable line is On **("High")**. Bit D0 – Parity Error check This bit is not applicable for this project, so it is set to '0' Note: This bit indicates current status of the line when this command has been sent.
Restriction	
Register Availability	Status / Availability Normal Mode On, Idle Mode Off, Sleep Out / Yes Normal Mode On, Idle Mode On, Sleep Out / Yes Sleep In or Booster Off / Yes
Default	Status / Default Value Power On Sequence / Refer to Description S/W Reset / Refer to Description H/W Reset / Refer to Description

Himax Confidential
Do Not Copy

Flow Chart



Himax Confidential
Do Not Copy

5.18.11 Get_power_mode (0Ah)

0AH	RDDPM (Read Display Power Mode)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	0	0	0	0	1	0	1	0	0A
1st parameter	1	0	D6	0	D4	D3	D2	0	0	xx

Description

This command indicates the current status of the display as described in the table below:

Bit	Description	Comment
D7	Not Defined	Set to '0'
D6	Idle Mode On/Off	-
D5	Partial Mode On/Off	Set to '0'
D4	Sleep In/Out	-
D3	Display Normal Mode On/Off	-
D2	Display On/Off	-
D1	Not Defined	Set to '0'
D0	Not Defined	Set to '0'

Bit D7 is for future use and set to '0'.
Bit D6 – Idle Mode On/Off
'0'=Idle Mode Off.
'1'=Idle Mode On.
Bit D5 – Partial Mode On/Off
'0'=Partial Mode Off.
'1'=Partial Mode On.
This bit is not applicable for this project, so it is set to '0'
Bit D4 – Sleep In/Out
'0'=Sleep In Mode.
'1'=Sleep Out Mode.
Bit D3 – Display Normal Mode On/Off
'0'=Display Normal Mode Off.
'1'=Display Normal Mode On.
Bit D2 – Display On/Off
'0'=Display is Off.
'1'=Display is On.

Restrictions

-

Register Availability

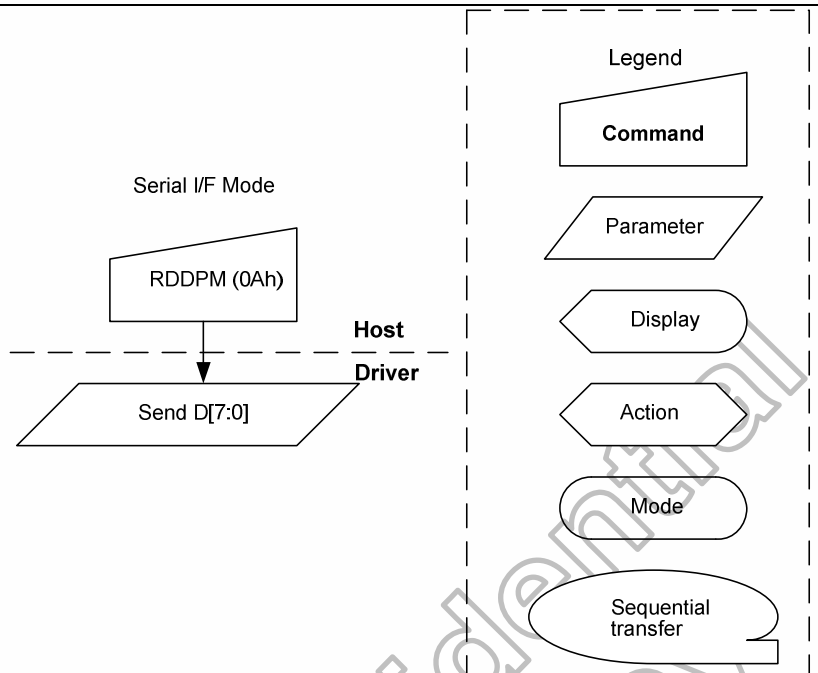
Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Sleep In or Booster Off	Yes

Default

Status	Default Value
Power On Sequence	08h
S/W Reset	08h
H/W Reset	08h

Himax Confidential
Do Not Copy

Flow Chart



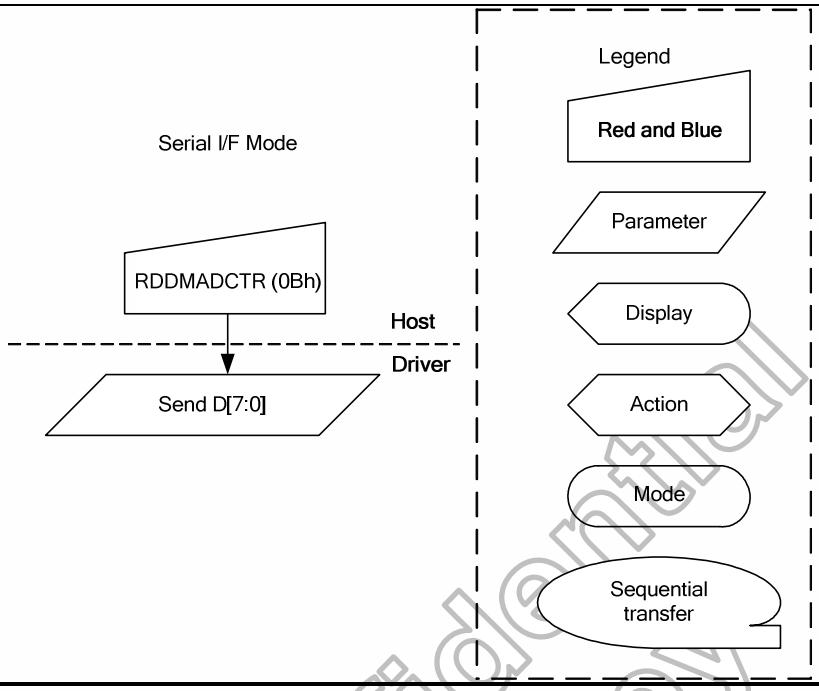
Himax Confidential
Do Not Copy

5.18.12 Read display MADCTL (0Bh)

<table>
<tr><th rowspan="2">0BH</th><th colspan="10">RDDMADCTL (Read Display MADCTL)</th></tr>
<tr><th>DCX</th><th>D7</th><th>D6</th><th>D5</th><th>D4</th><th>D3</th><th>D2</th><th>D1</th><th>D0</th><th>HEX</th></tr>
<tr><td>Command</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td><td>1</td><td>1</td><td>0B</td></tr>
<tr><td>1st parameter</td><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td><td>D3</td><td>0</td><td>D1</td><td>D0</td><td>xx</td></tr>
<tr><td>Description</td><td colspan="10">This command indicates the current status of the display as described in the table below:
<table>
<tr><th>Bit</th><th>Description</th><th>Comment</th></tr>
<tr><td>D7</td><td>Page Address Order</td><td>Set to ‘0’</td></tr>
<tr><td>D6</td><td>Column Address Order</td><td>Set to ‘0’</td></tr>
<tr><td>D5</td><td>Page/Column Order</td><td>Set to ‘0’</td></tr>
<tr><td>D4</td><td>Line Address Order</td><td>Set to ‘0’</td></tr>
<tr><td>D3</td><td>RGB/BGR Order</td><td>-</td></tr>
<tr><td>D2</td><td>Display Data Latch Order</td><td>Set to ‘0’</td></tr>
<tr><td>D1</td><td>Source scan sequence</td><td>-</td></tr>
<tr><td>D0</td><td>Gate scan sequence</td><td>-</td></tr>
</table>
Bit D7 – Page Address Order
This bit is not applicable for this project, so it is set to “0”.
Bit D6 – Column Address Order
This bit is not applicable for this project, so it is set to “0”.
Bit D5 – Page/Column Order
This bit is not applicable for this project, so it is set to “0”.
Bit D4 – Line Address Order
This bit is not applicable for this project, so it is set to “0”.
Bit D3 – RGB/BGR Order
‘0’=RGB (When MADCTL B3=’0’).
‘1’=BGR (When MADCTL B3=’1’).
Bit D2 – Display Data Latch Data Order
This bit is not applicable for this project, so it is set to “0”.
Bit D1 – Source scan sequence
‘0’=Normal (When MADCTL B1=’0’).
‘1’=Flipped (When MADCTL B1=’1’).
Bit D0 – Gate scan sequence
‘0’=Normal (When MADCTL B0=’0’).</td></tr>
<tr><td>Restrictions</td><td colspan="10">-</td></tr>
<tr><td>Register Availability</td><td colspan="10">
<table>
<tr><th>Status</th><th>Availability</th></tr>
<tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr>
<tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr>
<tr><td>Sleep In or Booster Off</td><td>Yes</td></tr>
</table></td></tr>
<tr><td>Default</td><td colspan="10">
<table>
<tr><th>Status</th><th>Default Value</th></tr>
<tr><td>Power On Sequence</td><td>00h</td></tr>
<tr><td>S/W Reset</td><td>Keep</td></tr>
<tr><td>H/W Reset</td><td>00h</td></tr>
</table></td></tr>
</table>

Himax Confidential Do Not Copy

Flow Chart



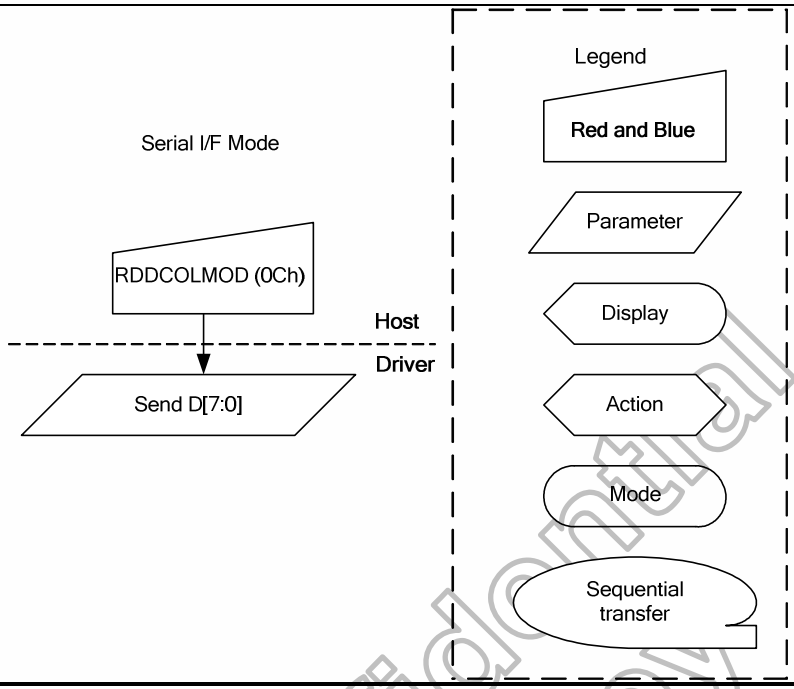
Himax Confidential
Do Not Copy

5.18.13 Get_pixel_format (0Ch)

<table>
<tr><th rowspan="2">0CH</th><th colspan="10">RDDCOLMOD (Read Display COLMOD)</th></tr>
<tr><th>DCX</th><th>D7</th><th>D6</th><th>D5</th><th>D4</th><th>D3</th><th>D2</th><th>D1</th><th>D0</th><th>HEX</th></tr>
<tr><td>Command</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>1</td><td>0</td><td>0</td><td>0C</td></tr>
<tr><td>1st parameter</td><td>1</td><td>0</td><td>D6</td><td>D5</td><td>D4</td><td>0</td><td>0</td><td>0</td><td>0</td><td>xx</td></tr>
<tr><td>Description</td><td colspan="10">This command indicates the current status of the display as described in the table below:
<table>
<tr><th>Bit</th><th>Description</th><th>Comment</th></tr>
<tr><td>D7</td><td>Reserved</td><td>Set to ‘0’</td></tr>
<tr><td>D6</td><td rowspan="3">DPI Interface Pixel format</td><td>-</td></tr>
<tr><td>D5</td><td>-</td></tr>
<tr><td>D4</td><td>-</td></tr>
<tr><td>D3</td><td>Reserved</td><td>Set to ‘0’</td></tr>
<tr><td>D2</td><td>Reserved</td><td>Set to ‘1’</td></tr>
<tr><td>D1</td><td>Reserved</td><td>Set to ‘1’</td></tr>
<tr><td>D0</td><td>Reserved</td><td>Set to ‘1’</td></tr>
</table>
Bits D6, D5, D4 – DPI Interface Colour Pixel Format Definition
Bits D2, D1, D0 – DBI Interface Colour Pixel Format Definition.
This bit is not applicable for this project, and it is set to “1”.
<table>
<tr><th>Interface Colour Format</th><th>D6</th><th>D5</th><th>D4</th></tr>
<tr><td>Not Defined</td><td>0</td><td>0</td><td>0</td></tr>
<tr><td>Not Defined</td><td>0</td><td>0</td><td>1</td></tr>
<tr><td>Not Defined</td><td>0</td><td>1</td><td>0</td></tr>
<tr><td>Not Defined</td><td>0</td><td>1</td><td>1</td></tr>
<tr><td>Not Defined</td><td>1</td><td>0</td><td>0</td></tr>
<tr><td>16-bit/pixel</td><td>1</td><td>0</td><td>1</td></tr>
<tr><td>18-bit/pixel</td><td>1</td><td>1</td><td>0</td></tr>
<tr><td>24-bit/pixel</td><td>1</td><td>1</td><td>1</td></tr>
</table>
If the setting is not used then the corresponding bits in the parameter returned from</td></tr>
<tr><td>Restrictions</td><td colspan="10">-</td></tr>
<tr><td>Register Availability</td><td colspan="10">
<table>
<tr><th>Status</th><th>Availability</th></tr>
<tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr>
<tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr>
<tr><td>Sleep In or Booster Off</td><td>Yes</td></tr>
</table></td></tr>
<tr><td>Default</td><td colspan="10">
<table>
<tr><th>Status</th><th>Default Value</th></tr>
<tr><td>Power On Sequence</td><td>77h</td></tr>
<tr><td>S/W Reset</td><td>77h</td></tr>
<tr><td>H/W Reset</td><td>77h</td></tr>
</table></td></tr>
</table>

Himax Confidential Do Not Copy

Flow Chart



Himax Confidential
Do Not Copy

5.18.14 Get_display_mode (0Dh)

0DH	RDDIM (Read Display Image Mode)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	0	0	0	0	1	1	0	1	0D
1st parameter	1	0	0	D5	D4	D3	D2	D1	D0	xx

Description

This command indicates the current status of the display as described in the table below:
Bit D7 – Vertical Scrolling On/Off
This bit is not applicable for this project, so it is set to '0'
Bit D6 – Horizontal Scrolling Status
This bit is not applicable for this project, so it is set to '0'
Bit D5 – Inversion On/Off
'0'=Inversion is Off.
'1'=Inversion is On.
Bit D4 – All Pixels On
'0'=Normal Display
'1'=White Display
Bit D3 – All Pixels Of
'0'=Normal Display
'1'=Black Display
Bits D2, D1, D0 – Gamma Curve Selection

Gamma Curve Selected	D2	D1	D0	Gamma Set **(26h)** Parameter
Gamma Curve1	0	0	0	GC0
Gamma Curve2	0	0	1	GC1
Gamma Curve3	0	1	0	Reload disable
Gamma Curve4	0	1	1	Setting Inhibit
Not Defined	1	0	0	Not Defined
Not Defined	1	0	1	Not Defined
Not Defined	1	1	0	Not Defined
Not Defined	1	1	1	Not Defined

Restrictions

-

Register Availability

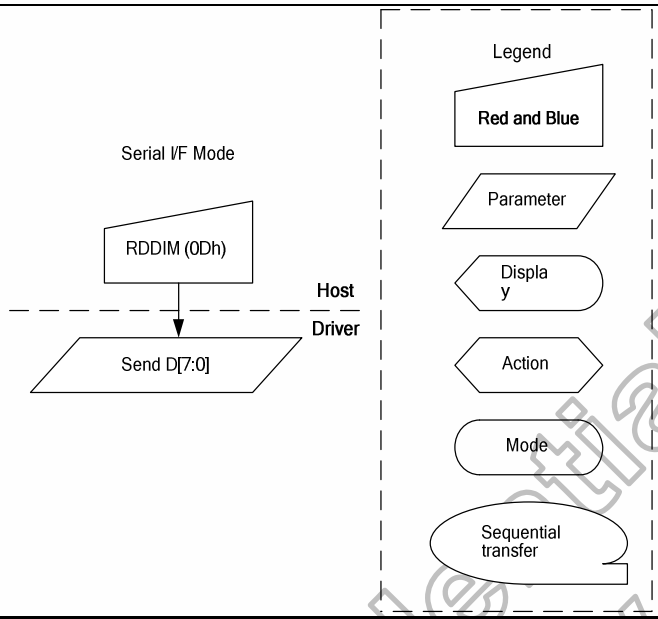
Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Sleep In or Booster Off	Yes

Default

Status	Default Value
Power On Sequence	00h
S/W Reset	00h
H/W Reset	00h

Himax Confidential Do Not Copy

Flow Chart



Himax Confidential
Do Not Copy

5.18.15 Get_signal_mode (0Eh)

0EH	RDDSM (Read Display Signal Mode)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	0	0	0	0	1	1	1	0	0E
1st parameter	1	D7	D6	D5	D4	D3	D2	0	0	xx

Description

This command indicates the current status of the display described in the table as below:
Bit D7 – Tearing Effect Line On/Off
'0'=Tearing Effect Line Off.
'1'=Tearing Effect On.
Bit D6 – Tearing Effect Line Output Mode, see section 5.1 for mode definitions.
'0'=Mode 1.
'1'=Mode 2.
Bit D5 – Horizontal Sync. **(DPI I/F)** On/Off.
'0'=Horizontal Sync. Line is Off **("Low")**.
'1'=Horizontal Sync. Line is On **("High")**.
Bit D4 – Vertical Sync. **(DPI I/F)** On/Off.
'0'=Vertical Sync. Line is Off **("Low")**.
'1'=Vertical Sync. Line is On **("High")**.
Bit D3 – Pixel Clock **(PCLK, DPI I/F)** On/Off.
'0'=PCLK line is Off **("Low")**.
'1'=PCLK line is On **("High")**.
Bit D2 – Data Enable **(DE, DPI I/F)** On/Off.
'0'=DE line is Off **("Low").**
'1'=DE line is On **("High")**.
Bits D[1:0] – Reserved.

Restrictions

-

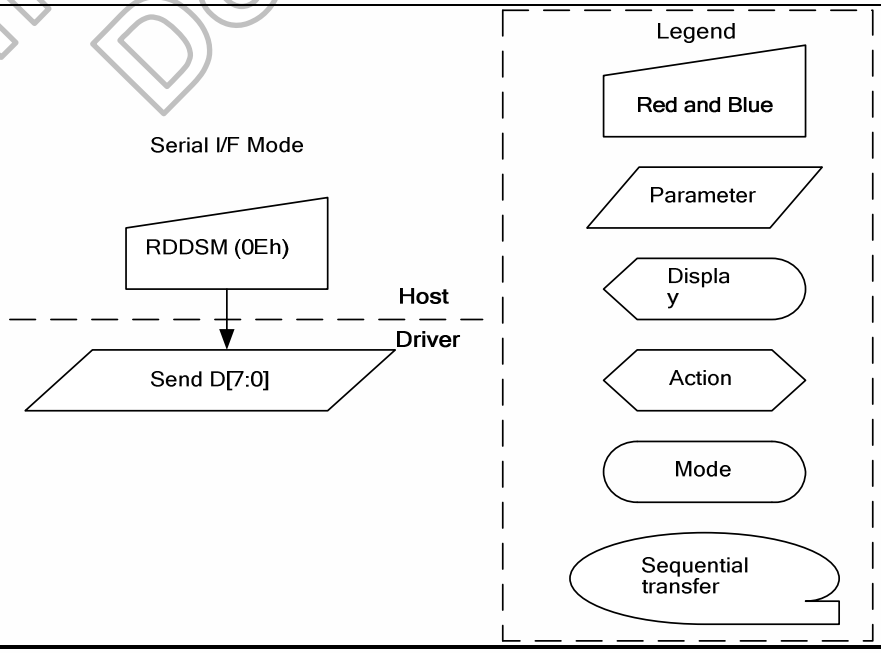
Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Sleep In or Booster Off	Yes

Default

Status	Default Value
Power On Sequence	00h
S/W Reset	00h
H/W Reset	00h

Flow Chart



Himax Confidential
Do Not Copy

5.18.16 Get_diagnostic_result (0Fh)

0FH	RDDSDR (Read Display Self-Diagnostic Result)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	0	0	0	0	1	1	1	1	0F
1st parameter	1	D7	D6	D5	D4	0	0	0	0	xx
Description	The display module returns the self-diagnostic results following a Sleep Out command. See section 5.10 for a description of the status results. Bit D7 – Register Loading Detection Bit D6 – Functionality Detection Bit D5 – Chip Attachment Detection Set to '0' if feature unimplemented. Bit D4 – Display Glass Break Detection Set to '0' if feature unimplemented. Bits D[3:0] – Reserved.									
Restrictions	-									

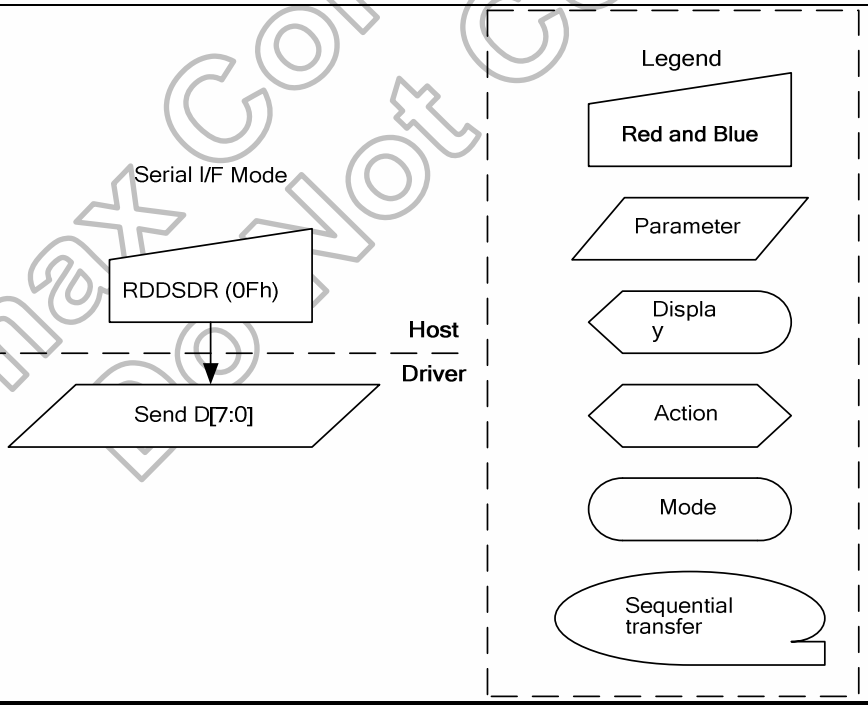
Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Sleep In or Booster Off	Yes

Default

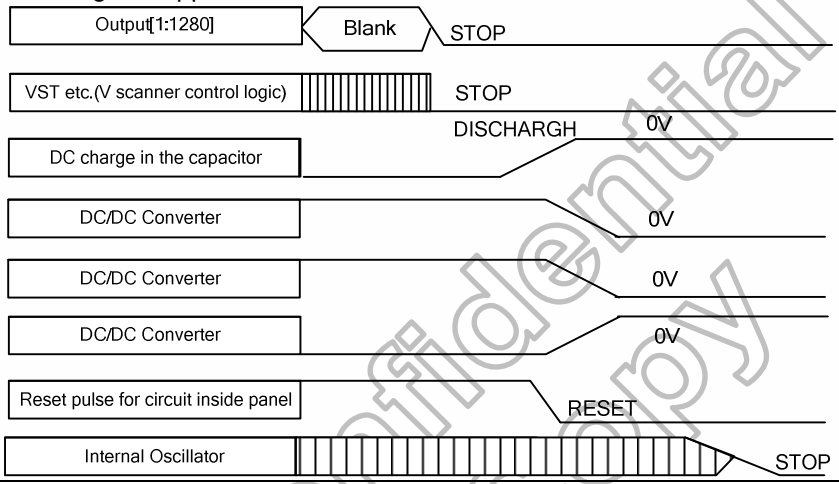
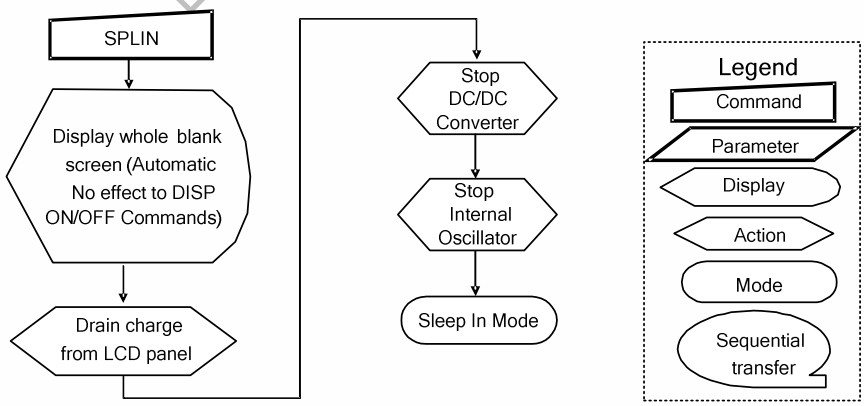
Status	Default Value
Power On Sequence	00h
S/W Reset	00h
H/W Reset	00h

Flow Chart



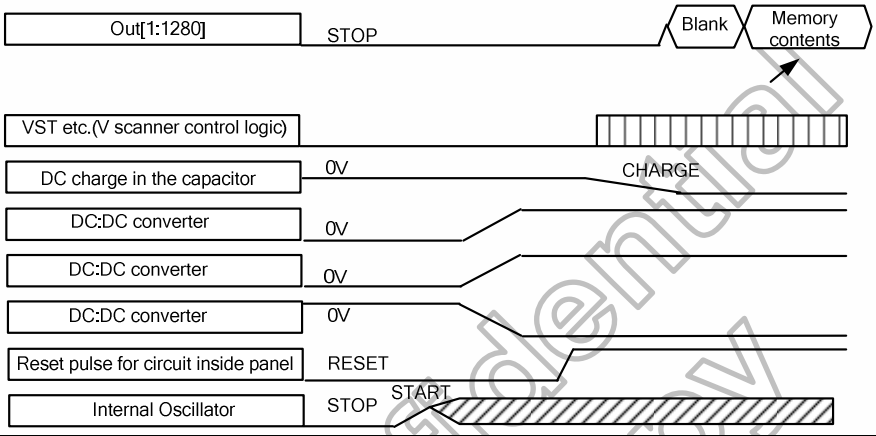
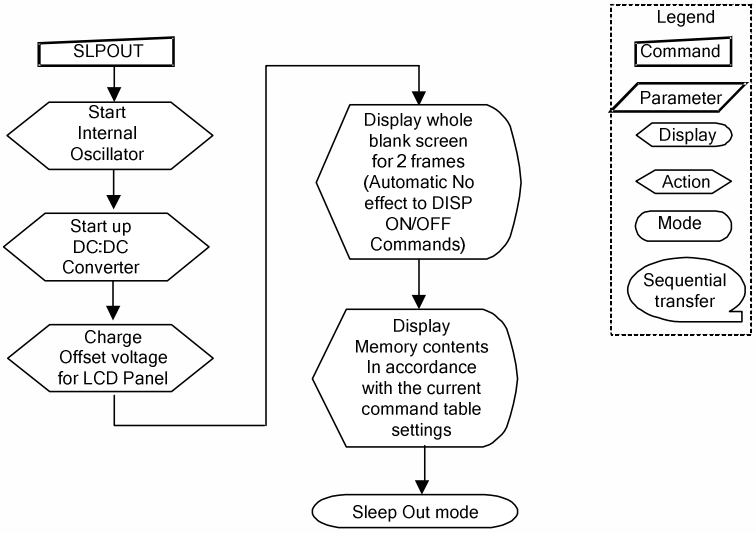
Himax Confidential Do Not Copy

5.18.17 Enter_sleep_mode (10h)

10H	SLPIN (Sleep In)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	0	0	0	1	0	0	0	0	10
Parameter	NO PARAMETER									
Description	This command causes the LCD module to enter the minimum power consumption mode. In this mode the DC/DC converter is stopped, Internal oscillator is stopped, and panel scanning is stopped.  									
Restriction	This command has no effect when module is already in sleep in mode.Sleep In Mode can only be left by the Sleep Out Command **(11h)**. It will be necessary to wait 5msec before sending next command, this is to allow time for the supply voltages and clock circuits to stabilize. It will be necessary to wait 120msec after sending Sleep Out command **(when in Sleep In Mode)** before Sleep In command can be sent. The host processor continues to send PCLK, HSYNC, and VSYNC and DE signals to HX8394-F for two frames after this command is sent.									
Register Availability	Status / Availability Normal Mode On, Idle Mode Off, Sleep Out: Yes Normal Mode On, Idle Mode On, Sleep Out: Yes Sleep In or Booster Off: Yes									
Default	N/A									
Flow Chart	It takes 120msec to get into Sleep In mode after SLPIN command issued.  									

Himax Confidential
DO Not Copy

5.18.18 Exit_sleep_omde (11h)

11H	SLPOUT (Sleep Out)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	0	0	0	1	0	0	0	1	11
Parameter	NO PARAMETER									
Description	This command turns off sleep mode. In this mode the DC/DC converter is enabled, Internal oscillator is started, and panel scanning is started. 									
Restriction	This command has no effect when module is already in sleep out mode. Sleep Out Mode can only be left by the Sleep In Command **(10h)**. It will be necessary to wait 5msec before sending next command, this is to allow time for the supply voltages and clock circuits to stabilize. The display module loads all display supplier's factory default values to the registers during this 5msec and there cannot be any abnormal visual effect on the display image if factory default and register values are same when this load is done and when the display module is already Sleep Out –mode. The display module is doing self-diagnostic functions during this 5msec. It will be necessary to alit 120msec after sending Sleep In command **(when in Sleep Out mode)** before Sleep Out command can be sent. The host processor continues to send PCLK, HSYNC, and VSYNC and DE signals to HX8394-F for two frames after this command is sent.									
Register Availability	Status | Availability Normal Mode On, Idle Mode Off, Sleep Out | Yes Normal Mode On, Idle Mode On, Sleep Out | Yes Sleep In or Booster Off | Yes									
Default	N/A									
Flow Chart	It takes 120msec to become Sleep Out mode after SLPOUT command issued. 									

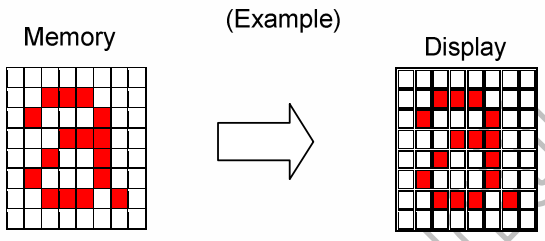
Himax Confidential
Do Not Copy

5.18.19 Enter_normal_mode (13h)

13H	NORON (Normal Display Mode On)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	0	0	0	1	0	0	1	1	13
Parameter	NO PARAMETER									
Description	This command returns the display to normal mode.									
Restriction	This command has no effect when Normal Display mode is active.									
Register Availability	Status: Normal Mode On, Idle Mode Off, Sleep Out — Availability: Yes Status: Normal Mode On, Idle Mode On, Sleep Out — Availability: Yes Status: Sleep In or Booster Off — Availability: Yes									
Default	Status: Power On Sequence — Default Value: Normal display mode Status: S/W Reset — Default Value: Normal display mode Status: H/W Reset — Default Value: Normal display mode									
Flow Chart	-									

Himax Confidential
Do Not Copy

5.18.20 Exit_inversion_mode (20h)

20H	INVOFF (Display Inversion Off)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	0	0	1	0	0	0	0	0	20
Parameter	NO PARAMETER									
Description	This command is used to recover from display inversion mode. This command does not change any other status. 									
Restriction	This command has no effect when module is already in inversion off mode.									

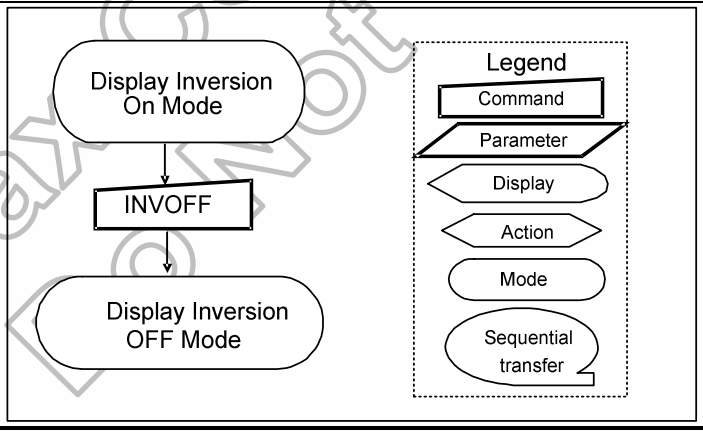
Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Sleep In or Booster Off	Yes

Default

Status	Default Value
Power On Sequence	Normal display mode
S/W Reset	Normal display mode
H/W Reset	Normal display mode

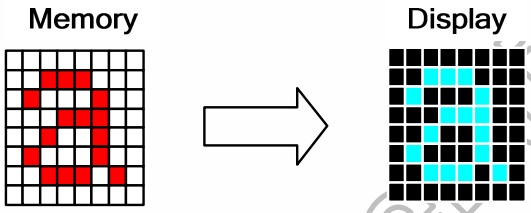
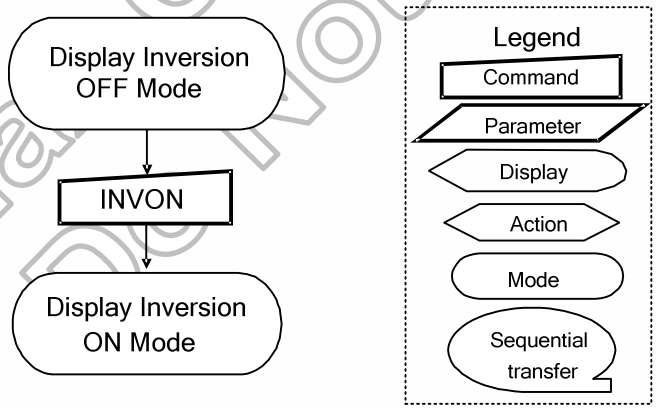
Flow Chart



Himax Confidential Do Not Copy

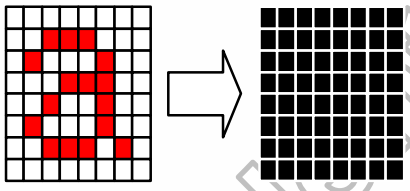
5.18.21 Enter_inversion_mode (21h)

<table>
<tr><th rowspan="2">21H</th><th colspan="10">INVON (Display Inversion On)</th></tr>
<tr><th>DCX</th><th>D7</th><th>D6</th><th>D5</th><th>D4</th><th>D3</th><th>D2</th><th>D1</th><th>D0</th><th>HEX</th></tr>
<tr><td>Command</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>21</td></tr>
<tr><td>Parameter</td><td colspan="10">NO PARAMETER</td></tr>
<tr><td>Description</td><td colspan="10">This command is used to enter into display inversion mode.
This command does not change any other status.

(Example)
 
</td></tr>
<tr><td>Restriction</td><td colspan="10">This command has no effect when module is already in inversion on mode.</td></tr>
<tr><td>Register Availability</td><td colspan="10"><table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In or Booster Off</td><td>Yes</td></tr></table></td></tr>
<tr><td>Default</td><td colspan="10"><table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power On Sequence</td><td>Normal display mode</td></tr><tr><td>S/W Reset</td><td>Normal display mode</td></tr><tr><td>H/W Reset</td><td>Normal display mode</td></tr></table></td></tr>
<tr><td>Flow Chart</td><td colspan="10"> 
</td></tr>
</table>

Himax Confidential Do Not Copy

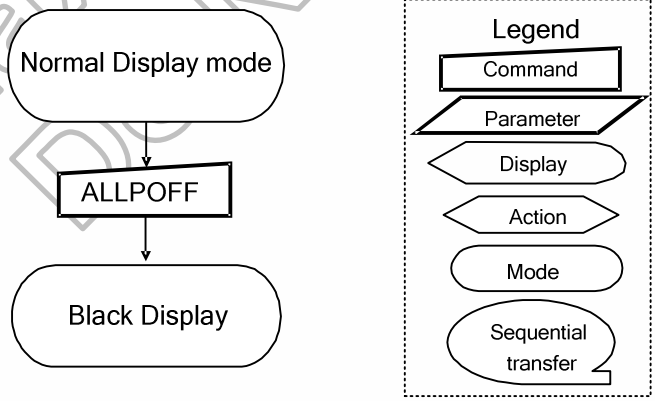
5.18.22 All_Pixel_Off (22h)

22H	ALLPOFF (All Pixel Off)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	0	0	1	0	0	0	1	0	22
Parameter	NO PARAMETER									
Description	This command turns the display panel black in 'Sleep Out' –mode and a status of the 'Display On/Off' –register can be 'on' or 'off'. This command does not change any other status. (Example) Memory Display  'All Pixels On' or 'Normal Display Mode On' – commands are used to leave this mode. The display is showing the content of the frame memory after 'Normal Display Mode On' command.									
Restriction	This command has no effect when module is already in inversion on mode.									

Register Availability	Status	Availability
	Normal Mode On, Idle Mode Off, Sleep Out	Yes
	Normal Mode On, Idle Mode On, Sleep Out	Yes
	Sleep In or Booster Off	Yes

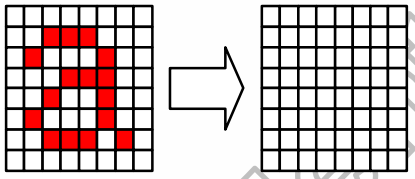
Default	Status	Default Value
	Power On Sequence	Normal display mode
	S/W Reset	Normal display mode
	H/W Reset	Normal display mode

Flow Chart



Himax Confidential Do Not Copy

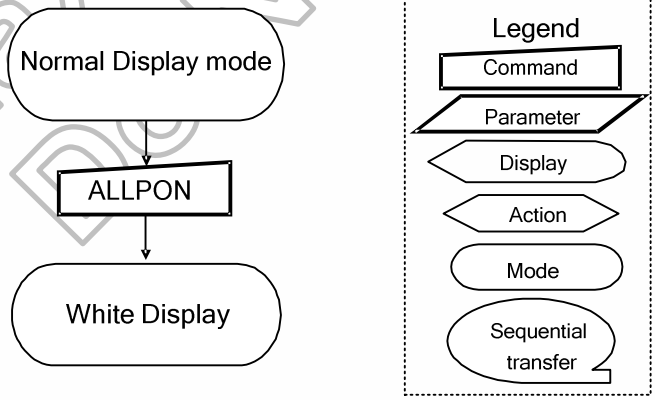
5.18.23 All_Pixel_On (23h)

23H	ALLPON(All Pixel On)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	0	0	1	0	0	0	1	1	23
Parameter	NO PARAMETER									
Description	This command turns the display panel white in 'Sleep out ' –mode and a status of the 'Display On/Off' –register can be 'on' or 'off'. This command does not change any other status. (Example) Memory Display  'All Pixels Off' or 'Normal Display Mode On' – commands are used to leave this mode. The display is showing the content of the frame memory after 'Normal Display Mode On' command.									
Restriction	This command has no effect when module is already in inversion on mode.									

Register Availability	Status	Availability
	Normal Mode On, Idle Mode Off, Sleep Out	Yes
	Normal Mode On, Idle Mode On, Sleep Out	Yes
	Sleep In or Booster Off	Yes

Default	Status	Default Value
	Power On Sequence	Normal display mode
	S/W Reset	Normal display mode
	H/W Reset	Normal display mode

Flow Chart



Himax Confidential Do Not Copy

5.18.24 Set_gamma_curve (26h)

26H	GAMSET (Gamma Set)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	0	0	1	0	0	1	1	0	26
Parameter	1	GC7	GC6	GC5	GC4	GC3	GC2	GC1	GC0	1..08

Description

This command is used to select the desired Gamma curve for the current display. A maximum of 4 fixed gamma curves can be selected. The curves are defined in Curve Correction Power Supply Circuit. The curve is selected by setting the appropriate bit in the parameter as described in the Table:

GC[7:0]	Parameter	Curve selected	OTP index
01h	GC0	Gamma Curve 1 (reload from OTP gamma set1)	45h~FDh
02h	GC1	Gamma Curve 2 (reload from OTP gamma set2)	100h~1B8h
04h	GC2	Reload disable	-
08h	GC3	Setting Inhibit	-

Note: All other values are undefined.

Restriction

Values of GC[7:0] not shown in table above are invalid and will not change the current selected Gamma curve until valid value is received.

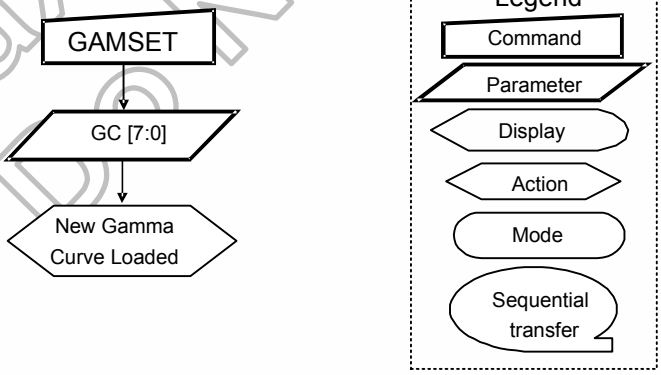
Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Sleep In or Booster Off	Yes

Default

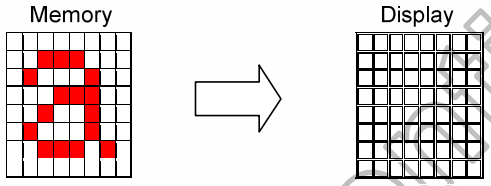
Status	Default Value
Power On Sequence	01h
S/W Reset	01h
H/W Reset	01h

Flow Chart



Himax Confidential Do Not Copy

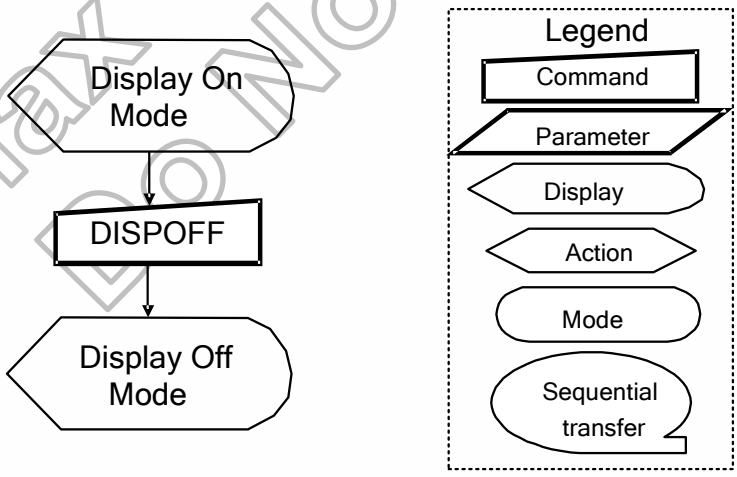
5.18.25 Set_display_off (28h)

28H	DISPOFF (Display Off)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	0	0	1	0	1	0	0	0	28
Parameter	NO PARAMETER									
Description	This command is used to enter into DISPLAY OFF mode. In this mode, the output from DPI I/F is disabled and blank page inserted. This command does not change any other status. There will be no abnormal visible effect on the display. Example Memory → Display 									
Restriction	This command has no effect when module is already in display off mode.									

Register Availability	Status	Availability
	Normal Mode On, Idle Mode Off, Sleep Out	Yes
	Normal Mode On, Idle Mode On, Sleep Out	Yes
	Sleep In or Booster Off	Yes

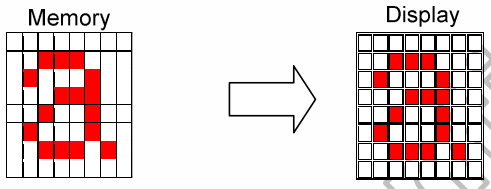
Default	Status	Default Value
	Power On Sequence	Display off
	S/W Reset	Display off
	H/W Reset	Display off

Flow Chart



Himax Confidential Do Not Copy

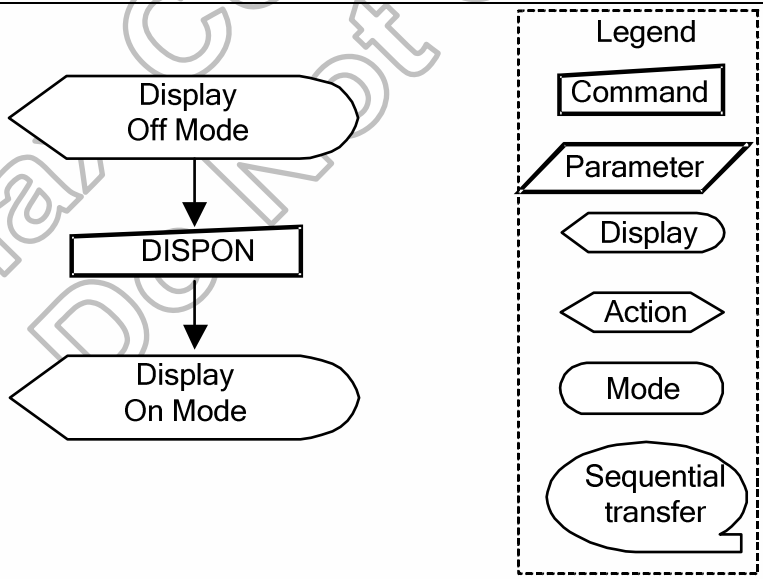
5.18.26 Set_display_on (29h)

29H	DISPON (Display On)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	0	0	1	0	1	0	0	1	29
Parameter	NO PARAMETER									
Description	This command is used to recover from DISPLAY OFF mode. Output from DPI I/F is enabled. This command does not change any other status. (Example) 									
Restriction	This command has no effect when module is already in display on mode.									

Register Availability	Status	Availability
	Normal Mode On, Idle Mode Off, Sleep Out	Yes
	Normal Mode On, Idle Mode On, Sleep Out	Yes
	Sleep In or Booster Off	Yes

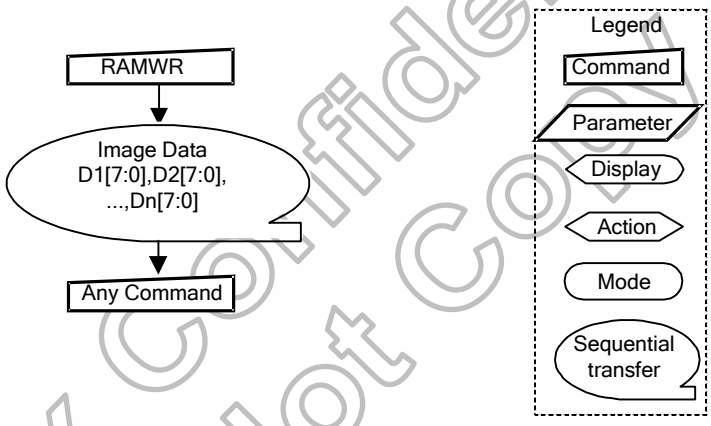
Default	Status	Default Value
	Power On Sequence	Display on
	S/W Reset	Display on
	H/W Reset	Display on

Flow Chart



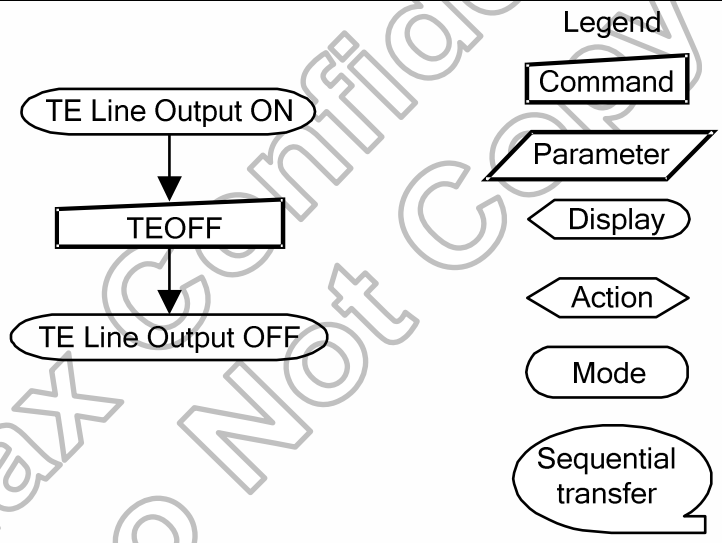
Himax Confidential Do Not Copy

5.18.27 Write_memory_start (2Ch)

2CH	RAMWR (Memory Write)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	0	0	1	0	1	1	0	0	2C
1st parameter	1	D17	D16	D15	D14	D13	D12	D11	D10	00..FF
:	1	Dx7	Dx6	Dx5	Dx4	Dx3	Dx2	Dx1	Dx0	00..FF
Nth parameter	1	Dn7	Dn6	Dn5	Dn4	Dn3	Dn2	Dn1	Dn0	00..FF
Description	This command transfers image data from the host processor to the display module's frame memory. The start pointer is (0,0).Frame memory pointer will auto increment when data is written.									
Restriction	The transferred data must be line based.									
Register Availability	Status: Normal Mode On, Idle Mode Off, Sleep Out — Availability: Yes Status: Normal Mode On, Idle Mode On, Sleep Out — Availability: Yes									
Default	Contents of memory is set randomly and not cleared.									
Flow Chart										

Himax Confidential
Do Not Copy

5.18.28 Tearing effect line off (34h)

34H	TEOFF (Tearing Effect Line OFF)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	0	0	1	1	0	1	0	0	34
Parameter	NO PARAMETER									
Description	This command is used to turn OFF **(Active Low)** the Tearing Effect output signal from the TE signal line.									
Restriction	This command has no effect when Tearing Effect output is already OFF.									
Register Availability	Status: Normal Mode On, Idle Mode Off, Sleep Out — Availability: Yes Status: Normal Mode On, Idle Mode On, Sleep Out — Availability: Yes Status: Sleep In or Booster Off — Availability: Yes									
Default	Status: Power On Sequence — Default Value: Off Status: S/W Reset — Default Value: Off Status: H/W Reset — Default Value: Off									
Flow Chart										

Himax Confidential
Do Not Copy

5.18.29 Set_tear_on (35h)

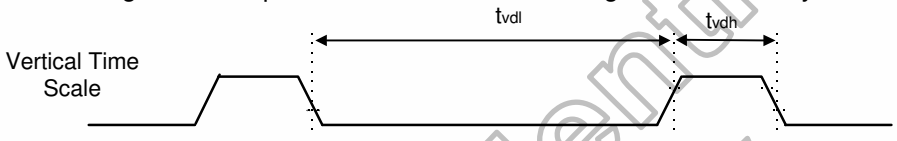
35H	TEON (Tearing Effect Line ON)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	0	0	1	1	0	1	0	1	35
Parameter	1	X	X	X	X	X	X	X	M	xx

Description

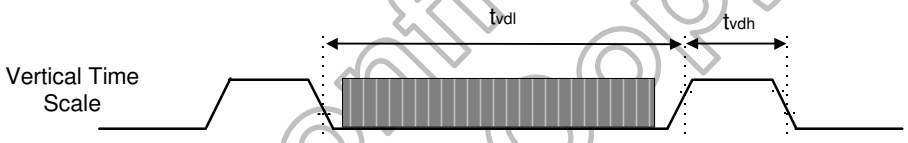
This command is used to turn ON the Tearing Effect output signal from the TE signal line.

The Tearing Effect Line On has one parameter which describes the mode of the Tearing Effect Output Line. **(X=Don't Care)**.

When M=0:
The Tearing Effect Output line consists of V-Blanking information only:



When M=1:
The Tearing Effect Output Line consists of both V-Blanking and H-Blanking information:



Note: M=1 is not applicable for this project, so it is set to "0".

Restriction

This command has no effect when Tearing Effect output is already ON.

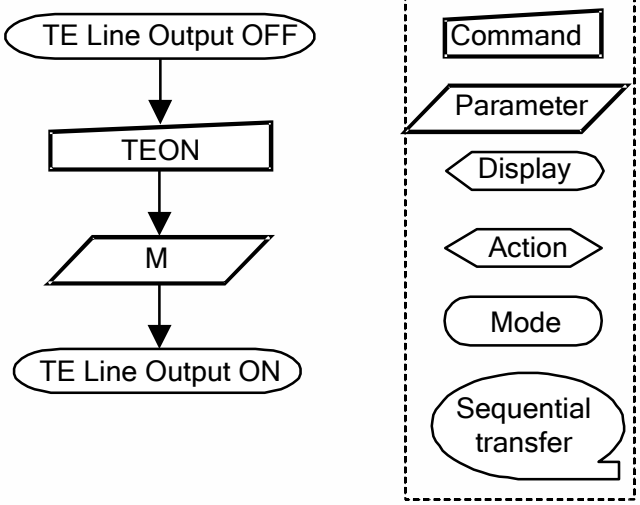
Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Sleep In or Booster Off	Yes

Default

Status	Default Value
Power On Sequence	Off
S/W Reset	Off
H/W Reset	Off

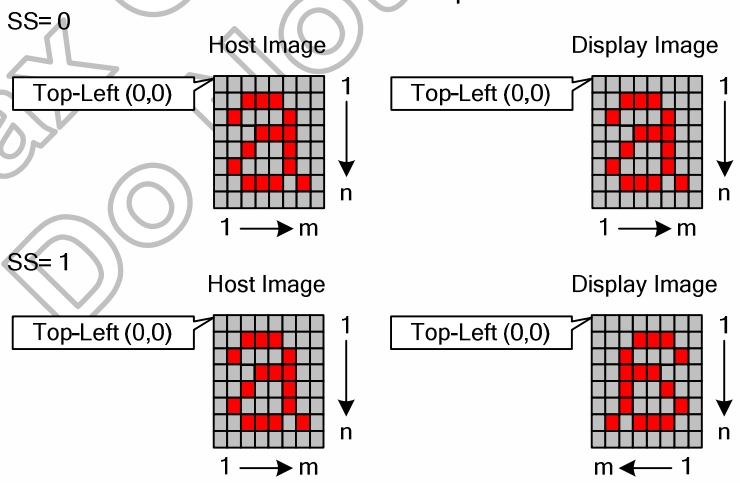
Flow Chart



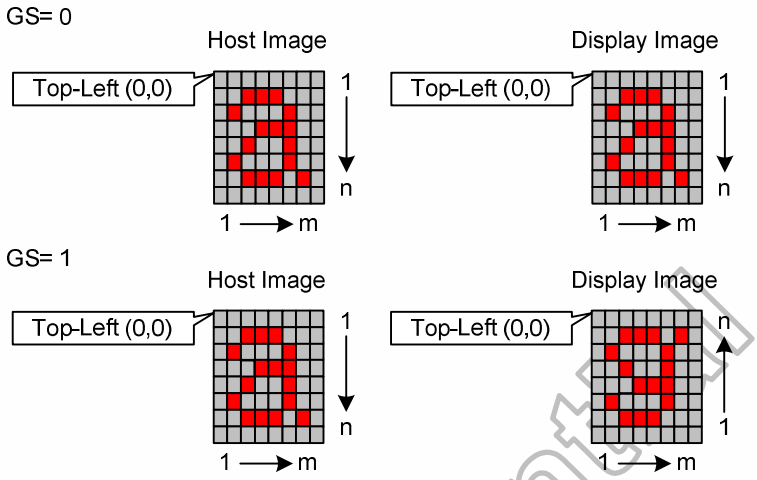
Himax Confidential Do Not Copy

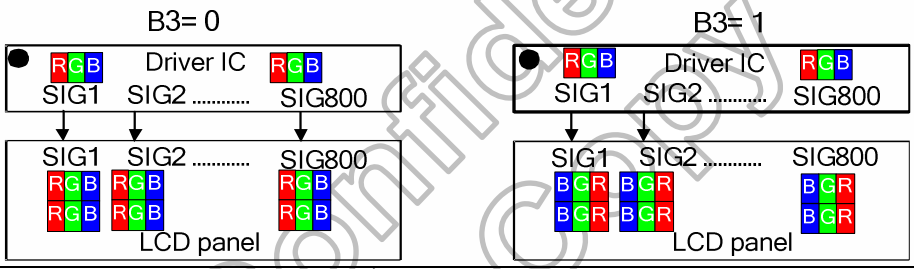
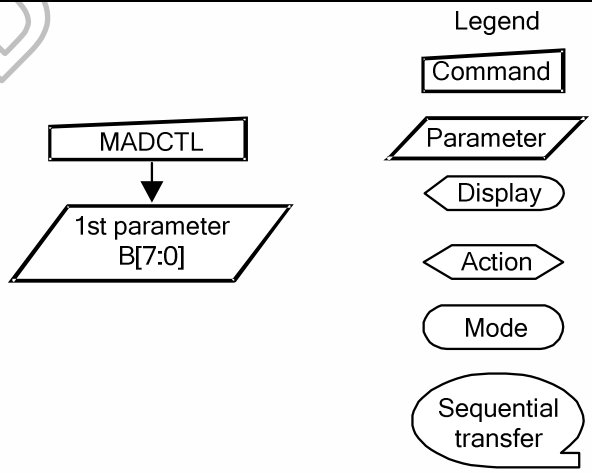
5.18.30 Set_address_mode (36h)

<table>
<tr><th rowspan="2">36H</th><th colspan="10">MADCTL (Memory Access Control)</th></tr>
<tr><th>DCX</th><th>D7</th><th>D6</th><th>D5</th><th>D4</th><th>D3</th><th>D2</th><th>D1</th><th>D0</th><th>HEX</th></tr>
<tr><td>Command</td><td>0</td><td>0</td><td>0</td><td>1</td><td>1</td><td>0</td><td>1</td><td>1</td><td>0</td><td>36</td></tr>
<tr><td>1st parameter</td><td>1</td><td>X</td><td>X</td><td>X</td><td>X</td><td>B3</td><td>X</td><td>B1</td><td>B0</td><td>xx</td></tr>
<tr><td>Description</td><td colspan="10">This command defines the display scanning direction of LCD.
This command makes no change on the other driver status.

Bit Assignment
<table>
<tr><th>Bit</th><th>Name</th><th>Description</th></tr>
<tr><td>B7</td><td>Page Address Order</td><td>This bit is not applicable for this project, so it is set to “0”.</td></tr>
<tr><td>B6</td><td>Column Address Order</td><td>This bit is not applicable for this project, so it is set to “0”.</td></tr>
<tr><td>B5</td><td>Page/Column Selection</td><td>This bit is not applicable for this project, so it is set to “0”.</td></tr>
<tr><td>B4</td><td>Display Device Line Refresh Order</td><td>This bit is not applicable for this project, so it is set to “0”.</td></tr>
<tr><td>B3</td><td>RGB-BGR Order (BGR)</td><td>Colour selector switch control (0=RGB colour filter panel, 1=BGR colour filter panel)</td></tr>
<tr><td>B2</td><td>Display Data Latch Data Order</td><td>This bit is not applicable for this project, so it is set to “0”.</td></tr>
<tr><td>B1</td><td>Flip Horizontal (Source scan sequence)</td><td>Select the Source driver scan direction on panel module</td></tr>
<tr><td>B0</td><td>Flip Vertical (Gate scan sequence)</td><td>Select the Gate driver scan direction on panel module</td></tr>
</table>
SS - Source scan sequence

</td></tr>
</table>

Himax Confidential
Do not Copy

<table>
<tr><td></td><td>GS - Gate scan sequence
 

RGB-BGR Order
 
</td></tr>
<tr><td>Restriction</td><td>D7, D6, D5, D4, and D2 of the 1st parameter are set to ‘0’ internally.</td></tr>
<tr><td>Register Availability</td><td><table>
<tr><th>Status</th><th>Availability</th></tr>
<tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr>
<tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr>
<tr><td>Sleep In or Booster Off</td><td>Yes</td></tr>
</table></td></tr>
<tr><td>Default</td><td><table>
<tr><th>Status</th><th>Default Value</th></tr>
<tr><td>Power On Sequence</td><td>00h</td></tr>
<tr><td>S/W Reset</td><td>No Change</td></tr>
<tr><td>H/W Reset</td><td>00h</td></tr>
</table></td></tr>
<tr><td>Flow Chart</td><td> 
</td></tr>
</table>

Himax Confidential
Do Not Copy!

5.18.31 Idle mode off (38h)

38H	IDMOFF (Idle mode off)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	0	0	1	1	1	0	0	0	38
Parameter	NO PARAMETER									
Description	This command is used to recover from Idle mode on. In the idle off mode, LCD can display maximum 16.7M colours.									
Restriction	This command has no effect when module is already in idle off mode.									

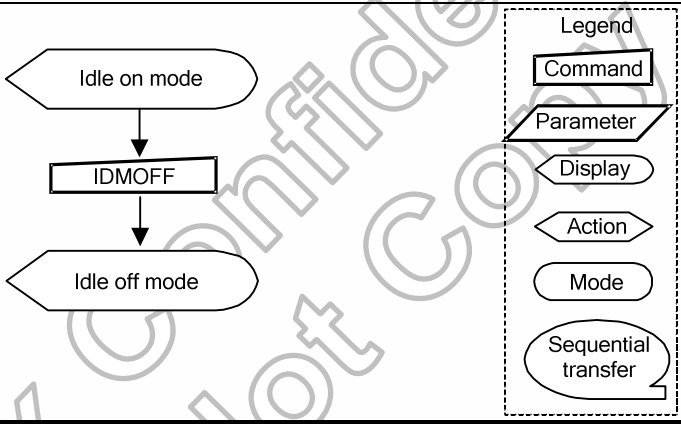
Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Sleep In or Booster Off	Yes

Default

Status	Default Value
Power On Sequence	Off
S/W Reset	Off
H/W Reset	Off

Flow Chart



Himax Confidential
Do Not Copy

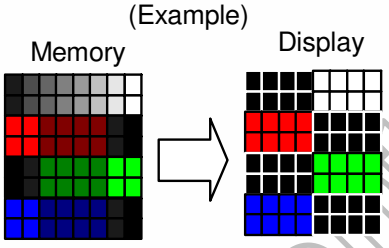
5.18.32 Enter_Idle_mode (39h)

39H	IDMON (Idle mode on)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	0	0	1	1	1	0	0	1	39
Parameter	NO PARAMETER									

Description

This command is used to enter into Idle mode on. In the idle on mode, colour expression is reduced. The primary and the secondary colours using MSB of each R, G and B, 8 colour depth data is displayed.

(Example)



Display Colour

	R7 – R0	G7 – G0	B7 – B0
Black	0XXXXX	0XXXXX	0XXXXX
Blue	0XXXXX	0XXXXX	1XXXXX
Red	1XXXXX	0XXXXX	0XXXXX
Magent	1XXXXX	0XXXXX	1XXXXX
Green	0XXXXX	1XXXXX	0XXXXX
Cyan	0XXXXX	1XXXXX	1XXXXX
Yellow	1XXXXX	1XXXXX	0XXXXX
White	1XXXXX	1XXXXX	1XXXXX

X=don't care

Restriction

This command has no effect when module is already in idle on mode.

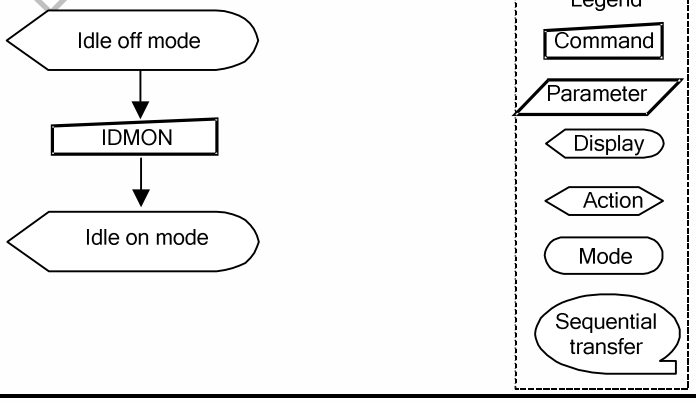
Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Sleep In or Booster Off	Yes

Default

Status	Default Value
Power On Sequence	Off
S/W Reset	Off
H/W Reset	Off

Flow Chart



Himax Confidential Do Not Copy

5.18.33 Set_pixel_format (3Ah)

3A H	COLMOD (Interface Pixel Format)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	0	0	1	1	1	0	1	0	3A
1st parameter	1	X	D6	D5	D4	X	X	X	X	XX

Description

This command is used to define the format of RGB picture data.
D6~D4: DPI Pixel format Definition.
Bits D7, D[3:0]: Reserved.
The formats are shown in the table:

Pixel Format	D6	D5	D4
Not Defined	0	0	0
Not Defined	0	0	1
Not Defined	0	1	0
Not Defined	0	1	1
Not Defined	1	0	0
16 Bit/Pixel	1	0	1
18 Bit/Pixel	1	1	0
24 Bit/Pixel	1	1	1

If the setting is not used then the corresponding bits in the parameter returned

Restriction

There is no visible effect until the image data is written.

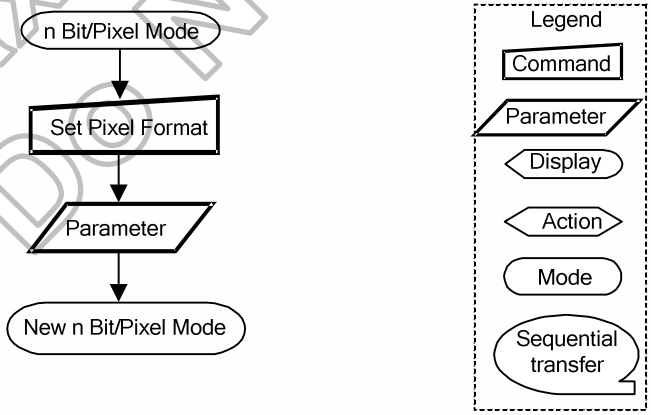
Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Sleep In or Booster Off	Yes

Default

Status	Default Value
Power On Sequence	70h
S/W Reset	70h
H/W Reset	70h

Flow Chart



Himax Confidential Do Not Copy

5.18.34 Write_memory_contiune (3Ch)

3CH	Write_memory_contiune									
	D/CX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	0	0	1	1	1	1	0	0	3C
1st parameter	1	D17	D16	D15	D14	D13	D12	D11	D10	00..FF
:	1	Dx7	Dx6	Dx5	Dx4	Dx3	Dx2	Dx1	Dx0	00..FF
Nth parameter	1	Dn7	Dn6	Dn5	Dn4	Dn3	Dn2	Dn1	Dn0	00..FF
Description	This command transfers image data from the host processor to the display module's frame memory continuing from the pixel location following the previous write_memory_continue or write_memory_start command. Sending any other command can stop frame Write.									
Restriction	The transferred data must be line based.									

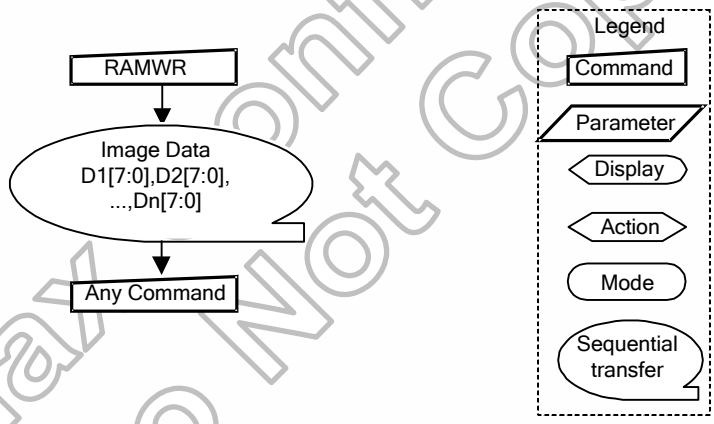
Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes

Default

Status	Default value
Power On Sequence	Contents of memory is set randomly
S/W Reset	Contents of memory is set randomly

Flow Chart



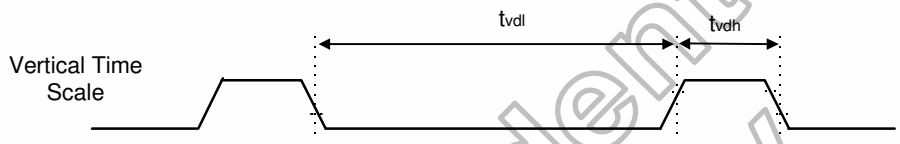
Himax Confidential
Do Not Copy

5.18.35 Set tear scan lines (44h)

44H	TESL (Tear Effect Scan Lines)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	0	1	0	0	0	1	0	0	44
1st parameter	1	TELINE[15:8] **(8'b0)**								00..FF
2nd parameter	1	TELINE[7:0] **(8'b0)**								00..FF

Description

This command is turns on the display module's Tearing Effect output signal on the TE signal
Line when the display module reacfes line TELINE. The TE signal is not affected by changing MADCTL bit B4.
The Tearing Effect Line On has one parameter which describes the mode of the Tearing Effect Output Line.
The Tearing Effect Output line consists of V-Blanking information only:



Note: That TELINE=0 is equivalent to TEMODE=0. The Tearing Effect Output Line shall be active low when the display module is in Sleep mode.

Restriction

The command has no effect when Tearing Effect output is already ON.

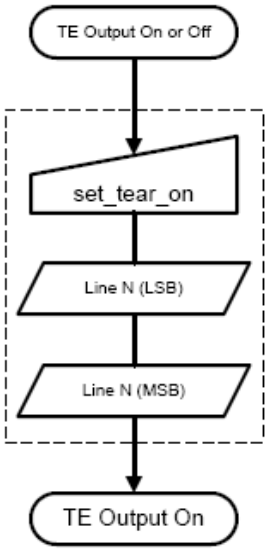
Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Sleep In or Booster Off	Yes

Default

Status	Default Value
Power On Sequence	0000h
S/W Reset	0000h
H/W Reset	0000h

Flow Chart



Himax Confidential
Do Not Copy

5.18.36 Get the current scanline(45h)

45H	GETSCAN (Get the current scanline)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	0	1	0	0	0	1	0	1	45
1st parameter	1	SLN[15:8] **(8'b0)**								00..FF
2nd parameter	1	SLN[7:0] **(8'b0)**								00..FF
Description	The display module returns the current scanline, N, used to update the display device. The total number of scanlines on a display device is defined as VSYNC + VBP + VACT + VFP. The first scanline is defined as the first line of V Sync and is denoted as Line 0. When in Sleep Mode, the value returned by get_scanline is undefined.									
Restriction	-									

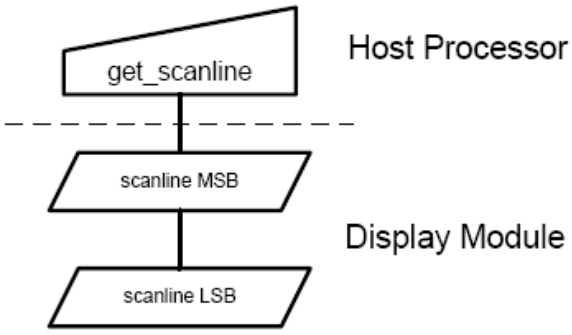
Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Sleep In or Booster Off	Yes

Default

Status	Default Value
Power On Sequence	0000h
S/W Reset	0000h
H/W Reset	0000h

Flow Chart



Himax Confidential Do not copy

5.18.37 Write display brightness (51h)

51H	WRDISBV (Write Display Brightness)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	0	1	0	1	0	0	0	1	51
1st parameter	1	DBV[7:0]								00..FF
Description	This command is used to adjust the brightness value of the display. It should be checked what the relationship between this written value and output brightness of the display is. This relationship is defined on the display module specification. In principle relationship is that 00h value means the lowest brightness and FFh value means the highest brightness. See chapter “5.13.3 Brightness Control Block”.									
Restriction	-									

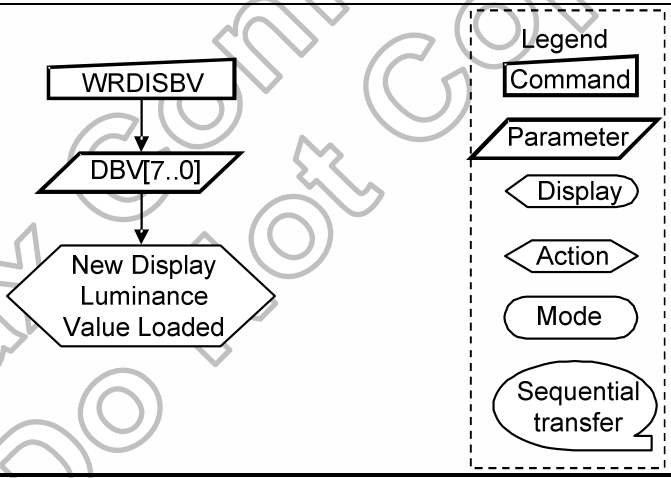
Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Sleep In or Booster Off	Yes

Default

Status	Default Value
Power On Sequence	00h
S/W Reset	00h
H/W Reset	00h

Flow Chart



Himax Confidential Do Not Copy

5.18.38 Read display brightness value (52h)

52H	RDDISBV (Read Display Brightness Value)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	0	1	0	1	0	0	1	0	52
1st parameter	1	DBV[7:0]								xx
Description	This command returns the brightness value of the display. It should be checked what the relationship between this returned value and output brightness of the display. This relationship is defined on the display modulespecification is. In principle the relationship is that 00h value means the lowest brightness and FFh value means the highest brightness. See chapters: "5.13.3 Brightness Control Block", and "6.2.33 Write Display Brightness **(51h)**" DBV[7:0] is reset when display is in sleep-in mode. DBV[7:0] is '00h' when bit BCTRL of "6.2.35 Write CTRL Display **(53h)**" command is '0'. DBV[7:0] is manual set brightness specified with "6.2.35 Write CTRL Display **(53h)**" command when bit BCTRL is '1'. When bit BCTRL of "6.2.35 Write CTRL Display (53h)" command is '1' and bit C1/C0 of "6.2.37 Write Content Adaptive Brightness Control **(55h)**" are '0', DBV[7:0] output is the brightness value specified with "6.2.33 Write Display Brightness **(51h)**" command.									
Restriction	-									

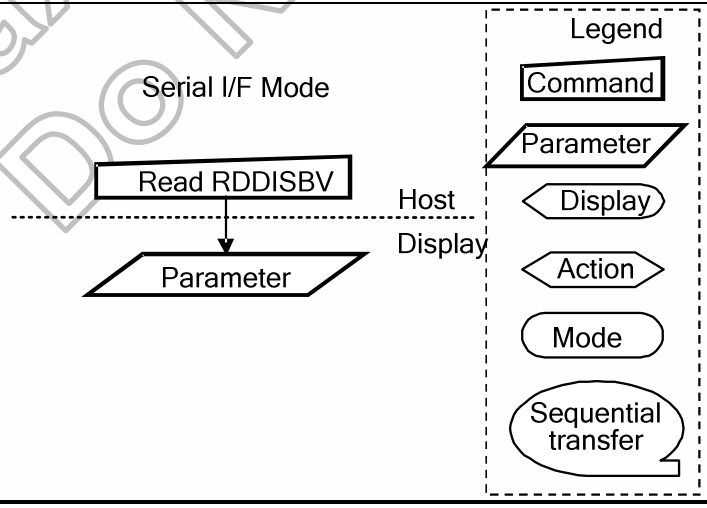
Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Sleep In or Booster Off	Yes

Default

Status	Default Value
Power On Sequence	00h
S/W Reset	00h
H/W Reset	00h

Flow Chart



Himax Confidential
Do Not Copy

5.18.39 Write CTRL display (53h)

53H	WRCTRLD (Write Control Display)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	0	1	0	1	0	0	1	1	53
1st parameter	1	X	X	BCTRL	X	DD	BL	X	X	00..FF
Description	This command is used to control display brightness. BCTRL: Brightness Control Block On/Off, This bit is always used to switch brightness for display. 0=Off **(Brightness registers are 00h, DBV[7..0])** 1=On **(Brightness registers are active, according to the other parameters.)** Display Dimming **(DD): (Only for manual brightness setting)** DD=0: Display Dimming is off DD=1: Display Dimming is on BL: Backlight Control On/Off 0=Off **(Completely turn off backlight circuit. Control lines must be low.)** 1=On Dimming function is adapted to the brightness registers for display when bit BCTRL is changed at DD=1, e.g. BCTRL: 0 -> 1 or 1-> 0. When BL bit change from “On” to “Off”, backlight is turned off without gradual dimming, even if dimming-on **(DD=1)** are selected. X=Don’t care.									
Restriction	-									

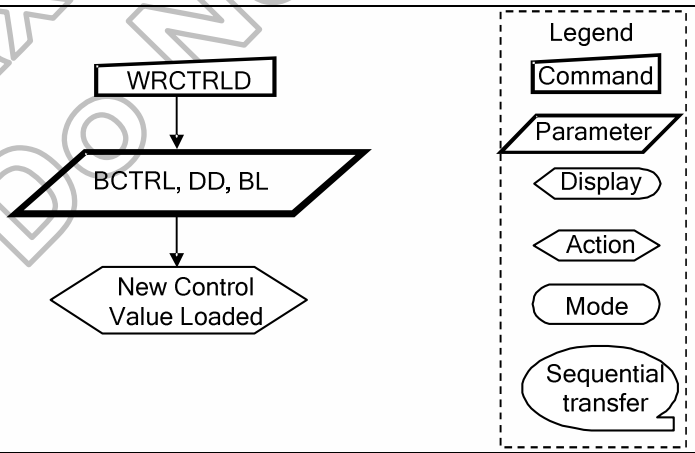
Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Sleep In or Booster Off	Yes

Default

Status	Default Value
Power On Sequence	00h
S/W Reset	00h
H/W Reset	00h

Flow Chart



Himax Confidential Do Not Copy

5.18.40 Read CTRL value display (54h)

54H	RDCTRLD (Read Control Value Display)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	0	1	0	1	0	1	0	0	54
1st parameter	1	X	X	BCTRL	X	DD	BL	X	X	xx
Description	This command returns ambient light and brightness control values, see chapter: "6.2.35 Write CTRL Display **(53h)**". BCTRL: Brightness Control Block On/Off, This bit is always used to switch brightness for display. 0=Off 1=On Display Dimming **(DD)**: DD=0: Display Dimming is off DD=1: Display Dimming is on BL: Backlight Control On/Off 0=Off **(Completely turn off backlight circuit)** 1=On									
Restriction	-									

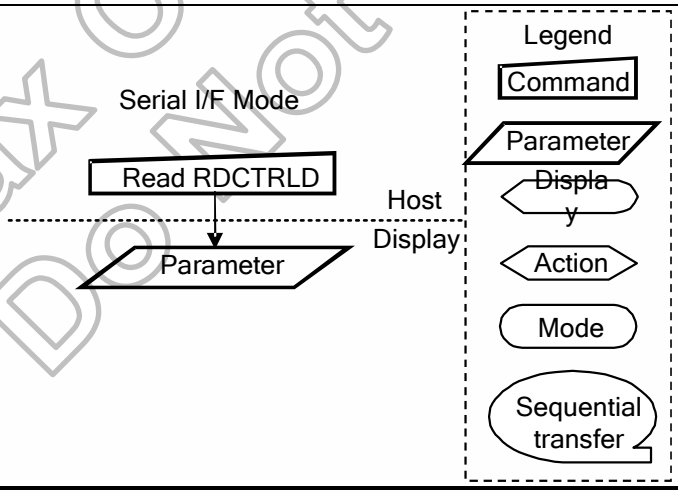
Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Sleep In or Booster Off	Yes

Default

Status	Default Value
Power On Sequence	00h
S/W Reset	00h
H/W Reset	00h

Flow Chart



Himax Confidential
Do Not Copy

5.18.41 Write content adaptive brightness control (55h)

55 H	WRCABC (Write Content Adaptive Brightness Control)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	0	1	0	1	0	1	0	1	55
1st parameter	1	X	X	X	X	X	X	CABC[1:0]		xx

Description

This command is used to set parameters for image content based adaptive brightness control functionality.
There is possible to use 4 different modes for content adaptive image functionality, which are defined on a table below. See chapter "5.13 Content Adaptive Brightness Control **(CABC)**".

CABC[1]	CABC[0]	Function
0	0	Off
0	1	User Interface Image
1	0	Still Picture
1	1	Moving Image

Restriction

-

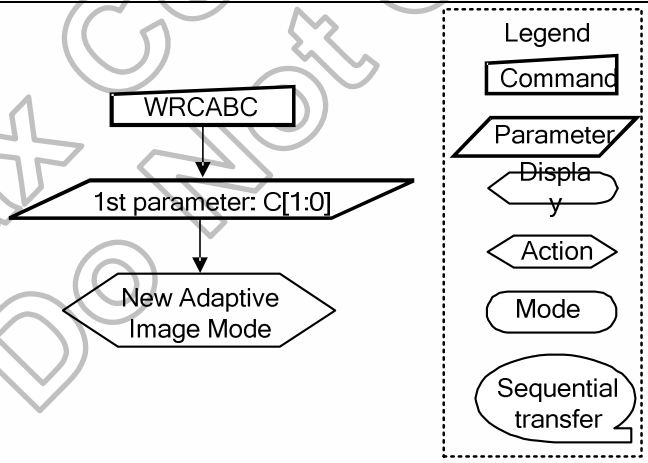
Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Sleep In or Booster Off	Yes

Default

Status	Default Value
Power On Sequence	00h
S/W Reset	00h
H/W Reset	00h

Flow Chart



Himax Confidential Do Not Copy

5.18.42 Read content adaptive brightness control (56h)

56H	RDCABC (Read Content Adaptive Brightness Control)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	0	1	0	1	0	1	1	0	56
1st parameter	1	0	0	0	X	0	0	CABC[1:0]		xx

Description

This command is used to set parameters for image content based adaptive brightness control functionality.
There is possible to use 4 different modes for content adaptive image functionality, which are defined on a table below. See chapter "5.13 Content Adaptive Brightness Control **(CABC)**".

CABC[1]	CABC[0]	Function
0	0	Off
0	1	User Interface Image
1	0	Still Picture
1	1	Moving Image

Restriction

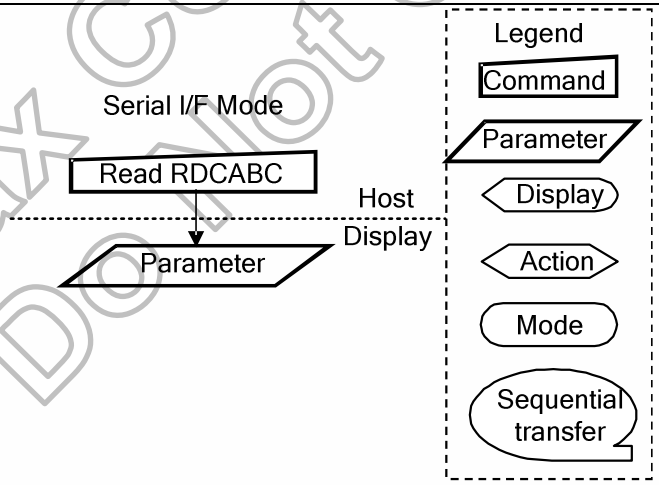
Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Sleep In or Booster Off	Yes

Default

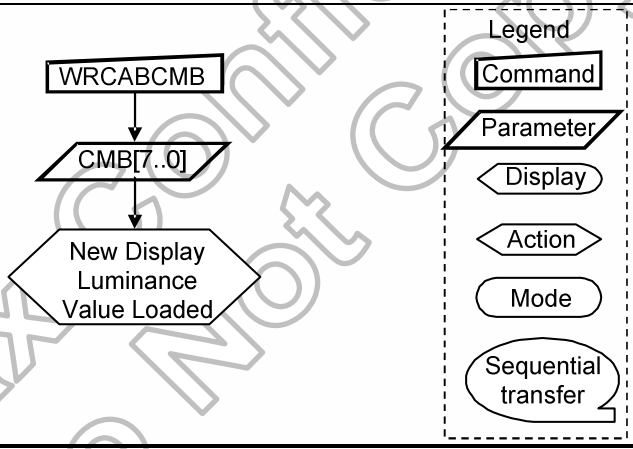
Status	Default Value
Power On Sequence	00h
S/W Reset	00h
H/W Reset	00h

Flow Chart



Himax Confidential Do Not Copy

5.18.43 Write CABC minimum brightness (5Eh)

5E H	WRCABCMB (Write CABC minimum brightness)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	0	1	0	1	1	1	1	0	5E
1st parameter	1	CMB[7:0]								00..FF
Description	This command is used to set the minimum brightness value of the display for CABC function. In principle relationship is that 00h value means the lowest brightness for CABC and FFh value means the highest brightness for CABC. See chapter “5.13.4 Minimum brightness setting of CABC function”.									
Restriction	-									
Register Availability	Status: Normal Mode On, Idle Mode Off, Sleep Out — Availability: Yes Status: Normal Mode On, Idle Mode On, Sleep Out — Availability: Yes Status: Sleep In or Booster Off — Availability: Yes									
Default	Status: Power On Sequence — Default Value: 00h Status: S/W Reset — Default Value: 00h Status: H/W Reset — Default Value: 00h									
Flow Chart										

Himax Confidential
Do Not Copy

5.18.44 Read CABC minimum brightness (5Fh)

5FH	RDCABCMB (Read CABC minimum brightness)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	0	1	0	1	1	1	1	1	5F
1st parameter	1	CMB[7:0]								XX
Description	This command returns the minimum brightness value of CABC function. In principle the relationship is that 00h value means the lowest brightness and FFh value means the highest brightness. See chapter "5.13.4 Minimum brightness setting of CABC function". CMB[7:0] is CABC minimum brightness specified with "6.2.39 Write CABC minimum brightness **(5Eh)**" command.									
Restriction	-									

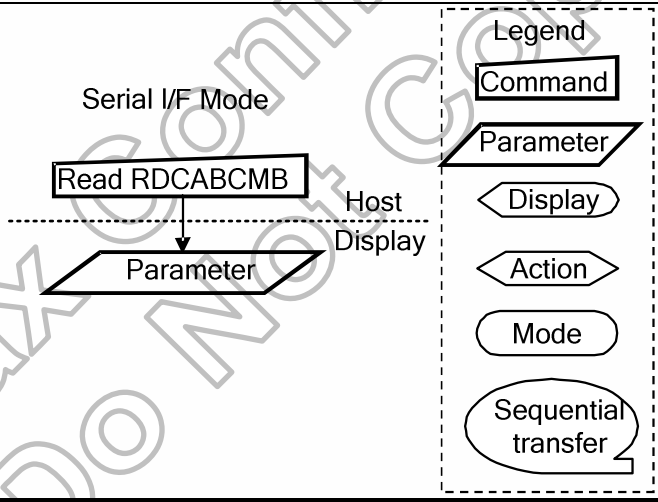
Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Sleep In or Booster Off	Yes

Default

Status	Default Value
Power On Sequence	00h
S/W Reset	00h
H/W Reset	00h

Flow Chart



Himax Confidential
Do Not Copy

5.18.45 Read automatic brightness control self-diagnostic result (68h)

68H	RDABCSDR (Read Automatic Brightness Control Self-Diagnostic Result)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	0	1	1	0	1	0	0	0	68
1st parameter	1	D[7:6]		0	0	0	0	0	0	xx
Description	This command indicates the status of the display self-diagnostic results for automatic brightness control after Sleep Out –command as described in the table below: Bit D7 – Register Loading Detection. See section “5.10.1 Register loading Detection”. Bit D6 – Functionality Detection. See section “5.10.2 Functionality Detection “. Bits D[5:0] are for future use and are set to ‘0’.									
Restriction	-									

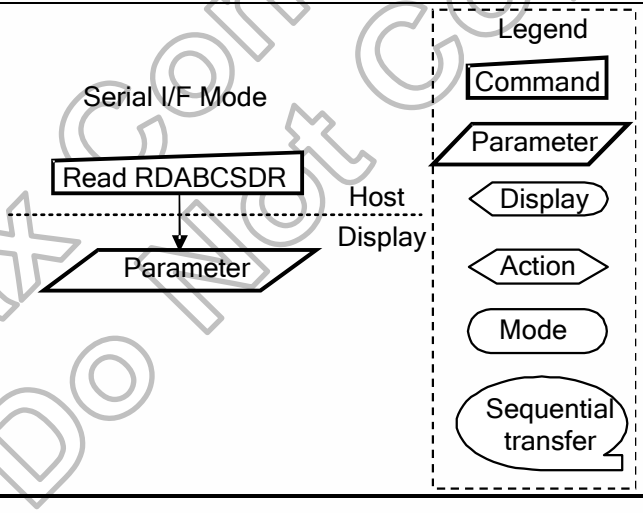
Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Sleep In or Booster Off	Yes

Default

Status	Default Value
Power On Sequence	00h
S/W Reset	00h
H/W Reset	00h

Flow Chart



Himax Confidential
Do Not Copy

5.18.46 Write Idle Mode Color(80h)

80 H	WRIMCOL (Write Idle Mode Color)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	0	0	0	0	0	0	0	80
1st parameter	1	X	X	X	X	X	R	G	B	07

Description

This command can be used to select color 1bbp Idle Mode. Color selection is configured with bits [2:0], where:
D2: R Component
D1: G Component
D0: B Component

Color selection is defined in the following table:

1bpp Idle Mode Color Selection	R	G	B
Black	0	0	0
Blue	0	0	1
Green	0	1	0
Cyan	0	1	1
Red	1	0	0
Magenta	1	0	1
Yellow	1	1	0
White	1	1	1

Default setting for 1bpp color selection for “Normal Black” panel is ‘White’; R=G=B=’1’

X=Reserved

Restriction

-

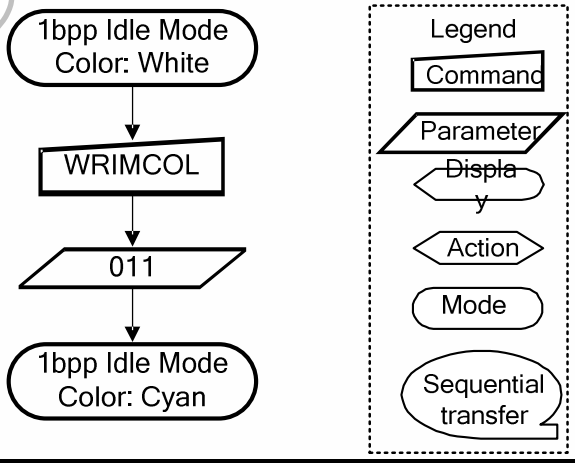
Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Sleep In or Booster Off	Yes

Default

Status	Default Value
Power On Sequence	07h
S/W Reset	07h
H/W Reset	07h

Flow Chart



Himax Confidential Do Not Copy

5.18.47

5.18.48 Read Idle Mode Color(81h)

81 H	RDIMCOL (Read Idle Mode Color)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	0	0	0	0	0	0	1	81
1st parameter	1	0	0	0	0	0	R	G	B	07

Description

This command returns the current color selection of 1bbp Idle Mode. Bits [2:0] are defined as:
D2: R Component
D1: G Component
D0: B Component

Color selection is defined in the following table:

1bpp Idle Mode Color Selection	R	G	B
Black	0	0	0
Blue	0	0	1
Green	0	1	0
Cyan	0	1	1
Red	1	0	0
Magenta	1	0	1
Yellow	1	1	0
White	1	1	1

Default setting for 1bpp color selection for “Normal Black” panel is ‘White’; R=G=B=’1’

Restriction

-

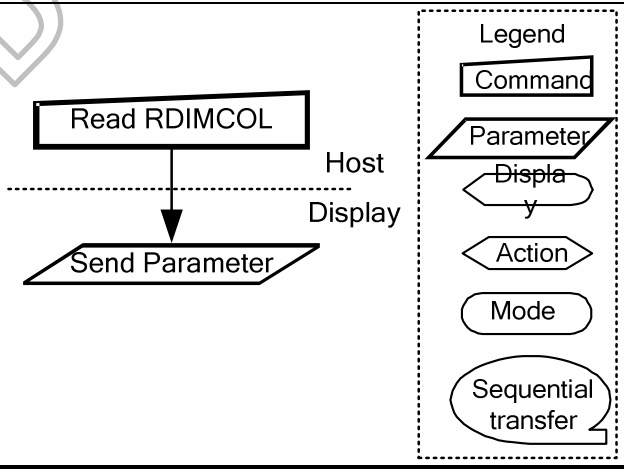
Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Sleep In or Booster Off	Yes

Default

Status	Default Value
Power On Sequence	07h
S/W Reset	07h
H/W Reset	07h

Flow Chart



Himax Confidential
Do Not Copy

5.18.49 Read_DDB_start (A1h)

A1H	Read_DDB_start									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	0	1	0	0	0	0	1	A1
1st parameter	1	xx	xx	xx	xx	xx	xx	xx	xx	xx
2nd parameter	1	xx	xx	xx	xx	xx	xx	xx	xx	xx
:	1	xx	xx	xx	xx	xx	xx	xx	xx	xx
Nth parameter	1	xx	xx	xx	xx	xx	xx	xx	xx	xx

Description

This command reads identifying and descriptive information from the peripheral. This information is organized in the Device Descriptor Block **(DDB)** stored on the peripheral. The response to this command returns a sequence of bytes that may be any length up to 64K bytes. Note that the returned sequence of bytes does not necessarily correspond to the entire DDB; it may be a portion of a larger block of data.

The format of returned data is as follows:

Parameter 1: LS **(least significant)** byte of Supplier ID. Supplier ID is a unique value assigned to each peripheral supplier by the MIPI organization.

Parameter 2: MS **(most significant)** byte of Supplier ID.

Parameter 3: LS **(least significant)** byte of Supplier Elective Data. This is a byte of information that is determined by the supplier. It could include model number or revision information, for example.

Parameter 4: MS **(most significant)** byte of Supplier Elective Data

Parameter 5: single-byte *Escape or Exit Code* **(EEC)**. The code is interpreted as follows:

- FFh – Exit code – there is no more data in the Descriptor Block
- 00h – Escape code – there is supplier-proprietary data in the Descriptor Block **(does not conform to any MIPI standard)**
- Any other value – there is DDB data in the Descriptor Block. The format and interpretation of this data is documented in *MIPI Alliance Standard for Device Descriptor Block* ***(DDB)***.

DDBs may contain many more data fields providing information about the peripheral.

In a DSI system, read activity takes the form of two separate transactions across the bus: first the read command read_DDB_start from host processor to peripheral, which includes the bus turn-around token.

The peripheral then takes control of the bus and returns the requested data. The peripheral response to read_DDB_start is a Long Packet type, so its length may be up to 64K bytes unless limited by a previous set_max_return_size command.

The response to a read_DDB_start command always starts at the beginning of the Device Descriptor Block. After receiving the first packet and processing the returned DDB data, the host processor may initiate a read_DDB_continue command to access the next portion of the DDB. A read_DDB_continue command begins the next read at the location following the last byte of the previous data read from the DDB.

Subsequent read_DDB_continue commands can be used to read a DDB or supplier-proprietary block of arbitrary size. There is, however, no obligation to read the entire block. The host processor may choose to stop reading after completion of any read_DDB_xxx command.

Restrictions

-

Register Availability

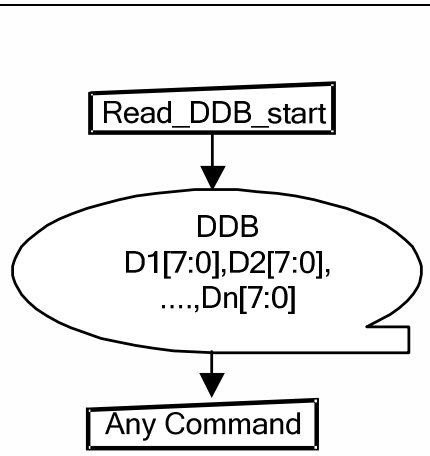
Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Sleep In or Booster Off	Yes

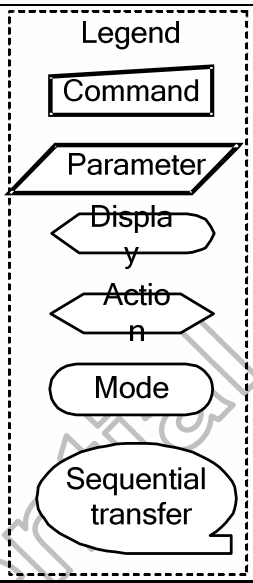
Default

Status	Default Value
Power On Sequence	PA1st~4th is OTP value, PA5th is FFh
S/W Reset	PA1st~4th is OTP value, PA5th is FFh
H/W Reset	PA1st~4th is OTP value, PA5th is FFh

Himax Confidential
Do Not Copy

Flow Chart





Himax Confidential
Do Not Copy

5.18.50 Read_DDB_continue (A8h)

A8H	Read_DDB_continue									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	0	1	0	1	0	0	0	A8
1st parameter	1	xx	xx	xx	xx	xx	xx	xx	xx	xx
2nd parameter	1	xx	xx	xx	xx	xx	xx	xx	xx	xx
:	1	xx	xx	xx	xx	xx	xx	xx	xx	xx
Nth parameter	1	xx	xx	xx	xx	xx	xx	xx	xx	xx
Description	A read_DDB_start command should be executed at least once before a read_DDB_continue command to define the read location. Otherwise, data read with a read_DDB_continue command is undefined.									
Restrictions	-									

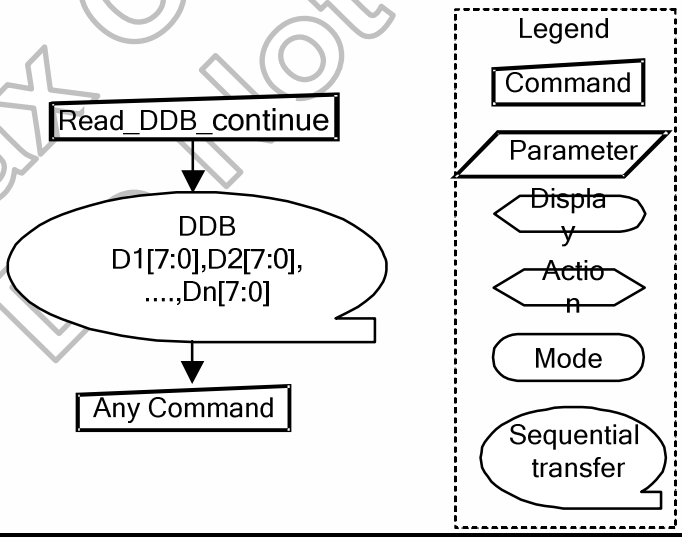
Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Sleep In or Booster Off	Yes

Default

Status	Default Value
Power On Sequence	Without A1h read, 1st~4th read is the same as A8h 1st~4th OTP value, after 5th read is FFh.
S/W Reset	Without A1h read, 1st~4th read is the same as A8h 1st~4th OTP value, after 5th read is FFh.
H/W Reset	Without A1h read, 1st~4th read is the same as A8h 1st~4th OTP value, after 5th read is FFh.

Flow Chart



Himax Confidential Do Not Copy

5.18.51 Read ID1 (DAh)

DAH	RDID1 (Read ID1)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	1	0	1	1	0	1	0	DA
1st parameter	1	module's manufacturer[7:0]								xx
Description	This read byte identifies the LCD module's manufacturer. It is specified by display supplier and for xx is defined as xxHEX.									
Restriction	-									

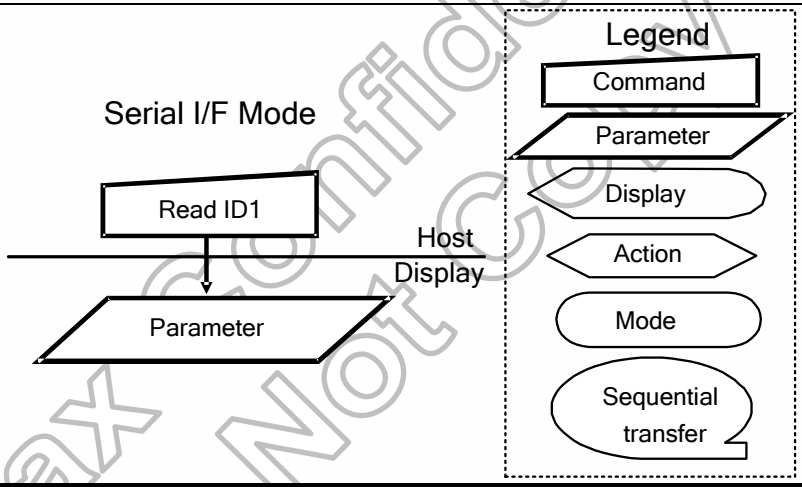
Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Sleep In or Booster Off	Yes

Default

Status	Default Value
Power On Sequence	OTP value
S/W Reset	OTP value
H/W Reset	OTP value

Flow Chart



Himax Confidential
Do Not Copy

5.18.52 Read ID2 (DBh)

DBH	RDID2 (Read ID2)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	1	0	1	1	0	1	1	DB
1st parameter	1	1	LCD module/driver version [6:0]							xx

Description

This read byte is used to track the LCD module/driver version. It is defined by display supplier and changes each time a revision is made to the display, material or construction specifications. See Table:

ID Byte Value V[7:0]	Version	Changes
80h		
81h		
82h		
83h		
84h		
85h		

X= Don't care

Restrictions

-

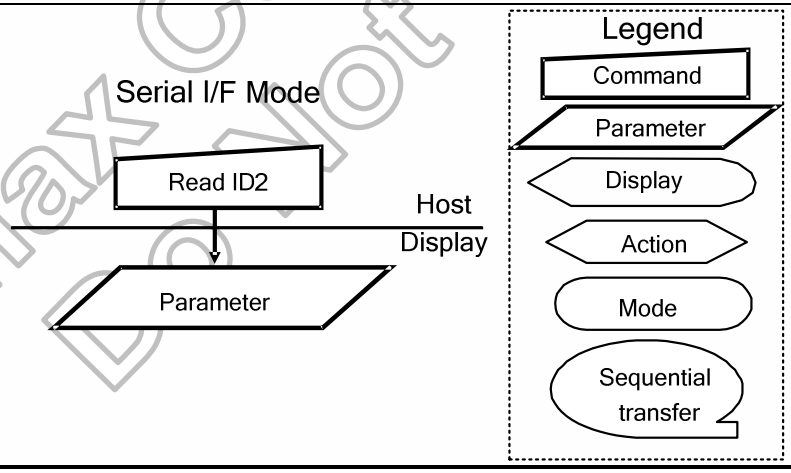
Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Sleep In or Booster Off	Yes

Default

Status	Default Value
Power On Sequence	OTP value
S/W Reset	OTP value
H/W Reset	OTP value

Flow Chart



Himax Confidential Do Not Copy

5.18.53 Read ID3 (DCh)

DCH	RDID3 (Read ID3)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	1	0	1	1	1	0	0	DC
1st parameter	1	LCD module/driver ID[7:0]								xx
Description	This read byte identifies the LCD module/driver. It is specified by display supplier and for this LCD project module is defined as xxHEX.									
Restrictions	-									

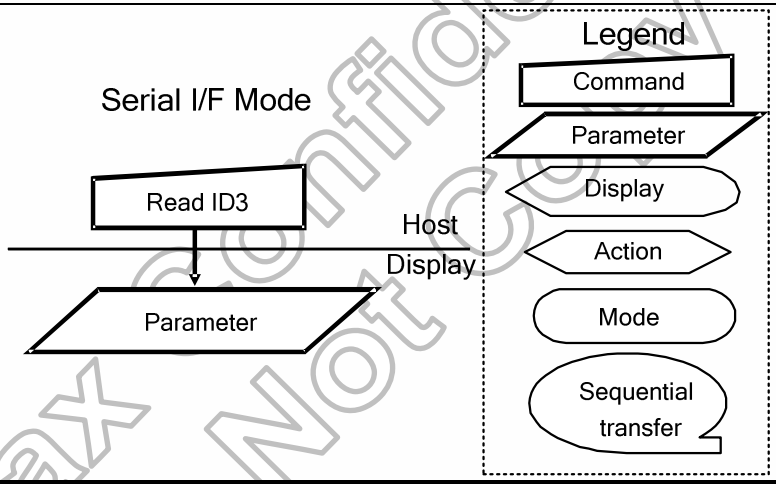
Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Sleep In or Booster Off	Yes

Default

Status	Default Value
Power On Sequence	OTP value
S/W Reset	OTP value
H/W Reset	OTP value

Flow Chart



Himax Confidential Do Not Copy

5.19User define command description

5.19.1 SETSEQUENCE: Set sequence (B0h)

B0H	SETSEQUENCE(Set Sequence)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	0	1	1	0	0	0	0	B0
1st parameter	1	-	-	-	-	-	-	-	-	-
2nd parameter	1	-	-	-	-	-	-	-	OSC_EN	-
3rd parameter	1	-	VSN_EN	VSP_EN	VGL_EN	VGH_EN	VCL_EN	-	STB	-
4th parameter	1	-	-	-	-	GON	DTE	D[1:0]		-

Description

This command is used for DCS command auto sequence and manual mode debug use, please don't access this command in initial code.

OSC_EN: Enable internal oscillator. “1”: Enable; “0”: Disable.

STB: When STB = “1”, the HX8394-F enters the standby mode, where all display operation stops, suspend all the internal operations. But the internal R-C oscillator stop or not is determined by OSC_EN bit. To minimize the standby power, OSC_EN is set to 0. During the standby mode, only the following process can be executed.

a. Exit the Standby mode (STB = “0”)
b. Enable or disable the oscillator
c. Software reset

VCL_EN : ON/OFF the operation of VCL charge bump circuit in manual power on mode. Using state machine SLPOUT CMD to power on this bit must set “1”.

VCL_EN	Operation of VCL charge bump circuit
0	OFF
1	ON

VGH_EN: ON/OFF the operation of VGH charge bump circuit in manual power on mode. Using state machine SLPOUT CMD to power on this bit must set “1”.

VGH_EN	Operation of VGH charge bump circuit
0	OFF
1	ON

VGL_EN : ON/OFF the operation of VGL charge bump circuit in manual power on mode. Using state machine SLPOUT CMD to power on this bit must set “1”.

VGL_EN	Operation of VGL charge bump circuit
0	OFF
1	ON

VSP_EN: ON/OFF the operation of VSP circuit in manual power on mode. Using state machine SLPOUT CMD to power on this bit must set “1”.

VSP_EN	Operation of VSP DC/DC circuit
0	OFF
1	ON

VSN_EN: ON/OFF the operation of VSN circuit in manual power on mode. Using state machine SLPOUT CMD to power on this bit must set “1”.

VSN_EN	Operation of VSN DC/DC circuit
0	OFF
1	ON

GON: GIP control signal enable selection

Himax Confidential
Do Not Copy

GON	GIP Enable
0	GIP Off
1	GIP On

DTE: Source output enable selection

DTE	Source Output
0	Source output off
1	Source output on

D[1:0]: Setting Source driver output

D1	D0	Source Output	HX8394-F Internal Operations
0	0	Halt	Halt
0	1	Blanking or GND	GIP initial sequence
1	0	V255 or V0	Blanking
1	1	Normal	Operate

Restrictions	SETEXTC turn on to enable this command.

Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Sleep In or Booster Off	Yes

Himax Confidential
Do Not Copy

5.19.2 SETPOWER: Set power (B1h)

B1H	SETPOWER(Set power related setting)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	0	1	1	0	0	0	1	B1
Bank0 1st parameter	1	-	POWMOD[1:0]		AP[2:0]			DSTBY_OPT	DSTB	-
2nd parameter	1	VCLS[2:0]			VRHP[4:0]					-
3rd parameter	1	VPPS[2:0]			VRHN[4:0]					-
4th parameter	1	EQ_VCLEN	-	-	-	VSP_FBOFF	XDK[2:0]			-
5th parameter	1	-	-	CLK_OPT[2:1]		FS0[3:0]				-
6th parameter		FS1[3:0]				FS2[3:0]				-
7th parameter	1	VGH_RATIO[2:0]			BTP[4:0]					-
8th parameter	1	VGL_RATIO[2:0]			BTN[4:0]					-
9th parameter	1	-	VGHS[7:0]							-
10th parameter	1	-	VGLS[7:0]							-
11stparameter	1	DT1[1:0]		DT2[1:0]		DCDIV[3:0]				-
12ndparameter	1	-	DCS[2:0]			-	DC[2:0]			-
13rd parameter	1	-	DTPS[2:0]			-	DTP[2:0]			-
14th parameter	1	-	DTNS[2:0]			-	DTN[2:0]			-
Bank1 1st parameter	1	-	APF_EN	GASIOVCC_OPT[1:0]		-	GASVCI_OPT[2:0]			-
2nd parameter	1	GAS_CP_VSPVSN	GASVSN_OPT[2:0]			-	GASVSP_OPT[2:0]			-
3rd parameter	1	-	-	GASVGL_OPT[1:0]		-	-	GASVGH_OPT[1:0]		-

Description

This command is used to set related setting of power.

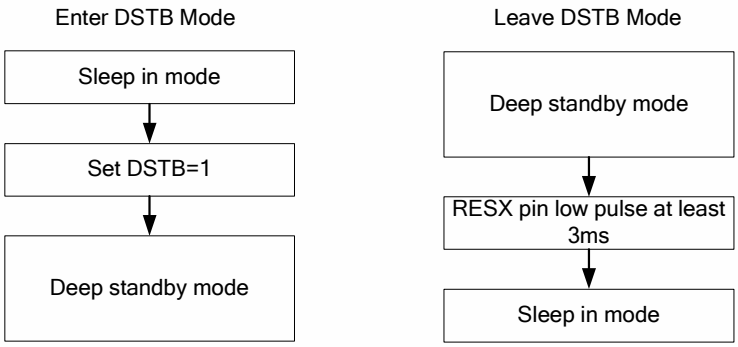
POWMOD[1:0]: Power mode selection

POWMOD[1]	POWMOD[0]	Power Mode
0	0	PFM Type A
0	1	PFM Type C
1	0	HX5186
1	1	inhibited

DSTBY_OPT: DSTB mode option. When DSTBY_OPT=0, logic power will be off and must HWRESET to leave deep standy mode.

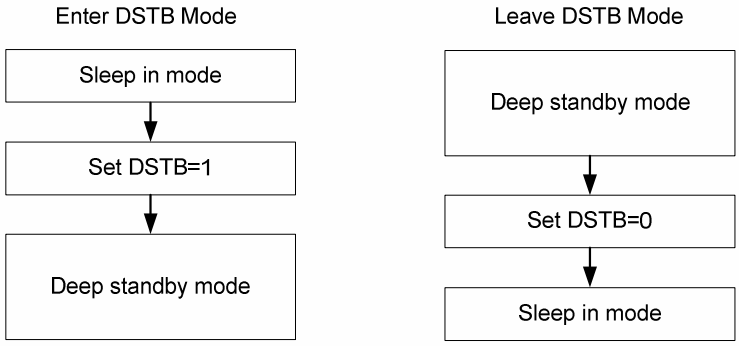
DSTB: Set '1' to enter deep standby mode for saving power in SLPIN mode. User must enter SLPIN mode before enter deep standby mode.

When DSTBY_OPT=0:



When DSTBY_OPT=1:

Himax Confidential Do not Copy



AP[2:0]: Adjust the amount of fixed current from the fixed current source for the operational amplifier in the power supply circuit. When the amount of fixed current is increased, the LCD driving capacity and the display quality are high, but the current consumption is increased. This is a tradeoff, Adjust the fixed current by considering both the display quality and the current consumption. During no display operation, when AP[2:0] = 000, the current consumption can be reduced by stopping the operations of operational amplifier and step-up circuit.

AP2	AP1	AP0	Constant Current of Operational Amplifier
0	0	0	Stop
0	0	1	0.5μA
0	1	0	1.0μA
0	1	1	1.5μA
1	0	0	2.0μA
1	0	1	2.5μA
1	1	0	3.0μA
1	1	1	3.5μA

VCLS[2:0]: Set VCL voltage

VCLS2	VCLS1	VCLS0	VCL voltage
0	0	0	-2.5V
0	0	1	-2.6V
0	1	0	-2.7V
0	1	1	-2.8V
1	0	0	-2.9V
1	0	1	-3.0V
1	1	0	-3.1V
1	1	1	-VDD3

VRHP[4:0]: VSPR regulator output control setting for source data output driving.

VRHP[4:0]					VSPR Voltage
0	0	0	0	0	VP_REF
0	0	0	0	1	3.1V
0	0	0	1	0	3.2V
0	0	0	1	1	3.3V
0	0	1	0	0	3.4V
0	0	1	0	1	3.5V
0	0	1	1	0	3.6V
0	0	1	1	1	3.7V
0	1	0	0	0	3.8V
0	1	0	0	1	3.9V
0	1	0	1	0	4.0V
0	1	0	1	1	4.1V
0	1	1	0	0	4.2V
0	1	1	0	1	4.3V
0	1	1	1	0	4.4V
0	1	1	1	1	4.5V

Himax Confidential Do Not Copy

This information contained herein is the exclusive property of Himax and shall not be distributed, reproduced, or disclosed in whole or in part without prior written permission of Himax.

1	0	0	0	0	4.6V
1	0	0	0	1	4.7V
1	0	0	1	0	4.8V
1	0	0	1	1	4.9V
1	0	1	0	0	5.0V
1	0	1	0	1	5.1V
1	0	1	1	0	5.2V
1	0	1	1	1	5.3V
1	1	0	0	0	5.4V
1	1	0	0	1	5.5V
1	1	0	1	0	5.6V
1	1	0	1	1	5.7V
1	1	1	0	0	5.8V
1	1	1	0	1	5.9V
1	1	1	1	0	6.0V
1	1	1	1	1	6.1V

VRHN[4:0]: VSNR regulator output control setting for source data output driving

VRHN[4:0]					VSNR Voltage
0	0	0	0	0	VN_REF
0	0	0	0	1	-3.1V
0	0	0	1	0	-3.2V
0	0	0	1	1	-3.3V
0	0	1	0	0	-3.4V
0	0	1	0	1	-3.5V
0	0	1	1	0	-3.6V
0	0	1	1	1	-3.7V
0	1	0	0	0	-3.8V
0	1	0	0	1	-3.9V
0	1	0	1	0	-4.0V
0	1	0	1	1	-4.1V
0	1	1	0	0	-4.2V
0	1	1	0	1	-4.3V
0	1	1	1	0	-4.4V
0	1	1	1	1	-4.5V
1	0	0	0	0	-4.6V
1	0	0	0	1	-4.7V
1	0	0	1	0	-4.8V
1	0	0	1	1	-4.9V
1	0	1	0	0	-5.0V
1	0	1	0	1	-5.1V
1	0	1	1	0	-5.2V
1	0	1	1	1	-5.3V
1	1	0	0	0	-5.4V
1	1	0	0	1	-5.5V
1	1	0	1	0	-5.6V
1	1	0	1	1	-5.7V
1	1	1	0	0	-5.8V
1	1	1	0	1	-5.9V
1	1	1	1	0	-6.0V
1	1	1	1	1	-6.1V

VPPS[2:0]: Set VPP output voltage for OTP.

VPPS[2:0]			VPP voltage
0	0	0	8.0V
0	0	1	8.1V
0	1	0	8.2V
0	1	1	8.25V

Himax Confidential Do Not Copy

1	0	0	8.3V
1	0	1	8.4V
1	1	0	8.5V
1	1	1	External from VGH

CLK_OPT1: The pumping clock of VGH will reset with Hsync when CLK_OPT1 = 1.

CLK_OPT2: The pumping clock of VGL will reset with Hsync when CLK_OPT2 = 1.

EQ_VCLEN: VCL enable bit. 0: Disable(sharing mode), 1: Enable(VCL level). Set to 0 for external VSP mode.

VSP_FBOFF: VSP voltage feedback to control VSP pumping clock operation. "1" no feecdback. It is effective only for HX5186 mode, not for PFM circuit.

XDK[2:0]: Setting VSP Voltage pumping ratio in HX5186 mode.

XDK2	XDK1	XDK0	VSP
0	0	0	inhibit
0	0	1	X2
0	1	0	X1.5
0	1	1	inhibit
1	0	0	inhibit
1	0	1	inhibit
1	1	0	X3
1	1	1	X3

FS0[3:0]: Set the operating frequency of the step-up circuit for VSP/VSN voltage generation. (Fosc_pump=5MHz)

FS0[3:0]				Operation Frequency of Step-up Circuit
0	0	0	0	Fosc_pump/2
0	0	0	1	Fosc_pump/4
0	0	1	0	Fosc_pump/8
0	0	1	1	Fosc_pump/16
0	1	0	0	Fosc_pump/32
0	1	0	1	Fosc_pump/48
0	1	1	0	Fosc_pump/64
0	1	1	1	Fosc_pump/80
1	0	0	0	Fosc_pump/96
1	0	0	1	Fosc_pump/112
1	0	1	0	Fosc_pump/128
1	0	1	1	Fosc_pump/144
1	1	0	0	Fosc_pump/160
1	1	0	1	Fosc_pump/176
1	1	1	0	Fosc_pump/192
1	1	1	1	Fosc_pump/208

FS1[3:0]: Set the operating frequency of the step-up circuit for VGH voltage generation. (Fosc_pump=5MHz)

FS1[3:0]				Operation Frequency of Step-up Circuit
0	0	0	0	Fosc_pump/72
0	0	0	1	Fosc_pump/96
0	0	1	0	Fosc_pump/128
0	0	1	1	Fosc_pump/160
0	1	0	0	Fosc_pump/192
0	1	0	1	Fosc_pump/224
0	1	1	0	Fosc_pump/256
0	1	1	1	Fosc_pump/336
1	0	0	0	Hsync*4

Himax Confidential Do Not Copy

1	0	0	1	Hsync*2
1	0	1	0	Hsync
1	0	1	1	Hsync/2
1	1	0	0	Hsync/4
1	1	0	1	Hsync/8
1	1	1	0	Hsync/16
1	1	1	1	Inhibited

FS2[3:0]: No function

BTP[4:0]: Switch the output factor for DC/DC circuit for VSP voltage generation. The LCD drive voltage level VSP can be selected according to the characteristic of liquid crystal which panel used.

BTP4	BTP3	BTP2	BTP1	BTP0	VSP Voltage
0	0	0	0	0	3.00V
0	0	0	0	1	3.15V
0	0	0	1	0	3.30V
0	0	0	1	1	3.45V
0	0	1	0	0	3.60V
:					:
1	0	0	0	0	5.40V
1	0	0	0	1	5.55V
1	0	0	1	0	5.70V
1	0	0	1	1	5.85V
1	0	1	0	0	6.00V
1	0	1	0	1	6.15V
1	0	1	1	0	6.30V
1	0	1	1	1	6.45V
1	1	0	0	0	6.60V
Others					Inhibited

BTN[4:0]: Switch the output factor of DC/DC circuit for VSN voltage generation. The LCD drive voltage level VSN can be selected according to the characteristic of liquid crystal which panel used.

BTN4	BTN3	BTN2	BTN1	BTN0	VSN Voltage
0	0	0	0	0	-3.00V
0	0	0	0	1	-3.15V
0	0	0	1	0	-3.30V
0	0	0	1	1	-3.45V
0	0	1	0	0	-3.60V
:					:
1	0	0	0	0	-5.40V
1	0	0	0	1	-5.55V
1	0	0	1	0	-5.70V
1	0	0	1	1	-5.85V
1	0	1	0	0	-6.00V
1	0	1	0	1	-6.15V
1	0	1	1	0	-6.30V
1	0	1	1	1	-6.45V
1	1	0	0	0	-6.60V
Others					Inhibited

VGH_RATIO[2:0]: Specify the VGH voltage source.

VGH_RATIO[2]	VGH_RATIO[1]	VGH_RATIO[0]	VGH Voltage pumping ratio
0	0	0	VDD3-VSN
0	0	1	VSP-VSN
0	1	0	VSP-VSN+VDD3
0	1	1	2*VSP-VSN
1	0	0	VDD3+VSP-2*VSN

Himax Confidential Do Not Copy

1	0	1	2*VSP-2*VSN
1	1	0	VDD3+2*VSP-2*VSN
1	1	1	3*VSP-2*VSN

VGHS[6:0]: VGH regulator output voltage setting. The LCD drive voltage level VGH can be selected according to the characteristic of liquid crystal which panel used.

VGHS[6:0]							VGH Voltage
0	0	0	0	0	0	0	7.3V
0	0	0	0	0	0	1	7.4V
0	0	0	0	0	1	0	7.5V
0	0	0	0	0	1	1	7.6V
:							:
0	1	1	0	0	1	1	12.4V
0	1	1	0	1	0	0	12.5V
0	1	1	0	1	0	1	12.6V
0	1	1	0	1	1	0	12.7V
:							:
1	1	1	1	1	0	1	19.8V
1	1	1	1	1	1	0	19.9V
1	1	1	1	1	1	1	20.0V

VGL_RATIO[2:0]: Specify the VGL voltage source.

VGL_RATIO[2]	VGL_RATIO[1]	VGL_RATIO[0]	VGL Voltage pumping ratio
0	0	0	VSN-VDD3
0	0	1	VSN-VSP
0	1	0	2*VSN-VDD3
0	1	1	2*VSN-VSP
1	0	0	2*VSN-VSP
1	0	1	2*VSN-VSP
1	1	0	2*VSN-VSP
1	1	1	2*VSN-VSP

VGLS[6:0]: VGL regulator output voltage setting. The LCD drive voltage level VGL can be selected according to the characteristic of liquid crystal which panel used.

VGLS[6:0]							VGL Voltage
0	0	0	0	0	0	0	-5.3V
0	0	0	0	0	0	1	-5.4V
0	0	0	0	0	1	0	-5.5V
0	0	0	0	0	1	1	-5.6V
:							:
1	0	0	0	1	1	1	-12.4V
1	0	0	1	0	0	0	-12.5V
1	0	0	1	0	0	1	-12.6V
1	0	0	1	0	1	0	-12.7V
:							:
1	1	1	1	1	0	1	-17.8V
1	1	1	1	1	1	0	-17.9V
1	1	1	1	1	1	1	-18.0V

DT1[1:0]: Delay time of power on and power off sequence.

DT1[1:0]		Delay time of power on and power off sequence
0	0	0ms
0	1	2.5ms
1	0	5ms
1	1	7.5ms

DT2[1:0]: Delay time of power on and power off sequence.

DT2[1:0]		Delay time of power on and power off sequence
0	0	0ms

Himax Confidential Do not Copy

0	1	2.5ms
1	0	5ms
1	1	7.5ms

DCDIV[3:0]: Set the normal operate frequency FoscD of DC/DC converter circuit during normal mode. (Fosc=45MHz)

DCDIV[3:0]				Normal operate frequency of DC/DC converter(foscD)
0	0	0	0	Fosc/1
0	0	0	1	Fosc/2
0	0	1	0	Fosc/3
0	0	1	1	Fosc/4
0	1	0	0	Fosc/5
0	1	0	1	Fosc/6
0	1	1	0	Fosc/7
0	1	1	1	Fosc/8
1	0	0	0	Fosc/9
1	0	0	1	Fosc/10
1	0	1	0	Fosc/11
1	0	1	1	Fosc/12
1	1	0	0	Fosc/13
1	1	0	1	Fosc/14
1	1	1	0	Fosc/15
1	1	1	1	Fosc/16

DCS[2:0]: Soft start VSP/VSN frequency of DC/DC clock.

DCS[2:0]			Operation Frequency of DC/DC Clock
0	0	0	foscD/4
0	0	1	foscD/5
0	1	0	foscD/6
0	1	1	foscD/7
1	0	0	foscD/8
1	0	1	foscD/10
1	1	0	foscD/11
1	1	1	foscD/12

DC[2:0]: Operation VSP/VSN frequency of DC/DC clock.

DC[2:0]			Operation Frequency of DC/DC Clock
0	0	0	foscD/4
0	0	1	foscD/5
0	1	0	foscD/6
0	1	1	foscD/7
1	0	0	foscD/8
1	0	1	foscD/10
1	1	0	foscD/11
1	1	1	foscD/12

DTPS[2:0]: For PFM circuit. Set the soft start operating duty cycle of VSP.

DTPS[2:0]			Soft start operation duty of VSP
0	0	0	1
0	0	1	2
0	1	0	3
0	1	1	4
1	0	0	5
1	0	1	6
1	1	0	7
1	1	1	8

DTP[2:0]: For PFM circuit:Set the operating duty cycle of VSP.

Himax Confidential Do Not Copy

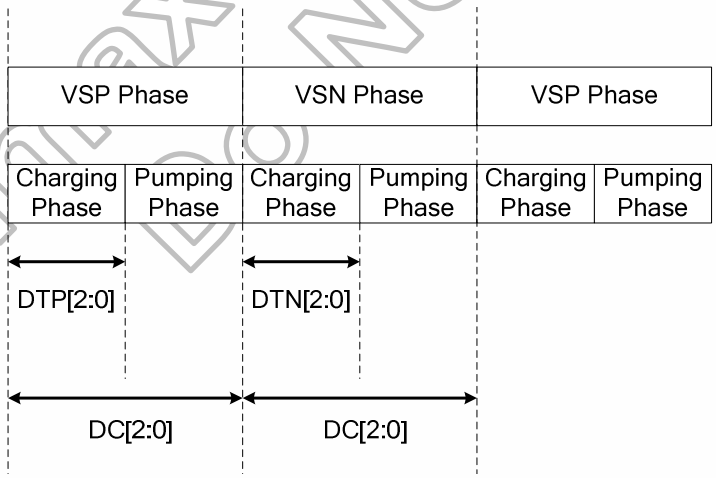
DTP[2:0]			Operation duty of VSP
0	0	0	1
0	0	1	2
0	1	0	3
0	1	1	4
1	0	0	5
1	0	1	6
1	1	0	7
1	1	1	8

DTNS[2:0]: For PFM circuit. Set the soft start operating duty cycle of VSN.

DTNS[2:0]			Soft start operation duty of VSN
0	0	0	1
0	0	1	2
0	1	0	3
0	1	1	4
1	0	0	5
1	0	1	6
1	1	0	7
1	1	1	8

DTN[2:0]: For PFM circuit:Set the operating duty cycle of VSN.

DTN[2:0]			Operation duty of VSN
0	0	0	1
0	0	1	2
0	1	0	3
0	1	1	4
1	0	0	5
1	0	1	6
1	1	0	7
1	1	1	8



APF_EN: Abnormal power-off detection enable **(GAS function)**. "1": Enable**.**

GASIOVCC_OPT[1:0]: Set VDD1 threshold voltage of GAS function.

GASIOVCC_OPT[1:0]		GAS threshold VDD1 voltage
0	0	1.2V
0	1	1.3V
1	0	1.4V
1	1	1.5V

GASVCI_OPT[2:0]: Set VDD3 threshold voltage of GAS function.

GASVCI_OPT[2:0]	GAS threshold VDD3 voltage

Himax Confidential Do Not Copy

0	0	0	1.9V
0	0	1	2.0V
0	1	0	2.1V
0	1	1	2.2V
1	0	0	2.3V
1	0	1	2.4V
1	1	0	2.5V
1	1	1	2.6V

GASVSN_OPT[2:0]: Set VSN threshold voltage of GAS function.

GASVSN_OPT[2:0]			GAS threshold VSN voltage
0	0	0	-2.8V
0	0	1	-3.0V
0	1	0	-3.2V
0	1	1	-3.4V
1	0	0	-3.6V
1	0	1	-3.8V
1	1	0	-4.0V
1	1	1	-4.2V

GASVSP_OPT[2:0]: Set VSP threshold voltage of GAS function.

GASVSP_OPT[2:0]			GAS threshold VSP voltage
0	0	0	2.8V
0	0	1	3.0V
0	1	0	3.2V
0	1	1	3.4V
1	0	0	3.6V
1	0	1	3.8V
1	1	0	4.0V
1	1	1	4.2V

GASVGL_OPT[1:0]: Set VGL threshold voltage of GAS function.

GASVGL_OPT[1:0]		GAS threshold VGL voltage
0	0	-6V
0	1	-7V
1	0	-8V
1	1	-9V

GASVGH_OPT[1:0]: Set VGH threshold voltage of GAS function.

GASVGH_OPT[1:0]		GAS threshold VGH voltage
0	0	6V
0	1	7V
1	0	8V
1	1	9V

Restrictions	SETEXTC turn on to enable this command.

Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Sleep In or Booster Off	Yes

Himax Confidential Do not copy

This information contained herein is the exclusive property of Himax and shall not be distributed, reproduced, or disclosed in whole or in part without prior written permission of Himax.

5.19.3 SETDISP: Set display related register (B2h)

B2H	SETDISP(Set display related register)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	0	1	1	0	0	1	0	B2
Bank 0 1nd parameter	1	-	ZZ_LR	ZZ_EO	ZZ_2PL	-	NW[2:0]			-
2rd parameter		MESSI _ENB	H_RES[2:0]			-	-	-	-	-
3th parameter	1	NL[7:0]								-
4th parameter	1	BP [7:0]								-
5th parameter	1	FP [7:0]								-
6th parameter	1	RTN[7:0]								-
7th parameter	1	-	-	-	-	-	-	-	-	-
8th parameter	1	-	-	-	-	-	-	-	-	-
9th parameter	1	-	-	-	-	-	-	-	-	-
10th parameter	1	-	-	-	-	-	-	-	-	-
11st parameter	1	FRM_PATTERN_CYCLE[3:0]				DISP_ BIST_E N	FRM_SCAN_CYCLE[2:0]			-

Description

This command is used to set display related register.

ZZ_LR: Zig-zag Left / Right mode selection.

ZZ_LR	Zig-zag Left / Right mode selection
0	S1~SR1
1	SL1~S2160

ZZ_EO: Zig-zag Odd / Even mode selection.

ZZ_EO	Zig-zag Odd / Even mode selection
0	Odd-line shift
1	Even-line shift

ZZ_2PL: Zig-zag 1H/2H selection when NW[2:0]=101

ZZ_2PL	Zig-zag 1H/2H selection
0	1H Zig-zag
1	2H Zig-zag

Himax Confidential Do not Copy

MESSI_ENB : Support Nokia DCS enable bit. "0" Enable.

0: support Nokia,don't support 44h CMD
1: support 44h CMD to adjust the position of TE(horizontal)

NW[2:0]: Inversion type setting.

NW2	NW1	NW0	Inversion type
0	0	0	Column inversion
0	0	1	1-dot inversion
0	1	0	2-dot inversion
0	1	1	4-dot inversion
1	0	0	8-dot inversion
1	0	1	Zig-zag
Others			Inhibited

H_RES[2:0]: Resolution selection.

H_RES[2:0]	Resolution	Source channels
000	720RGBx(480+ (8xNL))	S1~S2160
001	640RGBx(480+(8xNL))	S1~S960, S1201~S2160
010	600RGBx(480+(8xNL))	S1~S900,S1261~S2160
Others	inhibited	inhibited

NL[7:0]: Setting the number of lines to drive the LCD at an interval of 8 lines. The number of lines must be the same or more than the number of lines necessary for the size of the liquid crystal panel.

NL[7:0]								Line
0	0	0	0	0	0	0	0	480
0	0	0	0	0	0	0	1	488
0	0	0	0	0	0	1	0	496
0	0	0	0	0	0	1	1	504
				. . .				. . .
				. . .				. . .
0	1	0	1	0	0	1	0	1136
				. . .				. . .
0	1	1	0	0	1	0	0	1280
				. . .				. . .
1	0	1	1	0	1	0	0	1920
Others								Inhibited

BP[7:0] : Specify the amount of scan line for back porch(BP) in blanking.
FP[7:0]: Specify the amount of scan line for front porch (FP) in blanking.

FP[7:0] / BP[7:0]	Number of front porch/ back porch Lines
8h'00	2 lines
8h'01	3 lines
8h'02	4 lines
8h'03	5 lines
8h'04	6 lines
8h'05	7 lines
:	:
8h'FB	253 lines
8h'FC	254 lines
8h'FD	255 lines
8h'FE	256 lines
8h'FF	257 lines

Note: Set BP[7:0] = VS + VBP – 2, and FP[7:0] = VFP – 2.

Himax Confidential
Do Not Copy

RTN[7:0]: A cycle time of line width in blanking.
(1 TCON clock period= 4/45MHz)

RTN[7:0]	Clock per Line
8h'00	97 TCON CLK
8h'01	98 TCON CLK
8h'02	99 TCON CLK
:	:
8'hFD	350 TCON CLK
8'hFE	351 TCON CLK
8'hFF	352 TCON CLK

DISP_BIST_EN: Set "1" enable SW free running mode.

FRM_PATTERN_CYCLE[3:0]: Number of Free-running mode pattern.

FRM_PATTERN_CYCLE[3:0]	Free-running mode pattern number
0000	White
0001	White, Black
0010	White, Black, Red
0011	White, Black, Red, Blue
0100	White, Black, Red, Blue, Green
0101	White, Black, Red, Blue, Green, Gray127
0110	White, Black, Red, Blue, Green, Gray127, Color Bar
0111	White, Black, Red, Blue, Green, Gray127, Color Bar H-Grayscale
1000	White, Black, Red, Blue, Green, Gray127, Color Bar H-Grayscale, V-Grayscale
1001	White, Black, Red, Blue, Green, Gray127, Color Bar H-Grayscale, V-Grayscale, White pattern with black outline
1010	White, Black, Red, Blue, Green, Gray127, Color Bar H-Grayscale, V-Grayscale, White pattern with black outline, Checker 1x1
1011	White, Black, Red, Blue, Green, Gray127, Color Bar H-Grayscale, V-Grayscale, White pattern with black outline, Checker 1x1, Gray128
1100	White, Black, Red, Blue, Green, Gray127, Color Bar H-Grayscale, V-Grayscale, White pattern with black outline, Checker 1x1, Gray128, Black
Others	Inhibited

FRM_SCAN_CYCLE[2:0]: Free-running each pattern keeps time.

FRM_ SCAN_CYCLE[2:0]	Free-running mode pattern keep time
000	1 s
001	2 s
010	4 s
011	8 s
Others	Inhibited

Restrictions: SETEXTC turn on to enable this command

Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Sleep In or Booster Off	Yes

Himax Confidential Do Not Copy

5.19.4 SETCYC: Set display waveform cycles (B4h)

B4H	SETCYC(Set panel driving timing)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	0	1	1	0	1	0	0	B4
1st parameter	1	SPON[7:0]								-
2nd parameter	1	SPOFF[7:0]								-
3rd parameter	1	CON[7:0]								-
4th parameter	1	COFF[7:0]								-
5th parameter	1	CON1[7:0]								-
6th parameter	1	COFF1[7:0]								-
7th parameter	1	EQON1[7:0]								-
8th parameter	1	SON[7:0]								-
9th parameter	1	SOFF[7:0]								-
10th parameter	1	SAP1_P[3:0]				-	SAP2[2:0]			-
11st parameter	1	-	DX2_EN	EQ_ON	EQ_MIX	-	-	-	-	-
12nd parameter	1	DX2OFF[7:0]								-
13rd parameter	1	SPON_MPU[7:0]								-
14th parameter	1	SPOFF_MPU[7:0]								-
15th parameter	1	CON_MPU[7:0]								-
16th parameter	1	COFF_MPU[7:0]								-
17th parameter	1	CON1_MPU[7:0]								-
18th parameter	1	COFF1_MPU[7:0]								-
19th parameter	1	EQON1_MPU[7:0]								-
20th parameter	1	SON_MPU[7:0]								-
21st parameter	1	SOFF_MPU[7:0]								-
22nd parameter	1	DX2OFF_MPU[7:0]								-

Description

This command is used to set display waveform cycles.

SPON[7:0]: Fine tune the Start and End signal delay from original starting point.
SPON_MPU[7:0]: Fine tune the Start and End signal delay from original starting point for blanking frame.

(1 TCON CLK period = 4/45MHz)

SPON[7:0]/ SPON_MPU[7:0]	Start / END signal output start delay
0x00h	0 TCON CLK
0x01h	1TCON CLK
0x02h	2 TCON CLK
0x03h	3 TCON CLK
:	
0xFEh	254 TCON CLK
0xFFh	255 TCON CLK

SPOFF[7:0]: Fine tune the Start and End signal ending point.
SPOFF_MPU[7:0]: Fine tune the Start and End signal ending point for blanking frame.

SPOFF[7:0]/ SPOFF_MPU[7:0]	Start / END signal output end delay
0x00h	0 TCON CLK
0x01h	1 TCON CLK
0x02h	2 TCON CLK
0x03h	3 TCON CLK
:	
0xFEh	254 TCON CLK
0xFFh	255 TCON CLK

Note: When output Start / End signal width is 1- Hsync only, set SPON[7:0] < SPOFF[7:0]

CON[7:0]/CON1[7:0]: Fine tune the Clock signal delay from original starting point.
CON_MPU[7:0]/CON1_MPU[7:0]: Fine tune the Clock signal delay from original starting point for blanking frame.

Himax Confidential Do Not Copy

CON[7:0]/ CON_MPU[7:0] CON1[7:0]/ CON1_MPU[7:0]	Clock signal output start delay
0x00h	0 TCON CLK
0x01h	1 TCON CLK
0x02h	2 TCON CLK
0x03h	3 TCON CLK
:	
0xFEh	254 TCON CLK
0xFFh	255 TCON CLK

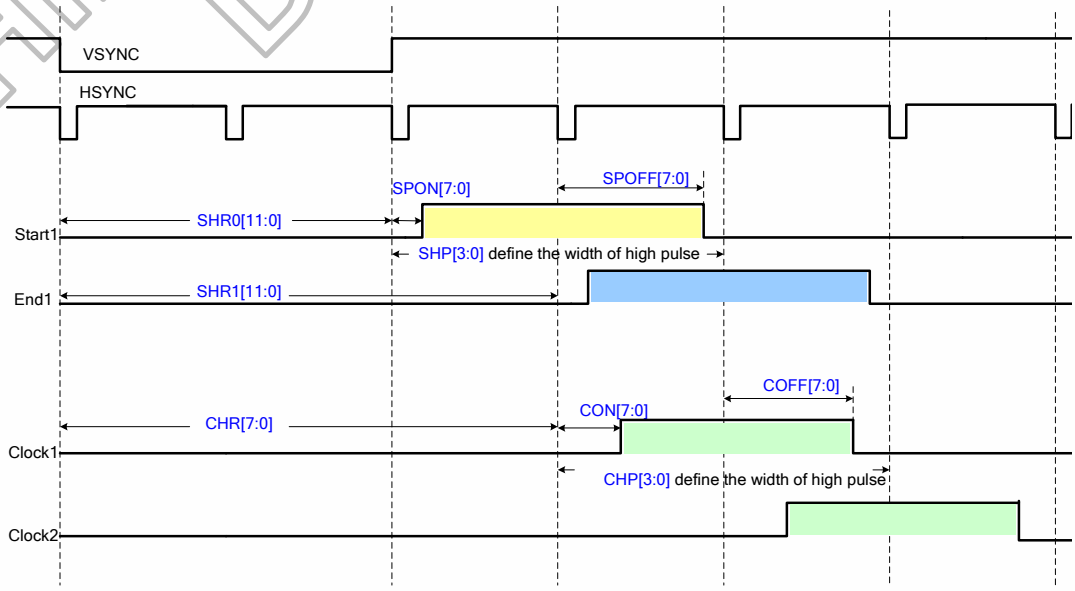
COFF[7:0]/COFF1[7:0]: Fine tune the Clock signal ending point.
COFF_MPU[7:0]/COFF1_MPU[7:0]: Fine tune the Clock signal ending point for blanking frame.

COFF[7:0]/ COFF_MPU[7:0] COFF1[7:0]/ COFF1_MPU[7:0]	Clock signal output end delay
0x00h	0 TCON CLK
0x01h	1 TCON CLK
0x02h	2 TCON CLK
0x03h	3 TCON CLK
:	
0xFEh	254 TCON CLK
0xFFh	255 TCON CLK

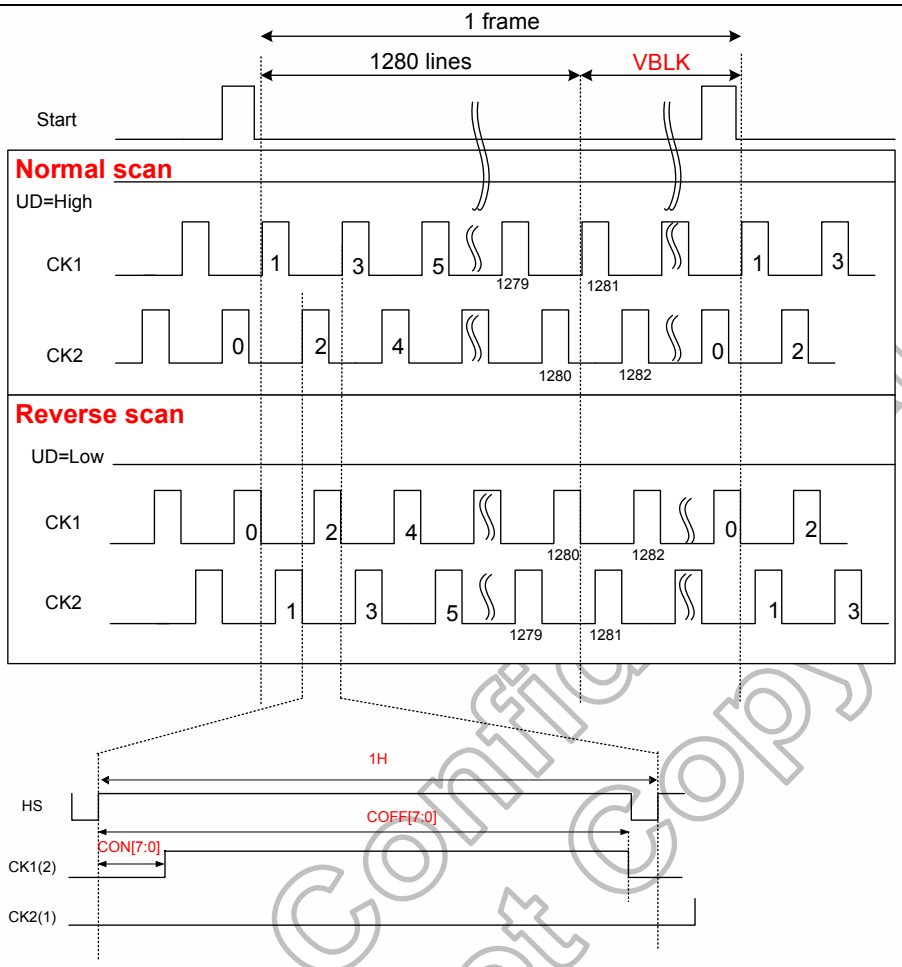
Note: When output Clock signal width is 1- Hsync only, set COFF[7:0] ≧ CON[7:0] + 2

SON[7:0]: Source OP turn on time.
SOFF[7:0]: Source OP turn off time.
SON_MPU[7:0]: Source OP turn on time for blanking frame.
SOFF_MPU[7:0]: Source OP turn off time for blanking frame.

SON[7:0]/SON_MPU[7:0] SOFF[7:0]/ SOFF_MPU[7:0]	Source OP turn on/off time
0x00h	0 TCON CLK
0x01h	1 TCON CLK
0x02h	2 TCON CLK
:	
0xFEh	254 TCON CLK
0xFFh	255 TCON CLK



Himax Confidential
Do Not Copy



EQON1[7:0]: Specify the source EQ period.
EQON1_MPU[7:0]: Specify the source EQ period for blanking frame.

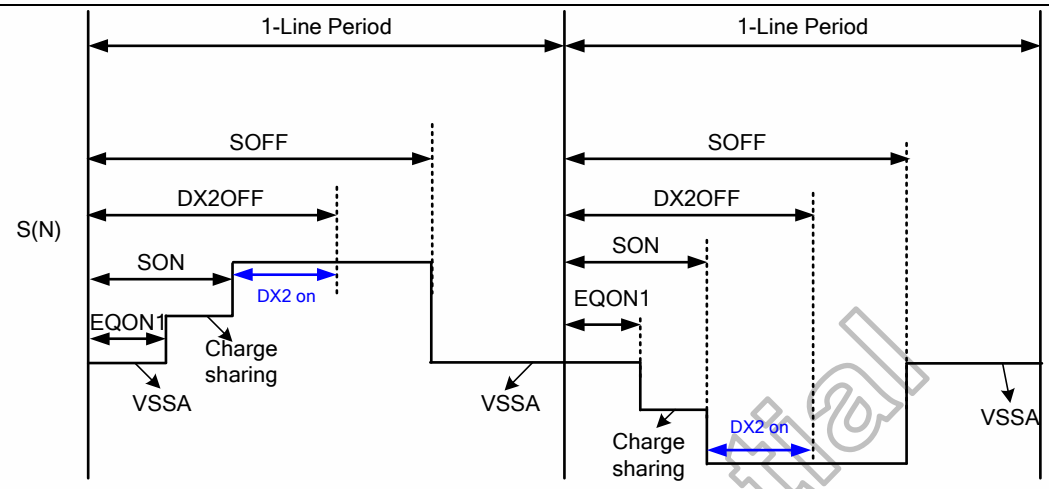
EQON1[7:0]/ EQON1_MPU[7:0]]	Source EQ time
0x00h	0 TCON CLK
0x01h	1 TCON CLK
0x02h	2 TCON CLK
0x03h	3 TCON CLK
:	
0xFEh	254 TCON CLK
0xFFh	255 TCON CLK

DX2OFF[7:0]: Source driving enhancement period setting.
DX2OFF_MPU[7:0]: Source driving enhancement period setting for blanking frame.

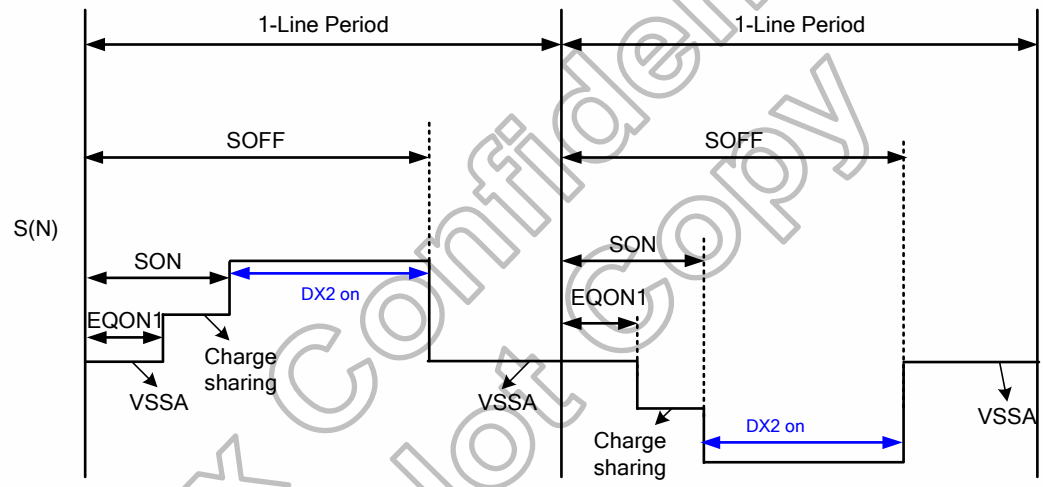
DX2OFF[7:0]/DX2OFF_MPU[7:0]	Set source enhancement time
0x00h	0 TCON CLK
0x01h	1 TCON CLK
0x02h	2 TCON CLK
0x03h	3 TCON CLK
:	
0xFEh	254 TCON CLK
0xFFh	255 TCON CLK

DX2_EN: DX2 on period selection
"1": SON ~ DX2OFF

Himax Confidential Do Not Copy



"0": SON ~ SOFF



SAP1_P[3:0]: 1st stage OP bias current adjust

SAP2[2:0]: class-AB (2nd) stage OP bias current adjust:

SAP2[2:0] (SAP1_P[3:0] =0011)			1st stage OP	Class AB OP	Total
0	0	0	0.757 uA	0.224uA	0.981 uA
0	0	1	0.778 uA	0.245uA	1.023 uA
0	1	0	0.815 uA	0.271uA	1.086 uA
0	1	1	0.857 uA	0.306uA	1.163 uA
1	0	0	0.906 uA	0.346uA	1.252 uA
1	0	1	0.954 uA	0.383uA	1.337uA
1	1	0	1.008 uA	0.423uA	1.431 uA
1	1	1	1.059 uA	0.463uA	1.522 uA

Total OP biase current = 1st OP current + class-AB biase current
Different SAP1_P[3:0] will have map to diffent class-AB current setting

Himax Confidential
Do Not Copy

SAP1_P[3:0] (SAP2[2:0]=011)				1st stage OP	Class AB OP	Total
0	0	0	0	0.222 uA	0.076 uA	0.298 uA
0	0	0	1	0.431 uA	0.149 uA	0.58 uA
0	0	1	0	0.657 uA	0.23 uA	0.887uA
0	0	1	1	0.857 uA	0.307 uA	1.164 uA
0	1	0	0	1.075 uA	0.392 uA	1.467 uA
0	1	0	1	1.269 uA	0.477 uA	1.746uA
0	1	1	0	1.484 uA	0.563 uA	2.047 uA
0	1	1	1	1.677 uA	0.644 uA	2.321 uA
1	0	0	0	1.888 uA	0.738 uA	2.626 uA
1	0	0	1	2.079 uA	0.823uA	2.902 uA
1	0	1	0	2.289 uA	0.916 uA	3.205 uA
1	0	1	1	2.476 uA	0.999 uA	3.475 uA
1	1	0	0	2.683 uA	1.096 uA	3.779 uA
1	1	0	1	2.874 uA	1.182 uA	4.056 uA
1	1	1	0	3.074 uA	1.277 uA	4.351 uA
1	1	1	1	3.258 uA	1.359 uA	4.617 uA

EQ_ON/ EQ_MIX:

EQ_ON	0	0	1	1
EQ_MIX	0	1	0	1
EQ to VCI/VCL	Sharing mode	Frist line EQ	All line EQ	All line EQ

Restrictions: -

Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Sleep In or Booster Off	Yes

Himax Confidential
Do Not Copy

5.19.5 SETVCOM: Set VCOM voltage (B6h)

B6 H	SETVCOM (Set VCOM Voltage)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	0	1	1	0	1	1	0	B6
1st parameter	1	VCMC_F[7:0]								-
2nd parameter	1	VCMC_B[7:0]								-
3rd parameter	1	VCOM_TIMES[2:0]			-	-	-	VCMC_B[8]	VCMC_F[8]	-

Description

This command is used to set VCOM Voltage.

VCMC_F[8:0]: Forward scan VCOM voltage control.

VCMC_B[8:0]: Backward scan VCOM voltage control.

VCMC_F[8:0]/VCMC_B[8:0]									VCOM
0	0	0	0	0	0	0	0	0	-0.30V
0	0	0	0	0	0	0	0	1	-0.31V
0	0	0	0	0	0	0	1	0	-0.32V
0	0	0	0	0	0	0	1	1	-0.33V
:									:
1	0	1	1	1	0	0	0	0	-3.98V
1	0	1	1	1	0	0	0	1	-3.99V
1	0	1	1	1	0	0	1	0	-4.00V
Others									Inhibited
1	1	1	1	1	1	1	1	0	VSSA
1	1	1	1	1	1	1	1	1	HZ

VCOM_TIMES[2:0]: Read the VCOM OTP programmed times.

VCOM_TIMES[2:0]	VCOM OTP programmed times
111	No programmed
011	VCOM has been programmed 1 time
001	VCOM has been programmed 2 times
000	VCOM has been programmed 3 times

Restrictions: SETEXTC turn on to enable this command.

Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Sleep In or Booster Off	Yes

Himax Confidential Do Not Copy

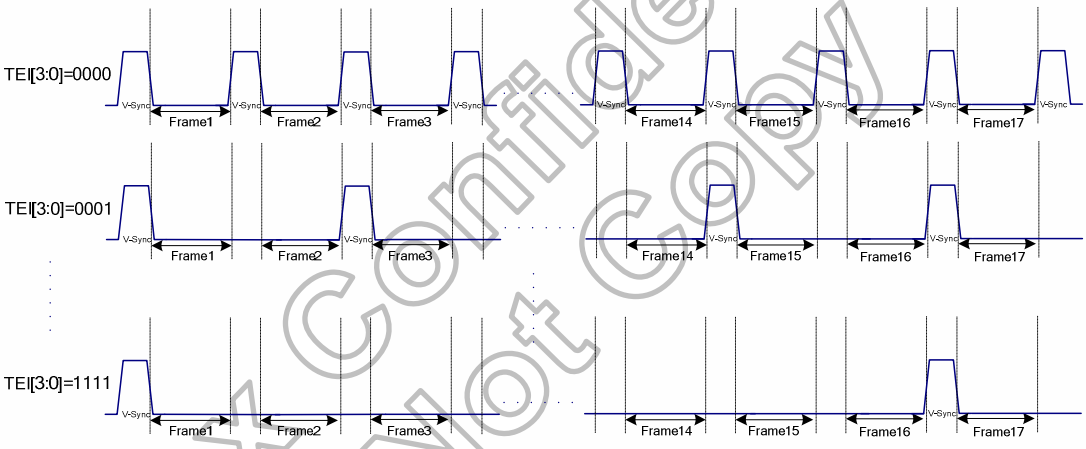
5.19.6 SETTE: Set internal TE function (B7h)

B7H	SETTE (Set internal TE function)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	0	1	1	0	1	1	1	B7
1st parameter	1	-	TEI[3:0]				TEP[10:8]			-
2nd parameter	1	TEP[7:0]								-
3rd parameter	1	TE_SEL[3:0]				TE1_SEL[3:0]				-

Description

TEI[3:0]: Set the output interval of TE signal according to the display data rewrite cycle and data transfer rate.

TEI3	TEI2	TEI1	TEI0	Output Interval
0	0	0	0	1 frame
0	0	0	1	2 frames
0	0	1	0	3 frames
:				:
1	1	1	0	15 frames
1	1	1	1	16 frames



TEP[10:0]: Set the output position of frame cycle signal. TE can be used as the trigger signal for frame synchronous write operation.
Make sure the setting restriction 11'h000 ≤ TEP[10:0] ≤ Numbers of Line-1.

TEP[10:0]	Output position
000h	0th line
001h	1st line
002h	2nd line
003h	3rd line
• • •	• • •
4FEh	1278th line
4FFh	1279th line
500h	1280th line

TE_SEL[3:0]: TE pin output signal select

TE_SEL[3:0]	TE pin Output signal
0000	Original TE
0001	V-blanking
0010	H-blanking
0011	Source driving period
Others	Inhibited

TE1_SEL[3:0]: TE1 pin output signal select

TE1_SEL[3:0]	TE1 pin Output signal
0000	Original TE
0001	V-blanking

Himax Confidential Do Not Copy

	0010 H-blanking 0011 Source driving period Others Inhibited
Restrictions	SETEXTC turn on to enable this command.
Register Availability	Status: Availability Normal Mode On, Idle Mode Off, Sleep Out: Yes Normal Mode On, Idle Mode On, Sleep Out: Yes Sleep In or Booster Off: Yes

Himax Confidential
Do Not Copy

5.19.7 SETSENSOR: Set temperature sensor (B8h)

B8H	SETSENSOR (Set temperature sensor)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	0	1	1	1	0	0	0	B8
Bank0 1st parameter	1	-	TSENSOR_EN	LT_EN	HT_EN	-	-	-	-	-
Description	**TSENSOR_EN:** Set "1" Enable Temperature sensor function. **LT_EN:** Set "1" Enable Low Temperature. **HT_EN:** Set "1" Enable High Temperature.									
Restrictions	SETEXTC turn on to enable this command.									
Register Availability	Status: Normal Mode On, Idle Mode Off, Sleep Out — Availability: Yes Status: Normal Mode On, Idle Mode On, Sleep Out — Availability: Yes Status: Sleep In or Booster Off — Availability: Yes									

Himax Confidential
Do Not Copy

5.19.8 SETEXTC: Set extension command (B9h)

<table>
<tr><th rowspan="2">B9H</th><th colspan="10">SETEXTC (Set extended command set)</th></tr>
<tr><th>DCX</th><th>D7</th><th>D6</th><th>D5</th><th>D4</th><th>D3</th><th>D2</th><th>D1</th><th>D0</th><th>HEX</th></tr>
<tr><td>Command</td><td>0</td><td>1</td><td>0</td><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td><td>1</td><td>B9</td></tr>
<tr><td>1st parameter</td><td>1</td><td colspan="8">EXTC1[7:0]</td><td>-</td></tr>
<tr><td>2nd parameter</td><td>1</td><td colspan="8">EXTC2[7:0]</td><td>-</td></tr>
<tr><td>3rd parameter</td><td>1</td><td colspan="8">EXTC3[7:0]</td><td>-</td></tr>
<tr><td>Description</td><td colspan="11">This command is used to set extended command set access enable.
<table>
<tr><th>Extend cmd</th><th>Command description</th></tr>
<tr><td>Enable</td><td>After command (B9h), write 3 parameters (FFh, 83h, 94h) by order</td></tr>
<tr><td>Disable (default)</td><td>After command (B9h), write 3 parameters (xxh,xxh,xxh) any value except (FFh, 83h, 94h)</td></tr>
</table></td></tr>
<tr><td>Restrictions</td><td colspan="11">-</td></tr>
<tr><td>Register Availability</td><td colspan="11">
<table>
<tr><th>Status</th><th>Availability</th></tr>
<tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr>
<tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr>
<tr><td>Sleep In or Booster Off</td><td>Yes</td></tr>
</table></td></tr>
</table>

Himax Confidential
Do Not Copy

5.19.9 SETMIPI: Set MIPI control (BAh)

BAH	SETMIPI(Set MIPI control)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	0	1	1	1	0	1	0	BA
1st parameter	1	-	DSISETUP0[6:0]							-
2nd parameter	1	DSISETUP1[7:0]								-
3rd parameter	1	DPHYCMD0[7:0]								-
4th parameter	1	DPHYCMD1[7:0]								-

Description

This command is used to set MIPI DSI Related Setting.

DSISETUP0[6]	Tx Type: Define the LP-TX BTA behavior when there are error. 0: only BTA Error 1: BTA Read + Error
DSISETUP0[5]	CD_disable: Define the contention detection (LP-CD) function. 0 : LP-CD function enable 1: LP-CD function disable
DSISETUP0[4]	Tx_OscDiv: LP-TX clock (T_{LPX}) selection. 0 : 50ns (10MHz)(osc_clk) 1 :100ns (5MHz)(osc_clk/2)
DSISETUP0[3:2]	Internal option
DSISETUP0[1:0]	LAN_NUM: Define the DSI lane number 00 : 1-lane 01 : 2-lane 10 : 3-lane 11 : 4-lane

DSISETUP1[7]	RxCRC_enable: Enable RX CRC Check.
DSISETUP1[6]	ECC_ignore: Define the RX behavior when error occurring. 0 : the transmission will be broken when there are ECC or CRC error 1 : the transmission will keep when there are ECC or CRC error
DSISETUP1[5]	Internal option
DSISETUP1[4]	conti_w_en: 29h continuous write function 0:disable,1:enable
DSISETUP1[3:2]	Txe_Wait: BTA from Tx into Rx overlap waiting time counter(T_{TA-GO}) 00: TA-go = 2 Tlpx 01: TA-go = 4 Tlpx 10: TA-go = 6 Tlpx 11: TA-go = 8 Tlpx
DSISETUP1[1:0]	Txs_Wait: BTA from Rx into Tx overlap waiting time counter(T_{TA-GET}) 00: Disable, no wait time 01: TA-get = 2 Tlpx 10: TA-get = 4 Tlpx 11: TA-get = 6 Tlpx

DPHYCMD0[7:6]	LP-RX LPF delay capacitance setting, for clock path: 00=4ns, 01=8ns(default), 10=13ns, 11=2ns
DPHYCMD0[5:4]	LP-RX VHYS trimming: 00=92mV, 01=37mV, 10=66mV (default), 11=3mV (w/o hysteresis)
DPHYCMD0[3]	LP-TX Slew Rate trimming: 0=w/i S/R (default), 1=relief SR limit
DPHYCMD0[2]	LP-TX Slew Rate trimming: 0=fast S/R, 1=slow S/R(default)
DPHYCMD0[1]	TXTEN:Enable LP-TX: 0=disable (default), floating output, 1=enable
DPHYCMD0[0]	Force: Force LP-TX output logic '1' or '0': 0=output '0' (default), 1=output '1'. To enable this function, TXTEN must be set to 1

DPHYCMD1[7:5]	LDOHB: HS_LDO output voltage trimming for DSI: 000=1.31V, 001=1.48V, 010=1.52V, 011=1.57V (dsi default), 100=1.67V, 101=1.71V, 110=1.75V, 111=1.80V
DPHYCMD1[4:3]	IHSRB[1:0]:HS-RX current trimming: 00=8x, 01=7x, 10=6x (default),

Himax Confidential
Do not Copy

	11=5x
DPHYCMD1[2:1]	RTB: HS-RX Rterm terminator resistance value trimming for DSI: 00=123ohm, 01=100ohm, 1x=84ohm
DPHYCMD1[0]	ILPCB: LP-CD current trimming: 0=4x, 1=2x (default)

Restrictions

SETEXTC turn on to enable this command.

Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Sleep In or Booster Off	Yes

Himax Confidential
Do Not Copy

5.19.10 SETOTP: Set OTP (BBh)

BBH	SETOTP(Set OTP Related Setting)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	0	1	1	1	0	1	1	BB
1st parameter	1	INTVPP_EN	-	OTP_PROG_ALL	-	-	-	OTP_INDEX[9:8]		-
2nd parameter	1	OTP_INDEX[7:0]								-
3rd parameter	1	OTP_DATA[7:0]								-
4th parameter	1	OTP_POR	OTP_PWE	OTP_PWR_SEL	OTP_PTM[2:0]			OTP_TEST	OTP_PROG	-
5th parameter	1	OTP_DATA_READ[7:0]								-
6th parameter	1	OTP_KEY0[7:0]								-
7th parameter	1	OTP_KEY1[7:0]								-
8th parameter	1	OTP_MASK[7:0]								-
9th parameter	1	-	-	-	-	-	-	OTP_BIT[1:0]		-

Description

This command is used to set OTP related setting.

OTP_KEY0[7:0], OTP_KEY1[7:0]	Description	Note
OTP_KEY0[7:0] = 0xAAh OTP_KEY1[7:0] = 0x55h	Enter OTP program mode	
OTP_KEY0[7:0] = 0x00h OTP_KEY1[7:0] = 0x00h	Leave OTP program mode	
Other value	Invalid	If HX8394-F operate on OTP program mode, Then keep on OTP program mode. If HX8394-F operate on non-OTP program mode, Then keep on non-OTP program mode.

INTVPP_EN: OTP_PWR power selected.
"0" : External OTP_PWR is selected when programmed.
"1" : Internal OTP_PWR is selected when programmed.

OTP_PROG_ALL: When set to "1', all OTP index is programmed, except for CAh, E5h, E6h.

OTP_INDEX[9:0]: Set index of OTP table for programming.

OTP_PROG: When set to "1', the register content of OTP index is programmed.

OTP_PWR_SEL: When written to "1", OTP power voltage is fed to OTP circuit.

OTP_PTM[2:0]: For test mode using. Not open.

OTP_PWE: OTP program write enable, "1" means OTP is able to be programmed.

OTP_POR: Pulse for OTP data read operation.

OTP_TEST: "0", setting OTP_PROG high will trigger internal state machine.
"1", setting OTP_PROG high will not trigger internal state machine.

OTP_DATA[7:0]: Write data of OTP index.

OTP_DATA_READ[7:0]: Read back the OTP index data.

OTP_MASK[7:0]: Bit programming mask, "1" means this bit can't be programmed.

Himax Confidential Do Not Copy

This information contained herein is the exclusive property of Himax and shall not be distributed, reproduced, or disclosed in whole or in part without prior written permission of Himax.

	OTP_BIT[1:0]: Internal programming bit number per cycle. 00: 1 bit/cycle 01: 2 bits/cycle 10: 4 bits/cycle 11: 8 bits/cycle
Restrictions	SETEXTC turn on to enable this command.
Register Availability	Status / Availability Normal Mode On, Idle Mode Off, Sleep Out / Yes Normal Mode On, Idle Mode On, Sleep Out / Yes Sleep In or Booster Off / Yes

Himax Confidential
Do Not Copy

5.19.11 Set register bank (BDh)

BDH	SET_BANK(Set register bank)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	0	1	1	1	1	0	1	BD
1st parameter	1	-	-	-	-	-	-	BANK_INDEX [1:0]		-
Description	Set the register bank for some Commands. This command is active only for RB1h/RB2h/RB8h/RC1h/RD3h/RD8h. For example: Write RC1h Bank1 Step1: write RBDh = 01h Step2: write RC1h, PA1~42 Read RC1h Bank2 Step1: write RBDh = 02h Step2: read RC1h, PA1~42									
Restrictions	SETEXTC turn on to enable this command.									
Register Availability	Status: Normal Mode On, Idle Mode Off, Sleep Out — Availability: Yes Status: Normal Mode On, Idle Mode On, Sleep Out — Availability: Yes Status: Sleep In or Booster Off — Availability: Yes									

Himax Confidential
Do Not Copy

5.19.12 SETDGCLUT: Set DGC LUT (C1h)

C1H	SETDGCLUT (Set DGC LUT)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	1	0	0	0	0	0	1	C1
Bank0 1st parameter	1	-	-	-	-	-	-	DGC_INV	DGC_EN	-
2nd parameter	1	R_GAMMA0[9:2]								-
3rd parameter	1	R_GAMMA1[9:2]								-
4th parameter	1	R_GAMMA2[9:2]								-
5th parameter	1	R_GAMMA3[9:2]								-
6th parameter	1	R_GAMMA4[9:2]								-
7th parameter	1	R_GAMMA5[9:2]								-
8th parameter	1	R_GAMMA6[9:2]								-
9th parameter	1	R_GAMMA7[9:2]								-
10th parameter	1	R_GAMMA8[9:2]								-
11st parameter	1	R_GAMMA9[9:2]								-
12nd parameter	1	R_GAMMA10[9:2]								-
13rd parameter	1	R_GAMMA11[9:2]								-
14th parameter	1	R_GAMMA12[9:2]								-
15th parameter	1	R_GAMMA13[9:2]								-
16th parameter	1	R_GAMMA14[9:2]								-
17th parameter	1	R_GAMMA15[9:2]								-
18th parameter	1	R_GAMMA16[9:2]								-
19th parameter	1	R_GAMMA17[9:2]								-
20th parameter	1	R_GAMMA18[9:2]								-
21st parameter	1	R_GAMMA19[9:2]								-
22nd parameter	1	R_GAMMA20[9:2]								-
23rd parameter	1	R_GAMMA21[9:2]								-
24th parameter	1	R_GAMMA22[9:2]								-
25th parameter	1	R_GAMMA23[9:2]								-
26th parameter	1	R_GAMMA24[9:2]								-
27th parameter	1	R_GAMMA25[9:2]								-
28th parameter	1	R_GAMMA26[9:2]								-
29th parameter	1	R_GAMMA27[9:2]								-
30th parameter	1	R_GAMMA28[9:2]								-
31st parameter	1	R_GAMMA29[9:2]								-
32nd parameter	1	R_GAMMA30[9:2]								-
33rd parameter	1	R_GAMMA31[9:2]								-
34th parameter	1	R_GAMMA32[9:2]								-
35th parameter	1	R_GAMMA0[1:0]		R_GAMMA1[1:0]		R_GAMMA2[1:0]		R_GAMMA3[1:0]		-
36th parameter	1	R_GAMMA4[1:0]		R_GAMMA5[1:0]		R_GAMMA6[1:0]		R_GAMMA7[1:0]		-
37th parameter	1	R_GAMMA8[1:0]		R_GAMMA9[1:0]		R_GAMMA10[1:0]		R_GAMMA11[1:0]		-
38th parameter	1	R_GAMMA12[1:0]		R_GAMMA13[1:0]		R_GAMMA14[1:0]		R_GAMMA15[1:0]		-
39th parameter	1	R_GAMMA16[1:0]		R_GAMMA17[1:0]		R_GAMMA18[1:0]		R_GAMMA19[1:0]		-
40th parameter	1	R_GAMMA20[1:0]		R_GAMMA21[1:0]		R_GAMMA22[1:0]		R_GAMMA23[1:0]		-
41st parameter	1	R_GAMMA24[1:0]		R_GAMMA25[1:0]		R_GAMMA26[1:0]		R_GAMMA27[1:0]		-
42nd parameter	1	R_GAMMA28[1:0]		R_GAMMA29[1:0]		R_GAMMA30[1:0]		R_GAMMA31[1:0]		-

Himax Confidential Do Not Copy

43rd parameter	1	R_GAMMA32[1:0]	-	-	-	-	-	-	-
Bank1 1st parameter	1	G_GAMMA0[9:2]							-
2nd parameter	1	G_GAMMA1[9:2]							-
3rd parameter	1	G_GAMMA2[9:2]							-
4th parameter	1	G_GAMMA3[9:2]							-
5th parameter	1	G_GAMMA4[9:2]							-
6th parameter	1	G_GAMMA5[9:2]							-
7th parameter	1	G_GAMMA6[9:2]							-
8th parameter	1	G_GAMMA7[9:2]							-
9th parameter	1	G_GAMMA8[9:2]							-
10th parameter	1	G_GAMMA9[9:2]							-
11st parameter	1	G_GAMMA10[9:2]							-
12nd parameter	1	G_GAMMA11[9:2]							-
13rd parameter	1	G_GAMMA12[9:2]							-
14th parameter	1	G_GAMMA13[9:2]							-
15th parameter	1	G_GAMMA14[9:2]							-
16th parameter	1	G_GAMMA15[9:2]							-
17th parameter	1	G_GAMMA16[9:2]							-
18th parameter	1	G_GAMMA17[9:2]							-
19th parameter	1	G_GAMMA18[9:2]							-
20th parameter	1	G_GAMMA19[9:2]							-
21st parameter	1	G_GAMMA20[9:2]							-
22nd parameter	1	G_GAMMA21[9:2]							-
23rd parameter	1	G_GAMMA22[9:2]							-
24th parameter	1	G_GAMMA23[9:2]							-
25th parameter	1	G_GAMMA24[9:2]							-
26th parameter	1	G_GAMMA25[9:2]							-
27th parameter	1	G_GAMMA26[9:2]							-
28th parameter	1	G_GAMMA27[9:2]							-
29th parameter	1	G_GAMMA28[9:2]							-
30th parameter	1	G_GAMMA29[9:2]							-
31st parameter	1	G_GAMMA30[9:2]							-
32nd parameter	1	G_GAMMA31[9:2]							-
33rd parameter	1	G_GAMMA32[9:2]							-
34th parameter	1	G_GAMMA0[1:0]	G_GAMMA1[1:0]		G_GAMMA2[1:0]		G_GAMMA3[1:0]		-
35th parameter	1	G_GAMMA4[1:0]	G_GAMMA5[1:0]		G_GAMMA6[1:0]		G_GAMMA7[1:0]		-
36th parameter	1	G_GAMMA8[1:0]	G_GAMMA9[1:0]		G_GAMMA10[1:0]		G_GAMMA11[1:0]		-
37th parameter	1	G_GAMMA12[1:0]	G_GAMMA13[1:0]		G_GAMMA14[1:0]		G_GAMMA15[1:0]		-
38th parameter	1	G_GAMMA16[1:0]	G_GAMMA17[1:0]		G_GAMMA18[1:0]		G_GAMMA19[1:0]		-
39th parameter	1	G_GAMMA20[1:0]	G_GAMMA21[1:0]		G_GAMMA22[1:0]		G_GAMMA23[1:0]		-
40th parameter	1	G_GAMMA24[1:0]	G_GAMMA25[1:0]		G_GAMMA26[1:0]		G_GAMMA27[1:0]		-
41st parameter	1	G_GAMMA28[1:0]	G_GAMMA29[1:0]		G_GAMMA30[1:0]		G_GAMMA31[1:0]		-
42nd parameter	1	G_GAMMA32[1:0]	-	-	-	-	-	-	-
Bank2 1st parameter	1	G_GAMMA0[9:2]							-
2nd parameter	1	G_GAMMA1[9:2]							-

Himax Confidential
Do Not Copy

3rd parameter	1	G_GAMMA2[9:2]							-
4th parameter	1	G_GAMMA3[9:2]							-
5th parameter	1	G_GAMMA4[9:2]							-
6th parameter	1	G_GAMMA5[9:2]							-
7th parameter	1	G_GAMMA6[9:2]							-
8th parameter	1	G_GAMMA7[9:2]							-
9th parameter	1	G_GAMMA8[9:2]							-
10th parameter	1	G_GAMMA9[9:2]							-
11st parameter	1	G_GAMMA10[9:2]							-
12nd parameter	1	G_GAMMA11[9:2]							-
13rd parameter	1	G_GAMMA12[9:2]							-
14th parameter	1	G_GAMMA13[9:2]							-
15th parameter	1	G_GAMMA14[9:2]							-
16th parameter	1	G_GAMMA15[9:2]							-
17th parameter	1	G_GAMMA16[9:2]							-
18th parameter	1	G_GAMMA17[9:2]							-
19th parameter	1	G_GAMMA18[9:2]							-
20th parameter	1	G_GAMMA19[9:2]							-
21st parameter	1	G_GAMMA20[9:2]							-
22nd parameter	1	G_GAMMA21[9:2]							-
23rd parameter	1	G_GAMMA22[9:2]							-
24th parameter	1	G_GAMMA23[9:2]							-
25th parameter	1	G_GAMMA24[9:2]							-
26th parameter	1	G_GAMMA25[9:2]							-
27th parameter	1	G_GAMMA26[9:2]							-
28th parameter	1	G_GAMMA27[9:2]							-
29th parameter	1	G_GAMMA28[9:2]							-
30th parameter	1	G_GAMMA29[9:2]							-
31st parameter	1	G_GAMMA30[9:2]							-
32nd parameter	1	G_GAMMA31[9:2]							-
33rd parameter	1	G_GAMMA32[9:2]							-
34th parameter	1	G_GAMMA0[1:0]	G_GAMMA1[1:0]		G_GAMMA2[1:0]		G_GAMMA3[1:0]		-
35th parameter	1	G_GAMMA4[1:0]	G_GAMMA5[1:0]		G_GAMMA6[1:0]		G_GAMMA7[1:0]		-
36th parameter	1	G_GAMMA8[1:0]	G_GAMMA9[1:0]		G_GAMMA10[1:0]		G_GAMMA11[1:0]		-
37th parameter	1	G_GAMMA12[1:0]	G_GAMMA13[1:0]		G_GAMMA14[1:0]		G_GAMMA15[1:0]		-
38th parameter	1	G_GAMMA16[1:0]	G_GAMMA17[1:0]		G_GAMMA18[1:0]		G_GAMMA19[1:0]		-
39th parameter	1	G_GAMMA20[1:0]	G_GAMMA21[1:0]		G_GAMMA22[1:0]		G_GAMMA23[1:0]		-
40th parameter	1	G_GAMMA24[1:0]	G_GAMMA25[1:0]		G_GAMMA26[1:0]		G_GAMMA27[1:0]		-
41st parameter	1	G_GAMMA28[1:0]	G_GAMMA29[1:0]		G_GAMMA30[1:0]		G_GAMMA31[1:0]		-
42nd parameter	1	G_GAMMA32[1:0]	-	-	-	-	-	-	-

Himax Confidential
Do Not Copy

Description

This command is used to set Digital Gamma Curve Look-Up Table.

DGC_EN: Enable the DGC function. “1”: Enable; “0”: Disable.

DGC_INV: DGC mapping inversion.

R/G/B_GAMMA0[9:0] ~ R/G/B_GAMMA32[9:0]: MSB 8-bit is setting to mapping related gary level to which gray level voltage of real gamma. LSB 2-bit is for dithering.

LUT	Mapping Gray level
R_GAMMA0[9:0]	R0
R_GAMMA1[9:0]	R8
R_GAMMA2[9:0]	R16
:	:
R_GAMMA31[9:0]	R240
R_GAMMA32[9:0]	R255
G_GAMMA0[9:0]	G0
G_GAMMA1[9:0]	G8
G_GAMMA2[9:0]	G16
:	:
G_GAMMA31[9:0]	G240
G_GAMMA32[9:0]	G255
B_GAMMA0[9:0]	B0
B_GAMMA1[9:0]	B8
B_GAMMA2[9:0]	B16
:	:
B_GAMMA31[9:0]	B240
B_GAMMA32[9:0]	B255

Restrictions

SETEXTC turn on to enable this command.

Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Sleep In or Booster Off	Yes

Himax Confidential
Do Not Copy

5.19.13 SETID: Set ID (C3h)

C3H	SETID (Set ID)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	1	0	0	0	0	1	1	C3
1st parameter	1	ID1[7:0]								-
2nd parameter	1	ID2[7:0]								-
3rd parameter	1	ID3[7:0]								-
4th parameter	1	ID4[7:0]								-
5th parameter	1	ID_TIMES[2:0]			-	-	-	-	-	-

Description

ID1[7:0] is used to set ID RDAh value.

ID2[7:0] is used to set ID RDBh value.

ID3[7:0] is used to set ID RDCh value.

ID4[7:0] is used to set the fourth ID.

ID_TIMES[2:0]: Read the ID OTP programmed times.

ID_TIMES[2:0]	ID OTP Programmed Times
111	No programmed
011	ID has been programmed 1 time
001	ID has been programmed 2 times
000	ID has been programmed 3 times

Restrictions

SETEXTC turn on to enable this command.

Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Sleep In or Booster Off	Yes

Himax Confidential
Do Not Copy

5.19.14 SETDDB: Set DDB (C4h)

C4H	SETDDB (Set DDB)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	1	0	0	0	1	0	0	C4
1st parameter	1	DDB1[7:0]								-
2nd parameter	1	DDB2[7:0]								-
3rd parameter	1	DDB3[7:0]								-
4th parameter	1	DDB4[7:0]								-
5th parameter	1	DDB5[7:0]								-
6th parameter	1	DDB6[7:0]								-
Description	This command is used to set CMD RA1h DDB1~6 value.									
Restrictions	SETEXTC turn on to enable this command.									
Register Availability	Status	Availability								
	Normal Mode On, Idle Mode Off, Sleep Out	Yes								
	Normal Mode On, Idle Mode On, Sleep Out	Yes								
	Sleep In or Booster Off	Yes								

Himax Confidential
Do Not Copy

5.19.15 SETCABC: Set CABC control (C9h)

C9H	SETCABC (Set CABC Control)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	1	0	0	1	0	0	1	C9
1st Parameter	1	-	SEL_PWMCLK[2:0]			PWM_PERIOD[16]	BC_CTRL_EN	INVPULS	SEL_BLDUTY	-
2nd Parameter	1	PWM_PERIOD[15:8]								-
3rd Parameter	1	PWM_PERIOD[7:0]								-
4th Parameter	1	CABC_FSYNC	DIM_FRAME[6:0]							-
5th Parameter	1	CABC_DD	CABC_DD_I	-	-	CABC_FLM[3:0]				-

Description

This command is used to set CABC parameter.

SEL_BLDUTY: Backligh pwm output duty on/off control when CABC operation.
'0', The Backligh pwm output duty is 100%.
'1', The Backligh pwm output duty is calculate from CABC operation..

INVPULS: The backlight PWM output polarity select.
'0', The backlight PWM output is low level active.
'1', The backlight PWM output is high level active.

BC_CTRL_EN: Enable Backlight EN signal output

SEL_PWMCLK[2:0] : Internal PWM_CLK divider for CABC clock.

SEL_PWMCLK[2:0]			Brightness Control Clock frequency
0	0	0	PWM_CLK / 1
0	0	1	PWM_CLK / 2
0	1	0	PWM_CLK / 4
0	1	1	PWM_CLK / 6
1	0	0	PWM_CLK / 8
1	0	1	PWM_CLK / 10
1	1	0	PWM_CLK / 12
1	1	1	PWM_CLK / 14

PWM_PERIOD[16:0]: The backlight PWM output period setting.

When PWM_PERIOD[16]=0, PWM_PERIOD[15:6] setting inhibited.
If PWM_PERIOD[16]=0 and SEL_PWMCLK[2:0]=3'b001.The PWM_CLK frequency mapping table is as bellows:

PWM_PERIOD[5:0]	PWM_CLK
00h	40KHz
01h	39KHz
02h	38KHz
03h	37KHz
04h	36KHz
05h	35KHz
06h	34KHz
07h	33KHz
08h	32KHz
09h	31KHz
0Ah	30KHz
0Bh	29KHz
0Ch	28KHz
0Dh	27KHz
0Eh	26KHz
0Fh	25KHz

Himax Confidential Do Not Copy

This information contained herein is the exclusive property of Himax and shall not be distributed, reproduced, or disclosed in whole or in part without prior written permission of Himax.

10h	24KHz
11h	23KHz
12h	22KHz
13h	21KHz
14h	20KHz
15h	19KHz
16h	18KHz
17h	17KHz
18h	16KHz
19h	15KHz
1Ah	14KHz
1Bh	13KHz
1Ch	12KHz
1Dh	11KHz
1Eh	10KHz
1Fh	9KHz
20h	8KHz
21h	7KHz
22h	6KHz
23h	5KHz
24h	4KHz
25h	3KHz
26h	2KHz
27h	1KHz
28h	900Hz
29h	800Hz
2Ah	700Hz
2Bh	600Hz
2Ch	500Hz
2Dh	400Hz
2Eh	300Hz
2Fh	200Hz
30h	100Hz

When PWM_PERIOD[16]=1:
CABC_PWM_OUT frequency= (Fosc/4) / (SEL_PWMCLK[2:0]+1) / (PWM_PERIOD[15:0])
Note: PWM_PERIOD[15:0]=0000h is inhibited.

CABC_FSYNC: Reset CABC_PWM_OUT signal by each VSYNC.

DIM_FRAME[6:0] : Manual brightness setting dimming period.

CABC_DD: CABC dimming function enable bit.
'0', Disable CABC dimming.
'1', Enable CABC dimming.

CABC_DD_I: CABC dimming function enable bit in idle mode.

CABC_FLM[3:0]: CABC dimming frame number for each step.

Restriction: SETEXTC turn on to enable this command.

Register Availability:

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Sleep In or Booster Off	Yes

Himax Confidential
Do Not Copy

5.19.16 SETCABCGAIN (CAh)

CAH	SETCABCGAIN									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	1	0	0	1	0	1	0	CA
1st Parameter	1	-	DBG0[6:0]							-
2nd parameter	1	-	DBG1[6:0]							-
3rd parameter	1	-	DBG2[6:0]							-
4th parameter	1	-	DBG3[6:0]							-
5th parameter	1	-	DBG4[6:0]							-
6th parameter	1	-	DBG5[6:0]							-
7th parameter	1	-	DBG6[6:0]							-
8th parameter	1	-	DBG7[6:0]							-
9th parameter	1	-	DBG8[6:0]							-

Description

DBG0~8[6:0] : Gain select register 0~8.

DBG0~8[6:0]							CABC Gain	CABC Duty
0	0	x	x	x	x	x	Reserve	
0	1	0	0	0	0	0	1+0/32	100%
0	1	0	0	0	0	1	1+1/32	96.97%
0	1	0	0	0	1	0	1+2/32	94.12%
0	1	0	0	0	1	1	1+3/32	91.43%
0	1	0	0	1	0	0	1+4/32	88.89%
0	1	0	0	1	0	1	1+5/32	86.49%
0	1	0	0	1	1	0	1+6/32	84.21%
0	1	0	0	1	1	1	1+7/32	82.05%
0	1	0	1	0	0	0	1+8/32	80%
0	1	0	1	0	0	1	1+9/32	78.05%
0	1	0	1	0	1	0	1+10/32	76.19%
0	1	0	1	0	1	1	1+11/32	74.42%
0	1	0	1	1	0	0	1+12/32	72.73%
0	1	0	1	1	0	1	1+13/32	71.11%
0	1	0	1	1	1	0	1+14/32	69.57%
0	1	0	1	1	1	1	1+15/32	68.09%
0	1	1	0	0	0	0	1+16/32	66.67%
0	1	1	0	0	0	1	1+17/32	65.31%
0	1	1	0	0	1	0	1+18/32	64%
0	1	1	0	0	1	1	1+19/32	62.75%
0	1	1	0	1	0	0	1+20/32	61.54%
0	1	1	0	1	0	1	1+21/32	60.38%
0	1	1	0	1	1	0	1+22/32	59.26%
0	1	1	0	1	1	1	1+23/32	58.18%
0	1	1	1	0	0	0	1+24/32	57.14%
0	1	1	1	0	0	1	1+25/32	56.14%
0	1	1	1	0	1	0	1+26/32	55.17%
0	1	1	1	0	1	1	1+27/32	54.24%
0	1	1	1	1	0	0	1+28/32	53.33%
0	1	1	1	1	0	1	1+29/32	52.46%
0	1	1	1	1	1	0	1+30/32	51.61%
0	1	1	1	1	1	1	1+31/32	50.79%
1	0	0	0	0	0	0	1+32/32	50%

Restrictions: SETEXTC turn on to enable this command.

Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Sleep In or Booster Off	Yes

Himax Confidential Do Not Copy

5.19.17 SETPANEL (CCh)

CCH	SETPANEL(Set panel related register)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	1	0	0	1	1	0	0	CC
1st parameter	1	-	-	-	-	SS_PANEL	GS_PANEL	REV_PANEL	BGR_PANEL	00
Description0	This command is used to set setting of panel related register and make panel module meets below spec from viewpoint of user **BGR_PANEL:** The order of <R><G><B> dot color for module supplier, default value is stored in OTP. If color filter of panel is <B><G><R> type, setting BGR_PANEL = 1, if color filter of panel is <R><G><B> type, setting BGR_PANEL = 0. This bit is to make panel module look like a <R><G><B> type panel form the user viewpoint. **REV_PANEL:** The REV_PANEL setting is used to select the inversion of the display of all characters and graphics. This setting allows the display of the same data on both normally white and normally black panels. **GS_PANEL:** Specify the shift direction of gate driver output. When GS_PANEL = 0, the panel control signal is normal scan. When GS_PANEL = 1, the panel control signal is reverse scan. **SS_PANEL:** Specify the shift direction of source driver output. When SS_PANEL = 0, the shift direction from S1 to S3240 When SS_PANEL = 1, the shift direction from S3240 to S1.									
Restrictions	SETEXTC turn on to enable this command									
Register Availability	Status: Normal Mode On, Idle Mode Off, Sleep Out — Availability: Yes Status: Normal Mode On, Idle Mode On, Sleep Out — Availability: Yes Status: Sleep In or Booster Off — Availability: Yes									

Himax Confidential
Do Not Copy

5.19.18 SETOFFSET (D2h)

D2H	SETOFFSET									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	1	0	1	0	0	1	0	D2
1st parameter	1	VN_REFS[3:0]				VP_REFS[3:0]				-

Description

This command is used to set VP_REF and VN_REF voltage.
Please set VSPR≦ P_REFS ≦ VSP-0.3V, |VSNR|≦ VN_REFS|≦ |VSN|-0.3V

VP_REFS[3:0]: Positive reference voltage VP_REF setting.

VP_REFS[3:0]				VP_REF Voltage
0	0	0	0	4.0V
0	0	0	1	4.3V
0	0	1	0	4.5V
0	0	1	1	4.6V
0	1	0	0	4.7V
0	1	0	1	4.8V
0	1	1	0	4.9V
0	1	1	1	5.0V
1	0	0	0	5.1V
1	0	0	1	5.2V
1	0	1	0	5.3V
1	0	1	1	5.4V
1	1	0	0	5.5V
1	1	0	1	5.6V
1	1	1	0	5.8V
1	1	1	1	6.1V

VN_REFS[3:0]: Negative reference voltage VN_REF setting.

VN_REFS[3:0]				VN_REF Voltage
0	0	0	0	-4.0V
0	0	0	1	-4.3V
0	0	1	0	-4.5V
0	0	1	1	-4.6V
0	1	0	0	-4.7V
0	1	0	1	-4.8V
0	1	1	0	-4.9V
0	1	1	1	-5.0V
1	0	0	0	-5.1V
1	0	0	1	-5.2V
1	0	1	0	-5.3V
1	0	1	1	-5.4V
1	1	0	0	-5.5V
1	1	0	1	-5.6V
1	1	1	0	-5.8V
1	1	1	1	-6.1V

Restrictions: SETEXTC turn on to enable this command.

Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Sleep In or Booster Off	Yes

Himax Confidential Do Not Copy

5.19.19 SETGIP0: Set GIP Option0 (D3h)

D3H	SETGIP0(Set GIP Option0)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	1	0	1	0	0	1	1	D3
Bank0 1st parameter	1	-	-	GIP_EQ_OPT[1:0]		-	-	EQ_DELAY_HSYNC[1:0]		-
2nd parameter	1	-	-	-	-	-	EQ_VSEL	EQ_DISC[1:0]		-
3rd parameter	1	EQ_DELAY_ON1[7:0]								-
4th parameter	1	EQ_DELAY_OFF1[7:0]								-
5th parameter	1	GTO[7:0]								-
6th parameter	1	GNO[7:0]								-
7th parameter	1	USER_GIP_GATE[7:0]								-
8th parameter	1	USER_GIP_GATE1[7:0]								-
9th parameter		SHR0_3[3:0]					SHR0_2[3:0]			-
10th parameter	1	SHR0_1[3:0]					SHR0[11:8]			-
11st parameter	1	SHR0[7:0]								-
12nd parameter	1	-	-	-	-		SHR0_GS[11:8]			-
13rd parameter	1	SHR0_GS[7:0]								-
14th parameter	1	SHR1_3[3:0]					SHR1_2[3:0]			-
15th parameter	1	SHR1_1[3:0]					SHR1[11:8]			-
16th parameter	1	SHR1[7:0]								-
17th parameter	1	-	-	-	-		SHR1_GS[11:8]			-
18th parameter	1	SHR1_GS[7:0]								-
19th parameter	1	SHR2_3[3:0]					SHR2_2[3:0]			-
20th parameter	1	SHR2_1[3:0]					SHR2[11:8]			-
21st parameter	1	SHR2[7:0]								-
22nd parameter	1	-	-	-	-		SHR2_GS[11:8]			-
23rd parameter	1	SHR2_GS[7:0]								-
24th parameter	1	SHP[3:0]					SCP[3:0]			-
25th parameter	1	SHP2[3:0]					SHP1[3:0]			-
26th parameter	1	CHR0[7:0]								-
27th parameter	1	CHR0_GS[7:0]								-
28th parameter	1	CHP0[3:0]					CCP0[3:0]			-
29th parameter	1	CHR1[7:0]								-
30th parameter	1	CHR1_GS[7:0]								-
31st parameter	1	CHP1[3:0]					CCP1[3:0]			-

Description

This command is used for GIP signal setting.

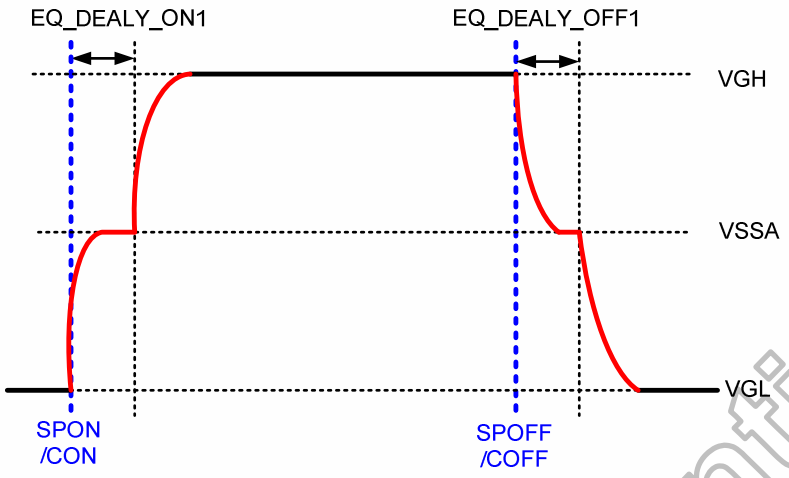
GIP_EQ_OPT[1:0]: GIP EQ (pre-charge) type selection

GIP_EQ_OPT 1	GIP_EQ_OPT 0	GIP EQ Type
0	0	Both rising / falling EQ
0	1	Only rising edge EQ
1	0	Only falling edge EQ
1	1	EQ Off

EQ_DELAY_ON1[7:0] / EQ_DELAY_OFF1[7:0]: Set GIP control signal EQ period (1 TCON CLK period = 4/45MHz)

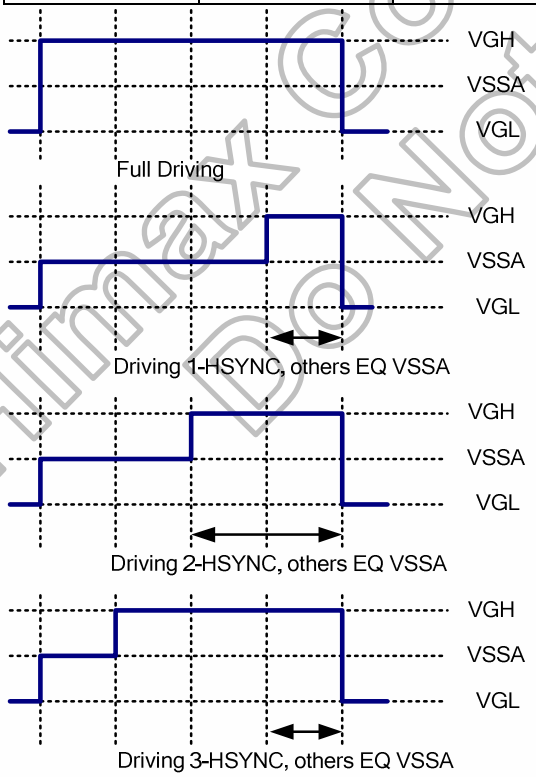
EQ_DELAY_ON1[7:0] / EQ_DELAY_OFF1[7:0]								GIP EQ Period
0	0	0	0	0	0	0	0	Inhibited
0	0	0	0	0	0	0	1	1 TCON clock cycle
0	0	0	0	0	0	1	0	2 TCON clock cycle
0	0	0	0	0	0	1	1	3 TCON clock cycle
:								:
1	1	1	1	1	1	0	1	253 TCON clock cycle
1	1	1	1	1	1	1	0	254 TCON clock cycle
1	1	1	1	1	1	1	1	255 TCON clock cycle

Himax Confidential Do Not Copy



EQ_DELAY_HSYNC[1:0]: Set the EQ period in HSYNC width.

EQ_DELAY_HSYNC[1:0]		EQ Period in HSYNC width
0	0	Normal Driving
0	1	1 x Hsync
1	0	2 x Hsync
1	1	3 x Hsync



EQ_VSEL: Select GIP falling EQ voltage level. 0: VSSA; 1: VSP.

GTO[7:0]: GPWR signal toggle frequency.

GTO[7:0]	GPWR toggle frequency
6'h00	256 x Frame
6'h01	1 x Frame
6'h02	2 x Frame

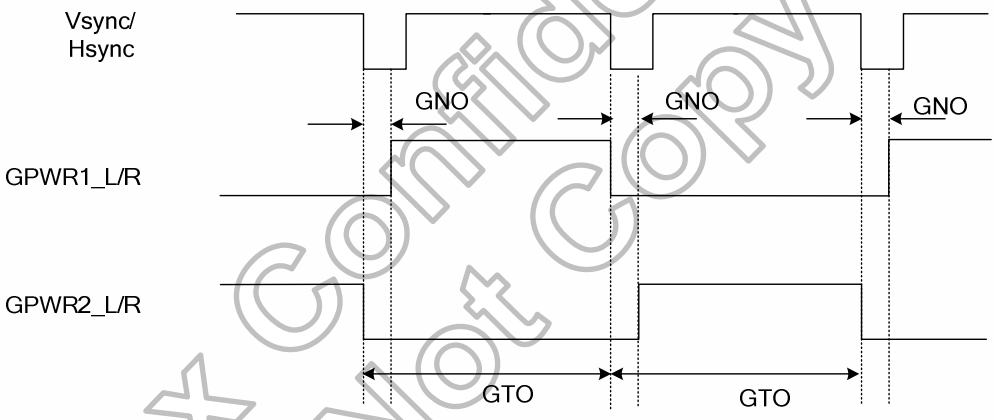
Himax Confidential Do Not Copy

6'h03	3 x Frame
:	:
6'h3D	253 x Frame
6'h3E	254 x Frame
6'h3F	255 x Frame

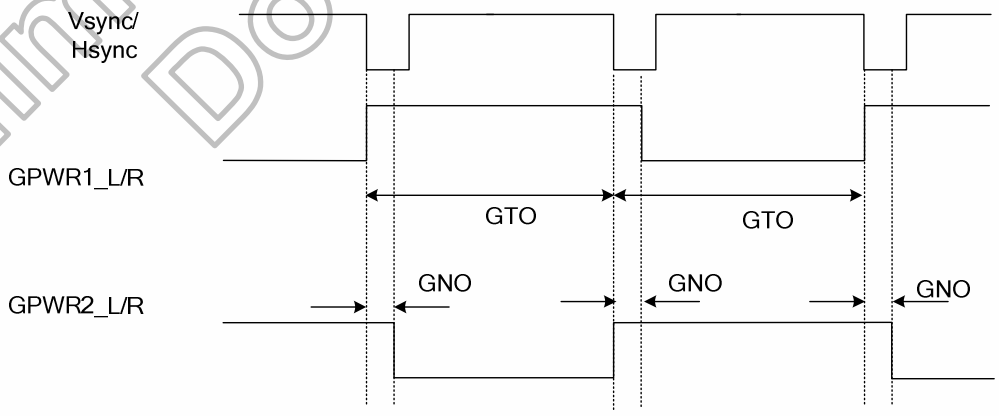
GNO[7:0]: GPWR signal non-overlap timing.

GNO[7:0]	GPWR non-overlap timing
8'h00	0
8'h01	1 TCON clock cycle
8'h02	2 TCON clock cycle
8'h03	3 TCON clock cycle
:	:
8'hFD	253 TCON clock cycle
8'hFE	254 TCON clock cycle
8'hFF	255 TCON clock cycle

GPWR1/2 non-overlap signal are as below:



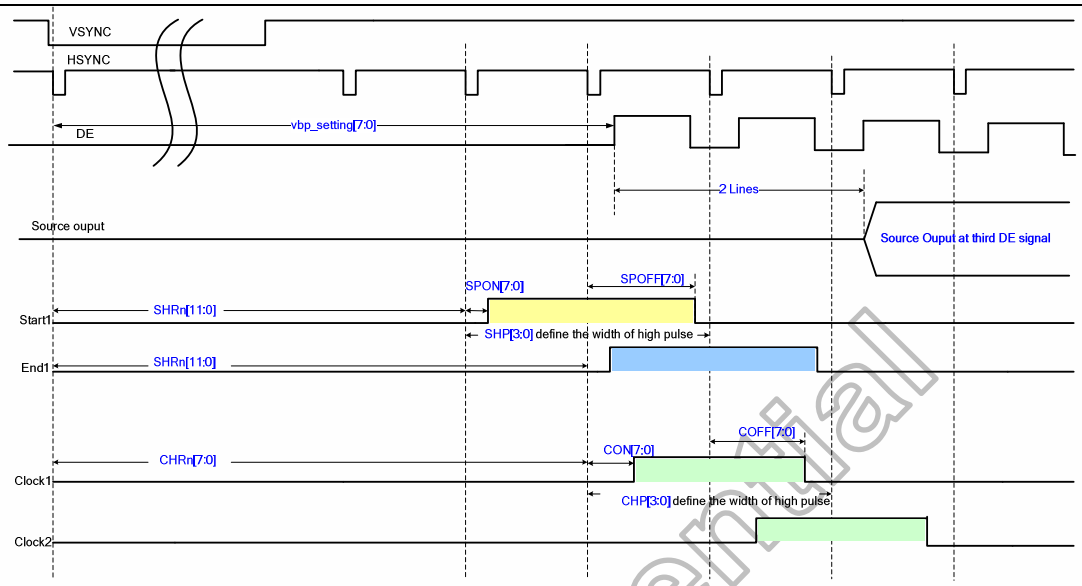
GPWR1/2 overlap signal are as below:



USER_GIP_Gate[7:0]: Set the GIP dummy clock numbers for first CKV.

USER_GIP_Gate1[7:0]: Set the GIP dummy clock numbers for second CKV.

Himax Confidential
Do Not Copy



Group0:
SHR0[11:0]: Set the 1st Start/End signal delay from VSYNC falling edge when GS=0.
SHR0_GS[11:0]: Set the 1st Start/End signal delay from VSYNC falling edge when GS=1.
SHR0_1 / SHR0_2 / SHR0_3[3:0]: Set the 2nd/3rd/4th Start / End signal delay from the 1st Start/End signal.

Group1:
SHR1[11:0]: Set the 1st Start/End siganl delay from VSYNC falling edge when GS=0.
SHR1_GS[11:0]: Set the 1st Start/End siganl delay from VSYNC falling edge when GS=1.
SHR1_1 / SHR1_2 / SHR1_3[3:0]: Set the 2nd/3rd/4th Start / End signal delay from the 1st Start/End signal.

Group2:
SHR2[11:0]: Set the 1st Start/End signal delay from VSYNC falling edge when GS=0.
SHR2_GS[11:0]: Set the 1st Start/End signal delay from VSYNC falling edge when GS=1.
SHR2_1 / SHR2_2 / SHR2_3[3:0]: Set the 2nd/3rd/4th Start / End signal delay from the 1st Start/End signal.

SHR0/SHR1/SHR2[11:0] SHR0_GS/SHR1_GS/SHR2_GS[11:0]	Start signal output delay
0x000h	2 x Hsync
0x001h	3 x Hsync
0x002h	4 x Hsync
:	:
0xFFEh	4096 x Hsync
0xFFFh	4097 x Hsync

SHR0_1 / SHR1_1 / SHR2_1[3:0]	2nd Start / End signal output delay
SHR0_2 / SHR1_2 / SHR2_2[3:0]	3th Start / End signal output delay
SHR0_3 / SHR1_3 / SHR2_3[3:0]	4th Start / End signal output delay
0000	0 x Hsync
0001	1 x Hsync
0010	2 x Hsync
:	:
1110	14 x Hsync
1111	15 x Hsync

SCP[3:0]: Numbers of output Start and End signal.

SCP3	SCP2	SCP1	SCP0	Start and End numbers
0	0	0	0	1
0	0	0	1	2

Himax Confidential Do Not Copy

0	0	1	0	3
:				:
1	1	0	1	14
1	1	1	0	15
1	1	1	1	16

SHP0/1/2[3:0]: Width of Start and End signal high pulse.

SHP3	SHP2	SHP1	SHP0	Start Pulse Width
0	0	0	0	1 x Hsync
0	0	0	1	2 x Hsync
0	0	1	0	3 x Hsync
:				:
1	1	1	0	15 x Hsync
1	1	1	1	16 x Hsync

CHR0[7:0]/CHR1[7:0]: Set the Clock signal delay from VSYNC falling edge when GS=0.
CHR0_GS[7:0]/CHR1_GS[7:0]: Set the Clock signal delay from VSYNC falling edge when GS=1.

CHR0[7:0]/CHR0_GS[7:0] CHR1[7:0]/CHR1_GS[7:0]	Clock signal output delay
0x00h	2 x HSYNC
0x01h	3 x HSYNC
0x02h	4 x HSYNC
:	:
0xFEh	256 x HSYNC
0xFFh	257 x HSYNC

CCP0[3:0]/CCP1[3:0]: Numbers of Output Clock signal.

CCP0[3:0]/CCP1[3:0]				Clock nunbers
0	0	0	0	1
0	0	0	1	2
0	0	1	0	3
:				:
1	1	1	0	15
1	1	1	1	16

CHP0[3:0]/CHP1[3:0]: Width of Clock signal high pulse.

CHP0[3:0]/CHP1[3:0]				Clock signal width
0	0	0	0	1 x Hsync
0	0	0	1	2 x Hsync
0	0	1	0	3 x Hsync
:				:
1	1	1	0	15 x Hsync
1	1	1	1	16 x Hsync

Restrictions	SETEXTC turn on to enable this command.

Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Sleep In or Booster Off	Yes

Himax Confidential Do Not Copy

5.19.20 Set GIP Option1 (D5h)

D5H	SETGIP1(Set GIP Option1)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	1	0	1	0	1	0	1	D5
1st parameter	1	-	CGTS_L_INV[1]	COS1_L[5:0]						-
2nd parameter	1	-	CGTS_R_INV[1]	COS1_R[5:0]						-
3rd parameter	1	-	CGTS_L_INV[2]	COS2_L[5:0]						-
4th parameter	1	-	CGTS_R_INV[2]	COS2_R[5:0]						-
5th parameter	1	-	CGTS_L_INV[3]	COS3_L[5:0]						-
6th parameter	1	-	CGTS_R_INV[3]	COS3_R[5:0]						-
7th parameter	1	-	CGTS_L_INV[4]	COS4_L[5:0]						-
8th parameter	1	-	CGTS_R_INV[4]	COS4_R[5:0]						-
9th parameter	1	-	CGTS_L_INV[5]	COS5_L[5:0]						-
10th parameter	1	-	CGTS_R_INV[5]	COS5_R[5:0]						-
11st parameter	1	-	CGTS_L_INV[6]	COS6_L[5:0]						-
12nd parameter	1	-	CGTS_R_INV[6]	COS6_R[5:0]						-
13rd parameter	1	-	CGTS_L_INV[7]	COS7_L[5:0]						-
14th parameter	1	-	CGTS_R_INV[7]	COS7_R[5:0						-
15th parameter	1	-	CGTS_L_INV[8]	COS8_L[5:0]						-
16th parameter	1	-	CGTS_R_INV[8]	COS8_R[5:0]						-
17th parameter	1	-	CGTS_L_INV[9]	COS9_L[5:0]						-
18th parameter	1	-	CGTS_R_INV[9]	COS9_R[5:0]						-
19th parameter	1	-	CGTS_L_INV[10]	COS10_L[5:0]						-
20th parameter	1	-	CGTS_R_INV[10]	COS10_R[5:0]						-
21st parameter	1	-	CGTS_L_INV[11]	COS11_L[5:0]						-
22nd parameter	1	-	CGTS_R_INV[11]	COS11_R[5:0]						-
23rd parameter	1	-	CGTS_L_INV[12]	COS12_L[5:0]						-
24th parameter	1	-	CGTS_R_INV[12]	COS12_R[5:0]						-
25th parameter	1	-	CGTS_L_INV[13]	COS13_L[5:0]						-
26th parameter	1	-	CGTS_R_INV[13]	COS13_R[5:0]						-
27th parameter	1	-	CGTS_L_INV[14]	COS14_L[5:0]						-
28th parameter	1	-	CGTS_R_INV[14]	COS14_R[5:0]						-
29th parameter	1	-	CGTS_L_INV[15]	COS15_L[5:0]						-
30th parameter	1	-	CGTS_R_INV[15]	COS15_R[5:0]						-

Himax Confidential
Do Not Copy

31st parameter	1	-	CGTS_L_INV[16]	COS16_L[5:0]	-
32nd parameter	1	-	CGTS_R_INV[16]	COS16_R[5:0]	-
33rd parameter	1	-	CGTS_L_INV[17]	COS17_L[5:0]	-
34th parameter	1	-	CGTS_R_INV[17]	COS17_R[5:0]	-
35th parameter	1	-	CGTS_L_INV[18]	COS18_L[5:0]	-
36th parameter	1	-	CGTS_R_INV[18]	COS18_R[5:0]	-
37th parameter	1	-	CGTS_L_INV[19]	COS19_L[5:0]	-
38th parameter	1	-	CGTS_R_INV[19]	COS19_R[5:0]	-
39th parameter	1	-	CGTS_L_INV[20]	COS20_L[5:0]	-
40th parameter	1	-	CGTS_R_INV[20]	COS20_R[5:0]	-
41st parameter	1	-	CGTS_L_INV[21]	COS21_L[5:0]	-
42nd parameter	1	-	CGTS_R_INV[21]	COS21_R[5:0]	-
43rd parameter	1	-	CGTS_L_INV[22]	COS22_L[5:0]	-
44th parameter	1	-	CGTS_R_INV[22]	COS22_R[5:0]	-

Description

CGTS_L_INV[22:1]: Set the corresponding signal CGOUTL_n output polarity,n=1~22
CGTS_R_INV[22:1]: Set the corresponding signal CGOUTR_n output polarity,n=1~22
0: Normal output
1: Invert output signal

COSn_L [5:0]: When GS_Panel=0, select CGOUTL_n output, n=1~22
COSn_R [5:0]: When GS_Panel=0, select CGOUTR_n output, n=1~22

COSn_L[5:0]~ COSn_R[5:0]~ n=1~22	Output Signal	Description
00_0000	CK[0]	GROUP0:Gate CLK
00_0001	CK[1]	Gate CLK
00_0010	CK[2]	Gate CLK
00_0011	CK[3]	Gate CLK
00_0100	CK[4]	Gate CLK
00_0101	CK[5]	Gate CLK
00_0110	CK[6]	Gate CLK
00_0111	CK[7]	Gate CLK
00_1000	CK[8]	Gate CLK
00_1001	CK[9]	Gate CLK
00_1010	CK[10]	Gate CLK
00_1011	CK[11]	Gate CLK
00_1100	CK[12]	Gate CLK
00_1101	CK[13]	Gate CLK
00_1110	CK[14]	Gate CLK
00_1111	CK[15]	Gate CLK
01_0000	CK_1[0]	GROUP1:Gate CLK
01_0001	CK_1 [1]	Gate CLK
01_0010	CK_1 [2]	Gate CLK
01_0011	CK_1 [3]	Gate CLK

Himax Confidential Do Not Copy

	01_0101	CK_1 [5]	Gate CLK
	01_0110	CK_1 [6]	Gate CLK
	01_0111	CK_1 [7]	Gate CLK
	01_1000	1'b0	VGL
	01_1001	1'b1	VGH
	01_1010	GPWR1	Frame or Line toggle signal (Non-overlap)
	01_1011	GPWR2	Frame or Line toggle signal (Non-overlap)
	01_1100	GPWR1	Frame or Line toggle signal (overlap)
	01_1101	GPWR2	Frame or Line toggle signal (overlap)
	01_1110	-	-
	01_1111	-	-
	10_0000	STV[0]	GROUP 0 SHR0[11:0]
	10_0001	STV[1]	SHR0[11:0]+SHR0_1[3:0]
	10_0010	STV[2]	SHR0[11:0]+SHR0_2[3:0]
	10_0011	STV[3]	SHR0[11:0]+SHR0_3[3:0]
	10_0100	STV[4]	GROUP 1 SHR1[11:0]
	10_0101	STV[5]	SHR1[11:0]+SHR1_1[3:0]
	10_0110	STV[6]	SHR1[11:0]+SHR1_2[3:0]
	10_0111	STV[7]	SHR1[11:0]+SHR1_3[3:0]
	10_1000	STV[8]	GROUP-2 SHR2[11:0]
	10_1001	STV[9]	SHR2[11:0]+SHR2_1[3:0]
	10_1010	STV[10]	SHR2[11:0]+SHR2_2[3:0]
	10_1011	STV[11]	SHR2[11:0]+SHR2_3[3:0]
	Others	inhibited	-
Restrictions	SETEXTC turn on to enable this command.		

Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Sleep In or Booster Off	Yes

Himax Confidential Do Not Copy

5.19.21 Set GIP Option2 (D6h)

D6H	SETGIP2(Set GIP Option2)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	1	0	1	0	1	1	0	D6
1st parameter	1	-	CGOUT_L_HIZ_IN[1]	COS1_L_GS[5:0]						-
2nd parameter	1	-	CGOUT_R_HIZ_IN[1]	COS1_R_GS[5:0]						-
3rd parameter	1	-	CGOUT_L_HIZ_IN[2]	COS2_L_GS[5:0]						-
4th parameter	1	-	CGOUT_R_HIZ_IN[2]	COS2_R_GS[5:0]						-
5th parameter	1	-	CGOUT_L_HIZ_IN[3]	COS3_L_GS[5:0]						-
6th parameter	1	-	CGOUT_R_HIZ_IN[3]	COS3_R_GS[5:0]						-
7th parameter	1	-	CGOUT_L_HIZ_IN[4]	COS4_L_GS[5:0]						-
8th parameter	1	-	CGOUT_R_HIZ_IN[4]	COS4_R_GS[5:0]						-
9th parameter	1	-	CGOUT_L_HIZ_IN[5]	COS5_L_GS[5:0]						-
10th parameter	1	-	CGOUT_R_HIZ_IN[5]	COS5_R_GS[5:0]						-
11st parameter	1	-	CGOUT_L_HIZ_IN[6]	COS6_L_GS[5:0]						-
12nd parameter	1	-	CGOUT_R_HIZ_IN[6]	COS6_R_GS[5:0]						-
13rd parameter	1	-	CGOUT_L_HIZ_IN[7]	COS7_L_GS[5:0]						-
14th parameter	1	-	CGOUT_R_HIZ_IN[7]	COS7_R_GS[5:0]						-
15th parameter	1	-	CGOUT_L_HIZ_IN[8]	COS8_L_GS[5:0]						-
16th parameter	1	-	CGOUT_R_HIZ_IN[8]	COS8_R_GS[5:0]						-
17th parameter	1	-	CGOUT_L_HIZ_IN[9]	COS9_L_GS[5:0]						-
18th parameter	1	-	CGOUT_R_HIZ_IN[9]	COS9_R_GS[5:0]						-
19th parameter	1	-	CGOUT_L_HIZ_IN[10]	COS10_L_GS[5:0]						-
20th parameter	1	-	CGOUT_R_HIZ_IN[10]	COS10_R_GS[5:0]						-
21st parameter	1	-	CGOUT_L_HIZ_IN[11]	COS11_L_GS[5:0]						-
22nd parameter	1	-	CGOUT_R_HIZ_IN[11]	COS11_R_GS[5:0]						-
23rd parameter	1	-	CGOUT_L_HIZ_	COS12_L_GS[5:0]						-

Himax Confidential Do Not Copy

This information contained herein is the exclusive property of Himax and shall not be distributed, reproduced, or disclosed in whole or in part without prior written permission of Himax.

			IN[12]		
24th parameter	1	-	CGOUT_R_HIZ_IN[12]	COS12_R_GS[5:0]	-
25th parameter	1	-	CGOUT_L_HIZ_IN[13]	COS13_L_GS[5:0]	-
26th parameter	1	-	CGOUT_R_HIZ_IN[13]	COS13_R_GS[5:0]	-
27th parameter	1	-	CGOUT_L_HIZ_IN[14]	COS14_L_GS[5:0]	-
28th parameter	1	-	CGOUT_R_HIZ_IN[14]	COS14_R_GS[5:0]	-
29th parameter	1	-	CGOUT_L_HIZ_IN[15]	COS15_L_GS[5:0]	-
30th parameter	1	-	CGOUT_R_HIZ_IN[15]	COS15_R_GS[5:0]	-
31st parameter	1	-	CGOUT_L_HIZ_IN[16]	COS16_L_GS[5:0]	-
32nd parameter	1	-	CGOUT_R_HIZ_IN[16]	COS16_R_GS[5:0]	-
33rd parameter	1	-	CGOUT_L_HIZ_IN[17]	COS17_L_GS[5:0]	-
34th parameter	1	-	CGOUT_R_HIZ_IN[17]	COS17_R_GS[5:0]	-
35th parameter	1	-	CGOUT_L_HIZ_IN[18]	COS18_L_GS[5:0]	-
36th parameter	1	-	CGOUT_R_HIZ_IN[18]	COS18_R_GS[5:0]	-
37th parameter	1	-	CGOUT_L_HIZ_IN[19]	COS19_L_GS[5:0]	-
38th parameter	1	-	CGOUT_R_HIZ_IN[19]	COS19_R_GS[5:0]	-
39th parameter	1	-	CGOUT_L_HIZ_IN[20]	COS20_L_GS[5:0]	-
40th parameter	1	-	CGOUT_R_HIZ_IN[20]	COS20_R_GS[5:0]	-
41st parameter	1	-	CGOUT_L_HIZ_IN[21]	COS21_L_GS[5:0]	-
42nd parameter	1	-	CGOUT_R_HIZ_IN[21]	COS21_R_GS[5:0]	-
43rd parameter	1	-	CGOUT_L_HIZ_IN[22]	COS22_L_GS[5:0]	-
44th parameter	1	-	CGOUT_R_HIZ_IN[22]	COS22_R_GS[5:0]	-
Description	**CGOUT_L_HIZ_IN[n]:** Set the CGOUTL_n output = Hi-Z, n=1~22 **CGOUT_R_HIZ_IN[n]:** Set the CGOUTR_n output = Hi-Z, n=1~22 0: Normal output 1: Hi-Z **COSn_L_GS[5:0]:** When GS_Panel=1, select CGOUTL_n output, n=1~22 **COSn_R_GS[5:0]:** When GS_Panel=1, select CGOUTR_n output, n=1~22				

Himax Confidential
Do Not Copy

COSn_L_GS[5:0]~ COSn_R_GS[5:0]~ n=1~22	Output Signal	Description
00_0000	CK[0]	GROUP0:Gate CLK
00_0001	CK[1]	Gate CLK
00_0010	CK[2]	Gate CLK
00_0011	CK[3]	Gate CLK
00_0100	CK[4]	Gate CLK
00_0101	CK[5]	Gate CLK
00_0110	CK[6]	Gate CLK
00_0111	CK[7]	Gate CLK
00_1000	CK[8]	Gate CLK
00_1001	CK[9]	Gate CLK
00_1010	CK[10]	Gate CLK
00_1011	CK[11]	Gate CLK
00_1100	CK[12]	Gate CLK
00_1101	CK[13]	Gate CLK
00_1110	CK[14]	Gate CLK
00_1111	CK[15]	Gate CLK
01_0000	CK_1[0]	GROUP1:Gate CLK
01_0001	CK_1 [1]	Gate CLK
01_0010	CK_1 [2]	Gate CLK
01_0011	CK_1 [3]	Gate CLK
01_0100	CK_1 [4]	Gate CLK
01_0101	CK_1 [5]	Gate CLK
01_0110	CK_1 [6]	Gate CLK
01_0111	CK_1 [7]	Gate CLK
01_1000	1'b0	VGL
01_1001	1'b1	VGH
01_1010	GPWR1	Frame or Line toggle signal (Non-overlap)
01_1011	GPWR2	Frame or Line toggle signal (Non-overlap)
01_1100	GPWR1	Frame or Line toggle signal (overlap)
01_1101	GPWR2	Frame or Line toggle signal (overlap)
01_1110	-	-
01_1111	-	-
10_0000	STV[0]	GROUP 0 SHR0[11:0]
10_0001	STV[1]	SHR0[11:0]+SHR0_1[3:0]
10_0010	STV[2]	SHR0[11:0]+SHR0_2[3:0]
10_0011	STV[3]	SHR0[11:0]+SHR0_3[3:0]
10_0100	STV[4]	GROUP 1 SHR1[11:0]
10_0101	STV[5]	SHR1[11:0]+SHR1_1[3:0]
10_0110	STV[6]	SHR1[11:0]+SHR1_2[3:0]
10_0111	STV[7]	SHR1[11:0]+SHR1_3[3:0]
10_1000	STV[8]	GROUP-2 SHR2[11:0]
10_1001	STV[9]	SHR2[11:0]+SHR2_1[3:0]
10_1010	STV[10]	SHR2[11:0]+SHR2_2[3:0]
10_1011	STV[11]	SHR2[11:0]+SHR2_3[3:0]
Others	inhibited	-

Restrictions	SETEXTC turn on to enable this command.

Himax Confidential Do Not Copy

Register Availability	

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Sleep In or Booster Off	Yes

Himax Confidential
Do Not Copy

5.19.22 SETGPO (D9h)

D9H	SETGPO									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	1	0	1	1	0	0	1	D9
1st parameter	1	ESD_DET	-	-	-	TE_GPO[3:0]				-
2nd parameter	1	-	-	-	-	TE1_GPO[3:0]				-
3rd parameter	1	-	-	-	-	CABC_GPO[3:0]				-
4th parameter	1	-	-	-	-	SDO_GPO[3:0]				-

Description

ESD_DET: ESD detection bit, read '0' for the following conditions: abnormal power off, STB=1
TE_GPO[3:0]: Set the output pin TE.
TE1_GPO[3:0]: Set the output pin TE1.
CABC_GPO[3:0]: : Set the output pin CABC_PWM_OUT.
SDO_GPO[3:0]: Set the output pin SDO.

TE_GPO[3:0]/ TE1_GPO[3:0]/ CABC_GPO[3:0]/ SDO_GPO[3:0]				TE/ TE1/ CABC_PWM_OUT/ SDO Output signal
0	0	0	0	TE
0	0	0	1	TE1
0	0	1	0	BC
0	0	1	1	HSYNC
0	1	0	0	VSYNC
0	1	0	1	DE
0	1	1	0	DSI_ERR_RPT
0	1	1	1	SDO
1	0	0	0	STB
1	0	0	1	CABC_LED_EN_IN
1	0	1	0	ESD_DET

Restrictions: SETEXTC turn on to enable this command

Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Sleep In or Booster Off	Yes

Himax Confidential Do Not Copy

5.19.23 SETSCALING (DDh)

DDH	SETSCALING									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	1	0	1	1	1	0	1	DD
1st parameter	1	-	-	-	-	-	-	SCALING_TYPE	SCALING_EN	-
Description	**SCALING_EN:** Set"1" enable scaling function. **SCALING_TYPE:** 0: 1.5x scaling: 480RGBx854 scales to 720RGBx1280 1: 1.33x scaling: 540RGBx960 scales to 720RGBx1280									
Restrictions	SETEXTC turn on to enable this command									
Register Availability	Status: Normal Mode On, Idle Mode Off, Sleep Out — Availability: Yes Status: Normal Mode On, Idle Mode On, Sleep Out — Availability: Yes Status: Sleep In or Booster Off — Availability: Yes									

Himax Confidential
Do Not Copy

5.19.24 SET1BPP (DFh)

DFH	SET1BPP									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	1	0	1	1	1	1	1	DF
1st parameter	1	-	-	-	-	-	NW_I[2:0]			-
2nd parameter	1	BP_I[7:0]								-
3rd parameter	1	FP_I[7:0]								-
4th parameter	1	RTN_I[7:0]								-
5th parameter	1	VCMC_F_I[7:0]								-
6h parameter	1	VCMC_B_I[7:0]								-
7th parameter	1	AP_I[2:0]			-	-	-	VCMC_B_I[8]	VCMC_F_I[8]	-

Description

Set Idle mode related setting.

NW_I[2:0]: Inversion type setting in idle mode.

NW_I[2:0]			Inversion type
0	0	0	Column inversion
0	0	1	1-dot inversion
0	1	0	2-dot inversion
0	1	1	4-dot inversion
1	0	0	8-dot inversion
1	0	1	Zig-zag inversion
1	1	0	2-dot-2 inversion
1	1	1	3-dot inversion

BP_I[7:0] : Specify the amount of scan line for back porch(BP) in idle mode.
FP_I[7:0]: Specify the amount of scan line for front porch (FP) in idle mode.

FP[7:0]_I / BP_I[7:0]	Number of front porch/ back porch Lines
8h'00	2 lines
8h'01	3 lines
8h'02	4 lines
8h'03	5 lines
8h'04	6 lines
8h'05	7 lines
:	:
8h'FB	253 lines
8h'FC	254 lines
8h'FD	255 lines
8h'FE	256 lines
8h'FF	257 lines

Note: Set BP_I[7:0] = VS + VBP – 2, and FP_I[7:0] = VFP – 2.

RTN_I[7:0]: A cycle time of line width in idle mode.
(1 TCON clock period= 1/22MHz)

RTN_I[7:0]	Clock per Line
8h'00	97 TCON CLK
8h'01	98 TCON CLK
8h'02	99 TCON CLK
:	:
8'hFD	350 TCON CLK
8'hFE	351 TCON CLK
8'hFF	352 TCON CLK

VCMC_F_I[8:0]: Forward scan VCOM voltage control in Idle mode.

Himax Confidential
Do Not Copy

VCMC_B_I[8:0]: Backward scan VCOM voltage control in Idle mode.

VCMC_F_I[8:0]/VCMC_B_I[8:0]									VCOM
0	0	0	0	0	0	0	0	0	VSSA
0	0	0	0	0	0	0	0	1	-4.00V
0	0	0	0	0	0	0	1	0	-4.00V
0	0	0	0	0	0	0	1	1	-4.00V
0	0	0	0	0	0	1	0	0	-4.00V
0	0	0	0	0	0	1	0	1	-4.00V
0	0	0	0	0	0	1	1	0	-4.00V
0	0	0	0	0	0	1	1	1	-4.00V
0	0	0	0	0	1	0	0	0	-4.00V
0	0	0	0	0	1	0	0	1	-3.99V
0	0	0	0	0	1	0	1	0	-3.98V
0	0	0	0	0	1	0	1	1	-3.97V
:									:
1	1	0	0	1	0	1	1	0	-0.02V
1	1	0	0	1	0	1	1	1	-0.01V
1	1	0	0	1	1	0	0	0	VSSA
1	1	0	0	1	1	0	0	1	0.01V
1	1	0	0	1	1	0	1	0	0.02V
1	1	0	0	1	1	0	1	1	0.03V
:									:
1	1	1	1	1	1	0	1	0	0.98V
1	1	1	1	1	1	0	1	1	0.99V
1	1	1	1	1	1	1	0	0	1V
1	1	1	1	1	1	1	0	1	1V
1	1	1	1	1	1	1	1	0	1V
1	1	1	1	1	1	1	1	1	HZ

AP_I[2:0]: Adjust the amount of fixed current from the fixed current source for the operational amplifier in idle mode.

AP_I[2:0]			Constant Current of Operational Amplifier
0	0	0	Stop
0	0	1	0.5μA
0	1	0	1.0μA
0	1	1	1.5μA
1	0	0	2.0μA
1	0	1	2.5μA
1	1	0	3.0μA
1	1	1	3.5μA

Restrictions	SETEXTC turn on to enable this command

Himax Confidential
Do Not Copy

Register Availability	Status	Availability
	Normal Mode On, Idle Mode Off, Sleep Out	Yes
	Normal Mode On, Idle Mode On, Sleep Out	Yes
	Sleep In or Booster Off	Yes

Himax Confidential
Do Not Copy

5.19.25 SETGAMMA: Set gamma curve related setting (E0h)

E0H	SETGAMMA (Set Gamma Curve Related Setting)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	1	1	0	0	0	0	0	E0
1st Parameter	1	-	VHP0[6:0]							-
2nd parameter	1	-	VHP1[6:0]							-
3rd parameter	1	-	VHP2[6:0]							-
4th parameter	1	-	VHP3[6:0]							-
5th parameter	1	-	VHP4[6:0]							-
6th parameter	1	-	VHP5[6:0]							-
7th parameter	1	-	VHP6[6:0]							-
8th parameter	1	-	VHP7[6:0]							-
9th parameter	1	VMP0[7:0]								-
10th parameter	1	VMP1[7:0]								-
11st parameter	1	VMP2[7:0]								-
12nd parameter	1	VMP3[7:0]								-
13rd parameter	1	VMP4[7:0]								-
14th parameter	1	VMP5[7:0]								-
15th parameter	1	VMP6[7:0]								-
16th parameter	1	VMP7[7:0]								-
17th parameter	1	VMP8[7:0]								-
18th parameter	1	VMP9[7:0]								-
19th parameter	1	VMP10[7:0]								-
20th parameter	1	VMP11[7:0]								-
21st Parameter	1	VMP12[7:0]								-
22nd parameter	1	-	VLP0[6:0]							-
23rd parameter	1	-	VLP1[6:0]							-
24th parameter	1	-	VLP2[6:0]							-
25th parameter	1	-	VLP3[6:0]							-
26th parameter	1	-	VLP4[6:0]							-
27th parameter	1	-	VLP5[6:0]							-
28th parameter	1	-	VLP6[6:0]							-
29th parameter	1	-	VLP7[6:0]							-
30th parameter	1	-	VHN0[6:0]							-
31st Parameter	1	-	VHN1[6:0]							-
32nd parameter	1	-	VHN2[6:0]							-
33rd parameter	1	-	VHN3[6:0]							-
34th parameter	1	-	VHN4[6:0]							-
35th parameter	1	-	VHN5[6:0]							-
36th parameter	1	-	VHN6[6:0]							-
37th parameter	1	-	VHN7[6:0]							-
38th parameter	1	VMN0[7:0]								-
39th parameter	1	VMN1[7:0]								-
40th parameter	1	VMN2[7:0]								-
41st Parameter	1	VMN3[7:0]								-
42nd parameter	1	VMN4[7:0]								-
43rd parameter	1	VMN5[7:0]								-
44th parameter	1	VMN6[7:0]								-
45th parameter	1	VMN7[7:0]								-
46th parameter	1	VMN8[7:0]								-
47th parameter	1	VMN9[7:0]								-
48th parameter	1	VMN10[7:0]								-
49th parameter	1	VMN11[7:0]								-
50th parameter	1	VMN12[7:0]								-
51st Parameter	1	-	VLN0[6:0]							-
52nd parameter	1	-	VLN1[6:0]							-

Himax Confidential Do Not Copy

53rd parameter	1	-	VLN2[6:0]	-
54th parameter	1	-	VLN3[6:0]	-
55th parameter	1	-	VLN4[6:0]	-
56th parameter	1	-	VLN5[6:0]	-
57th parameter	1	-	VLN6[6:0]	-
58th parameter	1	-	VLN7[6:0]	-

Description

This command is to set gamma register.

Register Groups	Positive Polarity	Negative Polarity	Description
Up Edge adjustment	VHP0 6-0	VHN0 6-0	128-to-1 selector (voltage level of grayscale 0)
	VHP1 6-0	VHN1 6-0	128-to-1 selector (voltage level of grayscale 1)
	VHP2 6-0	VHN2 6-0	128-to-1 selector (voltage level of grayscale 3)
	VHP3 6-0	VHN3 6-0	128-to-1 selector (voltage level of grayscale 5)
	VHP4 6-0	VHN4 6-0	128-to-1 selector (voltage level of grayscale 7)
	VHP5 6-0	VHN5 6-0	128-to-1 selector (voltage level of grayscale 9)
	VHP6 6-0	VHN6 6-0	128-to-1 selector (voltage level of grayscale 12)
	VHP7 6-0	VHN7 6-0	128-to-1 selector (voltage level of grayscale 15)
Center adjustment	VMP0 7-0	VMN0 7-0	255-to-1 selector (voltage level of grayscale 20)
	VMP1 7-0	VMN1 7-0	255-to-1 selector (voltage level of grayscale 28)
	VMP2 7-0	VMN2 7-0	255-to-1 selector (voltage level of grayscale 40)
	VMP3 7-0	VMN3 7-0	255-to-1 selector (voltage level of grayscale 52)
	VMP4 7-0	VMN4 7-0	255-to-1 selector (voltage level of grayscale 76)
	VMP5 7-0	VMN5 7-0	255-to-1 selector (voltage level of grayscale 100)
	VMP6 7-0	VMN6 7-0	255-to-1 selector (voltage level of grayscale 128)
	VMP7 7-0	VMN7 7-0	255-to-1 selector (voltage level of grayscale 156)
	VMP8 7-0	VMN8 7-0	255-to-1 selector (voltage level of grayscale 180)
	VMP9 7-0	VMN9 7-0	255-to-1 selector (voltage level of grayscale 204)
	VMP10 7-0	VMN10 7-0	255-to-1 selector (voltage level of grayscale 216)
	VMP11 7-0	VMN11 7-0	255-to-1 selector (voltage level of grayscale 228)
	VMP12 7-0	VMN12 7-0	255-to-1 selector (voltage level of grayscale 236)
Down Edge adjustment	VLP0 6-0	VLN0 6-0	128-to-1 selector (voltage level of grayscale 240)
	VLP1 6-0	VLN1 6-0	128-to-1 selector (voltage level of grayscale 243)
	VLP2 6-0	VLN2 6-0	128-to-1 selector (voltage level of grayscale 246)
	VLP3 6-0	VLN3 6-0	128-to-1 selector (voltage level of grayscale 248)
	VLP4 6-0	VLN4 6-0	128-to-1 selector (voltage level of grayscale 250)
	VLP5 6-0	VLN5 6-0	128-to-1 selector (voltage level of grayscale 252)
	VLP6 6-0	VLN6 6-0	128-to-1 selector (voltage level of grayscale 254)
	VLP7 6-0	VLN7 6-0	128-to-1 selector (voltage level of grayscale 255)

Restriction

SETEXTC turn on to enable this command.

Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Sleep In or Booster Off	Yes

Himax Confidential
Do not copy

5.19.26 SETCHEMODE_DYN (E4h)

E4H	SETCHEMODE_DYN (Set color enhancement mode, dynamic)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	1	1	0	0	1	0	0	E4
1st parameter	1	-	-	-	-	-	-	-	DYN_CEH_EN	-
2nd parameter	1	HUE_MODE[1:0]		SE_MODE[1:0]		BE_MODE[1:0]		CE_MODE[1:0]		-

Description

This command is to set color enhanment and dynamic mode.

DYN_CHE_EN: Select the color enhancement reload mode. 0: static mode, 1: dynamic mode.

CE_MODE[1:0]: Set color (saturation) enhancement.

BE_MODE[1:0]: Set brightness enhancement.

SE_MODE[1:0]: Set sharpness enhancement.

HUE_MODE[1:0]: Set hue enhancement.

In Static mode:
Enhancement level selection: when SE/BE/CE/HUE is turn on, the enhancement effect depends on the gobal gain curve set.

CE_MODE[1:0]/ BE_MODE[1:0]/ SE_MODE[1:0]/ HUE_MODE[1:0]		Enhance
0	0	Off
0	1	On
1	0	
1	1	

In Dynamic mode:
Enhance level selection. wWhen SE/BE/CE/HUE turn on, the enhancement gain setting will be read from the ROM table. Three sets of enhancement gain are provide for selection.

CE_MODE[1:0]/ BE_MODE[1:0]/ SE_MODE[1:0]/ HUE_MODE[1:0]		Enhance
0	0	Off
0	1	Low
1	0	Medium
1	1	High

Restrictions

SETEXTC turn on to enable this command.

Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Sleep In or Booster Off	Yes

Himax Confidential
Do Not Copy

This information contained herein is the exclusive property of Himax and shall not be distributed, reproduced, or disclosed in whole or in part without prior written permission of Himax.

5.19.27 SETCHE (E5h)

E5H	SETCHE (Set color enhancement, static/OTP)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	Def Val
Command	0	1	1	1	0	0	1	0	1	E5
1st parameter	1	-	-	-	SG0_F[4:0]					-
2nd parameter	1	-	-	-	SG64_F[4:0]					-
3rd parameter	1	-	-	-	SG128_F[4:0]					-
4th parameter	1	-	-	-	SG192_F[4:0]					-
5th parameter	1	-	-	-	SG255_F[4:0]					-
6th parameter	1	TH_SAT[7:0]								-
7th parameter	1	TH_WHITE[7:0]								-
8th parameter	1	TH_COLOR[7:0]								-
9th parameter	1	TH_GRAY[7:0]								-
10th parameter	1	-	-	-	CG0_F[4:0]					-
11st parameter	1	-	-	-	CG64_F[4:0]					-
12nd parameter	1	-	-	-	CG128_F[4:0]					-
13rd parameter	1	-	-	-	CG192_F[4:0]					-
14th parameter	1	-	-	-	CG255_F[4:0]					-
15th parameter	1	THC[7:0]								-
16th parameter	1	-	-	-	-	EG2_F[1:0]		EG1_F[1:0]		-
17th parameter	1	TH_RFL[7:0]								
18th parameter	1	-	-	-	SG16_F[4:0]					-
19th parameter	1	-	-	-	SG32_F[4:0]					-
20th parameter	1	-	-	-	SG48_F[4:0]					-
21st parameter	1	-	-	-	SG80_F[4:0]					-
22nd parameter	1	-	-	-	SG96_F[4:0]					-
23rd parameter	1	-	-	-	SG112_F[4:0]					-
24th parameter	1	-	-	-	SG144_F[4:0]					-
25th parameter	1	-	-	-	SG160_F[4:0]					-
26th parameter	1	-	-	-	SG176_F[4:0]					-
27th parameter	1	-	-	-	SG208_F[4:0]					-
28th parameter	1	-	-	-	SG224_F[4:0]					-
29th parameter	1	-	-	-	SG240_F[4:0]					-
30th parameter	1	-	-	-	CG16_F[4:0]					-
31st parameter	1	-	-	-	CG32_F[4:0]					-
32nd parameter	1	-	-	-	CG48_F[4:0]					-
33rd parameter	1	-	-	-	CG80_F[4:0]					-
34th parameter	1	-	-	-	CG96_F[4:0]					-
35th parameter	1	-	-	-	CG112_F[4:0]					-
36th parameter	1	-	-	-	CG144_F[4:0]					-
37th parameter	1	-	-	-	CG160_F[4:0]					-
38th parameter	1	-	-	-	CG176_F[4:0]					-
39th parameter	1	-	-	-	CG208_F[4:0]					-
40th parameter	1	-	-	-	CG224_F[4:0]					-
41st parameter	1	-	-	-	CG240_F[4:0]					-
Description	Parameters for CE function. **SG0_F[4:0]:** L:00h; M:04h; H:08h; **SG16_F[4:0]:** L:02h; M:05h; H:08h; **SG32_F[4:0]:** L:04h; M:08h; H:09h; **SG48_F[4:0]:** L:06h; M:0Ah; H:0Ah; **SG64_F[4:0]:** L:08h; M:0Bh; H:0Eh; **SG80_F[4:0]:** L:08h; M:0Bh; H:0Eh; **SG96_F[4:0]:** L:08h; M:0Ah; H:0Bh; **SG112_F[4:0]:** L:07h; M:09h; H:0Bh; **SG128_F[4:0]:** L:06h; M:08h; H:0Ah;									

Himax Confidential Do Not Copy

	SG144_F[4:0]: L:06h; M:08h; H:0Ah; **SG160_F[4:0]:** L:06h; M:08h; H:0Ah; **SG176_F[4:0]:** L:05h; M:07h; H:09h; **SG192_F[4:0]:** L:04h; M:06h;H:08h; **SG208_F[4:0]:** L:03h; M:05h;H:06h; **SG224_F[4:0]:** L:02h; M:04h;H:05h; **SG240_F[4:0]:** L:01h; M:02h;H:03h; **SG255_F[4:0]:** L:00h; M:00h; H:00h; **TH_SAT[7:0]:** L:20h; M:20h; H:20h; **TH_WHITE[7:0]:** L:20h; M:20h; H:20h; **TH_COLOR[7:0]:** L:20h; M:20h; H:20h; **TH_GRAY[7:0]:** L:10h; M:10h; H:10h; **TH_RFL[7:0]**: L:20h; M:20h; H:20h; Parameters for BE function **CG0_F[4:0]:** L:00h; M:04h; H:08h; **CG16_F[4:0]:** L:02h; M:05h; H:09h; **CG32_F[4:0]:** L:03h; M:07h; H:0Ah; **CG48_F[4:0]:** L:05h; M:09h; H:0Bh; **CG64_F[4:0]:** L:08h; M:0Ah H:0Ch; **CG80_F[4:0]:** L:08h; M:0Ah; H:0Ch; **CG96_F[4:0]:** L:07h; M:0Ah; H:0Bh; **CG112_F[4:0]:** L:07h; M:09h; H:0Bh; **CG128_F[4:0]:** L:06h; M:08h; H:0Ah; **CG144_F[4:0]:** L:06h; M:08h; H:0Ah; **CG160_F[4:0]:** L:05h; M:07h; H:0Ah; **CG176_F[4:0]:** L:05h; M:07h; H:09h; **CG192_F[4:0]:** L:04h; M:06h;H:08h; **CG208_F[4:0]:** L:03h; M:05h; H:07h; **CG224_F[4:0]:** L:02h; M:04h; H:06h; **CG240_F[4:0]:** L:01h; M:03h; H:03h; **CG255_F[4:0]:** L:00h; M:00h; H:00h; **THC[7:0]:** L:80h; M:80h; H:80h; Parameters for SE function **EG2_F[1:0], EG1_F[1:0]:** L:0Ah; M: 09h; H:08h;
Restrictions	SETEXTC turn on to enable this command.
Register Availability	Status / Availability Normal Mode On, Idle Mode Off, Sleep Out: Yes Normal Mode On, Idle Mode On, Sleep Out: Yes Sleep In or Booster Off: Yes

Himax Confidential
Do Not Copy

5.19.28 SETCESEL: Enable color enhance (E6h)

E6H	SETCESEL(Enable color enhancement)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	1	1	0	0	1	1	0	E6
1st parameter	1	REG_CE_vv_en	-	-	-	-	-	-	-	-
2nd parameter	1	-	-	-	REG_BE_16BN_EN	REG_CE_16BN_EN	-	-	-	-
3rd parameter	1	-								-
4th parameter	1	-								-
5th parameter	1	-								-
6th parameter	1	CS_RING(1)[7:0]								-
7th parameter	1	CS_RING(2) [7:0]								-
8th parameter	1	CS_RING(3) [7:0]								
9th parameter	1	CS_RING(4) [7:0]								-
10th parameter	1	CS_RING(5) [7:0]								-
11th parameter	1	CS_RING(6) [7:0]								-
12th parameter	1	CS_RING(7) [7:0]								-
13th parameter	1	CS_RING(8) [7:0]								-
14th parameter	1	CS_RING(9) [7:0]								-
15th parameter	1	CS_RING(10) [7:0]								-
16th parameter	1	CS_RING(11) [7:0]								-
17th parameter	1	CS_RING(12) [7:0]								-

Description

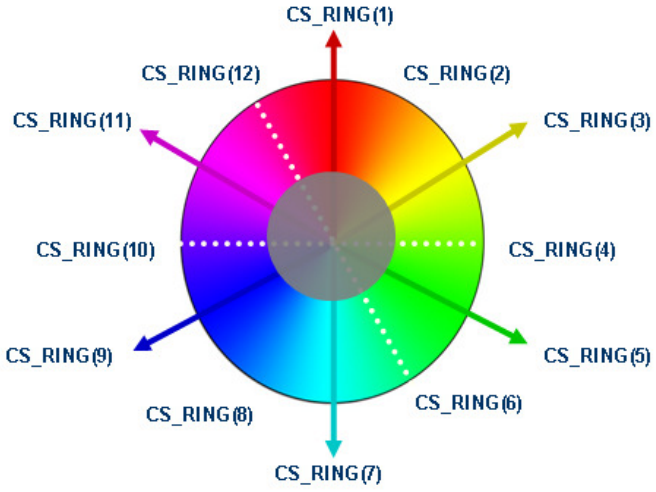
REG_CE_vv_en : 1:Enable TH_SAT, TH_WHITE, TH_RFL local gain function
0: Disable TH_SAT, TH_WHITE, TH_RFL local gain function

REG_BE_16BN_EN : 1:Enable 16-band global gain curve for BE
0: Disable 16-band global gain curve for BE

REG_CE_16BN_EN : 1:Enable 16-band global gain curve for CE
0: Disable 16-band global gain curve for CE

CS_RING(1)[7:0]~CS_RING(12)[7:0]: parameters for 12 hue gain adjustment. The available value for CS_RING(x)[7:0] is 00h~20h, other settings are not applicable.

12 hues can be adjusted to enhance the specified colors.



CS_RING(1): L:10h; M:13h; H:1Ch;

Himax Confidential
Do Not Copy

	CS_RING(2): L:0Ah; M:0Bh; H:18h; **CS_RING(3):** L:10h; M:13h; H:1Ch; **CS_RING(4):** L:20h; M:20h; H:20h; **CS_RING(5):** L:20h; M:20h; H:20h; **CS_RING(6):** L:20h; M:20h; H:20h; **CS_RING(7):** L:0Ch; M:12h; H:16h; **CS_RING(8):** L:08h; M:0Ch; H:12h; **CS_RING(9):** L:0Fh; M:13h; H:16h; **CS_RING(10):** L:20h; M:20h; H:20h; **CS_RING(11):** L:20h; M:20h; H:20h; **CS_RING(12):** L:20h; M:20h; H:20h;
Restrictions	SETEXTC turn on to enable this command.
Register Availability	Status / Availability Normal Mode On, Idle Mode Off, Sleep Out / Yes Normal Mode On, Idle Mode On, Sleep Out / Yes Sleep In or Booster Off / Yes

Himax Confidential
Do Not Copy

5.19.29 SET_SP_CMD (E9h)

E9H	GETHXID									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	1	1	1	0	1	0	0	F4
1st parameter	1	FORCE_OPT	RAND_WR_SERIAL_EN	RAND_WR_CNT[5:0]						-
Description	This command is used to directly set register value of specific parameter. FORCE_OPT: 0: disable, 1: enable RAND_WR_SERIAL_EN: Set to "1" for more than 1 parameter access. RAND_WR_CNT[5:0]: Specify the parameter index.									
Restrictions	SETEXTC turn on to enable this command.									
Register Availability	Status: Normal Mode On, Idle Mode Off, Sleep Out — Availability: Yes Status: Normal Mode On, Idle Mode On, Sleep Out — Availability: Yes Status: Sleep In or Booster Off — Availability: Yes									

Himax Confidential
Do Not Copy

5.19.30 SETREADINDEX: Set SPI Read Index (FEh)

FEH	SET SPI READ INDEX (Set SPI READ Command Address)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	1	1	1	1	1	1	0	FE
1st parameter	1	CMD_ADD[7:0]								-
Description	SET SPI Read Command Address for User Define Command.									
Restrictions	SETEXTC turn on to enable this command									
Register Availability	Status	Availability								
	Normal Mode On, Idle Mode Off, Sleep Out	Yes								
	Normal Mode On, Idle Mode On, Sleep Out	Yes								
	Sleep In or Booster Off	Yes								

Himax Confidential
Do Not Copy

This information contained herein is the exclusive property of Himax and shall not be distributed, reproduced, or disclosed in whole or in part without prior written permission of Himax.

5.19.31 GETSPIREAD: SPI Read Command Data (FFh)

FFH	GETSPIREAD (Read Command Data)									
	DCX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	1	1	1	1	1	1	1	FF
1st parameter	1	CMD_DATA1[7:0]								-
:	1	:								-
nth parameter	1	CMD_DATAn[7:0]								-
Description	Read SPI Command Data for User Define Command.									
Restrictions	SETEXTC turn on to enable this command.									
Register Availability	Status	Availability								
	Normal Mode On, Idle Mode Off, Sleep Out	Yes								
	Normal Mode On, Idle Mode On, Sleep Out	Yes								
	Sleep In or Booster Off	Yes								

Himax Confidential
Do Not Copy

This information contained herein is the exclusive property of Himax and shall not be distributed, reproduced, or disclosed in whole or in part without prior written permission of Himax.

6. Layout Recommendation

6.1 Maximum layout resistance

Name	Type	Maximum series resistance	Unit
VDD1	Power supply	5	Ω
VDD3	Power supply	5	Ω
VPP	Power supply	5	Ω
HS_VCC	Power supply	5	Ω
HS_VSS	Power supply	5	Ω
VSSD	Power supply	5	Ω
VSSA	Power supply	5	Ω
VSSAC	Power supply	20	Ω
BS[2:0]	Input	100	Ω
PCCS[1:0]	Input	100	Ω
CSX, DCX, RESX, SCL	Input	100	Ω
HSYNC, DE, VSYNC, PCLK	Input	100	Ω
SDI	Input	100	Ω
SDO	Output	100	Ω
DB[23:0]	Input	100	Ω
CABC_PWM_OUT, TE, TE1	Output	100	Ω
VCOM	Output	10	Ω
VCOMR	Input	100	Ω
HS_CP, HS_CN	Input	5	Ω
HS_D0P, HS_D0N	Input / Output	5	Ω
HS_D1P, HS_D1N	Input	5	Ω
HS_D2P, HS_D2N	Input	5	Ω
HS_D3P, HS_D3N	Input	5	Ω
VDDD	Capacitor Connection	5	Ω
VCL	Capacitor Connection	10	Ω
VSP, VSN	Capacitor Connection	10	Ω
VSPR, VSNR	Capacitor Connection	50	Ω
VREF	Capacitor Connection	20	Ω
VGH, VGL	Capacitor Connection	10	Ω
HS_LDO	Capacitor Connection	5	Ω
OSC	Input	100	Ω
C21P, C21N, C22P, C22N, C31P, C31N, C41P, C41N	Capacitor Connection	5	Ω
VCSW1, VCSW2	Output	10	Ω
GOUT_L1~22, GOUT_R1~22	Output	10	Ω
TEST[2:0]	Input	100	Ω
FRM	Input	100	Ω
VTESTOUTP, VTESTOUTN	Output	100	Ω

Table 6.1: Maximum layout resistance

Himax Confidential Do not Copy

6.2 External components connection

HX5186-A/B/C mode:

Pad Name	Symbol	Connection	Typical Component Value
VDD3	C1	Connect to Capacitor **(Max 10V)**: VDD3 ---(+)----| |--- (-)-----VSSA	2.2 μF
VDD1	C2	Connect to Capacitor **(Max 6V)**: VDD1 ---(+)----| |--- (-)-----VSSA	2.2 μF
C41P – C41N	C3	Connect to Capacitor **(Max 6V)**: C41P ---(+)----| |--- (-)-----C41N	1.0 μF
VCL	C4	Connect to Capacitor **(Max 6V)**: VCL ---(-)----| |--- (+)----- VSSA	1.0 μF
C21P – C21N	C5	Connect to Capacitor **(Max 16V)**: C21P ---(+)----| |--- (-)-----C21N	1.0 μF
C22P – C22N	C6	Connect to Capacitor **(Max 16V)**: C22P ---(+)----| |--- (-)-----C22N	1.0 μF
C31P – C31N	C7	Connect to Capacitor **(Max 16V)**: C31P ---(+)----| |--- (-)-----C31N	1.0 μF
VGL	C8	Connect to Capacitor **(Max 25V)**: VGL ---(-)----| |--- (+)----- VSSA	1.0 μF
	D1	Connect to Schottky Diode(VR≥30V): VSSA ---(-)----∫◄--- (+)---- VGL	VF < 0.4V / 20mA @ 25°C, VR ≥30V **(Recommended diode: RB521S-30)**
VGH	C9	Connect to Capacitor **(Max 25V)**: VGH ---(+)----| |--- (-)----- VSSA	1.0 μF
VREF	C10	Connect to Capacitor **(Max 6V)**: VREF ---(+)----| |--- (-)-----VSSA	1.0 μF
VCOM	C11	Connect to Capacitor **(Max 6V)**: VCOM ---(-)----| |--- (+)----- VSSA	2.2 μF
VSNR	C12	Connect to Capacitor **(Max 10V)**: VSNR ---(+)----| |--- (-)-----VSSA	1.0 μF
VSPR	C13	Connect to Capacitor **(Max 10V)**: VSPR ---(-)----| |--- (+)-----VSSA	1.0 μF
VDDD	C14	Connect to Capacitor **(Max 6V)**: VDDD ---(+)----| |--- (-)-----VSSA	2.2 μF
HS_LDO	C15	Connect to Capacitor **(Max 6V)**: HS_LDO ---(+)----| |--- (-)----HS_VSS	1.0 μF
VSP	C16	Connect to Capacitor **(Max 10V)**: VSP ---(+)----| |--- (-)-----VSSA	2.2 μF
VSN	C17	Connect to Capacitor **(Max 10V)**: VSN ---(-)----| |--- (+)-----VSSA	2.2 μF
HX5186-A/B/C	U1	Please refer HX5186-A/B/C datasheet	-
HX5186	C18	Please refer HX5186-A datasheet	1.0uF
HX5186	C19	Please refer HX5186-A datasheet	1.0uF
HX5186	C20	Please refer HX5186-A datasheet	1.0uF

Table 6.2: HX5186-A/B/C mode external components

Himax Confidential
Do Not Copy

PFM type A mode:

Pad Name	Symbol	Connection	Typical Component Value
VDD3	C1	Connect to Capacitor **(Max 10V)**: VDD3 ---(+)----| |--- (-)-----VSSA	2.2 μF
VDD1	C2	Connect to Capacitor **(Max 6V)**: VDD1 ---(+)----| |--- (-)-----VSSA	2.2 μF
C41P – C41N	C3	Connect to Capacitor **(Max 6V)**: C41P ---(+)----| |--- (-)-----C41N	1.0 μF
VCL	C4	Connect to Capacitor **(Max 6V)**: VCL ---(-)----| |--- (+)----- VSSA	1.0 μF
C21P – C21N	C5	Connect to Capacitor **(Max 16V)**: C21P ---(+)----| |--- (-)-----C21N	1.0 μF
C22P – C22N	C6	Connect to Capacitor **(Max 16V)**: C22P ---(+)----| |--- (-)-----C22N	1.0 μF
C31P – C31N	C7	Connect to Capacitor **(Max 16V)**: C31P ---(+)----| |--- (-)-----C31N	1.0 μF
VGL	C8	Connect to Capacitor **(Max 25V)**: VGL ---(-)----| |--- (+)----- VSSA	1.0 μF
	D1	Connect to Schottky Diode(VR≥30V): VSSA ---(-)----∫◄--- (+)---- VGL	VF < 0.4V / 20mA @ 25°C, VR ≥30V **(Recommended diode: RB521S-30)**
VGH	C9	Connect to Capacitor **(Max 25V)**: VGH ---(+)----| |--- (-)----- VSSA	1.0 μF
VREF	C10	Connect to Capacitor **(Max 6V)**: VREF ---(+)----| |--- (-)-----VSSA	1.0 μF
VCOM	C11	Connect to Capacitor **(Max 6V)**: VCOM ---(-)----| |--- (+)----- VSSA	2.2 μF
VSNR	C12	Connect to Capacitor **(Max 10V)**:VSNR ---(+)----| |--- (-)-----VSSA	1.0 μF
VSPR	C13	Connect to Capacitor **(Max 10V)**:VSPR ---(-)----| |--- (+)-----VSSA	1.0 μF
VDDD	C14	Connect to Capacitor **(Max 6V)**: VDDD ---(+)----| |--- (-)-----VSSA	2.2 μF
HS_LDO	C15	Connect to Capacitor **(Max 6V)**: HS_LDO ---(+)----| |--- (-)----HS_VSS	1.0 μF
VSN	C16	Connect to Capacitor **(Max 10V)**:VSN ---(-)----| |--- (+)-----VSSA	4.7 μF
VSP	C17	Connect to Capacitor **(Max 10V)**:VSP ---(+)----| |--- (-)-----VSSA	4.7 μF
PFM	L1	Inductance,please refer PFM type A connection.	10.0uH
PFM	D2	Schottky Diode , please refer PFM type A connection.	VF < 0.4V / 20mA @ 25°C, VR ≥30V **(Recommended diode: RB521S-30)**
PFM	D3	Schottky Diode , please refer PFM type A connection.	VF < 0.4V / 20mA @ 25°C, VR ≥30V **(Recommended diode: RB521S-30)**
PFM	SW1	MOS switch, please refer PFM type A connection.	-
PFM	SW2	MOS switch, please refer PFM type A connection.	-

Table 6.3: PFM type A external components

Himax Confidential
Do Not Copy

PFM type C mode:

Pad Name	Symbol	Connection	Typical Component Value
VDD3	C1	Connect to Capacitor **(Max 10V)**: VDD3 ---(+)----| |--- (-)-----VSSA	2.2 µF
VDD1	C2	Connect to Capacitor **(Max 6V)**: VDD1 ---(+)----| |--- (-)-----VSSA	2.2 µF
C41P – C41N	C3	Connect to Capacitor **(Max 6V)**: C41P ---(+)----| |--- (-)-----C41N	1.0 µF
VCL	C4	Connect to Capacitor **(Max 6V)**: VCL ---(-)----| |--- (+)----- VSSA	1.0 µF
C21P – C21N	C5	Connect to Capacitor **(Max 16V)**: C21P ---(+)----| |--- (-)-----C21N	1.0 µF
C22P – C22N	C6	Connect to Capacitor **(Max 16V)**: C22P ---(+)----| |--- (-)-----C22N	1.0 µF
C31P – C31N	C7	Connect to Capacitor **(Max 16V)**: C31P ---(+)----| |--- (-)-----C31N	1.0 µF
VGL	C8	Connect to Capacitor **(Max 25V)**: VGL ---(-)----| |--- (+)----- VSSA	1.0 µF
	D1	Connect to Schottky Diode(VR≥30V): VSSA ---(-)----∫◄--- (+)---- VGL	VF < 0.4V / 20mA @ 25°C, VR ≥30V **(Recommended diode: RB521S-30)**
VGH	C9	Connect to Capacitor **(Max 25V)**: VGH ---(+)----| |--- (-)----- VSSA	1.0 µF
VREF	C10	Connect to Capacitor **(Max 6V)**: VREF ---(+)----| |--- (-)-----VSSA	1.0 µF
VCOM	C11	Connect to Capacitor **(Max 6V)**: VCOM ---(-)----| |--- (+)----- VSSA	2.2 µF
VSNR	C12	Connect to Capacitor **(Max 10V)**: VSNR ---(+)----| |--- (-)-----VSSA	1.0 µF
VSPR	C13	Connect to Capacitor **(Max 10V)**: VSPR ---(-)----| |--- (+)-----VSSA	1.0 µF
VDDD	C14	Connect to Capacitor **(Max 6V)**: VDDD ---(+)----| |--- (-)-----VSSA	2.2µF
HS_LDO	C15	Connect to Capacitor **(Max 6V)**: HS_LDO ---(+)----| |--- (-)----HS_VSS	1.0 µF
VSN	C16	Connect to Capacitor **(Max 10V)**: VSN ---(-)----| |--- (+)-----VSSA	4.7 µF
VSP	C17	Connect to Capacitor **(Max 10V)**: VSP ---(+)----| |--- (-)-----VSSA	4.7 µF
PFM	L1	Inductance,please refer PFM type C connection.	10uH
PFM	D2	Schottky Diode, please refer PFM type C connection.	VF < 0.4V / 20mA @ 25°C, VR ≥30V **(Recommended diode: RB521S-30)**
PFM	D3	Schottky Diode, please refer PFM type C connection.	VF < 0.4V / 20mA @ 25°C, VR ≥30V **(Recommended diode: RB521S-30)**
PFM	D4	Schottky Diode, please refer PFM type C connection.	VF < 0.4V / 20mA @ 25°C, VR ≥30V **(Recommended diode: RB521S-30)**
PFM	SW1	MOS switch, please refer PFM type C connection.	-
PFM	C18	Capacitor, please refer PFM type C connection.	1.0 µF

Table 6.4: PFM type C external components

Himax Confidential
Do Not Copy

PFM type D mode:

Pad Name	Symbol	Connection	Typical Component Value
VDD3	C1	Connect to Capacitor **(Max 10V)**: VDD3 ---(+)----| |--- (-)-----VSSA	2.2 μF
VDD1	C2	Connect to Capacitor **(Max 6V)**: VDD1 ---(+)----| |--- (-)-----VSSA	2.2 μF
C41P – C41N	C3	Connect to Capacitor **(Max 6V)**: C41P ---(+)----| |--- (-)-----C41N	1.0 μF
VCL	C4	Connect to Capacitor **(Max 6V)**: VCL ---(-)----| |--- (+)----- VSSA	1.0 μF
C21P – C21N	C5	Connect to Capacitor **(Max 16V)**: C21P ---(+)----| |--- (-)-----C21N	1.0 μF
C22P – C22N	C6	Connect to Capacitor **(Max 16V)**: C22P ---(+)----| |--- (-)-----C22N	1.0 μF
C31P – C31N	C7	Connect to Capacitor **(Max 16V)**: C31P ---(+)----| |--- (-)-----C31N	1.0 μF
VGL	C8	Connect to Capacitor **(Max 25V)**: VGL ---(-)----| |--- (+)----- VSSA	1.0 μF
	D1	Connect to Schottky Diode(VR≥30V): VSSA ---(-)----∫◄--- (+)---- VGL	VF < 0.4V / 20mA @ 25°C, VR ≥30V **(Recommended diode: RB521S-30)**
VGH	C9	Connect to Capacitor **(Max 25V)**: VGH ---(+)----| |--- (-)----- VSSA	1.0 μF
VREF	C10	Connect to Capacitor **(Max 6V)**: VREF ---(+)----| |--- (-)-----VSSA	1.0 μF
VCOM	C11	Connect to Capacitor **(Max 6V)**: VCOM ---(-)----| |--- (+)----- VSSA	2.2 μF
VSNR	C12	Connect to Capacitor **(Max 10V)**: VSNR ---(+)----| |--- (-)-----VSSA	1.0 μF
VSPR	C13	Connect to Capacitor **(Max 10V)**: VSPR ---(-)----| |--- (+)-----VSSA	1.0 μF
VDDD	C14	Connect to Capacitor **(Max 6V)**: VDDD ---(+)----| |--- (-)-----VSSA	2.2 μF
HS_LDO	C15	Connect to Capacitor **(Max 6V)**: HS_LDO ---(+)----| |--- (-)----HS_VSS	1.0 μF
VSN	C16	Connect to Capacitor **(Max 10V)**: VSN ---(-)----| |--- (+)-----VSSA	4.7 μF
VSP	C17	Connect to Capacitor **(Max 10V)**: VSP ---(+)----| |--- (-)-----VSSA	4.7 μF
PFM	L1	Inductance,please refer PFM type D connection.	10.0uH
PFM	D2	Schottky Diode, please refer PFM type D connection.	VF < 0.4V / 20mA @ 25°C, VR ≥30V **(Recommended diode: RB521S-30)**
PFM	SW1	MOS switch, please refer PFM type Dconnection.	-

Table 6.5: PFM type D external components

Himax Confidential
Do Not Copy

External VDD1/ VSP / VSN / VGH / VGL mode:

Pad Name	Symbol	Connection	Typical Component Value
VSP	C1	Connect to Capacitor **(Max 10V)**: VSP ---(+)----| |--- (-)-----VSSA	2.2 μF
VSN	C2	Connect to Capacitor **(Max 10V)**: VSN ---(-)----| |--- (+)-----VSSA	2.2 μF
VDD1	C3	Connect to Capacitor **(Max 6V)**: VDD3 ---(+)----| |--- (-)-----VSSA	2.2 μF
VDD3	C4	Connect to Capacitor **(Max 10V)**: VDD3 ---(+)----| |--- (-)-----VSSA	2.2 μF
C41P – C41N	C5	Connect to Capacitor **(Max 6V)**: C41P ---(+)----| |--- (-)-----C41N	1.0 μF
VCL	C6	Connect to Capacitor **(Max 6V)**: VCL ---(-)----| |--- (+)----- VSSA	1.0 μF
VGL	C7	Connect to Capacitor **(Max 25V)**: VGL ---(-)----| |--- (+)----- VSSA	1.0 μF
	D1	Connect to Schottky Diode(VR≥30V): VSN ---(-)----∫◄--- (+)---- VGL	VF < 0.4V / 20mA @ 25°C, VR ≥30V **(Recommended diode: RB521S-30)**
VGH	C8	Connect to Capacitor **(Max 25V)**: VGH ---(+)----| |--- (-)----- VSSA	1.0 μF
VREF	C9	Connect to Capacitor **(Max 6V)**: VREF ---(+)----| |--- (-)-----VSSA	1.0 μF
VCOM	C10	Connect to Capacitor **(Max 6V)**: VCOM ---(-)----| |--- (+)----- VSSA	2.2 μF
VSNR	C11	Connect to Capacitor **(Max 10V)**: VSNR ---(+)----| |--- (-)-----VSSA	1.0 μF
VSPR	C12	Connect to Capacitor **(Max 10V)**: VSPR ---(-)----| |--- (+)-----VSSA	1.0 μF
VDDD	C13	Connect to Capacitor **(Max 6V)**: VDDD ---(+)----| |--- (-)-----VSSA	2.2 μF
HS_LDO	C14	Connect to Capacitor **(Max 6V)**: HS_LDO ---(+)----| |--- (-)----HS_VSS	1.0 μF

Table 6.6: External VDD1 / VSP / VSN / VGH / VGL external components

Himax Confidential
Do Not Copy

External VDD1/ VSP / VSN mode:

Pad Name	Symbol	Connection	Typical Component Value
VSP	C1	Connect to Capacitor **(Max 10V)**: VSP ---(+)----| |--- (-)-----VSSA	2.2 µF
VSN	C2	Connect to Capacitor **(Max 10V)**: VSN ---(-)----| |--- (+)-----VSSA	2.2 µF
VDD1	C3	Connect to Capacitor **(Max 6V)**: VDD1 ---(+)----| |--- (-)-----VSSA	2.2 µF
VDD3	C4	Connect to Capacitor **(Max 10V)**: VDD3 ---(+)----| |--- (-)-----VSSA	2.2 µF
C41P – C41N	C5	Connect to Capacitor **(Max 6V)**: C41P ---(+)----| |--- (-)-----C41N	1.0 µF
VCL	C6	Connect to Capacitor **(Max 6V)**: VCL ---(-)----| |--- (+)----- VSSA	1.0 µF
C21P – C21N	C7	Connect to Capacitor **(Max 16V)**: C21P ---(+)----| |--- (-)-----C21N	1.0 µF
C22P – C22N	C8	Connect to Capacitor **(Max 16V)**: C22P ---(+)----| |--- (-)-----C22N	1.0 µF
C31P – C31N	C9	Connect to Capacitor **(Max 16V)**: C31P ---(+)----| |--- (-)-----C31N	1.0 µF
VGL	C10	Connect to Capacitor **(Max 25V)**: VGL ---(-)----| |--- (+)----- VSSA	1.0 µF
	D1	Connect to Schottky Diode(VR≥30V): VSN ---(-)----∫◄--- (+)---- VGL	VF < 0.4V / 20mA @ 25°C, VR ≥30V **(Recommended diode: RB521S-30)**
VGH	C11	Connect to Capacitor **(Max 25V)**: VGH ---(+)----| |--- (-)----- VSSA	1.0 µF
	D2	Connect to Schottky Diode(VR≥30V): VGH ---(-)----∫◄--- (+)---- VSP	VF < 0.4V / 20mA @ 25°C, VR ≥30V **(Recommended diode: RB521S-30)**
VREF	C12	Connect to Capacitor **(Max 6V)**: VREF ---(+)----| |--- (-)-----VSSA	1.0 µF
VCOM	C13	Connect to Capacitor **(Max 6V)**: VCOM ---(-)----| |--- (+)----- VSSA	2.2 µF
VSNR	C14	Connect to Capacitor **(Max 10V)**: VSNR ---(+)----| |--- (-)-----VSSA	1.0 µF
VSPR	C15	Connect to Capacitor **(Max 10V)**: VSPR ---(-)----| |--- (+)-----VSSA	1.0 µF
VDDD	C16	Connect to Capacitor **(Max 6V)**: VDDD ---(+)----| |--- (-)-----VSSA	2.2 µF
HS_LDO	C17	Connect to Capacitor **(Max 6V)**: HS_LDO ---(+)----| |--- (-)----HS_VSS	1.0 µF

Table 6.7: External VDD1 / VSP / VSN external components

Himax Confidential
Do Not Copy

7. Electrical Characteristics

7.1 Absolute maximum ratings

The absolute maximum ratings are list on Table 7.1. When used out of the absolute maximum ratings, the LSI may be permanently damaged. Using the LSI within the following electrical characteristics limit is strongly recommended for normal operation. If these electrical characteristic conditions are exceeded during normal operation, the LSI will malfunction and cause poor reliability.

Parameter	Symbol	Spec.			Unit	Note
		Min.	Typ.	Max.		
Power Supply Voltage 1	VDD1~ VSSD	-0.3	-	+3.6	V	Note(1),(2)
Power Supply Voltage 2	VDD3 ~ VSSA	-0.3	-	+3.6	V	Note(1) (4)
Power Supply Voltage 3	HS_VCC ~ HS_VSS	-0.3	-	+3.6	V	Note(1) (5)
Power Supply Voltage 4	VSP ~ VSSA	-0.3	-	+6.6	V	Note(6)
Power Supply Voltage 5	VSSA ~ VSN	-6.6	-	0	V	Note(7)
Power Supply Voltage 6	VGH ~ VSSA	-0.3	-	+19.6	V	Note(8)
Power Supply Voltage 7	VSSA ~ VGL	-16	-	0	V	Note(9)
Operating Temperature	Topr	-40	-	+85	°C	Note(10)
Storage Temperature	Tstg	-55	-	+110	°C	Note(11)
Input Voltage	V_{IN}	-0.3	-	VDD1+0.3	V	Note(12)
HS Input Voltage	V_{HSIN}	-0.3	-	+2	V	Note(13)

Note: (1) VDD1, VSSD must be maintained.
(2) To make sure VDD1 ≥ VSSD.
(4) To make sure VDD3≥ VSSA.
(5) To make sure HS_VCC ≥ HS_VSS.
(6) To make sure VSP ≥ VSSA.
(7) To make sure VSSA ≥ VSN
(8) To make sure VGH ≥ VSSA.
(9) To make sure VSSA ≥ VGL, VGH +|VGL| < 32V
(10) For die and wafer products, specified up to +85℃.
(11) This temperature specifications apply to the TCP package.
(12) This specifications include input signals but without following: CP, CN, D0P, D0N, D1P, D1N, D2P, D2N, D3P, D3N.
(13) This specifications include following signals: CP, CN, D0P, D0N, D1P, D1N, D2P, D2N, D3P, D3N.

Table 7.1: Absolute maximum rating

Himax Confidential Do Not Copy

This information contained herein is the exclusive property of Himax and shall not be distributed, reproduced, or disclosed in whole or in part without prior written permission of Himax.

7.2 DC characteristics

(VDD3=2.5 ~ 3.6V, VDD1=1.65~3.6V, T_A=-40 ~ 85 °C)

Parameter	Symbol	Test condition	Spec.			Unit	Note
			Min.	Typ.	Max.		
Input high voltage	V_{IH}	VDD1= 1.65 ~ 3.6V VDD3= 2.5 ~ 3.6V	0.7 V_{DD1}	-	VDD1	V	-
Input low voltage	V_{IL}		0	-	0.3 V_{DD1}	V	-
VPP	V_{IH}	VPP	8.0	8.25	8.5	V	(1)
	V_{IL}					V	
Output high voltage **(SDO, CABC_PWM_OUT)**	V_{OH1}	I_{OH} = -1.0 mA	0.8 V_{DD1}	-	VDD1	V	-
Output low voltage **(SDO, CABC_PWM_OUT)**	V_{OL1}	VDD1= 1.65 ~ 3.6V I_{OL} = 1.0 mA	0	-	0.2 V_{DD1}	V	-
Logic High level input current	I_{IH}	VSYNC, HSYNC	-	-	1	uA	-
		RESX, DCX, CSX, SCL	-	-	1		-
	I_{IHD}	DB[23:0], SDI, DCX	-	-	1	uA	-
		DB[23:0]	-	-	1		
Logic Low level input current	I_{IL}	VSYNC, HSYNC	-1	-	-	uA	-
		RESX, DCX, CSX, SCL	-1	-	-		-
	I_{ILD}	DB[23:0], SDI, DCX	-1	-	-	uA	
		DB[23:0]	-1	-	-		-
Current consumption standby mode **(VDD3-VSSA)**	$I_{ST(VDD)}$	VDD3/HS_VCC=2.8V, VDD1=1.8V T_A =25°C (DSI Ultra Low Power mode)	-	--	TBD	µA	-
Current consumption standby mode **(VDD1– VSSD)**	$I_{ST(VDD1)}$		-	--	TBD	µA	-
Current consumption standby mode which include HS_VCC **(HS_VCC-HS_VSS)**	$I_{ST(VDD1)}$		-	--	TBD	µA	-
Oscillator tolerance	△OSC	T_A =25℃	-5	-	5	%	-

Note: (1) The VPP pin is open on normal mode and in used while OTP programming condition.

Table 7.2: DC characteristic

Himax Confidential
Do Not Copy

7.3 AC characteristics

7.3.1 DBI Type C interface characteristics

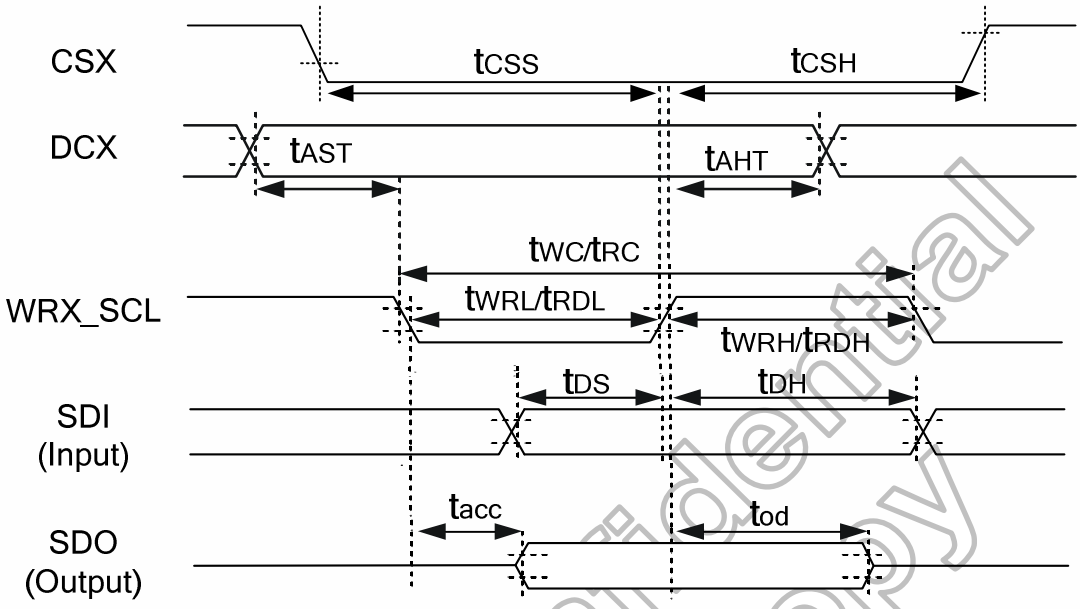


Figure 7.1: DBI Type C interface characteristics

(VSSA=0V, VDD1=1.8V, VDD2=2.8V, VDD3=2.8V, T_A = 25℃)

Parameter	Signal	Symbol	Description	Spec.		Unit
				Min.	Max.	
Chip select setup time **(Write)** Chip select setup time **(Read)**	CSX	t_{CSS} t_{CSH}	-	40 40	- -	ns
Address setup time Address hold time **(Write/Read)**	DCX	t_{AST} t_{AHT}	-	10 10	- -	ns
Write cycle Control pulse “H” duration Control pulse “L” duration	WRX_SCL **(Write)**	t_{WC} t_{WRH} t_{WRL}	-	100 40 40	- - -	ns
Read cycle Control pulse “H” duration Control pulse “L” duration	WRX_SCL **(Read)**	t_{RC} t_{RDH} t_{RDL}	-	150 60 60	- - -	ns
Data setup time Data hold time	SDI/SDO **(Input)**	t_{DS} t_{DT}	For maximum C_L=30pF For minimum C_L=8pF	30 30	- -	ns
Read access time Output disable time	SDI/SDO **(Output)**	t_{RACC} t_{OD}		10 10	- 50	ns

Note: The input signal rise time and fall time (tr, tf) is specified at 15 ns or less.
Logic high and low levels are specified as 30% and 70% of VDD1 for Input signals.

Table 7.3: DBI Type C interface characteristics

Himax Confidential
Do Not Copy

7.3.2 DSI D-PHY electrical characteristics

7.3.2.1 Electrical characteristics of D-PHY layer

In general, the DSI **D-**PHY may contain the following electrical functions: High-Speed Receiver **(HS-RX)**, Low Power Transmitter **(LP-TX)**, a Low-Power Receiver **(LP-RX)**, and the Low-Power Contention Detector **(LP-CD)**. Figure 7.5 shows the complete set of electrical functions required for a fully featured PHY transceiver.

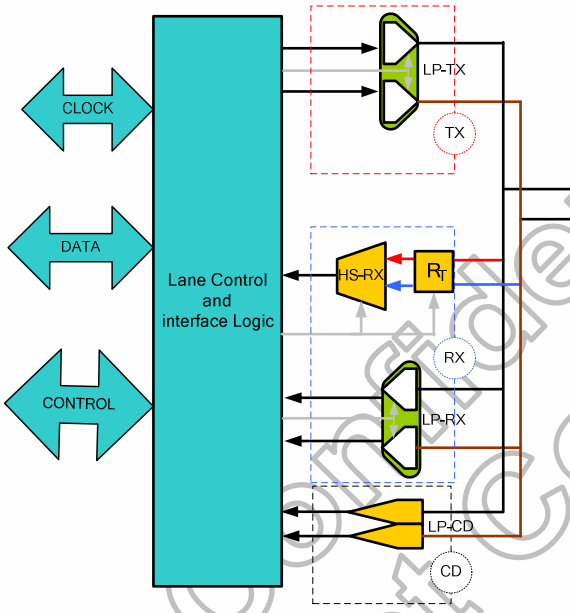


Figure 7.2: Electrical functions of a fully D-PHY transceiver

Where, the HS receiver utilize low-voltage swing differential signaling for signal transmission. The LP transmitter and LP receiver serve as a low power signaling mechanism. The Figure 7.6 shows both the HS and LP signal levels on the left and right sides, respectively.

Because the HS signaling levels are below the LP low-level input threshold, Lane switches between Low-Power and High-Speed mode during normal operation.

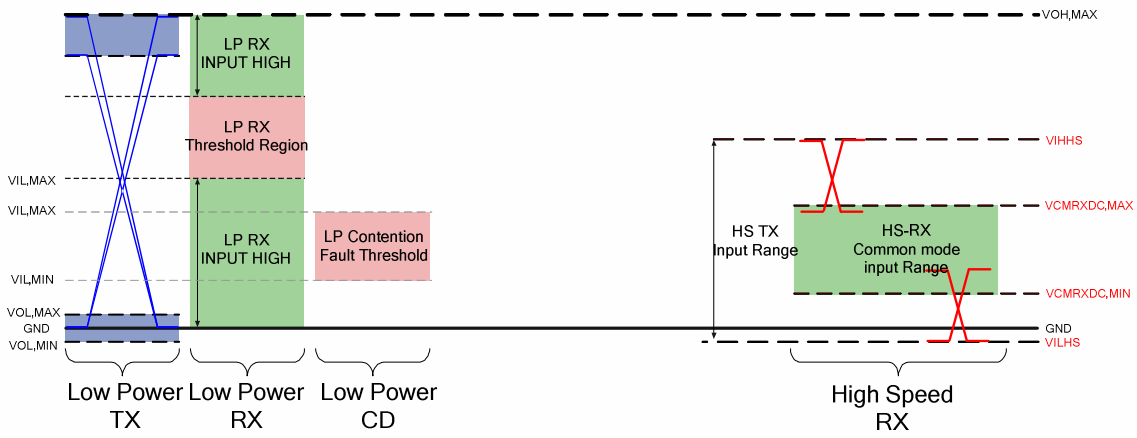


Figure 7.3: Shows both the HS and LP signal levels

Himax Confidential
Do Not Copy

7.3.2.2 Electrical characteristics of low-power transmitter

The Low-Power transmitter shall be a slew-rate controlled push-pull driver. It is used for driving the Lines in all Low-Power operating modes It is therefore important that the static power consumption of a LP transmitter be as low as possible. Under tables list DC and AC characteristic for LP-TX.

Parameter	Description	Spec.			Unit	Note
		Min.	Typ.	Max.		
V_{OL}	Thevenin output low level	-50	-	50	mV	-
V_{OH}	Thevenin output high level	1.1	1.2	1.3	V	-
Z_{OLP}	Output impedance of LP-TX	110	-	-	Ω	(1)

Note: (1)Though no maximum value for Z_{OLP} is specified, the LP transmitter output impedance shall ensure the t_{RLP}/t_{FLP} specification is met.

Table 7.4: LP transmitter DC specifications

Parameter	Description	Spec.			Unit	Note
		Min.	Typ.	Max.		
t_{RLP}/t_{FLP}	15%-85% rise time and fall time	-	-	25	ns	(1)
$\delta V/\delta t_{SR}$	Slew rate @ C_{LOAD}=0pF	-	-	500	mV/ns	(1),(3),(5),(6)
	Slew rate @ C_{LOAD}=5pF	-	-	300	mV/ns	(1),(3),(5),(6)
	Slew rate @ C_{LOAD}=20pF	-	-	250	mV/ns	(1),(3),(5),(6)
	Slew rate @ C_{LOAD}=70pF	-	-	150	mV/ns	(1),(3),(5),(6)
	Slew rate @ C_{LOAD}=0 to 70pF **(Falling Edge Only)**	30	-	-	mV/ns	(1),(2),(3)
	Slew rate @ C_{LOAD}=0 to 70pF **(Rising Edge Only)**	30	-	-	mV/ns	(1),(3),(7)
	Slew rate @ C_{LOAD}=0 to 70pF **(Rising Edge Only)**	30 – 0.075 * (VO,INST- 700)	-	-	mV/ns	(1),(8),(9)
C_{LOAD}	Load capacitance	0	-	70	pF	-

Note: (1) C_{LOAD} includes the low-frequency equivalent transmission line capacitance. The capacitance of TX and RX are assumed to always be <10pF. The distributed line capacitance can be up to 50pF for a transmission line with 2ns delay.
(2) When the output voltage is between 400 mV and 930 mV.
(3) Measured as average across any 50 mV segment of the output signal transition.
(4) This parameter value can be lower than TLPX due to differences in rise vs. fall signal slopes and trip levels and mismatches between Dp and Dn LP transmitters.
(5) This value represents a corner point in a piecewise linear curve.
(6) When the output voltage is in the range specified by VPIN(absmax).
(7) When the output voltage is between 400 mV and 700 mV.
(8) Where VO,INST is the instantaneous output voltage, VDP or VDN, in millivolts.
(9) When the output voltage is between 700 mV and 930 mV.

Table 7.5: LP transmitter AC specifications

Himax Confidential
Himax Not Copy

7.3.2.3 Electrical characteristics of receiver

This part will contain two parts which High-Speed Receiver and Low-Power Receiver. Because their have differential DC and AC characteristic, describe HS-RX first then describe LP-RX.

7.3.2.4 High-speed receiver

The HS receiver is a differential line receiver. It contains a switch-able parallel input termination, Z_{ID}, between the positive input pin Dp and the negative input pin Dn. Under Tables list DC and AC characteristic for HS-RX.

Parameter	Description	Spec.			Unit	Note
		Min.	Typ.	Max.		
V_{IDTH}	Differential input high threshold	-	-	70	mV	(1)
V_{IDTL}	Differential input low threshold	-70	-	-	mV	(1)
V_{ILHS}	Single-ended input low voltage	-40	-	-	mV	(2)
V_{IHHS}	Single-ended input high voltage	-	-	460	mV	(2)
V_{CMRXDC}	Common-mode voltage HS receive mode	70	-	330	mV	(2),(3)
Z_{ID}	Differential input impedance	80	100	125	Ω	-

Note: (1) +/-70mV only for reference, related to power and ground noise on system environment, this spec need to check on panel performance to fine tune.
(2) Excluding possible additional RF interference of 100mV peak sine wave beyond 450MHz.
(3) This table value includes a ground difference of 50mV between the transmitter and the receiver, the static common-mode level tolerance and variations below 450MHz

Table 7.6: HS receiver DC Specifications

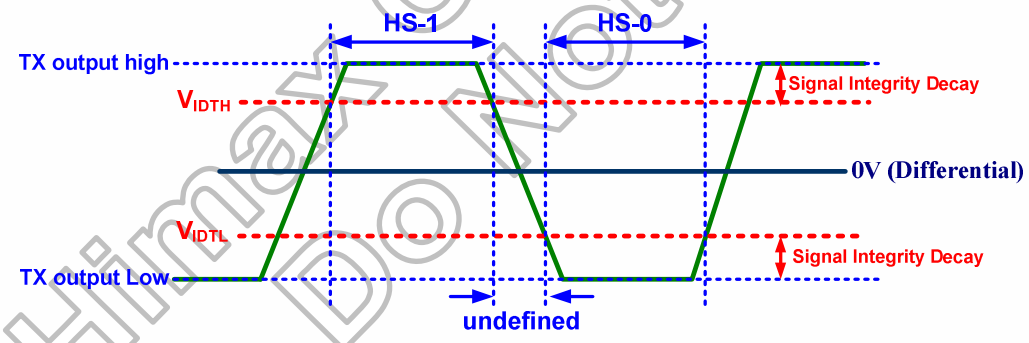


Figure 7.4: Differential HS signals for HS receiver

Parameter	Description	Spec.			Unit	Note
		Min.	Typ.	Max.		
$\Delta V_{CMRX(HF)}$	Common mode interference beyond 450 MHz	-	-	100	mV_{PP}	(1)
C_{CM}	Common mode termination	-	-	60	pF	(2)

Note: (1) $\Delta V_{CMRX(HF)}$ is the peak amplitude of a sine wave superimposed on the receiver inputs.
(2) For higher bit rates a 14pF capacitor will be needed to meet the common-mode return loss specification.

Table 7.7: HS receiver AC Specifications

Himax Confidential Do Not Copy

This information contained herein is the exclusive property of Himax and shall not be distributed, reproduced, or disclosed in whole or in part without prior written permission of Himax.

7.3.2.5 Low-power receiver

The low power receiver is an un-terminated, single-ended receiver circuit. The LP receiver is used to detect the Low-Power state on each pin. For high robustness, the LP receiver shall filter out noise pulses and RF interference. It is recommended the implementer optimize the LP receiver design for low power. The LP receiver shall reject any input glitch when the glitch is smaller than eSpike. The filter shall allow pulses wider than TMIN to propagate through the LP receiver. The related diagram shows as Figure 7.7 Input Glitch Rejection of Low-Power Receivers. Besides, under tables list DC and AC characteristic for LP-RX.

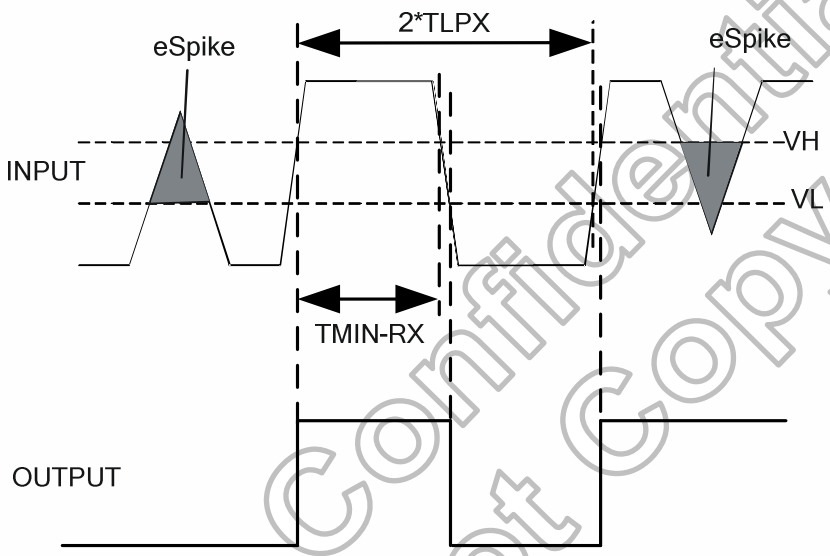


Figure 7.5: Input glitch rejections of low-power receivers

Parameter	Description	Spec.			Unit	Note
		Min.	Typ.	Max.		
V_{IL}	Logic 0 input threshold	-	-	550	mV	-
V_{IH}	Logic 1 input threshold	880	-	-	mV	-

Table 7.8: LP receiver DC specifications

Parameter	Description	Spec.			Unit	Note
		Min.	Typ.	Max.		
e_{SPIKE}	Input pulse rejection	-	-	300	V.ps	(1), (2), (3)
T_{MIN-RX}	Minimum pulse width response	20	-	-	ns	(4)
V_{INT}	Peak-to-peak interference voltage	-	-	200	mV	-
f_{INT}	Interference frequency	450	-	-	MHz	-

Note: (1) Time-voltage integration of a spike above VIL when being in LP-0 state or below VIH when being in LP-1 state
(2) An impulse less than this will not change the receiver state.
(3) In addition to the required glitch rejection, implementers shall ensure rejection of known RF-interferers.
(4) An input pulse greater than this shall toggle the output.

Table 7.9: LP receiver AC specifications

Himax Confidential Do Not Copy

This information contained herein is the exclusive property of Himax and shall not be distributed, reproduced, or disclosed in whole or in part without prior written permission of Himax.

7.3.2.6 Line contention detection

Contention can be inferred from any of the following conditions:

A. An LP high fault shall be detected when the LP transmitter is driving high and the pin voltage is less than V_{IL}.
B. An LP low fault shall be detected when the LP transmitter is driving low and the pad pin voltage is greater than V_{IL}.

Parameter	Description	Spec.			Unit	Note
		Min.	Typ.	Max.		
V_{IHCD}	Logic 1 contention threshold	450	-	-	mV	-
V_{ILCD}	Logic 0 contention threshold	-	-	200	mV	-

Table 7.10: Contention detector DC specifications

Himax Confidential
Do Not Copy

7.3.2.7 High-speed data-clock timing

This section specifies the required timings on the high-speed signaling interface independent of the electrical characteristics of the signal. The PHY is a source synchronous interface in the Forward direction.

The Master side of the Link shall send a differential clock signal to the Slave side to be used for data sampling. This signal shall be a DDR **(Half-rate)** clock and shall have one transition per data bit time. All timing relationships required for correct data sampling are defined relative to the clock transitions. Therefore, implementations may use frequency spreading modulation on the clock to reduce EMI.

The DDR clock signal shall maintain a quadrature phase relationship to the data signal. Data shall be sampled on both the rising and falling edges of the Clock signal. The term “rising edge” means “rising edge of the differential signal, i.e. CP – CN, and similarly for “falling edge”. Therefore, the period of the Clock signal shall be the sum of two successive instantaneous data bit times. This relationship is shown in Figure 7.6.

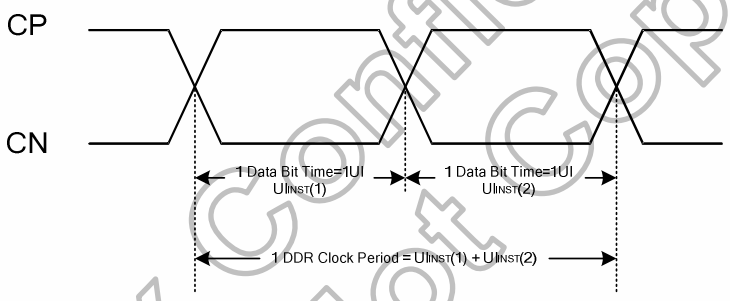


Figure 7.6: DDR clock definition

The same clock source is used to generate the DDR Clock and launch the serial data. Since the Clock and Data signals propagate together over a channel of specified skew, the Clock may be used directly to sample the Data lines in the receiver. Such a system can accommodate large instantaneous variations in UI.

The allowed instantaneous UI variation can cause large, instantaneous data rate variations. Therefore, devices shall either accommodate these instantaneous variations with appropriate FIFO logic outside of the PHY or provide an accurate clock source to the Lane Module to eliminate these instantaneous variations.

Himax Confidential
Do Not Copy

This information contained herein is the exclusive property of Himax and shall not be distributed, reproduced, or disclosed in whole or in part without prior written permission of Himax.

The UI_{INST} specifications for the Clock signal are summarized in Table 7.11.

Parameter	Symbol	Spec.			Unit	Note
		Min.	Typ.	Max.		
UI instantaneous	UI_{INST}	-	-	12.5	ns	(1), (2), (3)

Note: (1) This value corresponds to a minimum 80 Mbps data rate.
(2) The minimum UI shall not be violated for any single bit period, i.e., any DDR half cycle within a data burst.
(3) Maximum total bit rate is 2.2 Gbps of 4 data lanes 24-bit data format/ 1.5Gbps of 4 data lane 18-bit data format/ 1.33Gbps of 4 data lane 16-bit data format.

Table 7.11: Reverse HS data transmission timing parameters

The timing relationship of the DDR Clock differential signal to the Data differential signal is shown in Figure 7.7. Data is launched in a quadrature relationship to the clock such that the Clock signal edge may be used directly by the receiver to sample the received data.

The transmitter shall ensure that a rising edge of the DDR clock is sent during the first payload bit of a transmission burst such that the first payload bit can be sampled by the receiver on the rising clock edge, the second bit can be sampled on the falling edge, and all following bits can be sampled on alternating rising and falling edges.

All timing values are measured with respect to the actual observed crossing of the Clock differential signal. The effects due to variations in this level are included in the clock to data timing budget.

Receiver input offset and threshold effects shall be accounted as part of the receiver setup and hold parameters.

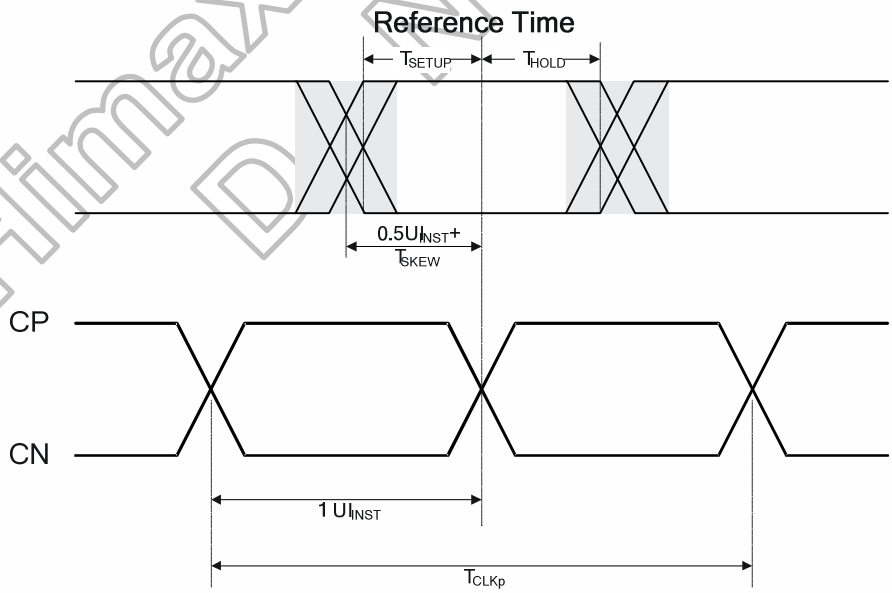


Figure 7.7: Data to clock timing definitions

Himax Confidential
Do Not Copy

7.3.2.8 Data-clock timing specifications

The Data-Clock timing specifications are shown in Table 7.12. Implementers shall specify a value UI_{INST},MIN that represents the minimum instantaneous UI possible within a High-Speed data transfer for a given implementation. Parameters in Table 7.12 are specified as a part of this value. The skew specification, $T_{SKEW[TX]}$, is the allowed deviation of the data launch time to the ideal ½UI_{INST} displaced quadrature clock edge. The setup and hold times, $T_{SETUP[RX]}$ and $T_{HOLD[RX]}$, respectively, describe the timing relationships between the data and clock signals. $T_{SETUP[RX]}$ is the minimum time that data shall be present before a rising or falling clock edge and $T_{HOLD[RX]}$ is the minimum time that data shall remain in its current state after a rising or falling clock edge. The timing budget specifications for a receiver shall represent the minimum variations observable at the receiver for which the receiver will operate at the maximum specified acceptable bit error rate.

The intent in the timing budget is to leave 0.4*U_{IINST}, i.e. ±0.2*UI_{INST} for degradation contributed by the interconnect.

Parameter	Symbol	Spec.			Unit	Note
		Min.	Typ.	Max.		
Data to Clock Setup Time [Receiver]	$T_{SETUP[RX]}$	0.15	-	-	UI_{INST}	(1)
Clock to Data Hold Time [Receiver]	$T_{HOLD[RX]}$	0.15	-	-	UI_{INST}	(1)

Note: (1) Total setup and hold window for receiver of 0.3*UI_{INST}.
(2) 0.15UI is only for reference, related to the signal jitter caused by the transmittion path, this spec need to check on panel performance to fine tune.

Table 7.12: Data to clock timing specifications

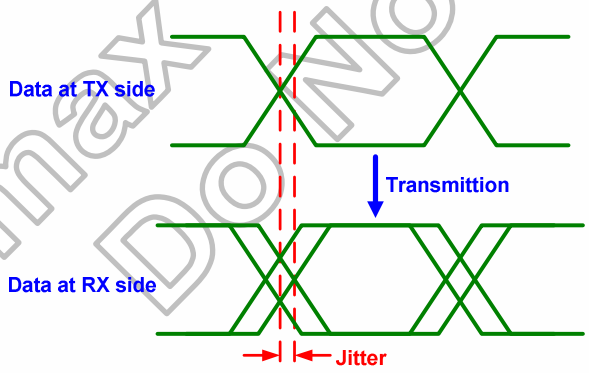


Figure 7.8: Skew window of transmittor and receiver

Himax Confidential DoNot Copy

7.3.3 Timings for DSI video mode

7.3.3.1 Vertical timings

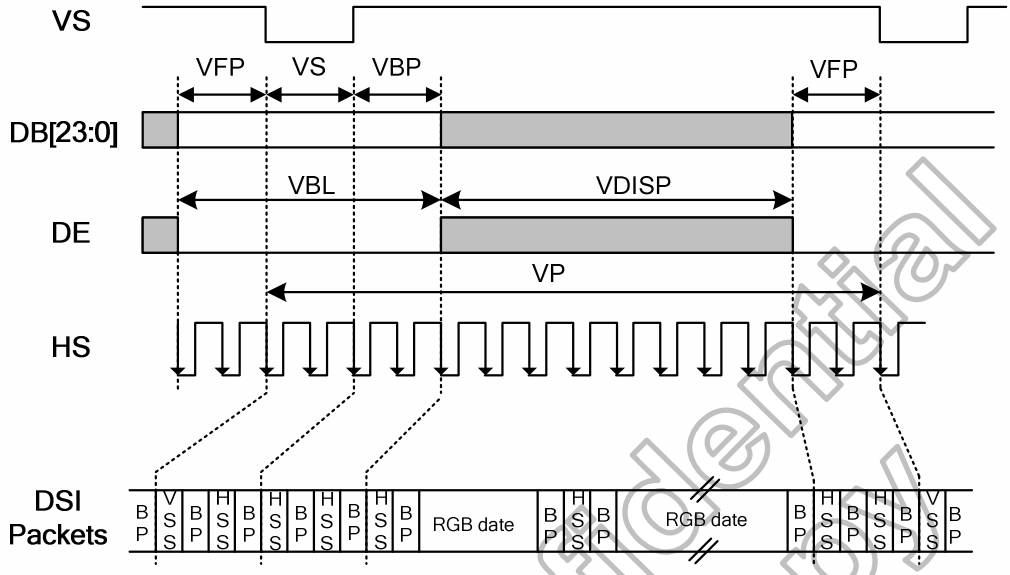


Figure 7.9: Vertical timings for DSI I/F

Resolution=720x1280(VSSA=0V, VDD1=1.8V, VDD2=2.8V, VDD3=2.8V, T_A=25℃)

Parameter	Symbol	Condition	Spec.			Unit
			Min.	Typ.	Max.	
Vertical cycle	VP	-	1286	-	-	Line
Vertical low pulse width	VS	-	2	-	Note(1)	Line
Vertical front porch	VFP	-	2	-	-	Line
Vertical back porch	VBP	-	2	-	Note(1)	Line
Vertical data start point	-	VS+VBP	4	-	Note(1)	Line
Vertical blanking period	VBL	VS+VBP+VFP	6	-	-	Line
Vertical active area	-	VDISP	-	1280	-	Line
Vertical Refresh rate	VRR	-	-	60	-	Hz

Note: (1) The VS/VBP/VFP pulse width are related to RD3h (Set GIP Option0). Please refer to the application note for VS/VBP/VFP setting.

Table 7.13: Vertical timings for DSI I/F

Himax Confidential Do Not Copy

7.3.3.2 Horizontal timings

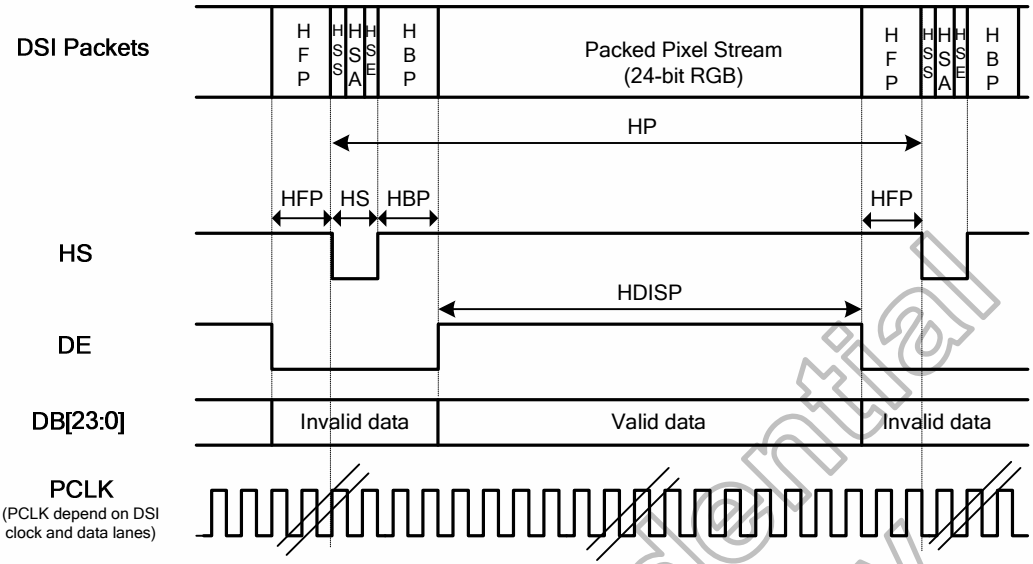


Figure 7.10: Horizontal timing for DSI video mode I/F

Resolution=720x1280 (VSSA=0V, VDD1=1.8V, VDD2=VDD3=VCC=2.8V, T_A=25℃)

Parameter	Symbol	Condition	Spec.			Unit
			Min.	Typ.	Max.	
Horizontal cycle	HP	-	762	-	-	DCK
HS low pulse width	HS	-	6	-	-	DCK
Horizontal back porch	HBP	-	24	-	-	DCK
Horizontal front porch	HFP	-	12	-	-	DCK
Horizontal data start point	-	HS+HBP	30	-	-	DCK
Horizontal blanking period	HBLK	HS+HBP+HFP	12	-	-	DCK
Horizontal active area	HDISP	-	-	720	-	DCK

Note: (1) HS > 0.2us (12-DCK @ 352.8Mbps /4-lane, Frame rate 60Hz, min DSI CLK rate)
(20-DCK @ 550Mbps /4-lane, Frame rate 60Hz, max DSI CLK rate)
(2) HBP > 0.4us (24-DCK @ 352.8Mbps /4-lane, Frame rate 60Hz, min DSI CLK rate)
(40-DCK @ 550Mbps /4-lane, Frame rate 60Hz, max DSI CLK rate)
(3) HFP > 0.2us (12-DCK @ 352.8Mbps /4-lane, Frame rate 60Hz, min DSI CLK rate)
(20-DCK @ 550Mbps /4-lane, Frame rate 60Hz, max DSI CLK rate)

Table 7.14: Horizontal timings for DSI video mode I/F

Himax Confidential
Do Not Copy

8. Ordering Information

Part no.	Package
HX8394-F PDxxx	PD: mean COG xxx: mean chip thickness **(μm), (default: 250 μm)**

Himax Confidential
Do Not Copy